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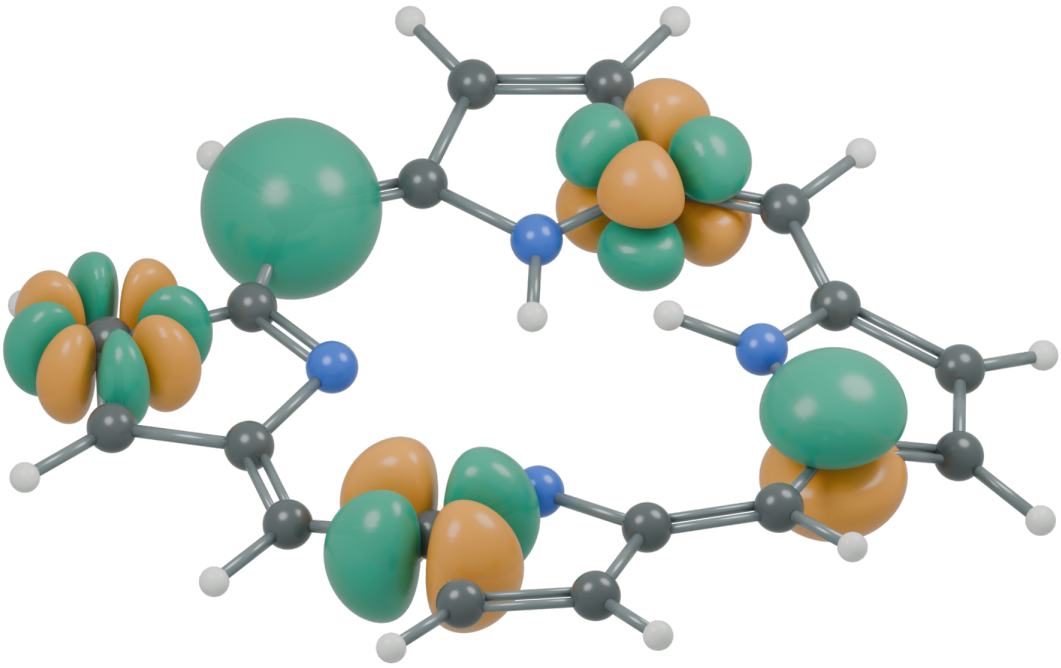
Many Electrons, One Determinant

Hartree–Fock Theory, Basis Sets, and Molecular Integrals



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Abstract | ESQC 2026

This chapter accompanies the lectures on Hartree–Fock theory and on basis sets and molecular integrals at the European Summer School of Quantum Chemistry (ESQC) 2026. Intended for doctoral students, it develops the single-determinant description of electronic structure from antisymmetry and the variational principle to the orbital equations and their finite-basis matrix representation. Detailed derivations connect the energy functional, the Fock operator, excited determinants, and the interpretation of orbital energies. Restricted, unrestricted, and generalized formulations show how the choice of variational space affects spin symmetry, symmetry breaking, and the limitations of the Hartree–Fock approximation.

The computational development covers orthogonalization, self-consistent-field iteration, convergence acceleration, overlap-based occupation selection, and orbital analysis, distinguishing orbital stationarity from stability. A connection to second-order Møller–Plesset perturbation theory explains the correlation-energy expression and the cost of transforming electron-repulsion integrals from atomic to molecular orbitals.

The basis-set and integral development examines radial and angular flexibility, contraction, basis-set incompleteness, and superposition error. The Gaussian product theorem leads to one- and two-electron integrals, recurrence relations, integral screening, and low-rank representations. Worked examples and exercises with detailed solutions connect these derivations to practical calculations. Historical perspectives and explicit notation conventions provide context for the approximations and algorithms that underpin molecular electronic-structure theory.

Contents

Historical perspectives **iii**

Notation and conventions **iv**

1 Electronic structure and the Hartree–Fock framework **1**
1.1 The electronic many-body problem . . . 3
1.2 The Hartree–Fock energy functional . . . 15
1.3 Spin-orbital and spatial-orbital formulations . . . 27
1.4 Excited determinants and Hamiltonian matrix elements . . . 32
1.5 Slater–Condon rules and Brillouin's theorem . . . 36
1.6 Spin symmetry and spin-adapted determinants . . . 41
1.7 The Fock operator and orbital energies . . . 44

2 Finite-basis Hartree–Fock and self-consistent-field methods **63**
2.1 The Roothaan–Hall equations . . . 63
2.2 The SCF algorithm and convergence . . . 76
2.3 Direct inversion in the iterative subspace (DIIS) . . . 81
2.4 Excited-state SCF and the maximum-overlap method . . . 84
2.5 Molecular properties and orbital analysis . . . 86

3 Open-shell and generalized Hartree–Fock methods **90**
3.1 Unrestricted Hartree–Fock theory . . . 90
3.2 Symmetry breaking in the dissociation of H_2 . . . 94
3.3 Generalized Hartree–Fock theory . . . 99
3.4 Classification of Hartree–Fock solutions . . . 102

4 One-electron basis sets **104**
4.1 Basis-set expansions and design principles . . . 104
4.2 Plane-wave basis sets . . . 105
4.3 Slater-type and Gaussian-type orbitals . . . 107
4.4 Gaussian basis-set design and selection . . . 109
4.5 Effective core potentials and pseudopotentials . . . 114

5 Molecular integrals over Gaussian functions **115**
5.1 The Gaussian product theorem and overlap integrals . . . 115
5.2 One-electron Gaussian integrals . . . 118
5.3 Four-center electron-repulsion integrals . . . 119
5.4 Integral bounds, screening, and direct SCF . . . 122

6 Resolution of the identity and low-rank ERI representations **125**
6.1 Fitting pair densities in the Coulomb metric . . . 125
6.2 RI, Cholesky decomposition, and practical contractions . . . 126

7 Concluding perspective **127**

Historical perspectives

Douglas Rayner Hartree (1897–1958) . . . 1
William Hartree (1870–1943) . . . 1
Vladimir Aleksandrovich Fock (1898–1974) . . . 2
Bertha Swirles, Lady Jeffreys (1903–1999) . . . 3
Born and Oppenheimer . . . 6
Wolfgang Pauli (1900–1958) . . . 9
John Clarke Slater (1900–1976) . . . 10
Edward Uhler Condon (1902–1974) . . . 36
Léon Nicolas Brillouin (1889–1969) . . . 40
Paul Adrien Maurice Dirac (1902–1984) . . . 49
Tjalling Charles Koopmans (1910–1985) . . . 54
Møller and Plesset . . . 60
Clemens C. J. Roothaan (1918–2019) . . . 63
George Garfield Hall (1925–2018) . . . 64
Per-Olov Löwdin (1916–2000) . . . 68
Roy McWeeny (1924–2021) . . . 69
Péter Pulay (b. 1941) . . . 81
Robert Sanderson Mulliken (1896–1986) . . . 87
Pople and Nesbet . . . 93
Gaston Berthier (1923–2009) . . . 93
Coulson and Fischer-Hjalmars . . . 97
Josef Paldus (1935–2023) . . . 102
Samuel Francis Boys (1911–1972) . . . 107
Sigeru Huzinaga . . . 108
Thom H. Dunning Jr. (b. 1943) . . . 111
Fernando Bernardi . . . 113

Notation and conventions

Table 1 summarizes the chapter's symbols and acronyms. Atomic units are used unless stated otherwise. Read each orbital sum with its spin-orbital or spatial-orbital convention, and each density matrix with its occupation normalization; local reuses are identified by topic. Acronyms and abbreviations are collected alphabetically at the end of the table.

Table 1: Symbols and acronyms, with formulation-specific local conventions.

Symbol or acronym	Meaning and convention
Particles, coordinates, and orbital indices	
N, K	Electron number; spatial basis size ($2K$ spin orbitals).
N^α, N^β	Collinear spin populations, with $N^\alpha + N^\beta = N$. Superscript spin labels are used throughout; N^σ denotes either population for $\sigma = \alpha, \beta$.
O, V	Closed-shell MP2 spatial counts: $O = N/2$ occupied and $V = K - O$ virtual orbitals.
$\boldsymbol{r}$, $\boldsymbol{x}$, ω	Spatial coordinate, combined coordinate $\boldsymbol{x} = (\omega, \boldsymbol{r})$, and spin coordinate. Integration over $\boldsymbol{x}$ includes the spin degree of freedom.
A, B, Z_A, $\boldsymbol{R}_A$	Nuclear labels, charges, and positions; M counts nuclei.
ϕ_μ, ψ_p, χ_p	Spatial AO, spatial MO, and spin orbital. χ_μ^{AO} is a spin-AO basis function in the general AO exercises.
$\bar{\psi}_p$	Shorthand for the β spin partner, not conjugation (denoted by a star).
$\mu, \nu, \lambda, \sigma$	AO indices, normally running over K spatial functions. In the general spin-orbital AO Fock/energy exercises, the basis label temporarily includes spin.
p, q, r, s	General spatial- or spin-orbital indices, as specified locally.
i, j, k, l; a, b, c, d	Occupied and virtual indices, relative to the chosen reference. Spin-orbital sums contain N occupied functions; closed-shell spatial sums contain $N/2$.
α, β; S, M_S	The two spin functions; total-spin and spin-projection quantum numbers. $\hat{S}^2$ and $\hat{S}_z$ are the associated operators.
Operators, states, and energies	
$\hat{\mathcal{H}}_{\mathrm{elec}}$, $\hat{h}$, $\hat{f}$	Electronic Hamiltonian, one-electron core Hamiltonian, and density-dependent Fock operator.
$\hat{\mathcal{J}}_i$, $\hat{\mathcal{K}}_i$	Coulomb and exchange operators generated by occupied orbital i. Their spatial versions follow the spin integration.
Ψ_{HF}, Ψ_0	HF determinant; the same determinant when used as the reference for substitutions or perturbation theory.
$\hat{\mathcal{A}}$, $\hat{\mathcal{P}}_\pi$	Antisymmetrizer weighted by $1/\sqrt{N!}$ and electron permutation operator. This $\hat{\mathcal{A}}$ is not an idempotent $1/N!$-weighted projector; Eq. (1.37).
Ψ_i^a, Ψ_{ij}^{ab}	Singly and doubly substituted determinants. Their phases follow the stated ordering of creation and annihilation operators, $\hat{a}_p^\dagger$ and $\hat{a}_p$.
ϵ_{ji}, $\boldsymbol{\epsilon}$	Hermitian occupied-space orthonormality multipliers before canonicalization; diagonalized by an occupied-space unitary transformation.
ϵ_p, $\boldsymbol{\epsilon}$	Canonical orbital energy and diagonal matrix of orbital energies. The matrix spans K spatial orbitals in RHF, or $2K$ spinors in GHF; $\boldsymbol{\epsilon}^\sigma$ denotes one UHF spin sector. $\boldsymbol{\epsilon}_{\mathrm{o}}$ and $\boldsymbol{\epsilon}_{\mathrm{v}}$ denote occupied and virtual blocks.
E_{HF}, U_I	Electronic HF energy; PES including nuclear repulsion. H_2 formulas specify $1/R$. E_{H} means the Hartree contribution in the stationarity exercise or atomic H energy in the dissociation example.
IP, EA	Removal energy $E^{N-1} - E^N$; attachment energy $E^N - E^{N+1}$. Koopmans comparisons fix geometry and basis.

Continued on the next page

Table 1: Symbols, acronyms, and local conventions (continued).

Symbol or acronym	Meaning and convention
$\widetilde{E}$, E^M_{corr}	A tilde on an ionic energy denotes frozen orbitals in the Koopmans discussion; E^M_{corr} is the exact minus optimized HF energy for electron number M.
$\hat{\mathcal{H}}^{(0)}$, $\hat{\mathcal{V}}$	MP zeroth-order Hamiltonian $\sum_i \hat{f}(i)$ and perturbation; Eq. (1.367).
$E^{(2)}_{\text{c}}$, Δ^{ab}_{ij}	MP2 correlation energy and denominator $\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b$, negative for the gapped Aufbau reference.
Integral brackets and complex conjugation	
$\langle \chi_i \vert \chi_j \rangle$	Ordinary one-particle overlap; orthonormal spin orbitals give δ_{ij}.
$\langle ij \vert kl \rangle$	Physicists' Coulomb integral: electron 1 carries i, k and electron 2 carries j, l; Eq. (1.92).
$(ij \mid kl)$	Chemists' Coulomb integral: electron 1 carries i, j and electron 2 carries k, l. Thus $\langle ij \vert kl \rangle = (ik \mid jl)$; Eq. (1.100).
$\langle ij \vert\vert kl \rangle$	Antisymmetrized integral $\langle ij \vert kl \rangle - \langle ij \vert lk \rangle$; Eq. (1.95).
h_{ij}, $\mathcal{J}_{ij}$, $\mathcal{K}_{ij}$	Spin-orbital core, Coulomb, and exchange matrix elements. Spatial versions use the indicated spatial orbitals and spin-summation factors.
$*$, $\dagger$, $\cdot$	Complex conjugation, Hermitian adjoint, and explicit matrix multiplication. Bra arguments are conjugated on forming the bra. Eightfold ERI symmetry requires real orbitals.
AO matrices and density normalizations	
$C_{\mu p}$, $\boldsymbol{C}$, $\boldsymbol{C}_{\text{o}}$	MO coefficient, coefficient matrix, and its occupied columns. The spatial expansion is $\psi_p = \sum_\mu C_{\mu p} \phi_\mu$.
$\boldsymbol{S}$, $\boldsymbol{X}$	AO overlap metric and orthogonalizer, satisfying $\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1}$ in the retained space.
$\boldsymbol{H}$, $\boldsymbol{F}$, $\boldsymbol{G}[\boldsymbol{P}]$	AO core-Hamiltonian, Fock, and two-electron matrices, with $\boldsymbol{F} = \boldsymbol{H} + \boldsymbol{G}[\boldsymbol{P}]$, including in GHF.
f_{pq}, $\boldsymbol{f}$	Fock matrix elements and blocks in the MO basis; distinct from the AO representation $F_{\mu\nu}$ and $\boldsymbol{F}$.
$\boldsymbol{J}$, $\boldsymbol{K}$	Coulomb and exchange matrices, with local density/spin prefactors; distinct from orbital operators $\hat{\mathcal{J}}_i$ and $\hat{\mathcal{K}}_i$.
$\boldsymbol{P}$ (RHF)	Spin-summed spatial density, $\boldsymbol{P} = 2\boldsymbol{C}_{\text{o}} \cdot \boldsymbol{C}_{\text{o}}^\dagger$ and $\text{Tr}(\boldsymbol{P} \cdot \boldsymbol{S}) = N$. A spin-orbital density counts each occupied spin orbital once instead.
$\boldsymbol{P}^\alpha$, $\boldsymbol{P}^\beta$ (UHF)	Separate single-spin densities, $\text{Tr}(\boldsymbol{P}^\sigma \cdot \boldsymbol{S}) = N^\sigma$ for $\sigma = \alpha, \beta$.
$\boldsymbol{P}^{\text{T}}$, $\boldsymbol{P}^{\text{S}}$ (UHF)	Charge and collinear spin-density matrices: the sum and difference of $\boldsymbol{P}^\alpha$ and $\boldsymbol{P}^\beta$.
$\boldsymbol{\mathcal{P}}$ (GHF)	Full $2K \times 2K$ spinor density, with blocks $P^{\sigma\sigma'}_{\mu\nu} = \sum_i C^\sigma_{\mu i} C^{\sigma' *}_{\nu i}$. Spin is the outer block index.
$\boldsymbol{P}$, $\boldsymbol{M}_x$, $\boldsymbol{M}_y$, $\boldsymbol{M}_z$ (GHF)	In the Pauli decomposition, $\boldsymbol{P} = \frac{1}{2}(\boldsymbol{P}^{\alpha\alpha} + \boldsymbol{P}^{\beta\beta})$ is *half* the charge-density matrix; $\boldsymbol{M}_k$ are the magnetization blocks. This $\boldsymbol{P}$ is not the spin-summed RHF/UHF density.
$\boldsymbol{C}$, $\boldsymbol{\mathcal{S}}$	Stacked spinor coefficients and block-diagonal overlap metric in GHF; $\text{Tr}(\boldsymbol{\mathcal{P}} \cdot \boldsymbol{\mathcal{S}}) = N$.
$\rho(\boldsymbol{r})$, $\rho^\sigma(\boldsymbol{r})$	Real-space total and single-spin electron densities, integrating to N and N^σ, respectively.
Iteration, orbital transformations, and numerical parameters	
$\boldsymbol{U}$, κ_{ai}	Unitary orbital transformation and infinitesimal occupied-virtual rotation parameter. Occupied-only rotations preserve the determinant's physical state, up to a phase.
$\boldsymbol{e}^{(n)}$, τ_{SCF}	Commutator residual at SCF iteration n and its convergence threshold. A zero residual establishes stationarity, not stability.
m, c_i, B_{ij} (DIIS)	History size, real weights ($\sum_i c_i = 1$), and residual inner products. Here i, j label stored iterations, not occupied orbitals.

Continued on the next page

Table 1: Symbols, acronyms, and local conventions (continued).

Symbol or acronym	Meaning and convention
w_p (MOM)	Overlap weight used to select the occupied space relative to the chosen reference. The MOM matrix $\boldsymbol{O}$ contains reference-to-new overlaps.
I, J (AO-to-MO)	Temporary four-index arrays in the successive one-index transformations; not identity or Coulomb operators.
Basis functions and Gaussian integrals	
$\boldsymbol{k}$, Ω, E_{cut}	Plane-wave vector, simulation-cell volume, and kinetic-energy cutoff.
$\alpha, \beta, \gamma, \delta$	Positive primitive Gaussian exponents in the integral sections, rather than spin labels. The combined exponents are $\zeta = \alpha + \beta$ and $\eta = \gamma + \delta$.
ζ, n, ℓ, m (STO/GTO)	In the radial/angular comparison, ζ is an STO exponent, n labels the radial form, and ℓ, m are angular-momentum labels. This use of ζ precedes its Gaussian-pair use.
$\boldsymbol{a} = (a_x, a_y, a_z)$	Cartesian polynomial powers, with total angular momentum $a = a_x + a_y + a_z$; not a virtual orbital. The same convention applies to $\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}$.
$\lvert\boldsymbol{a}]$, $\lvert\boldsymbol{a}\rangle$	Unnormalized primitive and contracted Cartesian Gaussian. Square and round integral brackets distinguish primitive and contracted Gaussian integrals in this context.
D_k, K_a	Contraction coefficient and number of primitives in contraction a. Primitive normalization factors are absorbed consistently; the local upper limit K in a contraction also counts primitives.
$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$	Gaussian centers, not coefficient matrices or Cartesian angular-momentum vectors.
$\boldsymbol{P}$, $\boldsymbol{Q}$, $K_{\zeta,\boldsymbol{P}}$	Gaussian product centers and a separation-dependent product prefactor. Here $\boldsymbol{P}$ is a position, not a density matrix.
ϑ, T, $F_m(T)$	Reduced pair exponent $\vartheta = \zeta\eta/(\zeta+\eta)$, dimensionless Boys argument $T = \vartheta\lvert\boldsymbol{P} - \boldsymbol{Q}\rvert^2$, and Boys function of order m; Section 5.3.3.
$\mathbf{0}$, $\mathbf{1}_i$ (recurrences)	Zero angular-momentum vector (an s function) and Cartesian unit vector. Here i labels x, y, z. In block matrices $\mathbf{0}$ denotes zero; $\mathbf{1}$ denotes an identity matrix.
Q_{ab}, τ, K_{sig}	Shell-pair screening bound, integral threshold, and number of retained significant integrals; Section 5.4.
Q_A, N_A^{core}	ECP ionic-core charge and removed core-electron count, with $Q_A = Z_A - N_A^{\text{core}}$; not a population-analysis charge q_A.
Auxiliary functions, fitting, and other local reuses	
φ_A^{aux}, K_{aux}	Auxiliary fitting function and auxiliary-basis size in RI/DF. Here A, B are auxiliary-function indices, not nuclear labels.
$\rho_{\mu\nu}$, $C_{\mu\nu}^A$	AO pair density and its auxiliary expansion coefficients in the real-function RI derivation; the latter are not MO coefficients $C_{\mu p}$.
V_{AB}, $\boldsymbol{V}$	Auxiliary Coulomb metric $V_{AB} = (A \mid B)$ in RI. In the Cholesky discussion $\boldsymbol{V}$ instead denotes the full ERI matrix in compound AO-pair indices.
$B_{\mu\nu}^P$, $L_{\mu\nu}^P$	Three-index RI and Cholesky factors. Here P labels an auxiliary direction, not a density matrix or Gaussian center.
N_{CD}, τ_{CD}	Number of Cholesky factors and decomposition threshold; Section 6.
S, $\boldsymbol{S}$	Scalar atomic overlap in the H_2 example versus AO overlap matrix; the scalar S elsewhere denotes total spin.
K, $\boldsymbol{K}$, $\hat{\mathcal{K}}$	Basis size, exchange matrix, and complex-conjugation operator in the HF classification table. These are distinct local uses.
Ω_{ER}, Ω_{BF}, Ω_{PM}	Localization criteria, distinct from the plane-wave volume Ω; their minimization or maximization convention is given with each functional.
Acronyms and abbreviations	
AO	Atomic orbital.
ASCW	Axial spin current wave.

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Table 1: Symbols, acronyms, and local conventions (continued).

Symbol or acronym	Meaning and convention
ASDW	Axial spin density wave.
ASW	Axial spin wave.
BF	Foster–Boys localization, labeled BF in Ω_{BF}.
BSSE	Basis-set superposition error.
CBS	Complete basis set.
CCW	Charge current wave.
CD	Cholesky decomposition.
CF	Coulson–Fischer.
CP	Counterpoise correction.
DF	Density fitting; also called RI.
DIIS	Direct inversion in the iterative subspace.
EA	Electron affinity.
ECP	Effective core potential.
ER	Edmiston–Ruedenberg localization.
ERI	Electron-repulsion integral.
ESQC	European Summer School of Quantum Chemistry.
GHF	Generalized Hartree–Fock.
GTO	Gaussian-type orbital.
HF	Hartree–Fock.
HFB	Hartree–Fock–Bogoliubov.
HOMO	Highest occupied molecular orbital.
HRR	Horizontal recurrence relation.
IMOM	Initial maximum-overlap method, with a fixed initial reference.
IP	Ionization potential.
LCAO	Linear combination of atomic orbitals.
LUMO	Lowest unoccupied molecular orbital.
MO	Molecular orbital.
MOM	Maximum-overlap method, with the previous iteration as reference.
MP	Møller–Plesset perturbation theory.
MP2	Second-order Møller–Plesset perturbation theory.
NAO	Numerical atomic orbital.
PES	Potential-energy surface.
PM	Pipek–Mezey localization.
PW	Plane wave.
RHF	Restricted Hartree–Fock.
RI	Resolution of the identity; also called DF.
ROHF	Restricted open-shell Hartree–Fock.
SAD	Superposition of atomic densities.
SCF	Self-consistent field.
STO	Slater-type orbital.
TICS	Time-reversal-invariant closed-shell.
TSCW	Torsional spin current wave.
TSDW	Torsional spin density wave.
TSW	Torsional spin wave.
UHF	Unrestricted Hartree–Fock.
VRR	Vertical recurrence relation.

1 Electronic structure and the Hartree–Fock framework

Predicting electronic structure from first principles is one of the central problems of quantum chemistry. Once the electronic structure of a molecular system is known, one can in principle determine its energy and rationalize a broad range of chemical phenomena, from bonding and molecular structure to reactivity and spectroscopy. At the non-relativistic level, the fundamental equations governing this problem are known. The difficulty lies not in writing down the Hamiltonian, but in solving the resulting many-electron Schrödinger equation.

The origin of this difficulty is the electron-electron interaction. Because the Coulomb repulsion couples the motion of all electrons, the electronic wavefunction depends simultaneously on all their coordinates and cannot, in general, be decomposed into independent one-electron contributions. The problem is therefore intrinsically many-body. Exact solutions are restricted to exceptionally small systems, and practical electronic structure theory must rely on approximations that retain the essential physics while reducing the complexity of the many-electron problem.

Hartree–Fock (HF) theory provides the most fundamental such reduction. Its central idea is to replace the exact many-electron wavefunction by the best wavefunction that can be written as a single Slater determinant. This restriction transforms the interacting many-electron problem into an effective independent-particle picture: each electron moves in a field generated by the nuclei and by the average distribution of the other electrons (see Fig. 1). Importantly, the antisymmetry of the electronic wavefunction is retained exactly within this approximation. The Pauli principle and the associated exchange interaction therefore emerge naturally, rather than being introduced as additional corrections [1–3].

Historically, Hartree's self-consistent-field construction dates from 1928. The incorporation of exchange through antisymmetric wavefunctions was developed independently by Fock and Slater in 1930 [3, 4].

Historical perspective: Douglas Rayner Hartree (1897–1958)

An English mathematician and physicist, Hartree worked at Manchester before returning to Cambridge as Plummer Professor of Mathematical Physics in 1946. His 1928 self-consistent-field method made atomic structure accessible through iterative numerical solutions of one-electron equations [1]. He also helped develop mechanical and electronic computing: for Hartree, theoretical approximations and the means of actually solving their equations were inseparable concerns.

Biography: University of St Andrews, MacTutor. Source portrait: unknown photographer, c. 1940; Centre for Computing History / Wikimedia Commons, listed as public domain. Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: William Hartree (1870–1943)

An English engineer and numerical researcher, William Hartree taught at Cambridge and later collaborated with the physiologist A. V. Hill. He was Douglas Hartree's father and an important collaborator in turning atomic self-consistent-field theory into numerical results. Their joint 1935 beryllium study included exchange explicitly [5]. His contribution illustrates a less visible part of the method's history: obtaining accurate atomic orbitals required sustained numerical work as well as the formulation of the equations.

Biography: A. V. Hill, Nature obituary, 1943; MacTutor Hartree biography. Source photograph: International Slide Rule Museum archive, late 1930s; photographer unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: Vladimir Aleksandrovich Fock (1898–1974)

Fock was a Russian theoretical physicist whose career was closely associated with the university in his native St Petersburg, later called Leningrad. In 1930 he incorporated exchange into the self-consistent-field equations by using an antisymmetric many-electron wavefunction [3]. His work ranged from wave mechanics and gravitation to quantum field theory; the Fock-space construction carries his name as well.

Biography: International Academy of Quantum Molecular Science (IAQMS) and MacTutor. Source portrait: IAQMS archive; photographer and date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

Figure 1: Illustration of the mean-field approximation using traffic on a roundabout. In the left panel, a short-exposure snapshot resolves the instantaneous positions of the individual cars, representing the fully interacting many-particle picture in which the configuration of all particles is explicitly retained. In the right panel, a long-exposure image of the same roundabout replaces the individual cars by smooth light trails, representing the mean-field description: the detailed instantaneous motion of the other particles is no longer followed explicitly, and each particle instead experiences an effective field generated by their average distribution. The comparison emphasizes that the mean field is an averaged environment rather than a blurred or uncertain version of the exact configuration.

The price of this simplification is equally fundamental. A single determinant cannot describe the full correlated motion of interacting electrons. Hartree–Fock theory accounts for exchange exactly within its single-determinant framework, but neglects electron correlation beyond exchange. Nevertheless, Hartree–Fock occupies a unique position in electronic structure theory. It provides a remarkably useful orbital picture of atoms and molecules, establishes the language of occupied and virtual molecular orbitals, and serves as the reference state for a large fraction of more advanced quantum-chemical methods. Understanding both its strengths and its limitations is therefore essential before moving to correlated theories.

There is, however, a second layer to the problem. The Hartree–Fock equations are formulated in terms of continuous one-electron functions, whereas an actual calculation requires a finite numerical representation of these orbitals. In molecular quantum chemistry, this is usually achieved by expanding the molecular orbitals in a finite set of basis functions. The choice of basis determines the flexibility with which the orbitals can be represented and has a direct impact on both accuracy and computational cost. It also reduces the continuous Hartree–Fock equations to a matrix problem whose elements are one- and two-electron molecular integrals. Basis sets and

molecular integrals are therefore not peripheral technical details: they are an integral part of the practical formulation of electronic structure theory.

This chapter develops these ideas from the ground up. We first formulate the electronic many-body problem and introduce the approximations that separate electronic and nuclear motion. We then impose the fermionic antisymmetry of the wavefunction, introduce Slater determinants, and derive the Hartree–Fock energy and equations in Section 1. Section 2 develops their finite-basis formulation, the self-consistent-field procedure, and the role of orbital rotations. Section 3 then examines unrestricted and generalized Hartree–Fock theory alongside the restricted description. We subsequently turn to the basis functions themselves and examine the principal representations used in practice (Section 4). Finally, Sections 5 and 6 develop the molecular integrals and low-rank representations required to construct the Hartree–Fock matrices.

Historical perspective: Bertha Swirles, Lady Jeffreys (1903–1999)

A British mathematician trained at Girton College, Cambridge, Swirles worked on quantum theory before returning to Girton as a Fellow in 1938. Her 1935 relativistic self-consistent-field treatment extended the atomic orbital framework beyond the non-relativistic approximation [6]. With Douglas and William Hartree, she also studied exchange and the superposition of configurations in a 1939 paper on oxygen [7]. These were distinct extensions of the ordinary single-determinant theory developed here. Her subsequent teaching and the book *Methods of Mathematical Physics*, written with Harold Jeffreys, influenced generations of mathematical physicists.

Biography: University of St Andrews, MacTutor. Source portrait: Dr John Davies, Girton College, Celebrating Our Benefactors booklet, 2017 entry; photograph date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

1.1 The electronic many-body problem

We begin by defining the framework in which Hartree–Fock theory will be developed. We consider the time-independent Schrödinger equation and seek stationary states of molecular systems. Unless stated otherwise, the treatment is non-relativistic and atomic units are employed. The separation between nuclear and electronic motion will be introduced through the Born–Oppenheimer approximation in the next section.

Within this framework, Hartree–Fock is an ab initio theory. Once the nuclear charges, nuclear positions, and number of electrons have been specified, the electronic Hamiltonian contains no parameters adjusted to experimental data. In practical calculations, the molecular orbitals are represented in a finite basis, and the quality of this representation becomes an additional approximation that will be examined in detail later in this chapter.

Within this electronic problem, the Hartree–Fock approximation restricts the wavefunction of N electrons to a single Slater determinant and optimizes its orbitals variationally. It does not turn off the electron-electron interaction: Coulomb repulsion remains present, together with the exchange contribution required by antisymmetry. Each electron experiences the nuclear attraction and the mean field generated by the occupied orbitals. Because that field depends on the orbitals being sought, the resulting one-electron equations are nonlinear and must be solved self-consistently. The optimal determinant is therefore a variational independent-particle description, not an exact correlated solution.

To make this variational construction precise, we first need a molecular Hamiltonian with the nuclei treated as fixed parameters. Antisymmetry then determines the allowed determinant form, from which the energy functional and orbital equations follow. The finite-basis and numerical formulations will retain these assumptions while changing how the orbitals are represented.

1.1.1 The Born–Oppenheimer approximation

A molecule is a quantum system composed of both electrons and nuclei. Before introducing any electronic structure approximation, the electronic and nuclear degrees of freedom are therefore treated on equal footing [8]. For a molecule containing N electrons and M nuclei, the molecular wavefunction depends on the electronic coordinates $\{\boldsymbol{r}_i\}$ and nuclear coordinates $\{\boldsymbol{R}_A\}$, and satisfies the time-independent Schrödinger equation

$$\hat{\mathcal{H}}\Phi(\{\boldsymbol{r}_i\},\{\boldsymbol{R}_A\}) = \mathcal{E}\Phi(\{\boldsymbol{r}_i\},\{\boldsymbol{R}_A\}) \tag{1.1}$$

The full non-relativistic molecular Hamiltonian contains five contributions,

$$\hat{\mathcal{H}} = \hat{\mathcal{T}}_\mathrm{n} + \hat{\mathcal{T}}_\mathrm{e} + \hat{\mathcal{V}}_\mathrm{ne} + \hat{\mathcal{V}}_\mathrm{ee} + \hat{\mathcal{V}}_\mathrm{nn} \tag{1.2}$$

Here, $\hat{\mathcal{T}}_\mathrm{n}$ and $\hat{\mathcal{T}}_\mathrm{e}$ are the nuclear and electronic kinetic-energy operators, respectively. The operator $\hat{\mathcal{V}}_\mathrm{ne}$ describes the Coulomb attraction between nuclei and electrons, while $\hat{\mathcal{V}}_\mathrm{ee}$ and $\hat{\mathcal{V}}_\mathrm{nn}$ describe the electron-electron and nucleus-nucleus Coulomb repulsions.

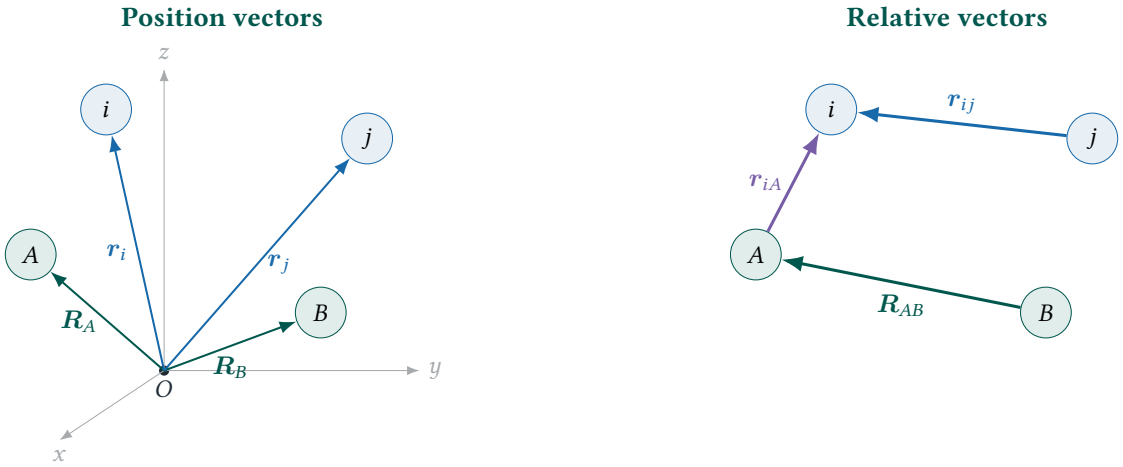


Figure 2: Electronic and nuclear coordinates entering the molecular Hamiltonian. Green circles denote nuclei A and B, and blue circles denote electrons i and j. In the left panel, O is the common origin: $\boldsymbol{R}_A$ and $\boldsymbol{R}_B$ are the nuclear position vectors, while $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ are the electronic position vectors. In the right panel, $\boldsymbol{r}_{iA} = \boldsymbol{r}_i - \boldsymbol{R}_A$ points from nucleus A to electron i, $\boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j$ points from electron j to electron i, and $\boldsymbol{R}_{AB} = \boldsymbol{R}_A - \boldsymbol{R}_B$ points from nucleus B to nucleus A. Their lengths $r_{iA} = \|\boldsymbol{r}_{iA}\|$, $r_{ij} = \|\boldsymbol{r}_{ij}\|$, and $R_{AB} = \|\boldsymbol{R}_{AB}\|$ are the corresponding electron-nucleus, electron-electron, and internuclear distances. Both panels use the same molecular geometry.

In atomic units, for which $m = e = \hbar = 1$ with m the electron mass, these operators read

$$\hat{\mathcal{T}}_\mathrm{n} = -\sum_{A=1}^{M} \frac{\nabla_A^2}{2M_A} \qquad \hat{\mathcal{T}}_\mathrm{e} = -\sum_{i=1}^{N} \frac{\nabla_i^2}{2} \tag{1.3a}$$

$$\hat{\mathcal{V}}_\mathrm{ne} = -\sum_{A=1}^{M}\sum_{i=1}^{N} \frac{Z_A}{r_{iA}} \qquad \hat{\mathcal{V}}_\mathrm{ee} = \sum_{i<j}^{N} \frac{1}{r_{ij}} \qquad \hat{\mathcal{V}}_\mathrm{nn} = \sum_{A<B}^{M} \frac{Z_A Z_B}{R_{AB}} \tag{1.3b}$$

The quantities M_A and Z_A are the mass and charge of nucleus A, r_{iA} is the distance between electron i and nucleus A, r_{ij} is the distance between electrons i and j, and R_{AB} is the distance between nuclei A and B (see Fig. 2).

Equation (1.2) defines a formidable quantum many-body problem. Even if the electron-electron interaction could somehow be treated exactly, the electronic and nuclear motions would remain coupled. The key observation behind the Born–Oppenheimer approximation is the large separation of mass scales between electrons and nuclei. In atomic units, the electron mass is unity whereas even the lightest nucleus is more than three orders of magnitude heavier. Correspondingly, the nuclear kinetic-energy operator contains the small factors $1/M_A$. Nuclear motion is therefore typically much slower than electronic motion.

This separation of time scales suggests a natural hierarchy. On the electronic time scale, the nuclei can be regarded as approximately fixed, and the electrons rapidly adjust to a given nuclear configuration. The first step is consequently to solve a clamped-nuclei electronic problem. For fixed nuclear positions $\{\boldsymbol{R}_A\}$, one defines the electronic Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{elec}} = \hat{\mathcal{T}}_{\mathrm{e}} + \hat{\mathcal{V}}_{\mathrm{ne}} + \hat{\mathcal{V}}_{\mathrm{ee}} \tag{1.4}$$

and solves

$$\hat{\mathcal{H}}_{\mathrm{elec}}\Phi^{(I)}_{\mathrm{elec}}(\{\boldsymbol{r}_i\};\{\boldsymbol{R}_A\}) = \mathcal{E}^{(I)}_{\mathrm{elec}}(\{\boldsymbol{R}_A\})\Phi^{(I)}_{\mathrm{elec}}(\{\boldsymbol{r}_i\};\{\boldsymbol{R}_A\}) \tag{1.5}$$

Here I labels an electronic state. The semicolon emphasizes the different roles played by electronic and nuclear coordinates: the electronic coordinates are genuine variables of $\Phi^{(I)}_{\mathrm{elec}}$, whereas the nuclear positions enter parametrically.

The nuclear repulsion $\hat{\mathcal{V}}_{\mathrm{nn}}$ does not appear in Eq. (1.4) because, for fixed nuclei, it is simply a constant with respect to the electronic coordinates. Once the electronic problem has been solved, it can be added to the electronic energy to define the energy surface

$$U_I(\{\boldsymbol{R}_A\}) = \mathcal{E}^{(I)}_{\mathrm{elec}}(\{\boldsymbol{R}_A\}) + \sum_{A<B}^{M} \frac{Z_A Z_B}{R_{AB}} \tag{1.6}$$

Repeating the electronic calculation for different nuclear geometries traces out a potential-energy surface (PES). Stable equilibrium structures correspond to local minima of this surface, while chemical reactions, molecular vibrations, and other nuclear motions can be interpreted in terms of motion over it.

It is important to distinguish this *clamped-nuclei approximation* from the Born–Oppenheimer approximation itself. The connection becomes clear by expanding the exact molecular wavefunction in the complete set of clamped-nuclei electronic states,

$$\Phi(\{\boldsymbol{r}_i\},\{\boldsymbol{R}_A\}) = \sum_I \Phi^{(I)}_{\mathrm{nucl}}(\{\boldsymbol{R}_A\})\Phi^{(I)}_{\mathrm{elec}}(\{\boldsymbol{r}_i\};\{\boldsymbol{R}_A\}) \tag{1.7}$$

This Born–Huang expansion is, in principle, exact if the electronic basis is complete. The electronic and nuclear motions remain coupled because $\Phi^{(I)}_{\mathrm{elec}}$ itself changes when the nuclei move. In particular, the nuclear kinetic-energy operator acts not only on $\Phi^{(I)}_{\mathrm{nucl}}$ but also on the parametric dependence of the electronic wavefunction on $\{\boldsymbol{R}_A\}$.

These additional terms generate couplings between different electronic states. Their leading contributions involve the derivative couplings

$$\boldsymbol{d}^A_{IJ} = \langle \Phi^{(I)}_{\mathrm{elec}} | \nabla_A | \Phi^{(J)}_{\mathrm{elec}} \rangle \tag{1.8}$$

which measure how rapidly the electronic wavefunctions change as nucleus A moves. The Born–Oppenheimer approximation consists in neglecting these nonadiabatic couplings and assuming that nuclear motion takes place on a single electronic potential-energy surface. In the nuclear equation below, we also neglect the diagonal correction arising when nuclear derivatives act on

that electronic state; the potential is therefore the clamped-nuclei energy plus nuclear repulsion, without a derivative correction.

For electronic state I, the resulting nuclear Schrödinger equation is

$$\left[\hat{\mathcal{T}}_{\mathrm{n}} + U_I(\{\boldsymbol{R}_A\})\right]\Phi^{(I)}_{\mathrm{nucl}}(\{\boldsymbol{R}_A\}) = \mathcal{E}_{\mathrm{tot}}\Phi^{(I)}_{\mathrm{nucl}}(\{\boldsymbol{R}_A\}) \tag{1.9}$$

Equivalently,

$$\hat{\mathcal{H}}_{\mathrm{nucl}} = \hat{\mathcal{T}}_{\mathrm{n}} + \hat{\mathcal{V}}_{\mathrm{nn}} + \mathcal{E}^{(I)}_{\mathrm{elec}}(\{\boldsymbol{R}_A\}) \tag{1.10}$$

Thus, within the Born–Oppenheimer picture, the electrons determine the potential on which the nuclei move.

For a single isolated electronic state, the molecular wavefunction is then approximated by the product

$$\Phi(\{\boldsymbol{r}_i\},\{\boldsymbol{R}_A\}) \approx \Phi_{\mathrm{nucl}}(\{\boldsymbol{R}_A\})\Phi_{\mathrm{elec}}(\{\boldsymbol{r}_i\};\{\boldsymbol{R}_A\}) \tag{1.11}$$

This familiar factorized form should therefore be viewed as the *consequence* of neglecting couplings between electronic states rather than as the starting point of the derivation.

The Born–Oppenheimer approximation is remarkably successful for most ground-state molecular problems because the electronic wavefunction usually changes smoothly with the nuclear coordinates and neighboring electronic states are sufficiently well separated in energy. It becomes less reliable when two or more electronic states approach one another. Near avoided crossings, conical intersections, or other regions of strong electronic near-degeneracy, the derivative couplings in Eq. (1.8) can become large and the electronic and nuclear motions can no longer be treated independently. Such nonadiabatic effects play a central role, for example, in photochemical processes and radiationless transitions.

Historical perspective: Born and Oppenheimer

Max Born (1882–1970). A German-born physicist, Born helped establish quantum mechanics at Göttingen and later taught at Edinburgh. His statistical interpretation of the wavefunction brought him a share of the 1954 Nobel Prize in Physics. His work with Oppenheimer addressed a different foundational question: how to organize the electronic and nuclear motions of a molecule.

J. Robert Oppenheimer (1904–1967). The American theoretical physicist completed his doctorate at Göttingen in 1927, the year of their joint molecular paper [8]. He later taught at Berkeley and Caltech and directed the Institute for Advanced Study. The approximation bearing their names supplies the electronic-structure framework used here; it is distinct from the subsequent Hartree–Fock approximation.

Biographies: Nobel Foundation; Institute for Advanced Study. Source portraits: Born, unknown photographer, via Wikimedia Commons, listed as public domain; Oppenheimer, 1944, US National Archives, record 558579, crop by Artem.G, tonal restoration by MyCatIsAChonk, public domain. Sketch: AI-assisted adaptation of the credited portrait.

The remainder of this chapter is concerned with the electronic problem. Unless stated otherwise, the nuclear positions will therefore be regarded as fixed, and we will seek approximate solutions of the clamped-nuclei electronic Schrödinger equation

$$\hat{\mathcal{H}}_{\mathrm{elec}}\Phi_{\mathrm{elec}} = \mathcal{E}_{\mathrm{elec}}\Phi_{\mathrm{elec}} \tag{1.12}$$

The next question is then not how the nuclei move, but what restrictions must be imposed on Φ_{elec} because the particles it describes are indistinguishable fermions. This brings us directly to the Pauli principle and the antisymmetry of the electronic wavefunction. In the electronic-only discussions that follow, $\hat{\mathcal{H}}$ without a subscript also denotes the clamped-nuclei electronic Hamiltonian $\hat{\mathcal{H}}_{\mathrm{elec}}$, rather than the full molecular Hamiltonian.

1.1.2 The Pauli principle and antisymmetry

Having separated the electronic and nuclear motions, we now turn to a second fundamental ingredient of the electronic problem: electrons are indistinguishable fermions. This statement imposes a strict symmetry requirement on the electronic wavefunction, independently of any approximation that will later be made to solve the electronic Schrödinger equation.

The non-relativistic electronic Hamiltonian introduced above is spin independent: it contains the electronic kinetic energy and Coulomb interactions, but no explicit spin operator. Nevertheless, a complete description of an electron must contain both its spatial coordinate and its intrinsic spin degree of freedom. Electrons have spin $s = 1/2$, and their spin functions satisfy

$$|\omega\rangle = |s, m_s\rangle \qquad s^2\,|s, m_s\rangle = s(s+1)\,|s, m_s\rangle \qquad s_z\,|s, m_s\rangle = m_s\,|s, m_s\rangle \tag{1.13}$$

The two one-electron spin functions are

$$|\alpha\rangle = \left|\tfrac{1}{2}, +\tfrac{1}{2}\right\rangle \qquad\qquad |\beta\rangle = \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle \tag{1.14}$$

corresponding respectively to spin-up and spin-down electrons. They form an orthonormal basis for the one-electron spin space,

$$\int \alpha^*(\omega)\beta(\omega)\,\mathrm{d}\omega = \int \beta^*(\omega)\alpha(\omega)\,\mathrm{d}\omega = 0 \qquad \int \alpha^*(\omega)\alpha(\omega)\,\mathrm{d}\omega = \int \beta^*(\omega)\beta(\omega)\,\mathrm{d}\omega = 1 \tag{1.15}$$

$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0 \qquad\qquad \langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1 \tag{1.16}$$

It is therefore convenient to combine the spin coordinate ω and the spatial coordinate $\boldsymbol{r}$ into a single composite coordinate,

$$\boldsymbol{x} = (\omega, \boldsymbol{r}) \tag{1.17}$$

An N-electron wavefunction is consequently a function of N such coordinates,

$$\Phi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) \tag{1.18}$$

Although the Hamiltonian is spin independent in the present treatment, the spin coordinates are indispensable because the symmetry of the full spin-spatial wavefunction determines whether the state is compatible with fermionic statistics.

Indistinguishability and permutation symmetry. The labels assigned to identical particles carry no physical information. Interchanging two electron labels must therefore leave every observable quantity unchanged. For two particles, this implies in particular that the probability density satisfies

$$|\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)|^2 = |\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1)|^2 \tag{1.19}$$

To obtain an exchange rule for the wavefunction, we additionally assume that exchange leaves the pure-state ray unchanged: the exchanged state differs from the original by a single, coordinate-independent phase. Equality of the coordinate probability densities alone would allow a coordinate-dependent phase and would not establish this stronger statement. With this state-ray assumption, the phase relation is

$$|\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)|^2 = |\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1)|^2 \quad \Rightarrow \quad \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) = e^{\mathrm{i}\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \tag{1.20}$$

Introducing the permutation operator $\hat{\mathcal{P}}_{12}$, which exchanges particles 1 and 2, gives

$$\begin{aligned} \hat{\mathcal{P}}_{12}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) &= \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) = e^{\mathrm{i}\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \\ \hat{\mathcal{P}}_{12}^2\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) &= e^{2\mathrm{i}\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \end{aligned} \tag{1.21}$$

Because exchanging the same pair twice restores the original labeling, $\hat{\mathcal{P}}_{12}^2 = 1$, and consequently

$$e^{2i\theta} = 1 \quad \Rightarrow \quad \theta = n\pi \quad \Rightarrow \quad e^{i\theta} = \pm 1 \tag{1.22}$$

The wavefunction can therefore belong to either a symmetric or an antisymmetric permutation sector. The plus sign defines bosons, for which

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) \tag{1.23}$$

whereas the minus sign defines fermions,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = -\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) \tag{1.24}$$

Electrons are fermions and must therefore be described by an antisymmetric wavefunction [9]. The connection between half-integer spin and fermionic statistics is the content of the spin-statistics theorem; within non-relativistic quantum mechanics, fermionic antisymmetry is imposed as a fundamental property of the electronic wavefunction [10].

A notable exception occurs for identical particles confined to two spatial dimensions. In this case, particle exchanges are described by the braid group rather than solely by the permutation group, and two successive exchanges need not be equivalent to the identity. The exchange phase is therefore not restricted to ± 1, allowing *anyonic* statistics. For Abelian anyons, an exchange produces an arbitrary phase $e^{i\theta}$, with bosons and fermions corresponding to $\theta = 0$ and $\theta = \pi$, respectively. In three spatial dimensions, only the bosonic and fermionic possibilities occur [11, 12].

For an arbitrary pair of electrons i and j, the antisymmetry requirement can be written compactly as

$$\hat{\mathcal{P}}_{ij}\Phi = -\Phi \tag{1.25}$$

The electronic Schrödinger equation and Pauli antisymmetry condition are therefore

$$\hat{\mathcal{H}}_{\text{elec}}\Phi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) = \mathcal{E}_{\text{elec}}\Phi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) \tag{1.26}$$

and

$$\Phi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_j, \ldots, \boldsymbol{x}_N) = -\Phi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_j, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_N) \tag{1.27}$$

Since the electronic Hamiltonian is invariant under the exchange of two electron labels,

$$\left[\hat{\mathcal{H}}_{\text{elec}}, \hat{\mathcal{P}}_{ij}\right] = 0 \tag{1.28}$$

this antisymmetry is preserved by the electronic dynamics.

The Pauli principle and the exchange hole. A particularly important consequence follows immediately from Eq. (1.27). If two fermions are assigned the same composite coordinate, $\boldsymbol{x} = \boldsymbol{x}_1 = \boldsymbol{x}_2$, then

$$\Psi(\boldsymbol{x}, \boldsymbol{x}) = -\Psi(\boldsymbol{x}, \boldsymbol{x}) \tag{1.29}$$

and hence

$$\Psi(\boldsymbol{x}, \boldsymbol{x}) = 0 \tag{1.30}$$

The probability of finding two electrons with the same spin at exactly the same spatial position therefore vanishes. This is one manifestation of the Pauli exclusion principle [13].

Pauli exclusion principle

Wolfgang Pauli

An electronic wavefunction is antisymmetric under exchange of two complete space–spin coordinates. Consequently, no two electrons can occupy the same spin orbital. In particular, the wavefunction vanishes when two electrons have identical space–spin coordinates [13].

Source portrait: Bettina Katzenstein / ETH-Bibliothek Zürich, Bildarchiv, Portr 01042 (cropped by Xabier), CC BY-SA 3.0. Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: Wolfgang Pauli (1900–1958)

Born in Vienna, Pauli became professor of theoretical physics at ETH Zürich in 1928. His exclusion principle, proposed in 1925, was decisive in understanding atomic electronic configurations [13] and earned him the 1945 Nobel Prize in Physics. In the many-electron formulation used here, exclusion follows from the antisymmetry of the electronic wavefunction rather than from an additional rule imposed on its orbitals.

Biography: CERN Pauli Archive; Nobel Foundation. Source portrait: Bettina Katzenstein / ETH-Bibliothek Zürich, Bildarchiv, Portr 01042, crop by Xabier, CC BY-SA 3.0. Sketch: AI-assisted adaptation of the credited portrait.

It is worth emphasizing that the fundamental statement is the *antisymmetry of the many-electron wavefunction*, not merely the familiar rule that "two electrons cannot occupy the same state." The latter follows from the former. If two columns or rows of an antisymmetric fermionic wavefunction become identical, the wavefunction must vanish.

Antisymmetry also has consequences beyond the special case in which the two electrons occupy exactly the same point. Conditioning on the presence of one electron reduces the probability of finding another electron of the same spin in its vicinity. This deficit in the same-spin pair probability is called the Fermi hole, or exchange hole. It arises entirely from fermionic antisymmetry and is present even in the absence of dynamical electron correlation.

We have therefore arrived at a central requirement: any acceptable many-electron electronic wavefunction must be antisymmetric under the exchange of two electrons. The next question is how to construct such a wavefunction from simpler one-electron objects.

Exercise 1: Two-electron antisymmetry

Given two orthonormal one-electron functions $\chi_1(\boldsymbol{x})$ and $\chi_2(\boldsymbol{x})$, construct a two-electron fermionic wavefunction $\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)$ and normalize it.

Solution 1: Two-electron antisymmetry

Starting from the two possible assignments of the one-electron functions to the two electrons, consider a linear combination with coefficients A and B,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = A\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) + B\chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2) \tag{1.31}$$

Exchanging the coordinates interchanges the two products. Requiring the result to be $-\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2)$ therefore gives $B = -A$. The two products are

$$\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) \qquad \text{and} \qquad \chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2) \tag{1.32}$$

Solution 1 (continued): Two-electron antisymmetry

the required antisymmetric combination is obtained by taking their difference,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) - \chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2)] \tag{1.33}$$

Indeed, exchanging the two electron coordinates immediately yields

$$\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) = -\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \tag{1.34}$$

For orthonormal χ_1 and χ_2, the prefactor $1/\sqrt{2}$ ensures normalization. Indeed, each product has unit norm and their overlap is $\langle\chi_1|\chi_2\rangle\langle\chi_2|\chi_1\rangle = 0$, so their difference has squared norm two. Choosing $A = 1/\sqrt{2}$ fixes the normalization; any additional global phase describes the same physical state. The minus sign enforces antisymmetry, while the prefactor fixes the norm: these are separate requirements.

Slater popularized the determinant representation of this combination [2],

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\begin{vmatrix}\chi_1(\boldsymbol{x}_1) & \chi_2(\boldsymbol{x}_1)\\ \chi_1(\boldsymbol{x}_2) & \chi_2(\boldsymbol{x}_2)\end{vmatrix} = \frac{1}{\sqrt{2}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) - \chi_2(\boldsymbol{x}_1)\chi_1(\boldsymbol{x}_2)] \tag{1.35}$$

The determinant is ideally suited to fermions because interchanging two electron coordinates corresponds to exchanging two rows of the determinant, which changes its sign automatically. Likewise, if two one-electron functions are identical, two columns become identical and the determinant vanishes. Antisymmetry and the Pauli exclusion principle are therefore built directly into the determinant structure.

Historical perspective: John Clarke Slater (1900–1976)

An American physicist with a long career at MIT and later at the University of Florida, Slater helped shape the quantum theory of atoms, molecules, and solids. His 1929 determinantal wavefunctions provide the antisymmetric building blocks used throughout this chapter [2]. His analytic atomic orbitals and screening rules also connect this many-electron construction with the Slater-type basis functions introduced in the second part [14].

Biography: P. M. Morse, NAS memoir. Source portrait detail: Larry C. Huston, 1967; identified on the right in the archive description. University of Chicago Photographic Archive, apf1-01019, Hanna Holborn Gray Special Collections Research Center, University of Chicago Library. Educational/scholarly reuse; commercial permission required. Sketch: AI-assisted adaptation of the credited portrait.

This construction is a Slater determinant. By contrast, the simple product

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2) \tag{1.36}$$

is a Hartree product. A Hartree product assigns a particular one-electron function to each labeled electron and is not, by itself, antisymmetric under particle exchange. It therefore cannot serve as a fermionic many-electron wavefunction.

The two-electron determinant provides the prototype for the general N-electron construction. Hartree–Fock theory adopts precisely this idea: the electronic wavefunction is approximated by a single Slater determinant built from N one-electron spin orbitals. We now turn to this single-determinant ansatz.

1.1.3 The single-determinant ansatz

The previous section established that an acceptable electronic wavefunction must be antisymmetric with respect to the exchange of any pair of electrons. A Slater determinant provides the simplest

construction that satisfies this requirement automatically. Hartree–Fock theory takes the decisive additional step of assuming that the entire N-electron wavefunction can be represented by a single such determinant. We denote this trial electronic wavefunction by Ψ_{HF} to distinguish it from the exact electronic wavefunction Φ_{elec}.

Thus,

$$\begin{aligned}\Psi_{\mathrm{HF}}(\boldsymbol{x}_1,\ldots,\boldsymbol{x}_N) &= \boxed{\frac{1}{\sqrt{N!}}\begin{vmatrix}\chi_1(\boldsymbol{x}_1) & \chi_2(\boldsymbol{x}_1) & \cdots & \chi_N(\boldsymbol{x}_1)\\ \chi_1(\boldsymbol{x}_2) & \chi_2(\boldsymbol{x}_2) & \cdots & \chi_N(\boldsymbol{x}_2)\\ \vdots & \vdots & \ddots & \vdots\\ \chi_1(\boldsymbol{x}_N) & \chi_2(\boldsymbol{x}_N) & \cdots & \chi_N(\boldsymbol{x}_N)\end{vmatrix}} \\ &\equiv |\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)\cdots\chi_N(\boldsymbol{x}_N)\rangle \\ &= \hat{\mathcal{A}}[\chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)\cdots\chi_N(\boldsymbol{x}_N)] = \hat{\mathcal{A}}\Pi(\boldsymbol{x}_1,\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N)\end{aligned} \tag{1.37}$$

Here,

$$\Pi(\boldsymbol{x}_1,\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N) = \chi_1(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_2)\cdots\chi_N(\boldsymbol{x}_N) \tag{1.38}$$

is the Hartree product introduced above, while $\hat{\mathcal{A}}$ is the antisymmetrizer.

The role of $\hat{\mathcal{A}}$ can be made explicit by writing

$$\hat{\mathcal{A}} = \frac{1}{\sqrt{N!}}\sum_{\pi\in S_N}(-1)^{\pi}\hat{\mathcal{P}}_{\pi} \tag{1.39}$$

where S_N is the group of all $N!$ permutations of the electron labels, $\hat{\mathcal{P}}_{\pi}$ is the operator associated with the permutation π, and $(-1)^{\pi}$ is $+1$ for an even permutation and -1 for an odd one. Applying $\hat{\mathcal{A}}$ to a Hartree product generates all possible assignments of the one-electron functions to the electrons, with the signs required by fermionic antisymmetry. The factor $1/\sqrt{N!}$ normalizes the resulting determinant when the spin orbitals are orthonormal. With this convention, $\hat{\mathcal{A}}$ is a normalization-weighted antisymmetrizer, rather than an idempotent projector; the latter would instead contain the factor $1/N!$.

The determinant notation is simply a compact representation of this antisymmetrized sum. Its usefulness follows directly from elementary properties of determinants. Exchanging two electron coordinates exchanges two rows and changes the sign of the determinant. Exchanging two spin orbitals exchanges two columns and likewise changes its sign. Finally, if two spin orbitals become identical, two columns coincide and the determinant vanishes. The Pauli principle is therefore built into the algebraic structure of the wavefunction.

The normalization factor $1/\sqrt{N!}$ in Eq. (1.37) assumes that the spin orbitals are orthonormal. Under this condition,

$$\langle\Psi_{\mathrm{HF}}|\Psi_{\mathrm{HF}}\rangle = 1 \tag{1.40}$$

A normalized Slater determinant is thus completely specified by its N occupied one-electron spin orbitals.

This is a dramatic reduction of the original many-electron problem. Instead of optimizing an arbitrary function of N electronic coordinates, Hartree–Fock theory optimizes a set of one-electron functions. The price of this simplification is that the allowed wavefunctions occupy only a very small subset of the full antisymmetric N-electron Hilbert space. The variational problem is therefore not to find the exact electronic wavefunction, but to find the *best single Slater determinant* for the Hamiltonian under consideration.

Spin orbitals and spatial orbitals. Each one-electron function $\chi_p(\boldsymbol{x})$ appearing in the determinant is a spin orbital: it depends on both the spatial coordinate $\boldsymbol{r}$ and the spin coordinate ω. In the collinear spin formulations considered initially, a spin orbital separates into spin and spatial factors,

$$\chi_p(\boldsymbol{x}) = \sigma(\omega)\psi_p^\sigma(\boldsymbol{r}) \quad \text{with} \quad \sigma(\omega) \in \{\alpha(\omega), \beta(\omega)\} \tag{1.41}$$

Thus, a spin orbital is obtained by associating a spatial orbital with either an α or a β spin function.

The distinction between *spin orbital* and *spatial orbital* will be important throughout the chapter. The Slater determinant itself is built from spin orbitals because antisymmetry applies to the complete spin-spatial coordinates of the electrons. In practical molecular calculations, however, it is usually the spatial functions $\psi_p^\sigma(\boldsymbol{r})$ that are expanded and optimized.

Finite one-electron basis. The spatial orbitals are not represented directly as arbitrary continuous functions. Instead, each is expanded in a finite set of K prescribed basis functions,

$$\boxed{\psi_p^\sigma(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu p}^\sigma \phi_\mu(\boldsymbol{r})} \tag{1.42}$$

The coefficients $C_{\mu p}^\sigma$ then become the quantities to be determined variationally.

This expansion is conventionally called the linear combination of atomic orbitals (LCAO) representation. The terminology is historical: although the basis functions ϕ_μ are most commonly centered on atoms in molecular calculations, the essential mathematical point is simply that the unknown orbitals are represented in a finite one-electron basis.

The finite expansion in Eq. (1.42) introduces a second, independent approximation on top of the single-determinant ansatz. Even within Hartree–Fock theory, the exact solution is recovered only in the limit of a complete one-electron basis. The construction, properties, and convergence of such basis sets will be examined in detail later in this chapter.

Orthonormality. The spin orbitals entering a Slater determinant are chosen to be orthonormal,

$$\langle \chi_p | \chi_q \rangle = \int \chi_p^*(\boldsymbol{x}) \chi_q(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} = \delta_{pq} = \begin{cases} 1, & \text{if } p = q \\ 0, & \text{otherwise} \end{cases} \tag{1.43}$$

where δ_{pq} is the Kronecker delta.

This does not imply that the underlying basis functions are themselves orthogonal. In fact, atom-centered basis functions on different atoms generally overlap. Their pairwise overlaps define the overlap matrix

$$\langle \phi_\mu | \phi_\nu \rangle = \int \phi_\mu^*(\boldsymbol{r}) \phi_\nu(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} = S_{\mu\nu} \tag{1.44}$$

Consequently, the orthonormality of the molecular orbitals imposes a constraint on the expansion coefficients rather than on the basis functions themselves. For example, for a given spin sector,

$$\langle \psi_p^\sigma | \psi_q^\sigma \rangle = \sum_{\mu\nu} C_{\mu p}^{\sigma *} S_{\mu\nu} C_{\nu q}^\sigma = \delta_{pq} \tag{1.45}$$

or, in matrix notation,

$$(\boldsymbol{C}^\sigma)^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C}^\sigma = \mathbf{1} \tag{1.46}$$

This distinction between an orthonormal set of molecular orbitals and a generally nonorthogonal atomic-orbital basis will become central when we derive the Roothaan–Hall equations.

Restricted and unrestricted Hartree–Fock. The single-determinant ansatz still leaves freedom in how the α and β spin orbitals are related. Different constraints on their spatial parts lead to different variants of Hartree–Fock theory.

In restricted Hartree–Fock (RHF) theory, appropriate for the usual closed-shell description, an α electron and a β electron in a pair share the same spatial orbital,

$$\chi_p^{\mathrm{RHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p(\boldsymbol{r}) \\ \beta(\omega)\psi_p(\boldsymbol{r}) \end{cases} = \begin{cases} \psi_p(\boldsymbol{r}) \\ \bar{\psi}_p(\boldsymbol{r}) \end{cases} \tag{1.47}$$

The barred notation will be used when it is convenient to distinguish the β spin orbital associated with a given spatial orbital from its α partner.

For an even number of electrons in a closed-shell RHF state, each occupied spatial orbital therefore contains two electrons with opposite spins. Only $N/2$ distinct occupied spatial orbitals are required to construct the N occupied spin orbitals.

The restriction that the two spin channels share the same spatial orbitals can be advantageous: it preserves the closed-shell spin symmetry and reduces the number of independent orbital coefficients. It is also a genuine constraint on the variational space. In situations where the optimal α and β electrons would prefer different spatial distributions, RHF cannot accommodate that relaxation.

The closed-shell RHF ansatz does not describe general open-shell systems. In unrestricted Hartree–Fock (UHF) theory, the α and β spin components are allowed to use different spatial orbitals (see Fig. 3),

$$\chi_p^{\mathrm{UHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) \\ \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \end{cases} \tag{1.48}$$

This additional variational freedom permits the treatment of open-shell systems and can also lower the energy of nominally closed-shell systems when the restricted solution becomes unstable. As we shall see later, however, this freedom comes at a price: a UHF determinant is generally an eigenfunction of $\hat{\mathcal{S}}_z$, but not necessarily of $\hat{\mathcal{S}}^2$. The resulting breaking of spin symmetry is the origin of spin contamination.

Restricted open-shell Hartree–Fock (ROHF) provides an alternative treatment of open-shell systems in which the α and β electrons retain a common set of spatial orbitals [15]. We mention this possibility for completeness, but do not develop the ROHF formalism further here.

For RHF, the spatial molecular orbitals form an orthonormal set,

$$\langle \psi_p | \psi_q \rangle = \int \psi_p^*(\boldsymbol{r})\psi_q(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = \delta_{pq} \tag{1.49}$$

For UHF, orthonormality holds separately within each spin sector,

$$\begin{aligned} \langle \psi_p^{\alpha} | \psi_q^{\alpha} \rangle &= \int \psi_p^{\alpha *}(\boldsymbol{r})\psi_q^{\alpha}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = \delta_{pq} \\ \langle \psi_p^{\beta} | \psi_q^{\beta} \rangle &= \int \psi_p^{\beta *}(\boldsymbol{r})\psi_q^{\beta}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = \delta_{pq} \end{aligned} \tag{1.50}$$

whereas opposite-spin spatial orbitals need not be orthogonal,

$$\langle \psi_p^{\alpha} | \psi_q^{\beta} \rangle = \int \psi_p^{\alpha *}(\boldsymbol{r})\psi_q^{\beta}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = S_{pq}^{\alpha\beta} \tag{1.51}$$

There is no contradiction with the orthonormality of the full spin orbitals: because

$$\langle \alpha | \beta \rangle = 0 \tag{1.52}$$

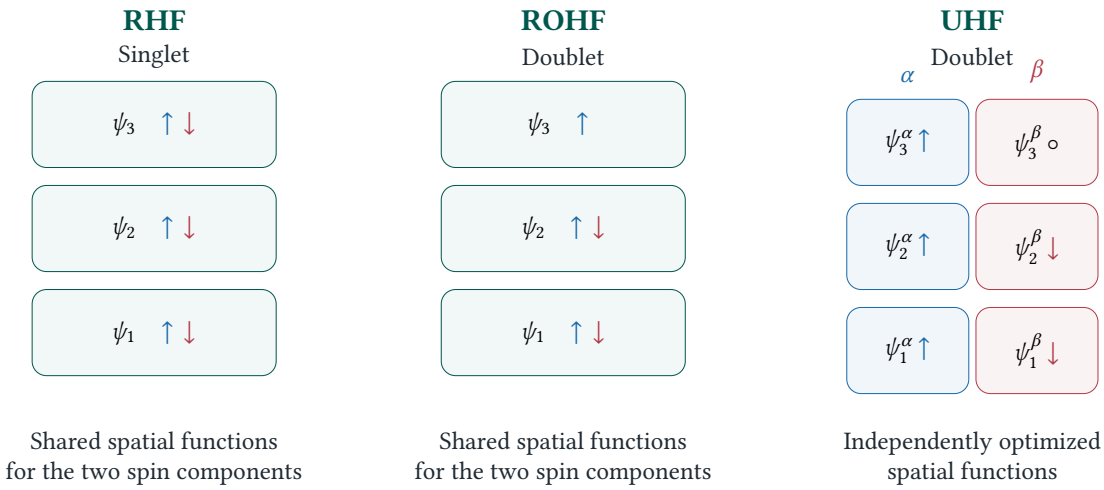


Figure 3: Restricted, restricted open-shell, and unrestricted choices for the spatial parts of the spin orbitals. In RHF, paired α and β electrons share the same spatial orbital, whereas UHF allows the two spin channels to relax independently. ROHF retains a common spatial-orbital set with singly occupied orbitals. The cards illustrate spatial-orbital sharing, not orbital-energy ordering.

an α spin orbital and a β spin orbital are orthogonal regardless of the overlap between their spatial parts. This distinction is useful to keep in mind. In UHF, one may have $\langle \psi_p^\alpha | \psi_q^\beta \rangle \neq 0$ but $\langle \alpha\psi_p^\alpha | \beta\psi_q^\beta \rangle = 0$.

Orbital spaces and index conventions. We conclude by fixing the notation used throughout the remainder of the chapter. The one-electron spaces are

$$\begin{aligned} \{\phi_\mu \mid \mu = 1, \ldots, K\} &: \quad \text{basis functions or atomic orbitals (AOs)} \\ \{\psi_p \mid p = 1, \ldots, K\} &: \quad \text{restricted spatial orbitals or molecular orbitals (MOs)} \\ \{\chi_p \mid p = 1, \ldots, 2K\} &: \quad \text{spin orbitals} \end{aligned} \tag{1.53}$$

Thus, a basis of K spatial functions generates K spatial molecular orbitals and, when the two spin functions are included, $2K$ spin orbitals.

Unless otherwise stated, the following index convention will be used:

$$\begin{aligned} \mu, \nu, \lambda, \sigma, \ldots &: \text{basis functions / AOs} \\ p, q, r, s, \ldots &: \text{general orbitals} \\ i, j, k, l, \ldots &: \text{occupied orbitals} \\ a, b, c, d, \ldots &: \text{unoccupied or virtual orbitals} \end{aligned} \tag{1.54}$$

Among the $2K$ spin orbitals, N are occupied in the HF determinant and are therefore labeled by $i, j, k, l, \ldots$. The remaining $2K - N$ spin orbitals are unoccupied. They are variously called vacant, unoccupied, or virtual orbitals and are labeled by $a, b, c, d, \ldots$.

The distinction between occupied and virtual orbitals is relative to the particular determinant under consideration. Only the occupied orbitals enter the Slater determinant explicitly, but the virtual orbitals provide the complementary one-electron space and will become essential when discussing orbital optimization, excited determinants, response, and correlated electronic-structure methods.

Finally, a system with two electrons in each occupied spatial orbital is called closed shell. Otherwise, it is open shell. For the ground state of a closed-shell system containing an even

number N of electrons, the first $N/2$ spatial orbitals are doubly occupied, $\psi_1^2\psi_2^2\cdots\psi_{N/2}^2$, while the remaining $K - N/2$ spatial orbitals are vacant.

The single-determinant ansatz, together with these orthonormality constraints, defines the variational space of Hartree–Fock theory. The next step is to determine which determinant within this space is optimal. This requires evaluating the expectation value of the electronic Hamiltonian as a functional of the occupied orbitals.

1.2 The Hartree–Fock energy functional

The variational content of Hartree–Fock theory is encoded in the expectation value of the electronic Hamiltonian over a normalized Slater determinant. The derivation below separates the one- and two-electron contributions and makes the Coulomb, exchange, and self-interaction terms explicit before introducing their integral representations.

1.2.1 Energy expectation value

We have now specified the class of wavefunctions allowed within Hartree–Fock theory: normalized single Slater determinants constructed from orthonormal spin orbitals. The remaining question is variational. Among all such determinants, which one provides the lowest possible energy?

For fixed nuclear positions, the electronic energy associated with a given Slater determinant is the expectation value of the electronic Hamiltonian,

$$E_{\mathrm{HF}} = \langle \Psi_{\mathrm{HF}} | \hat{\mathcal{H}}_{\mathrm{elec}} | \Psi_{\mathrm{HF}} \rangle \qquad \hat{\mathcal{H}}_{\mathrm{elec}} = \hat{\mathcal{T}}_{\mathrm{e}} + \hat{\mathcal{V}}_{\mathrm{ne}} + \hat{\mathcal{V}}_{\mathrm{ee}} \tag{1.55}$$

Since Ψ_{HF} depends on the occupied spin orbitals, this expression should be regarded as an energy functional,

$$E_{\mathrm{HF}} = E_{\mathrm{HF}}[\chi_1, \chi_2, \ldots, \chi_N] \tag{1.56}$$

Hartree–Fock theory seeks the stationary point—and, for the ground-state solution, the minimum—of this functional subject to the orthonormality constraints

$$\langle \chi_i | \chi_j \rangle = \delta_{ij} \tag{1.57}$$

The variational principle guarantees that the energy of any normalized trial wavefunction lies above or at the exact ground-state energy,

$$\langle \Psi_{\mathrm{trial}} | \hat{\mathcal{H}}_{\mathrm{elec}} | \Psi_{\mathrm{trial}} \rangle \geq \mathcal{E}_{\mathrm{elec}}^{(0)} \tag{1.58}$$

Hartree–Fock therefore finds the lowest energy accessible within the restricted manifold of single determinants. The remaining difference between this optimal single-determinant energy and the exact non-relativistic electronic energy is associated with electron correlation beyond the HF description.

One- and two-electron operators. It is useful to separate the electronic Hamiltonian into one- and two-electron parts,

$$\hat{\mathcal{H}}_{\mathrm{elec}} = \hat{O}_1 + \hat{O}_2 = \sum_{i=1}^{N} \hat{h}(\boldsymbol{x}_i) + \sum_{i<j}^{N} \frac{1}{r_{ij}} \tag{1.59}$$

The one-electron, or core, Hamiltonian is

$$\hat{h}(\boldsymbol{x}_i) = -\frac{\nabla_i^2}{2} - \sum_{A=1}^{M} \frac{Z_A}{r_{iA}} \tag{1.60}$$

so that

$$\hat{O}_1 = \sum_{i=1}^{N} \hat{h}(\boldsymbol{x}_i) \tag{1.61}$$

It contains the kinetic energy of electron i and its attraction to all nuclei. The explicit electron-electron interaction is contained in the two-electron operator

$$\hat{O}_2 = \hat{\mathcal{V}}_{\mathrm{ee}} = \sum_{i<j}^{N} \frac{1}{r_{ij}} \tag{1.62}$$

The distinction between one- and two-electron operators is important because their expectation values over a Slater determinant have particularly simple forms. This simplification is one of the main reasons why the single-determinant ansatz is computationally tractable.

One-electron contribution. For a normalized determinant constructed from orthonormal spin orbitals, the Slater–Condon rules reduce the expectation value of a one-electron operator to a sum over the occupied spin orbitals. We use their diagonal expectation values here; the few-electron examples below derive them explicitly, and the general off-diagonal rules are developed in Section 1.5. For the one-electron part,

$$\langle \Psi_{\mathrm{HF}} | \hat{O}_1 | \Psi_{\mathrm{HF}} \rangle = \sum_{i=1}^{N} \langle \chi_i | \hat{h} | \chi_i \rangle \tag{1.63}$$

Introducing

$$h_{ij} = \left\langle \chi_i \middle| \hat{h} \middle| \chi_j \right\rangle \tag{1.64}$$

we obtain simply

$$\langle \Psi_{\mathrm{HF}} | \hat{O}_1 | \Psi_{\mathrm{HF}} \rangle = \sum_{i=1}^{N} h_{ii} \tag{1.65}$$

Thus, the one-electron part of the HF energy is obtained by adding the kinetic and electron-nuclear energies associated with each occupied spin orbital.

The nucleus-nucleus repulsion does not belong to the electronic Hamiltonian because, within the clamped-nuclei problem, it contains no electronic coordinates. Nevertheless, it must be added when the total Born–Oppenheimer energy is required. For a normalized electronic wavefunction,

$$\langle \Psi_{\mathrm{HF}} | \hat{\mathcal{V}}_{\mathrm{nn}} | \Psi_{\mathrm{HF}} \rangle = V_{\mathrm{nn}} \langle \Psi_{\mathrm{HF}} | \Psi_{\mathrm{HF}} \rangle = V_{\mathrm{nn}} \qquad \text{if} \qquad \langle \Psi_{\mathrm{HF}} | \Psi_{\mathrm{HF}} \rangle = 1 \tag{1.66}$$

Hence,

$$E_{\mathrm{tot}}^{\mathrm{HF}} = E_{\mathrm{HF}} + V_{\mathrm{nn}} \tag{1.67}$$

In what follows, E_{HF} will denote the electronic Hartree–Fock energy unless stated otherwise.

Two-electron contribution: Coulomb and exchange. The effect of antisymmetry first appears explicitly in the expectation value of the electron-electron interaction. For each pair of occupied spin orbitals, two contributions arise,

$$\begin{aligned} \langle \Psi_{\mathrm{HF}} | \hat{O}_2 | \Psi_{\mathrm{HF}} \rangle &= \sum_{i<j}^{N} \Big[\left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \middle| r_{12}^{-1} \middle| \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \right\rangle \\ &\qquad - \left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2) \middle| r_{12}^{-1} \middle| \chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2) \right\rangle \Big] \\ &= \sum_{i<j}^{N} \Big(\underbrace{\mathcal{J}_{ij}}_{\text{Coulomb}} - \underbrace{\mathcal{K}_{ij}}_{\text{exchange}} \Big) \end{aligned} \tag{1.68}$$

The first term,

$$\mathcal{J}_{ij} = \left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\right\rangle \tag{1.69}$$

is the Coulomb integral. Written explicitly,

$$\mathcal{J}_{ij} = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.70}$$

It has a direct classical interpretation: it is the electrostatic repulsion between the charge distributions associated with orbitals i and j.

The second term,

$$\mathcal{K}_{ij} = \left\langle \chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\right\rangle \tag{1.71}$$

is the exchange integral,

$$\mathcal{K}_{ij} = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.72}$$

Unlike $\mathcal{J}_{ij}$, the exchange contribution has no classical electrostatic counterpart. It arises entirely from the antisymmetry of the fermionic wavefunction: it is the energetic signature of the exchange of electron labels between the two terms of the determinant.

This point is fundamental. The Coulomb interaction itself is the same $1/r_{12}$ operator as before; there is no separate "exchange force" in the Hamiltonian. Exchange appears because the expectation value of the Coulomb operator is evaluated with an antisymmetric wavefunction.

Because the spin orbitals factorize into spin and spatial parts, the exchange integral also carries an important spin selection rule. If χ_i and χ_j have opposite spin,

$$\langle\alpha|\beta\rangle = 0 \qquad \Longrightarrow \qquad \mathcal{K}_{ij} = 0 \tag{1.73}$$

Exchange therefore acts only between electrons of the same spin in the collinear RHF and UHF formulations considered here. Coulomb interactions, by contrast, are present between every pair of electrons regardless of spin.

Self-interaction cancellation. The pairwise form of the two-electron energy can equivalently be written as a double sum,

$$\langle\Psi_{\mathrm{HF}}|\hat{O}_2|\Psi_{\mathrm{HF}}\rangle = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right) \tag{1.74}$$

The factor $1/2$ avoids counting every distinct electron pair twice.

At first sight, the terms with $i = j$ in this expression might appear to describe the Coulomb interaction of an electron with itself. However, $\mathcal{J}_{ii} = \mathcal{K}_{ii}$ because

$$\mathcal{J}_{ii} = \mathcal{K}_{ii} = \iint \frac{|\chi_i(\boldsymbol{x}_1)|^2|\chi_i(\boldsymbol{x}_2)|^2}{r_{12}}\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.75}$$

Consequently,

$$\mathcal{J}_{ii} - \mathcal{K}_{ii} = 0 \tag{1.76}$$

The spurious Coulomb self-interaction is therefore cancelled exactly by exchange in Hartree–Fock theory.

Combining the one- and two-electron contributions gives the fundamental HF energy expression

$$E_{\mathrm{HF}} = \sum_{i=1}^{N} h_{ii} + \sum_{i<j}^{N}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right) \tag{1.77}$$

Equivalently,

$$\boxed{E_{\mathrm{HF}} = \sum_{i=1}^{N} h_{ii} + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right)} \tag{1.78}$$

This expression already reveals the central structure of HF theory. The energy depends on each occupied orbital individually through h_{ii}, but also on every pair of occupied orbitals through the Coulomb and exchange terms. The orbitals therefore cannot be optimized independently: changing one orbital changes the effective field experienced by all the others. This coupling is the origin of the self-consistent character of the Hartree–Fock equations.

Coulomb operator. It is useful to promote the Coulomb and exchange integrals to one-electron operators. For a fixed occupied spin orbital χ_j, the Coulomb operator $\hat{\mathcal{J}}_j$ acts on χ_i as

$$\hat{\mathcal{J}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle = \left\langle \chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_j(\boldsymbol{x}_2)\right\rangle |\chi_i(\boldsymbol{x}_1)\rangle \tag{1.79}$$

or explicitly,

$$\hat{\mathcal{J}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle = \left[\int \chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_2\right]|\chi_i(\boldsymbol{x}_1)\rangle \tag{1.80}$$

The quantity in square brackets is the electrostatic potential generated by the density of orbital j at the position of electron 1.

Its expectation value is

$$\begin{aligned}\mathcal{J}_{ij} &= \langle\chi_i(\boldsymbol{x}_1)|\hat{\mathcal{J}}_j(\boldsymbol{x}_1)|\chi_i(\boldsymbol{x}_1)\rangle \\ &= \left\langle\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\right\rangle \\ &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\end{aligned} \tag{1.81}$$

The Coulomb operator is local in the usual one-particle sense. Once the integration over $\boldsymbol{x}_2$ has been carried out, it acts on $\chi_i(\boldsymbol{x}_1)$ by multiplication with a function of $\boldsymbol{x}_1$,

$$\hat{\mathcal{J}}_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_1) = J_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_1) \tag{1.82}$$

Every orbital acted upon by $\hat{\mathcal{J}}_j$ therefore experiences the same multiplicative Coulomb potential generated by orbital j.

Exchange operator. The exchange operator has a qualitatively different structure. Its action is defined by

$$\hat{\mathcal{K}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle = \left\langle \chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_i(\boldsymbol{x}_2)\right\rangle \left|\chi_j(\boldsymbol{x}_1)\right\rangle \tag{1.83}$$

or explicitly,

$$\hat{\mathcal{K}}_j(\boldsymbol{x}_1)\,|\chi_i(\boldsymbol{x}_1)\rangle = \left[\int \chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_i(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_2\right]\left|\chi_j(\boldsymbol{x}_1)\right\rangle \tag{1.84}$$

Its matrix element is

$$\begin{aligned}\mathcal{K}_{ij} &= \langle\chi_i(\boldsymbol{x}_1)|\hat{\mathcal{K}}_j(\boldsymbol{x}_1)|\chi_i(\boldsymbol{x}_1)\rangle \\ &= \left\langle\chi_i(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_2)\middle|r_{12}^{-1}\middle|\chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\right\rangle \\ &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_j(\boldsymbol{x}_1)\chi_i(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\end{aligned} \tag{1.85}$$

Unlike the Coulomb operator, $\hat{\mathcal{K}}_j$ is nonlocal. Its value at $\boldsymbol{x}_1$ depends on the values of the orbital on which it acts, $\chi_i(\boldsymbol{x}_2)$, over all coordinates $\boldsymbol{x}_2$. Moreover, the resulting function is

proportional to $\chi_j(\boldsymbol{x}_1)$ rather than simply to $\chi_i(\boldsymbol{x}_1)$. Exchange therefore cannot, in general, be represented by a single multiplicative potential.

This distinction between the local Coulomb operator and the nonlocal exchange operator is one of the defining features of Hartree–Fock theory. Both arise from the same electron-electron Coulomb interaction, but the exchange term is generated solely by the antisymmetry of the Slater determinant.

The HF energy can therefore be regarded as a functional of the occupied orbitals,

$$E_{\mathrm{HF}}[\{\chi_i\}] = \sum_i \langle \chi_i|\hat{h}|\chi_i\rangle + \frac{1}{2}\sum_{ij}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right) \qquad \langle \chi_i|\chi_j\rangle = \delta_{ij} \tag{1.86}$$

The remainder of Hartree–Fock theory consists essentially in minimizing this functional with respect to the occupied orbitals while preserving their orthonormality. Before carrying out that minimization, however, it is useful to introduce a compact notation for the one- and two-electron integrals appearing throughout the theory.

1.2.2 One- and two-electron integrals

The Hartree–Fock energy derived in the previous section is expressed entirely in terms of matrix elements of one- and two-electron operators. These quantities occur throughout electronic-structure theory, so it is useful to introduce their notation carefully at this stage.

There are two widely used conventions for two-electron integrals: physicists' notation and chemists' notation. They describe exactly the same mathematical objects, but group the four orbital indices differently. Confusion between these conventions is a frequent source of index errors, particularly when moving between quantum-chemistry and many-body-theory literature.

One-electron integrals. For a general one-electron operator $\hat{h}$, the matrix element between spin orbitals χ_i and χ_j is

$$\langle i|\hat{h}|j\rangle = h_{ij} = \langle \chi_i|\hat{h}|\chi_j\rangle = \int \chi_i^*(\boldsymbol{x}_1)\hat{h}(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\,\mathrm{d}\boldsymbol{x}_1 \tag{1.87}$$

For the Hartree–Fock problem, $\hat{h}$ is the core Hamiltonian,

$$\hat{h}(\boldsymbol{x}) = -\frac{\nabla^2}{2} - \sum_A \frac{Z_A}{r_A} \tag{1.88}$$

so that h_{ij} contains both the electronic kinetic energy and the electron-nuclear attraction.

Since $\hat{h}$ is Hermitian,

$$h_{ij} = h_{ji}^* \tag{1.89}$$

For real orbitals and a real one-electron operator, this reduces to

$$h_{ij} = h_{ji} \tag{1.90}$$

In Mulliken notation, the same one-electron integral may equivalently be written

$$(i\,|\,\hat{h}\,|\,j) = h_{ij} = (\chi_i\,|\,\hat{h}\,|\,\chi_j) = \int \chi_i^*(\boldsymbol{x}_1)\hat{h}(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\,\mathrm{d}\boldsymbol{x}_1 \tag{1.91}$$

Thus, for one-electron quantities, the distinction between physicists' and chemists' notation is essentially cosmetic. The important difference appears for two-electron integrals.

Electron-repulsion integrals: Dirac or physicists' notation. The matrix element of the Coulomb interaction between two-electron product states defines an electron-repulsion integral (ERI). In Dirac (or physicists') notation,

$$\langle ij|kl\rangle = \big\langle \chi_i\chi_j\big|\chi_k\chi_l\big\rangle = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.92}$$

The ordering of the indices reflects the underlying braket structure: i and j label the two orbitals in the bra, whereas k and l label those in the ket. More importantly, the first and third indices refer to electron coordinate $\boldsymbol{x}_1$, while the second and fourth refer to $\boldsymbol{x}_2$:

$$\langle \underbrace{i}_{\boldsymbol{x}_1}\ \underbrace{j}_{\boldsymbol{x}_2}\ |\ \underbrace{k}_{\boldsymbol{x}_1}\ \underbrace{l}_{\boldsymbol{x}_2}\ \rangle \tag{1.93}$$

With this notation, the Coulomb and exchange integrals introduced in the previous section are simply

$$\mathcal{J}_{ij} = \langle ij|ij\rangle \qquad\qquad \mathcal{K}_{ij} = \langle ij|ji\rangle \tag{1.94}$$

The only difference between the two is therefore the permutation of the two indices in the ket.

This observation motivates the introduction of the antisymmetrized two-electron integral,

$$\boxed{\langle ij||kl\rangle = \langle ij|kl\rangle - \langle ij|lk\rangle} \tag{1.95}$$

Explicitly,

$$\begin{aligned}\langle ij||kl\rangle &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}[\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2) - \chi_l(\boldsymbol{x}_1)\chi_k(\boldsymbol{x}_2)]\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \\ &= \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}(1-\hat{\mathcal{P}}_{12})\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2\end{aligned} \tag{1.96}$$

The antisymmetrized integral combines the direct Coulomb contribution and its exchange counterpart into a single object. For example,

$$\langle ij||ij\rangle = \mathcal{J}_{ij} - \mathcal{K}_{ij} \tag{1.97}$$

The HF energy can therefore also be written compactly as

$$E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij}\langle ij||ij\rangle \tag{1.98}$$

As expected from its definition, the antisymmetrized integral changes sign upon exchanging either pair of fermionic indices,

$$\langle ij||kl\rangle = -\langle ji||kl\rangle = -\langle ij||lk\rangle = \langle ji||lk\rangle \tag{1.99}$$

This notation becomes particularly convenient in post-Hartree–Fock and many-body perturbation theories, where direct and exchange contributions repeatedly occur together.

Mulliken or chemists' notation. Quantum-chemistry literature very often uses a different grouping of the same four orbital indices. In Mulliken, or chemists', notation,

$$(ij \mid kl) = (\chi_i \chi_j \mid \chi_k \chi_l) = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.100}$$

Here, the first pair of indices belongs to electron coordinate $\boldsymbol{x}_1$ and the second pair belongs to electron coordinate $\boldsymbol{x}_2$:

$$(\underbrace{ij}_{\boldsymbol{x}_1} \mid \underbrace{kl}_{\boldsymbol{x}_2}) \tag{1.101}$$

The two notations therefore differ only by an index permutation. Comparing Eqs. (1.92) and (1.100) gives

$$\boxed{\langle ij|kl\rangle = (ik \mid jl)} \tag{1.102}$$

This simple correspondence is worth remembering:

$$\text{physicists: } \langle ij|kl\rangle \qquad \longleftrightarrow \qquad \text{chemists: } (ik \mid jl) \tag{1.103}$$

Consequently, the Coulomb and exchange integrals become

$$\mathcal{J}_{ij} = (ii \mid jj) \qquad \mathcal{K}_{ij} = (ij \mid ji) \qquad \text{(chemists' notation)} \tag{1.104}$$

These expressions also make the physical interpretation transparent: $\mathcal{J}_{ij}$ couples the densities $\chi_i^*\chi_i$ and $\chi_j^*\chi_j$, whereas $\mathcal{K}_{ij}$ couples the transition-like products $\chi_i^*\chi_j$ and $\chi_j^*\chi_i$.

The same notation will be used throughout this chapter for integrals over spin orbitals and over spatial orbitals; the surrounding context specifies which type is intended.

Spin selection rules. When the χ_p are collinear spin orbitals,

$$\chi_p(\boldsymbol{x}) = \sigma_p(\omega)\psi_p(\boldsymbol{r}) \tag{1.105}$$

integration over the spin coordinates gives

$$\langle ij|kl\rangle = \delta_{\sigma_i\sigma_k}\delta_{\sigma_j\sigma_l}\iint \psi_i^*(\boldsymbol{r}_1)\psi_j^*(\boldsymbol{r}_2)\frac{1}{r_{12}}\psi_k(\boldsymbol{r}_1)\psi_l(\boldsymbol{r}_2)\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\boldsymbol{r}_2 \tag{1.106}$$

A spin-orbital ERI therefore vanishes unless the spin on each electron line is conserved, $\sigma_i = \sigma_k$ and $\sigma_j = \sigma_l$. For the exchange integral,

$$\mathcal{K}_{ij} = \langle ij|ji\rangle \tag{1.107}$$

this requires $\sigma_i = \sigma_j$. Hence, $\mathcal{K}_{ij} = 0$ for opposite-spin orbitals. This is the integral form of the same-spin nature of Hartree–Fock exchange discussed in the previous section.

Permutation symmetries. Electron-repulsion integrals possess extensive permutation symmetries. These follow from three simple properties: the symmetry of the Coulomb operator under interchange of the two electrons, its Hermiticity, and, for real orbitals, the commutativity of the orbital products evaluated at a common coordinate.

For reference, the physicists' ERI is

$$\langle ij|kl\rangle = \big\langle \chi_i\chi_j\big|\chi_k\chi_l\big\rangle = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j^*(\boldsymbol{x}_2)\frac{1}{r_{12}}\chi_k(\boldsymbol{x}_1)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.108}$$

For complex-valued orbitals, physicists' ERIs obey

$$\langle ij|kl\rangle = \langle ji|lk\rangle = \langle kl|ij\rangle^* = \langle lk|ji\rangle^* \tag{1.109}$$

The first equality corresponds to exchanging the two electron coordinates, while the complex-conjugated relations follow from Hermiticity.

For real-valued orbitals, the additional within-coordinate interchanges give

$$\begin{aligned}\langle ij|kl\rangle &= \langle ji|lk\rangle = \langle kl|ij\rangle = \langle lk|ji\rangle \\ &= \langle kj|il\rangle = \langle jk|li\rangle = \langle il|kj\rangle = \langle li|jk\rangle\end{aligned} \tag{1.110}$$

In chemists' notation, the defining integral is

$$(ij\,|\,kl) = (\chi_i\chi_j\,|\,\chi_k\chi_l) = \iint \chi_i^*(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_l(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.111}$$

For real-valued orbitals, its symmetry relations take the familiar form

$$\begin{aligned}(ij\,|\,kl) &= (ji\,|\,kl) = (ij\,|\,lk) = (ji\,|\,lk) \\ &= (kl\,|\,ij) = (lk\,|\,ij) = (kl\,|\,ji) = (lk\,|\,ji)\end{aligned} \tag{1.112}$$

Thus, for real-valued orbitals, ERIs possess eightfold permutation symmetry. The condition concerns the orbital functions, not merely the numerical values of their integrals. Complex functions can yield real ERIs without possessing all eight permutation symmetries, as the plane-wave expressions in Section 4 illustrate.

This symmetry has an important computational consequence. A naive set of K one-electron basis functions generates K^4 four-index ERIs, $(\mu\nu\,|\,\lambda\sigma)$. However, many of these entries are identical by permutation symmetry. For four distinct indices, up to eight nominally different integrals represent the same numerical quantity. Modern integral algorithms exploit these symmetries extensively to reduce both the amount of computation and the storage required for the two-electron integrals.

The apparent K^4 growth nevertheless makes the treatment of ERIs one of the central computational challenges of conventional electronic-structure theory. Later in this chapter, when the basis functions themselves are introduced in detail, we will return to the evaluation of these integrals and to the algorithms that make their computation practical.

1.2.3 Worked examples for few-electron determinants

The general expressions derived above can appear rather compact when first encountered. It is therefore useful to recover them explicitly for systems containing only a few electrons. These examples illustrate how the orthonormality of the spin orbitals and the antisymmetry of the Slater determinant combine to produce the characteristic Hartree–Fock energy expression.

We begin with two electrons occupying the orthonormal spin orbitals χ_1 and χ_2. Throughout the following examples, the shorthand

$$\chi_p(i) \equiv \chi_p(\boldsymbol{x}_i) \tag{1.113}$$

will be used, where i labels the complete spin-spatial coordinate of electron i.

Exercise 2: Normalization of the HF wavefunction

Show that the HF wavefunction built with two normalized, orthogonal spin orbitals χ_1 and χ_2 is normalized.

Solution 2: Normalization of the HF wavefunction

Take two orthonormal spin orbitals, allowing them to be complex. The two-electron Slater determinant with the proposed normalization factor is

$$\Psi_{\mathrm{HF}} = \frac{1}{\sqrt{2}} \begin{vmatrix} \chi_1(1) & \chi_2(1) \\ \chi_1(2) & \chi_2(2) \end{vmatrix} \tag{1.114}$$

or, after expanding the determinant,

$$\Psi_{\mathrm{HF}} = \frac{\chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2)}{\sqrt{2}} \tag{1.115}$$

Its norm is

$$\begin{aligned} \langle \Psi_{\mathrm{HF}} | \Psi_{\mathrm{HF}} \rangle &= \frac{1}{2} \langle \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) | \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) \rangle \\ &= \frac{1}{2} \Big[\langle \chi_1(1)\chi_2(2) | \chi_1(1)\chi_2(2) \rangle - \langle \chi_1(1)\chi_2(2) | \chi_2(1)\chi_1(2) \rangle \\ &\quad - \langle \chi_2(1)\chi_1(2) | \chi_1(1)\chi_2(2) \rangle + \langle \chi_2(1)\chi_1(2) | \chi_2(1)\chi_1(2) \rangle \Big] \end{aligned} \tag{1.116}$$

Because the coordinates of the two electrons are independent, each two-electron overlap factorizes into products of one-electron overlaps. For example,

$$\langle \chi_1(1)\chi_2(2) | \chi_1(1)\chi_2(2) \rangle = \langle \chi_1(1) | \chi_1(1) \rangle \langle \chi_2(2) | \chi_2(2) \rangle = 1 \tag{1.117}$$

Likewise,

$$\begin{aligned} \langle \chi_1(1)\chi_2(2) | \chi_2(1)\chi_1(2) \rangle &= \langle \chi_1 | \chi_2 \rangle \langle \chi_2 | \chi_1 \rangle \\ &= 0 \end{aligned} \tag{1.118}$$

since the spin orbitals are orthogonal. The same argument applies to the other two terms, and therefore

$$\langle \Psi_{\mathrm{HF}} | \Psi_{\mathrm{HF}} \rangle = \frac{1}{2}[1 - 0 - 0 + 1] = \boxed{1} \tag{1.119}$$

This example explains the origin of the normalization factor $1/\sqrt{2}$ for a two-electron determinant. More generally, a determinant built from N orthonormal spin orbitals contains $N!$ mutually orthogonal Hartree products and is normalized by the factor $1/\sqrt{N!}$.

Exercise 3: Core Hamiltonian

Evaluate the one-electron energy of the normalized two-electron determinant and explain the general occupied-orbital sum

$$\langle \Psi_{HF} | \hat{O}_1 | \Psi_{HF} \rangle = \sum_{i=1}^{N} h_{ii}$$

Solution 3: Core Hamiltonian

For two electrons, the one-electron part of the Hamiltonian is

$$\hat{O}_1 = \hat{h}(1) + \hat{h}(2) \tag{1.120}$$

Solution 3 (continued): Core Hamiltonian

Using Eq. (1.115),

$$\begin{aligned}
&\langle \Psi_{\mathrm{HF}} | \hat{h}(1) + \hat{h}(2) | \Psi_{\mathrm{HF}} \rangle \\
&= \frac{1}{2} \langle \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) | \hat{h}(1) + \hat{h}(2) | \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2) \rangle \\
&= \frac{1}{2} \Big[\langle \chi_1(1)\chi_2(2) | \hat{h}(1) + \hat{h}(2) | \chi_1(1)\chi_2(2) \rangle - \langle \chi_1(1)\chi_2(2) | \hat{h}(1) + \hat{h}(2) | \chi_2(1)\chi_1(2) \rangle \\
&- \langle \chi_2(1)\chi_1(2) | \hat{h}(1) + \hat{h}(2) | \chi_1(1)\chi_2(2) \rangle + \langle \chi_2(1)\chi_1(2) | \hat{h}(1) + \hat{h}(2) | \chi_2(1)\chi_1(2) \rangle \Big]
\end{aligned} \tag{1.121}$$

Consider first the diagonal matrix element

$$\langle \chi_1(1)\chi_2(2) | \hat{h}(1) + \hat{h}(2) | \chi_1(1)\chi_2(2) \rangle \tag{1.122}$$

Because $\hat{h}(1)$ acts only on the coordinates of electron 1,

$$\begin{aligned}
\langle \chi_1(1)\chi_2(2) | \hat{h}(1) | \chi_1(1)\chi_2(2) \rangle &= \langle \chi_1 | \hat{h} | \chi_1 \rangle \langle \chi_2 | \chi_2 \rangle \\
&= h_{11}
\end{aligned} \tag{1.123}$$

Similarly,

$$\langle \chi_1(1)\chi_2(2) | \hat{h}(2) | \chi_1(1)\chi_2(2) \rangle = h_{22} \tag{1.124}$$

and hence

$$\langle \chi_1(1)\chi_2(2) | \hat{h}(1) + \hat{h}(2) | \chi_1(1)\chi_2(2) \rangle = h_{11} + h_{22} \tag{1.125}$$

The off-diagonal terms vanish. For example,

$$\langle \chi_1(1)\chi_2(2) | \hat{h}(1) | \chi_2(1)\chi_1(2) \rangle = \langle \chi_1 | \hat{h} | \chi_2 \rangle \langle \chi_2 | \chi_1 \rangle = h_{12} \times 0 = 0 \tag{1.126}$$

and

$$\langle \chi_1(1)\chi_2(2) | \hat{h}(2) | \chi_2(1)\chi_1(2) \rangle = \langle \chi_1 | \chi_2 \rangle \langle \chi_2 | \hat{h} | \chi_1 \rangle = 0 \times h_{21} = 0 \tag{1.127}$$

The cancellation therefore does *not* require $h_{12} = 0$; it follows from the orthogonality of the spectator orbital.
Collecting all terms gives

$$\begin{aligned}
\langle \Psi_{\mathrm{HF}} | \hat{O}_1 | \Psi_{\mathrm{HF}} \rangle &= \frac{1}{2} [h_{11} + h_{22} - 0 - 0 + h_{22} + h_{11}] \\
&= h_{11} + h_{22}
\end{aligned} \tag{1.128}$$

This is the two-electron instance of the general Slater–Condon rule

$$\langle \Psi_{\mathrm{HF}} | \hat{O}_1 | \Psi_{\mathrm{HF}} \rangle = \sum_{i=1}^{N} h_{ii} \tag{1.129}$$

Only diagonal matrix elements over the occupied spin orbitals survive. To see why the same argument holds for N electrons, expand the bra and ket determinants into their $N!$ Hartree products. A one-electron operator leaves $N-1$ spectator overlaps. These force the corresponding orbital assignments to agree; because each product contains the same occupied set, the remaining assignment must then agree as well. Thus only identical permutations survive. Each gives the sum of the N occupied diagonal matrix elements, and the $N!$ equal contributions cancel the normalization factor $1/N!$. No assumption that the occupied orbitals diagonalize $\hat{h}$ is needed.

Exercise 4: Two-electron Hamiltonian

Evaluate the two-electron interaction for the same normalized determinant, explain the general pair sum

$$\langle \Psi_{\mathrm{HF}} | \hat{O}_2 | \Psi_{\mathrm{HF}} \rangle = \sum_{i<j}^{N} (\mathcal{J}_{ij} - \mathcal{K}_{ij}) \tag{1.130}$$

and write down the HF energy.

Solution 4: Two-electron Hamiltonian

For a two-electron system there is only one distinct electron pair, so

$$\hat{O}_2 = \frac{1}{r_{12}} \tag{1.131}$$

Direct expansion gives

$$\begin{aligned} \langle \Psi_{\mathrm{HF}} | r_{12}^{-1} | \Psi_{\mathrm{HF}} \rangle &= \frac{1}{2} \langle \chi_1\chi_2 - \chi_2\chi_1 | r_{12}^{-1} | \chi_1\chi_2 - \chi_2\chi_1 \rangle \\ &= \frac{1}{2} \Big[\langle \chi_1\chi_2 | r_{12}^{-1} | \chi_1\chi_2 \rangle - \langle \chi_1\chi_2 | r_{12}^{-1} | \chi_2\chi_1 \rangle - \langle \chi_2\chi_1 | r_{12}^{-1} | \chi_1\chi_2 \rangle + \langle \chi_2\chi_1 | r_{12}^{-1} | \chi_2\chi_1 \rangle \Big] \end{aligned} \tag{1.132}$$

The first term is the direct, or Coulomb, contribution,

$$\langle \chi_1\chi_2 | r_{12}^{-1} | \chi_1\chi_2 \rangle = \mathcal{J}_{12} \tag{1.133}$$

whereas the second is the exchange contribution,

$$\langle \chi_1\chi_2 | r_{12}^{-1} | \chi_2\chi_1 \rangle = \mathcal{K}_{12} \tag{1.134}$$

By exchanging the dummy electron coordinates,

$$\langle \chi_2\chi_1 | r_{12}^{-1} | \chi_2\chi_1 \rangle = \langle \chi_1\chi_2 | r_{12}^{-1} | \chi_1\chi_2 \rangle = \mathcal{J}_{12} \tag{1.135}$$

and similarly,

$$\langle \chi_2\chi_1 | r_{12}^{-1} | \chi_1\chi_2 \rangle = \mathcal{K}_{12} \tag{1.136}$$

Consequently,

$$\begin{aligned} \langle \Psi_{\mathrm{HF}} | r_{12}^{-1} | \Psi_{\mathrm{HF}} \rangle &= \frac{1}{2} [\mathcal{J}_{12} - \mathcal{K}_{12} - \mathcal{K}_{12} + \mathcal{J}_{12}] \\ &= \mathcal{J}_{12} - \mathcal{K}_{12} \end{aligned} \tag{1.137}$$

This simple calculation contains the essential physics of Hartree–Fock theory. The Coulomb term would already be present for a product wavefunction. The exchange term appears solely because the wavefunction has been antisymmetrized. For N electrons, the same reasoning applies to each pair of occupied spin orbitals. The $N-2$ spectator overlaps fix all other assignments. The two active orbitals can then be matched directly, giving $\mathcal{J}_{ij}$, or exchanged, giving $-\mathcal{K}_{ij}$. Summing over $i<j$ counts each distinct pair once; a double sum over all i, j instead requires a factor of one half. The terms with $i=j$ vanish because $\mathcal{J}_{ii} = \mathcal{K}_{ii}$.
Combining the one- and two-electron contributions gives the electronic Hartree–Fock energy

$$E_{\mathrm{HF}} = h_{11} + h_{22} + \mathcal{J}_{12} - \mathcal{K}_{12} \tag{1.138}$$

Solution 4 (continued): Two-electron Hamiltonian

If the two occupied spin orbitals have opposite spin, as in a closed-shell two-electron system,

$$\chi_1(\boldsymbol{x}) = \psi(\boldsymbol{r})\alpha(\omega) \qquad \chi_2(\boldsymbol{x}) = \psi(\boldsymbol{r})\beta(\omega) \tag{1.139}$$

then their exchange integral vanishes because $\langle\alpha|\beta\rangle = 0$, i.e., $\mathcal{K}_{12} = 0$. This does not mean that exchange is absent from Hartree–Fock theory; rather, exchange couples only same-spin spin orbitals. The corresponding closed-shell spatial-orbital expression will be derived explicitly in the next subsection.

Exercise 5: Three-electron system

Find the HF energy of a three-electron system composed of the orthonormal spin orbitals χ_1, χ_2, and χ_3.

Solution 5: Three-electron system

Assume that χ_1, χ_2, and χ_3 are orthonormal and that the electronic energy is sought at a fixed nuclear geometry. The normalized determinant is

$$\Psi_{\mathrm{HF}} = \frac{1}{\sqrt{3!}} \begin{vmatrix} \chi_1(1) & \chi_2(1) & \chi_3(1) \\ \chi_1(2) & \chi_2(2) & \chi_3(2) \\ \chi_1(3) & \chi_2(3) & \chi_3(3) \end{vmatrix} \tag{1.140}$$

Its six Hartree products are mutually orthogonal, so their squared coefficients sum to $6/3! = 1$. We may therefore use the one- and two-electron expectation values established in the preceding exercises. The one-electron operator contains one contribution for each electron,

$$\hat{O}_1 = \hat{h}(\boldsymbol{x}_1) + \hat{h}(\boldsymbol{x}_2) + \hat{h}(\boldsymbol{x}_3) \tag{1.141}$$

while the two-electron interaction contains one term for every distinct electron pair,

$$\hat{O}_2 = r_{12}^{-1} + r_{13}^{-1} + r_{23}^{-1} \tag{1.142}$$

There are therefore three one-electron contributions,

$$h_{11} + h_{22} + h_{33} \tag{1.143}$$

and three distinct orbital pairs: (1,2), (1,3), and (2,3). Each pair contributes one Coulomb term and one exchange term. Applying the preceding two-electron result to each pair gives

$$E_{\mathrm{HF}} = h_{11} + h_{22} + h_{33} + \mathcal{J}_{12} + \mathcal{J}_{13} + \mathcal{J}_{23} - \mathcal{K}_{12} - \mathcal{K}_{13} - \mathcal{K}_{23} \tag{1.144}$$

Equivalently,

$$E_{\mathrm{HF}} = \sum_{i=1}^{3} h_{ii} + \sum_{i<j}^{3} (\mathcal{J}_{ij} - \mathcal{K}_{ij}) \tag{1.145}$$

The pattern is now clear: extending the result to an arbitrary number of electrons gives

$$E_{\mathrm{HF}} = \sum_{i=1}^{N} h_{ii} + \sum_{i<j}^{N} (\mathcal{J}_{ij} - \mathcal{K}_{ij}) \tag{1.146}$$

For three electrons, the interaction contains exactly $\binom{3}{2} = 3$ distinct pairs. This count is a useful check: there is neither a self-interaction term nor an additional factor of one half in the explicit pair expression. The spin dependence remains inside the spin-orbital exchange integrals; no particular spin occupation was assumed.

These few-electron examples provide an explicit derivation of the Slater–Condon expressions used above. They also reveal the structure that will remain throughout Hartree–Fock theory: one-electron contributions are associated with individual occupied spin orbitals, while the electron-electron interaction generates pairwise Coulomb and exchange terms.

So far, all expressions have been written in terms of spin orbitals. For closed-shell systems, however, the α and β electrons occur in pairs that share the same spatial orbital. Exploiting this structure leads to the more familiar restricted spatial-orbital formulation developed next.

1.3 Spin-orbital and spatial-orbital formulations

The spin-orbital energy expression is general, but closed-shell calculations are more naturally interpreted in terms of doubly occupied spatial orbitals. This subsection performs that reduction explicitly and then applies it to closed- and open-shell atomic examples.

1.3.1 From spin orbitals to spatial orbitals

The Hartree–Fock energy has so far been written in its most general spin-orbital form,

$$E_{\mathrm{HF}} = \sum_{i=1}^{N} \langle i|\hat{h}|i\rangle + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} [\langle ij|ij\rangle - \langle ij|ji\rangle] \tag{1.147}$$

This expression is compact and applies equally well to restricted and unrestricted determinants. In many molecular applications, however, it is more natural to work with spatial orbitals. The passage from one representation to the other is obtained simply by carrying out the spin integrations explicitly.

The essential ingredients are the orthonormality relations

$$\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1 \qquad\qquad \langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0 \tag{1.148}$$

As we shall see, the first two relations preserve the Coulomb interaction between all electron pairs, whereas the last two eliminate exchange between opposite-spin electrons.

1.3.2 Closed-shell H_2 in a minimal basis

The simplest example is closed-shell H_2 in a minimal basis. There is one doubly occupied spatial orbital, $\psi_1(\boldsymbol{r})$, from which the two occupied spin orbitals are constructed:

$$\begin{aligned} \chi_1(\boldsymbol{x}) &\equiv \psi_1(\boldsymbol{x}) = \psi_1(\boldsymbol{r})\alpha(\omega) \\ \chi_2(\boldsymbol{x}) &\equiv \bar{\psi}_1(\boldsymbol{x}) = \psi_1(\boldsymbol{r})\beta(\omega) \end{aligned} \tag{1.149}$$

The bar therefore denotes the β spin partner of the corresponding α spin orbital; both contain the same spatial function.

In the spin-orbital representation, the Hartree–Fock energy is

$$E_{\mathrm{HF}} = \langle\chi_1|\hat{h}|\chi_1\rangle + \langle\chi_2|\hat{h}|\chi_2\rangle + \langle\chi_1\chi_2|\chi_1\chi_2\rangle - \langle\chi_1\chi_2|\chi_2\chi_1\rangle \tag{1.150}$$

Using the barred notation, the same expression can be written in spatial-orbital-style notation as

$$E_{\mathrm{HF}} = (\psi_1 \,|\, \hat{h} \,|\, \psi_1) + (\bar{\psi}_1 \,|\, \hat{h} \,|\, \bar{\psi}_1) + (\psi_1\psi_1 \,|\, \bar{\psi}_1\bar{\psi}_1) - (\psi_1\bar{\psi}_1 \,|\, \bar{\psi}_1\psi_1) \tag{1.151}$$

We now evaluate each contribution by separating the spin and spatial coordinates.

One-electron contribution. The core Hamiltonian contains no spin dependence. Consequently, its matrix elements factorize into independent spin and spatial integrals. For the α spin orbital,

$$\begin{aligned}(\chi_1 \,|\, \hat{h} \,|\, \chi_1) &= \int \chi_1^*(\boldsymbol{x}) \hat{h}(\boldsymbol{x}) \chi_1(\boldsymbol{x}) \,\mathrm{d}\boldsymbol{x} \\ &= \iint \alpha^*(\omega) \psi_1^*(\boldsymbol{r}) \hat{h}(\boldsymbol{r}) \alpha(\omega) \psi_1(\boldsymbol{r}) \,\mathrm{d}\omega \,\mathrm{d}\boldsymbol{r} \\ &= \underbrace{\left[\int \alpha^*(\omega)\alpha(\omega) \,\mathrm{d}\omega\right]}_{=1} \underbrace{\left[\int \psi_1^*(\boldsymbol{r}) \hat{h}(\boldsymbol{r}) \psi_1(\boldsymbol{r}) \,\mathrm{d}\boldsymbol{r}\right]}_{(\psi_1 \,|\, \hat{h} \,|\, \psi_1)}\end{aligned} \tag{1.152}$$

Thus,

$$(\chi_1 \,|\, \hat{h} \,|\, \chi_1) = (\psi_1 \,|\, \hat{h} \,|\, \psi_1) \tag{1.153}$$

Exactly the same argument applies to the β spin orbital,

$$\begin{aligned}(\chi_2 \,|\, \hat{h} \,|\, \chi_2) &= \int \chi_2^*(\boldsymbol{x}) \hat{h}(\boldsymbol{x}) \chi_2(\boldsymbol{x}) \,\mathrm{d}\boldsymbol{x} \\ &= \iint \beta^*(\omega) \psi_1^*(\boldsymbol{r}) \hat{h}(\boldsymbol{r}) \beta(\omega) \psi_1(\boldsymbol{r}) \,\mathrm{d}\omega \,\mathrm{d}\boldsymbol{r} \\ &= \underbrace{\left[\int \beta^*(\omega)\beta(\omega) \,\mathrm{d}\omega\right]}_{=1} \underbrace{\left[\int \psi_1^*(\boldsymbol{r}) \hat{h}(\boldsymbol{r}) \psi_1(\boldsymbol{r}) \,\mathrm{d}\boldsymbol{r}\right]}_{(\psi_1 \,|\, \hat{h} \,|\, \psi_1)}\end{aligned} \tag{1.154}$$

and therefore

$$(\chi_2 \,|\, \hat{h} \,|\, \chi_2) = (\psi_1 \,|\, \hat{h} \,|\, \psi_1) \tag{1.155}$$

The two electrons thus contribute identically to the one-electron energy:

$$(\chi_1 \,|\, \hat{h} \,|\, \chi_1) + (\chi_2 \,|\, \hat{h} \,|\, \chi_2) = 2(\psi_1 \,|\, \hat{h} \,|\, \psi_1) \tag{1.156}$$

The factor of two is simply the occupation number of the doubly occupied spatial orbital.

Coulomb contribution. The Coulomb term contains only diagonal spin overlaps and therefore survives the spin integration. Explicitly,

$$\begin{aligned}(\chi_1\chi_1 \,|\, \chi_2\chi_2) &= \iint \chi_1^*(\boldsymbol{x}_1) \chi_1(\boldsymbol{x}_1) \frac{1}{r_{12}} \chi_2^*(\boldsymbol{x}_2) \chi_2(\boldsymbol{x}_2) \,\mathrm{d}\boldsymbol{x}_1 \,\mathrm{d}\boldsymbol{x}_2 \\ &= \iiiint \alpha^*(\omega_1) \psi_1^*(\boldsymbol{r}_1) \alpha(\omega_1) \psi_1(\boldsymbol{r}_1) \frac{1}{r_{12}} \\ &\quad \times \beta^*(\omega_2) \psi_1^*(\boldsymbol{r}_2) \beta(\omega_2) \psi_1(\boldsymbol{r}_2) \,\mathrm{d}\omega_1 \,\mathrm{d}\boldsymbol{r}_1 \,\mathrm{d}\omega_2 \,\mathrm{d}\boldsymbol{r}_2 \\ &= \underbrace{\left[\int \alpha^*(\omega_1)\alpha(\omega_1) \,\mathrm{d}\omega_1\right]}_{=1} \underbrace{\left[\int \beta^*(\omega_2)\beta(\omega_2) \,\mathrm{d}\omega_2\right]}_{=1} \\ &\quad \times \underbrace{\left[\iint \psi_1^*(\boldsymbol{r}_1)\psi_1(\boldsymbol{r}_1) \frac{1}{r_{12}} \psi_1^*(\boldsymbol{r}_2)\psi_1(\boldsymbol{r}_2) \,\mathrm{d}\boldsymbol{r}_1 \,\mathrm{d}\boldsymbol{r}_2\right]}_{(\psi_1\psi_1 \,|\, \psi_1\psi_1)}\end{aligned} \tag{1.157}$$

Hence,

$$(\chi_1\chi_1 \,|\, \chi_2\chi_2) = (\psi_1\psi_1 \,|\, \psi_1\psi_1) \tag{1.158}$$

This is the electrostatic repulsion between the two electrons occupying the same spatial orbital. The Coulomb interaction is insensitive to whether the electrons have parallel or antiparallel spins.

Exchange contribution. The exchange term behaves differently because the spin functions are crossed:

$$\begin{aligned}(\chi_1\chi_2 \mid \chi_2\chi_1) &= \iint \chi_1^*(\boldsymbol{x}_1)\chi_2(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_2^*(\boldsymbol{x}_2)\chi_1(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \\ &= \iiiint \alpha^*(\omega_1)\psi_1^*(\boldsymbol{r}_1)\beta(\omega_1)\psi_1(\boldsymbol{r}_1)\frac{1}{r_{12}} \\ &\times \beta^*(\omega_2)\psi_1^*(\boldsymbol{r}_2)\alpha(\omega_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\omega_1\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\omega_2\,\mathrm{d}\boldsymbol{r}_2 \\ &= \underbrace{\left[\int \alpha^*(\omega_1)\beta(\omega_1)\,\mathrm{d}\omega_1\right]}_{=0}\underbrace{\left[\int \beta^*(\omega_2)\alpha(\omega_2)\,\mathrm{d}\omega_2\right]}_{=0} \\ &\times \underbrace{\left[\iint \psi_1^*(\boldsymbol{r}_1)\psi_1(\boldsymbol{r}_1)\frac{1}{r_{12}}\psi_1^*(\boldsymbol{r}_2)\psi_1(\boldsymbol{r}_2)\,\mathrm{d}\boldsymbol{r}_1\,\mathrm{d}\boldsymbol{r}_2\right]}_{(\psi_1\psi_1 \mid \psi_1\psi_1)} \\ &= 0\end{aligned} \tag{1.159}$$

Thus,

$$(\chi_1\chi_2 \mid \chi_2\chi_1) = 0 \tag{1.160}$$

The spatial part of the integral is perfectly finite; it is the *orthogonality of the spin functions* that makes the complete spin-orbital exchange integral vanish.

This is an important point. Opposite-spin electrons still repel each other through the Coulomb interaction, but there is no HF exchange contribution between them. Exchange survives only for pairs of electrons with the same spin.

Combining the preceding results gives the spatial-orbital expression for minimal-basis H_2,

$$E_{\mathrm{HF}} = 2(\psi_1 \mid \hat{h} \mid \psi_1) + (\psi_1\psi_1 \mid \psi_1\psi_1) = 2(1 \mid \hat{h} \mid 1) + (11 \mid 11) \tag{1.161}$$

The simple H_2 example contains the entire logic needed for the general closed-shell case: every occupied spatial orbital contributes twice to the one-electron energy, Coulomb interactions occur between all spin combinations, and exchange occurs only between same-spin electrons.

1.3.3 General closed-shell determinants

Consider now a system containing an even number N of electrons described by a closed-shell RHF determinant. There are $N/2$ occupied spatial orbitals, and each spatial orbital ψ_i generates an α and a β spin orbital, $\psi_i(\boldsymbol{r})\alpha(\omega)$ and $\psi_i(\boldsymbol{r})\beta(\omega)$.

To avoid ambiguity, in the remainder of this derivation the indices $i, j = 1, \ldots, N/2$ label *occupied spatial orbitals*, while $\bar{i}$ denotes the β spin partner of spatial orbital i.

Starting from the general spin-orbital expression,

$$E_{\mathrm{HF}} = \sum_{i=1}^{N}\langle i|\hat{h}|i\rangle + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\langle ij|ij\rangle - \langle ij|ji\rangle\right] \tag{1.162}$$

we now sum explicitly over the two spin components associated with each occupied spatial orbital.

One-electron contribution. The spin-orbital sum can be separated into its α and β components:

$$\begin{aligned}\sum_{i=1}^{N}\langle i|\hat{h}|i\rangle &= \sum_{i=1}^{N/2}\langle i|\hat{h}|i\rangle + \sum_{i=1}^{N/2}\left\langle \bar{i}\middle|\hat{h}\middle|\bar{i}\right\rangle = \sum_{i=1}^{N/2}(i \mid \hat{h} \mid i) + \sum_{i=1}^{N/2}(i \mid \hat{h} \mid i) \\ &= 2\sum_{i=1}^{N/2}(i \mid \hat{h} \mid i)\end{aligned} \tag{1.163}$$

Again, the factor of two simply reflects the double occupation of every occupied spatial orbital.

Two-electron contribution. For every pair of occupied spatial orbitals i and j, four spin combinations are possible: $\alpha\alpha$, $\alpha\beta$, $\beta\alpha$, and $\beta\beta$. Separating these four possibilities gives

$$\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\langle ij|ij\rangle - \langle ij|ji\rangle\right] = \frac{1}{2}\Bigg\{\sum_{i=1}^{N/2}\sum_{j=1}^{N/2}\left(\langle ij|ij\rangle - \langle ij|ji\rangle\right) + \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}\left(\langle \bar{i}j|\bar{i}j\rangle - \langle \bar{i}j|j\bar{i}\rangle\right) + \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}\left(\langle i\bar{j}|i\bar{j}\rangle - \langle i\bar{j}|\bar{j}i\rangle\right) + \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}\left(\langle \bar{i}\bar{j}|\bar{i}\bar{j}\rangle - \langle \bar{i}\bar{j}|\bar{j}\bar{i}\rangle\right)\Bigg\} \tag{1.164}$$

The four Coulomb contributions all survive because they involve diagonal spin overlaps. After integration over spin,

$$\langle ij|ij\rangle = \langle \bar{i}j|\bar{i}j\rangle = \langle i\bar{j}|i\bar{j}\rangle = \langle \bar{i}\bar{j}|\bar{i}\bar{j}\rangle = (ii\,|\,jj) \tag{1.165}$$

Exchange behaves differently. The two same-spin combinations survive,

$$\langle ij|ji\rangle = \langle \bar{i}\bar{j}|\bar{j}\bar{i}\rangle = (ij\,|\,ji) \tag{1.166}$$

whereas the two opposite-spin exchange integrals vanish,

$$\langle \bar{i}j|j\bar{i}\rangle = \langle i\bar{j}|\bar{j}i\rangle = 0 \tag{1.167}$$

For each pair of spatial orbitals, the four spin combinations therefore produce

$$\frac{1}{2}[4(ii\,|\,jj) - 2(ij\,|\,ji)] = 2(ii\,|\,jj) - (ij\,|\,ji) \tag{1.168}$$

Consequently,

$$\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}[\langle ij|ij\rangle - \langle ij|ji\rangle] = \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}[2(ii\,|\,jj) - (ij\,|\,ji)] \tag{1.169}$$

Combining the one- and two-electron contributions gives the familiar closed-shell RHF energy,

$$E_{\mathrm{HF}} = 2\sum_{i=1}^{N/2}(i\,|\,\hat{h}\,|\,i) + \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}[2(ii\,|\,jj) - (ij\,|\,ji)] \tag{1.170}$$

Introducing the spatial-orbital Coulomb and exchange integrals

$$J_{ij} \equiv (ii\,|\,jj) \qquad\qquad K_{ij} \equiv (ij\,|\,ji) \tag{1.171}$$

the same expression takes the compact form

$$\boxed{E_{\mathrm{HF}} = 2\sum_{i=1}^{N/2} h_{ii} + \sum_{i=1}^{N/2}\sum_{j=1}^{N/2}\left(2J_{ij} - K_{ij}\right)} \tag{1.172}$$

The origin of the numerical factors is now transparent. Each occupied spatial orbital contains two electrons, giving the factor of two in the one-electron contribution. For each pair of spatial orbitals, all four spin combinations contribute to the Coulomb interaction, whereas only the two same-spin combinations contribute to exchange. After the overall factor $1/2$ removing double counting, the result is precisely

$$2J_{ij} - K_{ij} \tag{1.173}$$

Notice also that the terms with $i = j$ are perfectly well behaved:

$$2J_{ii} - K_{ii} = J_{ii} \tag{1.174}$$

since $J_{ii} = K_{ii}$. Thus, a doubly occupied spatial orbital contains one physical Coulomb repulsion between its α and β electrons; the spurious same-spin self-interactions cancel through exchange.

Exercise 6: HF energy of the Li atom

Find the HF energy of the Li atom with occupation $\psi_1^2\psi_2^1$ in terms of spatial orbitals.

Solution 6: HF energy of the Li atom

For the determinant considered here, the three occupied spin orbitals are

$$\chi_1 = \alpha\psi_1 \qquad \chi_2 = \beta\psi_1 \qquad \chi_3 = \alpha\psi_2 \tag{1.175}$$

The first spatial orbital is therefore doubly occupied, whereas the second contains a single α electron. Thus $N^\alpha = 2$ and $N^\beta = 1$, with orthonormal spatial orbitals ψ_1 and ψ_2. The closed-shell core uses the same spatial function for its two spins.
The one-electron contribution is immediately

$$E_1 = 2h_{11} + h_{22} \tag{1.176}$$

There are three electron pairs. The two electrons in ψ_1 have opposite spin and contribute J_{11}. The α electron in ψ_2 interacts Coulombically with both electrons in ψ_1, giving $2J_{12}$. Exchange occurs only with the α electron in ψ_1, and therefore contributes $-K_{12}$. Explicitly, the three spin-orbital pair contributions reduce to

$$\mathcal{J}_{12} - \mathcal{K}_{12} = J_{11} \tag{1.177}$$
$$\mathcal{J}_{13} - \mathcal{K}_{13} = J_{12} - K_{12} \tag{1.178}$$
$$\mathcal{J}_{23} - \mathcal{K}_{23} = J_{12} \tag{1.179}$$

The indices on $\mathcal{J}$ and $\mathcal{K}$ label the three spin orbitals, whereas those on J and K label spatial orbitals. In particular, the surviving J_{11} is the repulsion between two different, opposite-spin electrons, not a one-electron self-interaction. The total energy is consequently

$$E_{\mathrm{HF}} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12} \tag{1.180}$$

This example provides a useful way of reading spatial-orbital HF energies: Coulomb terms count interacting electron pairs, whereas exchange terms count same-spin pairs.

Exercise 7: HF energy of the B atom

Find the HF energy of the B atom with occupation $\psi_1^2\psi_2^2\psi_3^1$ in terms of spatial orbitals.

Solution 7: HF energy of the B atom

For the occupation considered here, the first two spatial orbitals are doubly occupied and the third contains a single α electron, corresponding schematically to

$$\psi_1^2\psi_2^2\psi_3^1 \tag{1.181}$$

Solution 7 (continued): HF energy of the B atom

The five occupied spin orbitals are $\alpha\psi_1$, $\beta\psi_1$, $\alpha\psi_2$, $\beta\psi_2$, and $\alpha\psi_3$, so $N^\alpha = 3$ and $N^\beta = 2$. As in the lithium exercise, the doubly occupied orbitals share their spatial parts between the two spins. The one-electron contribution is therefore

$$E_1 = 2h_{11} + 2h_{22} + h_{33} \tag{1.182}$$

The two electrons occupying ψ_1 contribute J_{11}, while those in ψ_2 contribute J_{22}. Between the two doubly occupied orbitals there are four Coulomb pairs but only two same-spin exchange pairs, giving

$$4J_{12} - 2K_{12} \tag{1.183}$$

The singly occupied ψ_3 orbital interacts with both electrons in each doubly occupied orbital. Hence, $2J_{13} - K_{13}$ arises from the interaction between ψ_1 and ψ_3, and $2J_{23} - K_{23}$ from the interaction between ψ_2 and ψ_3.
Collecting all terms gives

$$E_{\mathrm{HF}} = 2h_{11} + 2h_{22} + h_{33} + J_{11} + 4J_{12} + J_{22} - 2K_{12} + 2J_{13} + 2J_{23} - K_{13} - K_{23} \tag{1.184}$$

There are $\binom{5}{2} = 10$ Coulomb pairs: one within each doubly occupied orbital, four between these two orbitals, and two between each of them and ψ_3. Exchange survives for $\binom{N^\alpha}{2} + \binom{N^\beta}{2} = 3 + 1 = 4$ pairs, explaining the coefficients 2, 1, and 1 of K_{12}, K_{13}, and K_{23}. The pair counts provide an independent check on the spatial-orbital result.

The Li and B examples illustrate that the reduction from spin orbitals to spatial orbitals is not restricted to closed shells. The closed-shell formula in Eq. (1.170) benefits from the particularly simple double-occupation pattern of RHF, but for a general determinant the same procedure can always be carried out by keeping track of which spatial orbitals are occupied by α and β electrons.

We now return to spin orbitals so that individual substitutions can be specified without ambiguity. The spin selection rules just derived continue to determine which exchange matrix elements survive when occupied orbitals are replaced by virtual ones.

1.4 Excited determinants and Hamiltonian matrix elements

The Hartree–Fock determinant also provides a reference from which alternative occupation patterns can be generated. The excitation rank of these determinants determines which Hamiltonian matrix elements can be nonzero, leading directly to the Slater–Condon rules and Brillouin's theorem.

1.4.1 Excited determinants

The Hartree–Fock solution provides not only an energy but also a complete orthonormal set of occupied and virtual spin orbitals. Once this orbital basis has been obtained, many other Slater determinants can be generated by replacing one or more occupied spin orbitals of the Hartree–Fock determinant by virtual ones. These determinants play a central role in configuration-interaction [16], coupled-cluster [17, 18], perturbation [19, 20], and excited-state theories.

We therefore introduce the language of orbital substitutions, or excitations, relative to the Hartree–Fock reference determinant.

Reference determinant. For a conventional ground-state HF solution, the electrons occupy the N lowest spin orbitals according to the Aufbau principle. The reference determinant is

$$|\Psi_0\rangle = \left|\chi_1 \cdots \chi_i \chi_j \cdots \chi_N\right\rangle \tag{1.185}$$

Here and in the following, $i, j, k, l, \ldots$ and $a, b, c, d, \ldots$ denote occupied and virtual spin orbitals, respectively.

The distinction is always made *relative to the reference determinant*: an occupied orbital contains an electron in $|\Psi_0\rangle$, whereas a virtual orbital does not. Figure 4 illustrates this partition of the spin-orbital space into occupied and virtual levels.

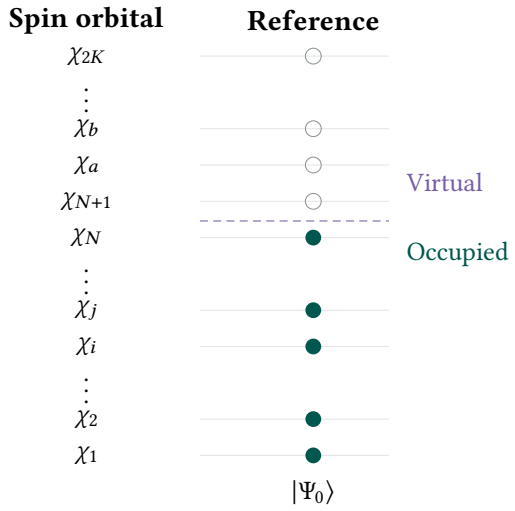


Figure 4: Reference Hartree–Fock determinant constructed from the occupied spin orbitals. Filled green circles indicate occupied spin orbitals; empty circles indicate virtual ones. Adapted from Szabo and Ostlund [21].

Because the spin orbitals form an orthonormal set, the occupied and virtual subspaces are mutually orthogonal,

$$\langle \chi_i | \chi_a \rangle = 0 \tag{1.186}$$

This simple property will have important consequences for matrix elements between different determinants.

Singly excited determinants. A singly excited determinant is obtained by removing an electron from an occupied spin orbital i and placing it in a virtual spin orbital a. Symbolically,

$$i \longrightarrow a \tag{1.187}$$

The resulting determinant is denoted

$$\left|\Psi_i^a\right\rangle \tag{1.188}$$

In determinant notation,

$$\left|\Psi_i^a\right\rangle = \left|\chi_1 \cdots \chi_a \chi_j \cdots \chi_N\right\rangle \tag{1.189}$$

Thus, $\left|\Psi_i^a\right\rangle$ differs from the reference determinant by exactly one spin orbital.

The same substitution can be expressed particularly compactly using creation and annihilation operators,

$$\left|\Psi_i^a\right\rangle = \hat{a}_a^\dagger \hat{a}_i \, |\Psi_0\rangle \tag{1.190}$$

The annihilation operator $\hat{a}_i$ first removes the electron occupying spin orbital i, while the creation operator $\hat{a}_a^\dagger$ places an electron in the previously unoccupied spin orbital a.

It is useful to interpret this substitution as the simultaneous creation of a hole in the occupied orbital i and a particle in the virtual orbital a:

$$i \longrightarrow a \qquad \Longleftrightarrow \qquad \text{one hole } i + \text{one particle } a \tag{1.191}$$

This particle-hole language will reappear naturally in excited-state and many-body formulations.

Because one occupied orbital has been replaced by an orthogonal virtual orbital,

$$\left\langle \Psi_0 \middle| \Psi_i^a \right\rangle = 0 \tag{1.192}$$

Thus, every singly excited determinant is orthogonal to the Hartree–Fock reference.

Doubly excited determinants. A doubly excited determinant is obtained by replacing two occupied spin orbitals, i and j, by two virtual spin orbitals, a and b:

$$ij \longrightarrow ab \tag{1.193}$$

In determinant notation,

$$|\Psi_{ij}^{ab}\rangle = |\chi_1 \cdots \chi_a \chi_b \cdots \chi_N\rangle \tag{1.194}$$

In second-quantized notation,

$$|\Psi_{ij}^{ab}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i \, |\Psi_0\rangle \tag{1.195}$$

The determinant now differs from the reference by two spin orbitals and contains two holes, i and j, together with two particles, a and b.

The ordering of the fermionic creation and annihilation operators matters: interchanging two such operators changes the sign because

$$\hat{a}_p^\dagger \hat{a}_q^\dagger = -\hat{a}_q^\dagger \hat{a}_p^\dagger \qquad\qquad \hat{a}_p \hat{a}_q = -\hat{a}_q \hat{a}_p \tag{1.196}$$

A consistent ordering convention therefore fixes the overall phase of each excited determinant. This phase has no effect on diagonal expectation values, but it is essential when evaluating off-diagonal matrix elements between different determinants.

Figure 5 compares the holes and particles created by single and double substitutions relative to the reference occupation.

Excitation rank. The construction extends immediately to higher substitutions. A triple excitation replaces three occupied orbitals,

$$ijk \longrightarrow abc \tag{1.197}$$

and may be written

$$|\Psi_{ijk}^{abc}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j \hat{a}_i \, |\Psi_0\rangle \tag{1.198}$$

More generally, an m-fold excited determinant differs from the reference by m occupied-to-virtual substitutions. The integer m is called the excitation rank:

$$\begin{aligned} m = 0 &: \text{reference determinant} \\ m = 1 &: \text{single excitation} \\ m = 2 &: \text{double excitation} \\ m = 3 &: \text{triple excitation} \\ &\vdots \end{aligned} \tag{1.199}$$

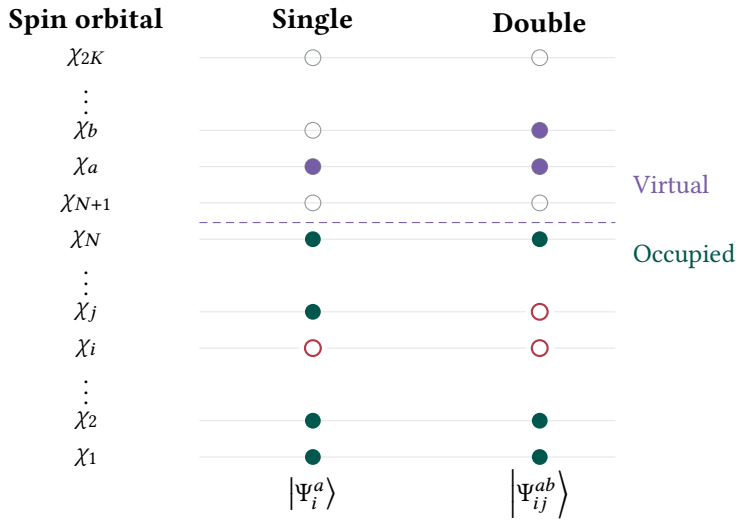


Figure 5: Singly excited (left) and doubly excited (right) determinants obtained by replacing occupied spin orbitals of the Hartree–Fock reference by virtual ones. Filled green circles indicate unchanged occupations, violet circles promoted electrons, and red outlines holes. Empty circles are unoccupied; unshown occupations remain unchanged. Adapted from Szabo and Ostlund [21].

With N occupied spin orbitals and $2K - N$ virtual spin orbitals, the number of singly excited determinants is

$$N_{\mathrm{S}} = N(2K - N) \tag{1.200}$$

whereas the number of doubly excited determinants is

$$N_{\mathrm{D}} = \binom{N}{2}\binom{2K - N}{2} \tag{1.201}$$

The number of possible determinants therefore grows extremely rapidly with the size of the orbital space. This combinatorial growth is one of the fundamental computational difficulties encountered by correlated wavefunction methods.

Excited determinants are not necessarily excited states. An important distinction must be made at this point. The terminology "singly excited determinant" describes how a determinant is constructed *relative to the HF reference*; it does not imply that this determinant is itself an eigenfunction of the electronic Hamiltonian.

In general,

$$\hat{\mathcal{H}}_{\mathrm{elec}} |\Psi_i^a\rangle \neq E_i^a |\Psi_i^a\rangle \tag{1.202}$$

and similarly for doubly and higher excited determinants. The exact excited states are, in general, linear combinations of many determinants,

$$|\Psi_I\rangle = C_0^{(I)} |\Psi_0\rangle + \sum_{ia} C_i^{a,(I)} |\Psi_i^a\rangle + \frac{1}{4}\sum_{ijab} C_{ij}^{ab,(I)} |\Psi_{ij}^{ab}\rangle + \cdots \tag{1.203}$$

The determinants introduced here should therefore be viewed as a convenient many-electron basis built from the Hartree–Fock orbitals.

This distinction is central to post-Hartree–Fock theory: the reference and its excited determinants provide the building blocks, while the electronic Hamiltonian determines how these configurations interact and combine.

Why excitation rank matters. Two Slater determinants may differ by zero, one, two, or more spin orbitals. For Hamiltonians containing only one- and two-electron operators, the matrix element between two determinants depends very strongly on this number.

Schematically,

number of orbital substitutions	$\Longrightarrow$	structure of the Hamiltonian matrix element

A one-electron operator can connect determinants differing by at most one spin orbital, whereas a two-electron operator can connect determinants differing by at most two. Consequently, determinants separated by three or more substitutions have a vanishing Hamiltonian matrix element.

These selection rules are precisely the content of the Slater–Condon rules, to which we now turn.

1.5 Slater–Condon rules and Brillouin's theorem

The determinants introduced in the previous subsection form an orthonormal many-electron basis constructed from a common set of orthonormal spin orbitals. The next question is therefore how the electronic Hamiltonian acts in this determinant basis.

At first sight, evaluating a matrix element between two N-electron Slater determinants might appear to require an explicit expansion of both determinants into their $N!$ Hartree products. The Slater–Condon rules [2, 22] avoid this combinatorial expansion. They express matrix elements of one- and two-electron operators directly in terms of one- and two-electron orbital integrals.

Historical perspective: Edward Uhler Condon (1902–1974)

Condon was an American theoretical physicist whose work connected the new quantum mechanics with atomic and molecular spectroscopy. His treatment of complex spectra contributed to the determinant matrix-element rules now named after Slater and Condon [22]. He worked at Princeton and Westinghouse before directing the US National Bureau of Standards from 1945 to 1951, and later held academic appointments at Washington University and the University of Colorado.

Biography: American Institute of Physics, Condon papers; NIST historical record. Source portrait: National Bureau of Standards, 1947, via Wikimedia Commons; US federal-government photograph, public domain. Sketch: AI-assisted adaptation of the credited portrait.

The central principle is simple: the value of a matrix element depends only on the number of spin orbitals by which the two determinants differ. This number is often called their degree of substitution. Because a one-electron operator acts on only one electron at a time, it can connect determinants differing by at most one spin orbital. Similarly, a two-electron operator can connect determinants differing by at most two spin orbitals.

Throughout this subsection, we assume that the determinants are written with a consistent orbital ordering so that their relative fermionic phase has already been fixed.

1.5.1 Matrix elements of one-electron operators

Consider a general one-electron operator,

$$\hat{O}_1 = \sum_{i=1}^{N} \hat{h}(\boldsymbol{x}_i) \tag{1.204}$$

The Slater–Condon rule depends on how many spin orbitals differ between the two determinants.

Case 1: determinants differ by zero spin orbitals. If the two determinants are identical,

$$|K\rangle = |\cdots ij\cdots\rangle \tag{1.205}$$

then the matrix element is simply the sum of the one-electron expectation values over all occupied spin orbitals:

$$\langle K|\hat{O}_1|K\rangle = \sum_{i\in K} \langle i|\hat{h}|i\rangle \tag{1.206}$$

For the Hartree–Fock reference,

$$\langle \Psi_0|\hat{O}_1|\Psi_0\rangle = \sum_{i=1}^{N} h_{ii} \tag{1.207}$$

This is the result already encountered in the derivation of the HF energy: each occupied spin orbital contributes one diagonal matrix element of the one-electron operator.

Case 2: determinants differ by one spin orbital. Consider

$$|K\rangle = |\cdots ij\cdots\rangle \qquad\qquad |L\rangle = |\cdots kj\cdots\rangle \tag{1.208}$$

All occupied spin orbitals are common to the two determinants except for the replacement

$$i \longleftrightarrow k \tag{1.209}$$

The spectator orbitals integrate to unity, while orthogonality eliminates all terms in which the one-electron operator acts on a common orbital. Consequently,

$$\langle K|\hat{O}_1|L\rangle = \langle i|\hat{h}|k\rangle \tag{1.210}$$

For example, between the HF reference and a singly excited determinant,

$$\langle \Psi_0|\hat{O}_1|\Psi_i^a\rangle = h_{ia} \tag{1.211}$$

Thus a one-electron operator can directly remove an electron from one spin orbital and place it in another.

Case 3: determinants differ by two or more spin orbitals. Let

$$|K\rangle = |\cdots ij\cdots\rangle \qquad\qquad |L\rangle = |\cdots kl\cdots\rangle \tag{1.212}$$

so that at least two occupied spin orbitals differ. Since $\hat{O}_1$ can act on only one electron, it cannot simultaneously repair both substitutions. At least one overlap between different orthogonal spin orbitals therefore remains, and the matrix element vanishes:

$$\langle K|\hat{O}_1|L\rangle = 0 \tag{1.213}$$

In particular,

$$\langle \Psi_0|\hat{O}_1|\Psi_{ij}^{ab}\rangle = 0 \tag{1.214}$$

More generally,

$$\boxed{\hat{O}_1 \quad \text{connects determinants differing by at most one spin orbital}} \tag{1.215}$$

1.5.2 Matrix elements of two-electron operators

Consider now the electron-electron interaction,

$$\hat{O}_2 = \sum_{i<j}^{N} \frac{1}{r_{ij}} \tag{1.216}$$

Because this operator acts simultaneously on two electrons, its connectivity extends one excitation rank further than that of a one-electron operator.

Case 1: determinants differ by zero spin orbitals. For identical determinants,

$$\langle K|\hat{O}_2|K\rangle = \frac{1}{2}\sum_{i\in K}\sum_{j\in K} \langle ij||ij\rangle \tag{1.217}$$

The antisymmetrized integral

$$\langle ij||ij\rangle = \langle ij|ij\rangle - \langle ij|ji\rangle \tag{1.218}$$

contains the Coulomb and exchange contributions associated with the occupied pair (i, j). The factor $1/2$ removes the double counting of electron pairs.

For the HF reference,

$$\langle \Psi_0|\hat{O}_2|\Psi_0\rangle = \frac{1}{2}\sum_{ij}^{N} \langle ij||ij\rangle \tag{1.219}$$

Case 2: determinants differ by one spin orbital. Consider again

$$|K\rangle = |\cdots ij\cdots\rangle \qquad\qquad |L\rangle = |\cdots kj\cdots\rangle \tag{1.220}$$

One electron undergoes the substitution $i \longrightarrow k$, while the second electron acted upon by $\hat{O}_2$ may occupy any spin orbital common to the two determinants. Summing over these spectator occupied orbitals gives

$$\langle K|\hat{O}_2|L\rangle = \sum_{j\in K\cap L} \langle ij||kj\rangle \tag{1.221}$$

For the reference and a singly excited determinant,

$$\langle \Psi_0|\hat{O}_2|\Psi_i^a\rangle = \sum_{k}^{N} \langle ik||ak\rangle \tag{1.222}$$

Physically, the promoted electron is coupled through the two-electron interaction to each of the remaining occupied electrons.

Case 3: determinants differ by two spin orbitals. Let

$$|K\rangle = |\cdots ij\cdots\rangle \qquad\qquad |L\rangle = |\cdots kl\cdots\rangle \tag{1.223}$$

The two-electron operator can now act precisely on the two orbitals that differ, giving

$$\langle K|\hat{O}_2|L\rangle = \langle ij||kl\rangle \tag{1.224}$$

Thus,

$$\langle \Psi_0|\hat{O}_2|\Psi_{ij}^{ab}\rangle = \langle ij||ab\rangle \tag{1.225}$$

This result is particularly important in correlated wavefunction theory: the two-electron interaction couples the HF reference directly to doubly excited determinants.

Case 4: determinants differ by three or more spin orbitals. A two-electron operator cannot change three independent orbital occupations simultaneously. Therefore, $\langle K|\hat{O}_2|L\rangle = 0$ if K and L differ by three or more spin orbitals.

Hence,

$$\boxed{\hat{O}_2 \quad \text{connects determinants differing by at most two spin orbitals}} \tag{1.226}$$

1.5.3 Matrix elements of the electronic Hamiltonian

Because

$$\hat{\mathcal{H}}_{\text{elec}} = \hat{O}_1 + \hat{O}_2 \tag{1.227}$$

the preceding results may be summarized directly for the electronic Hamiltonian.

For identical determinants,

$$\langle K|\hat{\mathcal{H}}_{\text{elec}}|K\rangle = \sum_{i\in K} h_{ii} + \frac{1}{2}\sum_{i,j\in K} \langle ij||ij\rangle \tag{1.228}$$

For determinants differing by a single substitution $i \longrightarrow a$,

$$\langle \Psi_0|\hat{\mathcal{H}}_{\text{elec}}|\Psi_i^a\rangle = h_{ia} + \sum_{k}^{N} \langle ik||ak\rangle \tag{1.229}$$

For determinants differing by a double substitution $ij \longrightarrow ab$,

$$\langle \Psi_0|\hat{\mathcal{H}}_{\text{elec}}|\Psi_{ij}^{ab}\rangle = \langle ij||ab\rangle \tag{1.230}$$

Finally,

$$\langle \Psi_0|\hat{\mathcal{H}}_{\text{elec}}|\Psi_{ijk}^{abc}\rangle = 0 \tag{1.231}$$

and similarly for all higher excitation ranks.

The Hamiltonian matrix in a determinant basis is therefore highly structured rather than dense: two determinants can interact directly only if they differ by at most two spin orbitals. This sparsity is a direct consequence of the fact that the nonrelativistic electronic Hamiltonian contains at most two-electron operators.

Excited determinant versus excited HF solution. The terminology "excited determinant" must be distinguished carefully from a separately optimized excited-state Hartree–Fock solution.

An excited determinant is an orbital substitution within a fixed set of reference orbitals. It is a valid many-electron basis function and serves as a building block for configuration-interaction and coupled-cluster expansions [16, 18]. In general, however, an excited determinant is not necessarily a separately optimized HF state, a stationary point of the HF energy, or an eigenfunction of $\hat{S}^2$. Thus,

$$\boxed{\text{orbital excitation} \neq \text{state-specific orbital optimization}} \tag{1.232}$$

In the first procedure, the reference orbitals remain frozen and only their occupation pattern changes. In the second, the orbitals themselves are reoptimized for the targeted electronic state. The distinction is important because orbital relaxation can be substantial, particularly for states with a qualitatively different electronic distribution from the ground state.

1.5.4 Brillouin's theorem

For a reference and a singly excited determinant, the Slater–Condon rules give

$$\left\langle \Psi_0 \middle| \hat{\mathcal{H}}_{\text{elec}} \middle| \Psi_i^a \right\rangle = h_{ia} + \sum_k^N \langle ik||ak\rangle \tag{1.233}$$

The right-hand side is precisely an occupied-virtual matrix element of the Fock operator, introduced here in anticipation of its variational derivation in Section 1.7.1:

$$f_{ia} = h_{ia} + \sum_k^N \langle ik||ak\rangle \tag{1.234}$$

Brillouin's theorem

Léon Brillouin

Source portrait: Benjamin Couprie, Institut International de Physique Solvay, 1927 Solvay photograph (crop: Setreset), public domain. Sketch: AI-assisted adaptation of the credited portrait.

At a stationary Hartree–Fock determinant, the Hamiltonian matrix elements to singly substituted determinants vanish for the occupied-virtual rotations admitted by the chosen variational space. Stationarity, not the requirement of an energy minimum, is the essential assumption.

At a stationary Hartree–Fock solution, occupied-virtual orbital rotations cannot lower the energy to first order. The corresponding Fock matrix elements therefore vanish:

$$f_{ia} = 0 \tag{1.235}$$

Hence,

$$\left\langle \Psi_0 \middle| \hat{\mathcal{H}}_{\text{elec}} \middle| \Psi_i^a \right\rangle = h_{ia} + \sum_k^N \langle ik||ak\rangle = f_{ia} = 0 \tag{1.236}$$

This result is Brillouin's theorem [23]. The derivation of the HF equations below will also establish the stationarity condition used here.

Historical perspective: Léon Nicolas Brillouin (1889–1969)

A French theoretical physicist, Brillouin held chairs in Paris before continuing his career in the United States. His research encompassed wave propagation, the electronic structure of solids, and the relation between information and thermodynamics. His work on self-consistent fields led to the stationarity result discussed here [23]; his name is also associated with Brillouin zones and the semiclassical WKB approximation.

Biography: L. Hilleth Thomas, NAS memoir. Source portrait: Benjamin Couprie, Institut International de Physique Solvay, 1927 Solvay photograph, crop by Setreset; public domain. Sketch: AI-assisted adaptation of the credited portrait.

Brillouin's theorem does not state that singly excited determinants are unimportant. Rather, it states that a *stationary HF determinant does not couple directly to singles through the Hamiltonian.* This has several important consequences. In a nondegenerate perturbative expansion about a canonical HF reference, the direct first-order mixing of singles with the reference vanishes. Singles can nevertheless enter indirectly through coupling to other configurations; the theorem does not remove them from correlated wavefunctions.

The theorem also provides a useful physical interpretation of HF stationarity: the orbitals have already been optimized so that infinitesimal occupied-virtual mixing produces no first-order lowering of the energy.

1.6 Spin symmetry and spin-adapted determinants

1.6.1 Spin quantum numbers and collinear determinants

Excited determinants also provide a useful setting in which to distinguish the spin projection M_S from the total spin S.

Within the nonrelativistic spin-free framework, the electronic Hamiltonian commutes with both the total-spin operator and its projection along the chosen quantization axis,

$$\left[\hat{\mathcal{H}}, \hat{\mathcal{S}}^2\right] = 0 \qquad \left[\hat{\mathcal{H}}, \hat{\mathcal{S}}_z\right] = 0 \tag{1.237}$$

The exact eigenfunctions of $\hat{\mathcal{H}}$ may therefore be chosen as simultaneous eigenfunctions of $\hat{\mathcal{S}}^2$ and $\hat{\mathcal{S}}_z$:

$$\hat{\mathcal{S}}^2 \left|S, M_S\right\rangle = S(S+1) \left|S, M_S\right\rangle \qquad \hat{\mathcal{S}}_z \left|S, M_S\right\rangle = M_S \left|S, M_S\right\rangle \tag{1.238}$$

The quantum number S determines the spin multiplicity $2S + 1$. Thus,

$$S = 0, \quad \frac{1}{2}, \quad 1, \quad \frac{3}{2}, \ldots \tag{1.239}$$

correspond to singlets, doublets, triplets, quartets, and so forth. For a given value of S, the spin projection takes the values

$$M_S = -S, -S+1, \ldots, S-1, S \tag{1.240}$$

For a collinear determinant containing N^α spin-up and N^β spin-down electrons,

$$\hat{\mathcal{S}}_z \left|\Psi\right\rangle = M_S \left|\Psi\right\rangle \qquad M_S = \frac{N^\alpha - N^\beta}{2} \tag{1.241}$$

Here, *collinear* means that all occupied spin orbitals are expressed with respect to the same spin quantization axis.

Every RHF or collinear UHF determinant therefore has a definite value of M_S. Having a definite M_S, however, does *not* imply having a definite total spin S.

1.6.2 Spin purity and spin contamination

Closed-shell RHF. For a closed-shell RHF determinant,

$$N^\alpha = N^\beta = \frac{N}{2} \qquad M_S = 0 \tag{1.242}$$

Because every spatial orbital is occupied by one α and one β electron, the determinant is a pure singlet:

$$\hat{\mathcal{S}}^2 \left|\Psi_{\mathrm{RHF}}\right\rangle = 0 \tag{1.243}$$

Thus, a closed-shell RHF determinant is simultaneously an eigenfunction of $\hat{\mathcal{S}}_z$ and $\hat{\mathcal{S}}^2$.

Open-shell UHF. For UHF, the α and β electrons are allowed to occupy different spatial orbitals. The determinant remains an eigenfunction of $\hat{S}_z$, but, in general,

$$\hat{S}^2 \left|\Psi_{\mathrm{UHF}}\right\rangle \neq S(S+1) \left|\Psi_{\mathrm{UHF}}\right\rangle \tag{1.244}$$

Hence, every collinear RHF or UHF determinant has definite M_S, but only special determinants have definite S.

For $N^\alpha \geq N^\beta$, the expectation value of $\hat{S}^2$ for a UHF determinant is

$$\left\langle \hat{S}^2 \right\rangle_{\mathrm{UHF}} = M_S(M_S+1) + N^\beta - \sum_{i=1}^{N^\alpha} \sum_{j=1}^{N^\beta} \left| S_{ij}^{\alpha\beta} \right|^2 \qquad \underbrace{S_{ij}^{\alpha\beta}}_{\text{overlap}} = \left\langle \psi_i^\alpha \middle| \psi_j^\beta \right\rangle \tag{1.245}$$

If the α and β spatial orbitals coincide pairwise, the overlap terms take their maximal values and the spin-pure restricted result is recovered. As the two spin manifolds relax independently, the value of $\langle \hat{S}^2 \rangle$ may depart from the desired eigenvalue.

For a target high-spin state with $S = M_S$, the spin contamination may be quantified as

$$\Delta S^2 = \left\langle \hat{S}^2 \right\rangle_{\mathrm{UHF}} - \left\langle \hat{S}^2 \right\rangle_{\mathrm{exact}} \geq 0 \tag{1.246}$$

More explicitly, a singlet may be contaminated by triplets, quintets, and higher-spin states; a doublet may be contaminated by quartets, sextets, and higher-spin states; and a triplet may be contaminated by quintets, septets, and higher-spin states. Restoring spin symmetry generally requires spin projection or a linear combination of determinants. The last point is particularly important: a linear combination of determinants can possess a well-defined total spin even when the individual determinants do not.

1.6.3 Spin adaptation of singly excited determinants

Consider a closed-shell singlet reference and promote one electron from the doubly occupied spatial orbital i to the virtual spatial orbital a. There are two distinct $M_S = 0$ determinants,

$$\left|D_\alpha\right\rangle = \hat{a}_{a\alpha}^\dagger \hat{a}_{i\alpha} \left|\Psi_0\right\rangle \qquad \left|D_\beta\right\rangle = \hat{a}_{a\beta}^\dagger \hat{a}_{i\beta} \left|\Psi_0\right\rangle \tag{1.247}$$

The first promotes the α electron and leaves the β electron in orbital i, whereas the second promotes the β electron. Both determinants contain one α and one β electron in the two open-shell orbitals. Consequently,

$$\hat{S}_z \left|D_\sigma\right\rangle = 0 \tag{1.248}$$

and each determinant has

$$\langle D_\sigma | \hat{S}^2 | D_\sigma \rangle = 1 \tag{1.249}$$

The value 1 is neither the singlet eigenvalue 0 nor the triplet eigenvalue 2. Therefore, neither determinant is individually an eigenfunction of $\hat{S}^2$.

Figure 6 compares the occupations of the two open-shell determinants.

Figure 6: Two spin-conserving single substitutions from a closed-shell reference. The determinants $|D_\alpha\rangle$ and $|D_\beta\rangle$ each have $M_S = 0$ but are not individually eigenfunctions of $\hat{S}^2$.

Each determinant has $M_S = 0$, but contains equal singlet and triplet character. This provides a concrete demonstration that being an eigenfunction of $\hat{S}_z$ does not imply being an eigenfunction of $\hat{S}^2$. In particular, an open-shell determinant need not be a pure spin function.

In the determinant basis $\{|D_\alpha\rangle, |D_\beta\rangle\}$, the matrix representation of $\hat{S}^2$ is

$$[\hat{S}^2]_{\{D_\alpha, D_\beta\}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{1.250}$$

The determinant basis is therefore not spin adapted. Diagonalizing this 2×2 matrix produces eigenvectors with definite values of S. The symmetric combination has eigenvalue zero,

$$|{}^1\Psi_i^a\rangle = \frac{|D_\alpha\rangle + |D_\beta\rangle}{\sqrt{2}} \qquad \hat{S}^2 |{}^1\Psi_i^a\rangle = 0 \tag{1.251}$$

and is therefore a pure singlet $S = 0$. The antisymmetric combination has eigenvalue two,

$$|{}^{3,0}\Psi_i^a\rangle = \frac{|D_\alpha\rangle - |D_\beta\rangle}{\sqrt{2}} \tag{1.252}$$

with

$$\hat{S}^2 |{}^{3,0}\Psi_i^a\rangle = 2 |{}^{3,0}\Psi_i^a\rangle \tag{1.253}$$

and therefore $S = 1$. It is the $M_S = 0$ component of a triplet. Thus, spin adaptation is the diagonalization of $\hat{S}^2$ in a determinant space.

Spatial and spin symmetry. The same result can be understood directly from the permutation symmetry of the two open-shell electrons. The following factors display only these two electrons; a common closed-shell core and antisymmetrization with that core are implicit for a larger system. Here ψ_i and ψ_a denote the two spatial molecular orbitals, not members of the AO basis.

For two open-shell spatial orbitals i and a, the singlet may be written as

$$|{}^1\Psi_i^a\rangle = \frac{1}{2} \underbrace{[|\psi_i(1)\psi_a(2)\rangle + |\psi_a(1)\psi_i(2)\rangle]}_{\text{spatially symmetric}} \underbrace{[\alpha(1)\beta(2) - \beta(1)\alpha(2)]}_{\text{spin antisymmetric}} \tag{1.254}$$

The spatial factor is symmetric under exchange of the two electrons, whereas the spin factor is antisymmetric. Their product is therefore antisymmetric, as required for fermions. For the $M_S = 0$ triplet,

$$|{}^{3,0}\Psi_i^a\rangle = \frac{1}{2} \underbrace{[|\psi_i(1)\psi_a(2)\rangle - |\psi_a(1)\psi_i(2)\rangle]}_{\text{spatially antisymmetric}} \underbrace{[\alpha(1)\beta(2) + \beta(1)\alpha(2)]}_{\text{spin symmetric}} \tag{1.255}$$

Here the spatial factor is antisymmetric and the spin factor is symmetric. Again,

$$\text{the total electronic wavefunction is antisymmetric} \tag{1.256}$$

This complementary symmetry between spatial and spin parts is a direct consequence of the Pauli principle introduced earlier in the chapter.

The three triplet components. A state with $S = 1$ contains three components corresponding to $M_S = +1, 0, -1$. For the two open-shell orbitals i and a, these are

$$\left|{}^{3,+1}\Psi_i^a\right\rangle = |\cdots i^\alpha a^\alpha \cdots\rangle \qquad \left|{}^{3,0}\Psi_i^a\right\rangle = \frac{|D_\alpha\rangle - \left|D_\beta\right\rangle}{\sqrt{2}} \qquad \left|{}^{3,-1}\Psi_i^a\right\rangle = \left|\cdots i^\beta a^\beta \cdots\right\rangle \tag{1.257}$$

All three belong to the same spin multiplet and satisfy

$$\hat{S}^2 \,|1, M_S\rangle = 2\,|1, M_S\rangle \qquad\qquad \hat{S}_z \,|1, M_S\rangle = M_S\,|1, M_S\rangle \tag{1.258}$$

The components are connected by the spin ladder operators,

$$\underbrace{\hat{S}_\pm}_{\text{ladder operator}} \quad |S, M_S\rangle = \sqrt{S(S+1) - M_S(M_S \pm 1)}\,|S, M_S \pm 1\rangle \tag{1.259}$$

The $M_S = \pm 1$ triplet components are each represented by a single spin-pure determinant. By contrast, the $M_S = 0$ component requires a linear combination of two determinants.

This example illustrates a general lesson that will recur throughout electronic-structure theory: determinants are a convenient antisymmetric basis, but states with the correct physical symmetries often require specific linear combinations of determinants.

1.7 The Fock operator and orbital energies

The preceding discussion used the occupied-virtual Fock matrix to express stationarity. We now derive that operator by varying the Hartree–Fock energy under the orbital orthonormality constraints. Its eigenfunctions define the canonical orbitals and their orbital energies, whose physical interpretation is developed here together with Koopmans' theorem.

1.7.1 The Fock operator

The Hartree–Fock approximation has so far been formulated as an energy functional of a set of orthonormal occupied spin orbitals,

$$E_{\mathrm{HF}}[\{\chi_i\}] = \sum_{i=1}^{N} h_{ii} + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} \langle ij||ij\rangle \tag{1.260}$$

The Hartree–Fock problem consists in finding the set of spin orbitals that makes this energy stationary while preserving their orthonormality,

$$\left\langle \chi_i \middle| \chi_j \right\rangle = \delta_{ij} \tag{1.261}$$

This is a constrained variational problem. Since the orthonormality conditions couple every pair of occupied spin orbitals, a single Lagrange multiplier is not sufficient. Instead, one introduces a complete matrix of multipliers ϵ and defines the Hartree–Fock Lagrangian

$$\mathcal{L}[\{\chi_i\}] = E_{\mathrm{HF}}[\{\chi_i\}] - \sum_{ij} \underbrace{\epsilon_{ji}}_{\substack{\text{Lagrange}\\ \text{multipliers}}} \underbrace{\left(\left\langle \chi_i \middle| \chi_j \right\rangle - \delta_{ij}\right)}_{\substack{\text{orthonormality}\\ \text{constraint}}} \tag{1.262}$$

The orbitals and their complex conjugates may be treated as independent variables in the variation. Stationarity with respect to $\chi_i^*(\boldsymbol{x})$ requires

$$\frac{\delta \mathcal{L}}{\delta \chi_i^*(\boldsymbol{x})} = 0 \tag{1.263}$$

Functional derivative of the HF energy. The one-electron part is straightforward:

$$\frac{\delta}{\delta\chi_i^*(\boldsymbol{x})}\sum_j \langle\chi_j|\hat{h}|\chi_j\rangle = \hat{h}(\boldsymbol{x})\chi_i(\boldsymbol{x}) \tag{1.264}$$

The derivative of the two-electron contribution generates the Coulomb and exchange operators. Using the symmetry of the double sum, the factor $1/2$ disappears because the orbital χ_i may occur in either member of an interacting pair:

$$\frac{\delta}{\delta\chi_i^*(\boldsymbol{x})}\left[\frac{1}{2}\sum_{jk}\langle jk||jk\rangle\right] = \sum_{j=1}^{N}\left[\hat{\mathcal{J}}_j(\boldsymbol{x}) - \hat{\mathcal{K}}_j(\boldsymbol{x})\right]\chi_i(\boldsymbol{x}) \tag{1.265}$$

Consequently,

$$\frac{\delta E_{\mathrm{HF}}}{\delta\chi_i^*(\boldsymbol{x})} = \left\{\hat{h}(\boldsymbol{x}) + \sum_{j=1}^{N}\left[\hat{\mathcal{J}}_j(\boldsymbol{x}) - \hat{\mathcal{K}}_j(\boldsymbol{x})\right]\right\}\chi_i(\boldsymbol{x}) \tag{1.266}$$

This motivates the definition of the Fock operator,

$$\boxed{\hat{f}(\boldsymbol{x}) = \hat{h}(\boldsymbol{x}) + \underbrace{\sum_{j=1}^{N}\left[\hat{\mathcal{J}}_j(\boldsymbol{x}) - \hat{\mathcal{K}}_j(\boldsymbol{x})\right]}_{\hat{v}^{\mathrm{HF}}(\boldsymbol{x})\ =\ \text{Hartree–Fock potential}}} \tag{1.267}$$

The Fock operator is therefore an effective one-electron operator composed of the core Hamiltonian and the average interaction with all occupied spin orbitals. The Coulomb term represents the classical electrostatic interaction with the electronic distribution, whereas the exchange term is a purely fermionic contribution arising from antisymmetry. As discussed previously, the Coulomb operator is local while the exchange operator is nonlocal. The Fock operator is therefore, in general, a nonlocal one-electron operator.

Notice also that an electron does not interact with itself. When the Fock operator acts on χ_i,

$$\left(\hat{\mathcal{J}}_i - \hat{\mathcal{K}}_i\right)|\chi_i\rangle = 0 \tag{1.268}$$

because the Coulomb and exchange self-interaction contributions are identical and cancel exactly.

The Hartree–Fock equations. Taking the functional derivative of the constraint term in Eq. (1.262) gives

$$\frac{\delta}{\delta\chi_i^*(\boldsymbol{x})}\sum_{jk}\epsilon_{kj}\langle\chi_j|\chi_k\rangle = \sum_j \epsilon_{ji}\chi_j(\boldsymbol{x}) \tag{1.269}$$

The stationarity condition therefore becomes

$$\hat{f}(\boldsymbol{x})\chi_i(\boldsymbol{x}) = \sum_j \epsilon_{ji}\chi_j(\boldsymbol{x}) \tag{1.270}$$

These are the general Hartree–Fock equations. An important feature is that the Fock operator acting on a given occupied spin orbital need not return that same orbital. It is only required to remain within the space spanned by the occupied spin orbitals.

Multiplying Eq. (1.270) from the left by $\langle\chi_k|$ and using orthonormality gives

$$\epsilon_{ki} = \langle\chi_k|\hat{f}|\chi_i\rangle \tag{1.271}$$

Since $\hat{f}$ is Hermitian,

$$\langle \chi_k | \hat{f} | \chi_i \rangle = \langle \chi_i | \hat{f} | \chi_k \rangle^* \tag{1.272}$$

and hence

$$\epsilon_{ki} = \epsilon_{ik}^* \tag{1.273}$$

The matrix of Lagrange multipliers is therefore Hermitian,

$$\boldsymbol{\epsilon}^\dagger = \boldsymbol{\epsilon} \tag{1.274}$$

This property is crucial because every Hermitian matrix can be diagonalized by a unitary transformation.

Why the HF equations are self-consistent. Equation (1.270) resembles an ordinary one-electron eigenvalue problem, but there is a fundamental difference. The Fock operator itself depends on the occupied orbitals through the Coulomb and exchange operators:

$$\hat{f} = \hat{f}[\chi_1, \chi_2, \ldots, \chi_N] \tag{1.275}$$

The orbitals are therefore required to satisfy an equation containing an operator constructed from those same orbitals. Schematically,

$$\{\chi_i\} \quad \longrightarrow \quad \hat{f}[\{\chi_i\}] \quad \longrightarrow \quad \{\chi_i\} \tag{1.276}$$

The Hartree–Fock equations are consequently nonlinear and must be solved self-consistently. This is the origin of the term self-consistent field (SCF).

In practice, one starts from a trial set of orbitals, constructs the corresponding Fock operator, obtains a new set of orbitals, and repeats the procedure until the orbitals, density, and energy are self-consistent. The variational equations establish a stationary point of the Hartree–Fock energy. They do not, by themselves, guarantee that a particular SCF solution is the global minimum: multiple HF stationary solutions may exist.

Exercise 8 supplies the functional-derivative steps behind this stationarity condition, including the separate variations of the two electron coordinates in the Coulomb and exchange terms.

Exercise 8: Stationarity of the Hartree–Fock energy

Derive the Hartree–Fock equations by making the spin-orbital energy stationary under the orbital orthonormality constraints.

Solution 8: Stationarity of the Hartree–Fock energy

We vary the occupied spin orbitals without imposing restricted spatial pairing. Thus the derivation establishes the general spin-orbital stationarity condition; restricted variations retain only the rotations allowed by their additional constraints. The nuclei and one-electron Hamiltonian are fixed throughout the variation.

Write the energy as the sum of one-electron, Coulomb, and exchange contributions,

$$E_{\mathrm{HF}} = E_1 + E_{\mathrm{H}} + E_{\mathrm{X}} \tag{1.277}$$

with

$$E_1 = \sum_j \int \chi_j^*(\boldsymbol{x}_1) \hat{h}(\boldsymbol{x}_1) \chi_j(\boldsymbol{x}_1)\, \mathrm{d}\boldsymbol{x}_1 \tag{1.278}$$

Solution 8 (continued): Stationarity of the Hartree–Fock energy

$$E_{\mathrm{H}} = \frac{1}{2}\sum_{jk}\iint \chi_j^*(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_k(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.279}$$

$$E_{\mathrm{X}} = -\frac{1}{2}\sum_{jk}\iint \chi_j^*(\boldsymbol{x}_1)\chi_k(\boldsymbol{x}_1)\frac{1}{r_{12}}\chi_k^*(\boldsymbol{x}_2)\chi_j(\boldsymbol{x}_2)\,\mathrm{d}\boldsymbol{x}_1\,\mathrm{d}\boldsymbol{x}_2 \tag{1.280}$$

The sums run over occupied spin orbitals. For complex functions we take χ_i and χ_i^* as independent variables, with the conjugate variation supplying the conjugate equation. The elementary differentiation rule is

$$\frac{\delta\chi_j^*(\boldsymbol{x}_1)}{\delta\chi_i^*(\boldsymbol{x})} = \delta_{ij}\delta(\boldsymbol{x}_1 - \boldsymbol{x}) \tag{1.281}$$

Here the delta function in the combined coordinate denotes a spatial Dirac delta and the identity on the spin coordinate. It selects both the orbital label and the electron coordinate at which the variation is taken.

For the one-electron term only $j = i$ contributes, so

$$\begin{aligned}\frac{\delta E_1}{\delta\chi_i^*(\boldsymbol{x})} &= \sum_j \int \delta_{ij}\delta(\boldsymbol{x}_1 - \boldsymbol{x})\hat{h}(\boldsymbol{x}_1)\chi_j(\boldsymbol{x}_1)\,\mathrm{d}\boldsymbol{x}_1 \\ &= \hat{h}(\boldsymbol{x})\chi_i(\boldsymbol{x})\end{aligned} \tag{1.282}$$

In the Coulomb term the varied conjugate can occur at either coordinate. Differentiating each occurrence gives

$$\begin{aligned}\frac{\delta E_{\mathrm{H}}}{\delta\chi_i^*(\boldsymbol{x})} &= \frac{1}{2}\sum_k \int \frac{\chi_i(\boldsymbol{x})\chi_k^*(\boldsymbol{x}')\chi_k(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{x}' \\ &\quad + \frac{1}{2}\sum_j \int \frac{\chi_j^*(\boldsymbol{x}')\chi_j(\boldsymbol{x}')\chi_i(\boldsymbol{x})}{|\boldsymbol{r}' - \boldsymbol{r}|}\,\mathrm{d}\boldsymbol{x}' \\ &= \sum_j \left[\int \frac{|\chi_j(\boldsymbol{x}')|^2}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{x}'\right]\chi_i(\boldsymbol{x}) \\ &= \sum_j \hat{\mathcal{J}}_j(\boldsymbol{x})\chi_i(\boldsymbol{x})\end{aligned} \tag{1.283}$$

The first contribution comes from $j = i$ at electron 1 and the second from $k = i$ at electron 2. Relabeling the dummy orbital index makes them identical. Their sum cancels the original factor of one half. The denominator involves the spatial coordinates $\boldsymbol{r}$ and $\boldsymbol{r}'$, not the spin coordinate. The exchange term has the same two possible conjugate occurrences, but the unstarred orbitals are crossed between the coordinates:

$$\begin{aligned}\frac{\delta E_{\mathrm{X}}}{\delta\chi_i^*(\boldsymbol{x})} &= -\frac{1}{2}\sum_k \int \frac{\chi_k(\boldsymbol{x})\chi_k^*(\boldsymbol{x}')\chi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{x}' \\ &\quad - \frac{1}{2}\sum_j \int \frac{\chi_j^*(\boldsymbol{x}')\chi_i(\boldsymbol{x}')\chi_j(\boldsymbol{x})}{|\boldsymbol{r}' - \boldsymbol{r}|}\,\mathrm{d}\boldsymbol{x}' \\ &= -\sum_j \chi_j(\boldsymbol{x}) \int \frac{\chi_j^*(\boldsymbol{x}')\chi_i(\boldsymbol{x}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{x}' \\ &= -\sum_j \hat{\mathcal{K}}_j(\boldsymbol{x})\chi_i(\boldsymbol{x})\end{aligned} \tag{1.284}$$

Solution 8 (continued): Stationarity of the Hartree–Fock energy

Again, the two halves are equal, but the overall exchange minus sign remains. Unlike the Coulomb potential, the resulting operator samples the varied orbital at $\boldsymbol{x}'$ and therefore acts nonlocally. Combining the three pieces recovers

$$\frac{\delta E_{\mathrm{HF}}}{\delta \chi_i^*(\boldsymbol{x})} = \left[\hat{h}(\boldsymbol{x}) + \sum_j \left(\hat{\mathcal{J}}_j(\boldsymbol{x}) - \hat{\mathcal{K}}_j(\boldsymbol{x})\right)\right]\chi_i(\boldsymbol{x}) = \hat{f}(\boldsymbol{x})\chi_i(\boldsymbol{x}) \tag{1.285}$$

The energy derivative alone cannot be set to zero because arbitrary orbital variations do not preserve orthonormality. Using the Lagrangian of Eq. (1.262), differentiation of the constraint term yields

$$\begin{aligned}\frac{\delta}{\delta \chi_i^*(\boldsymbol{x})} \sum_{jk} \epsilon_{kj}\left(\langle \chi_j | \chi_k \rangle - \delta_{jk}\right) &= \sum_{jk} \epsilon_{kj}\delta_{ij}\chi_k(\boldsymbol{x}) \\ &= \sum_j \epsilon_{ji}\chi_j(\boldsymbol{x})\end{aligned} \tag{1.286}$$

Variation with respect to the multipliers independently restores the overlap constraints. Variation with respect to the conjugate orbitals gives

$$\frac{\delta \mathcal{L}}{\delta \chi_i^*(\boldsymbol{x})} = \hat{f}(\boldsymbol{x})\chi_i(\boldsymbol{x}) - \sum_j \epsilon_{ji}\chi_j(\boldsymbol{x}) = 0 \tag{1.287}$$

Thus the Fock operator maps every occupied orbital back into the occupied space. Projecting onto an occupied χ_k identifies the multipliers,

$$\epsilon_{ki} = \langle \chi_k | \hat{f} | \chi_i \rangle = \epsilon_{ik}^* \tag{1.288}$$

Their Hermiticity allows the occupied-space unitary canonicalization developed next; it is not an extra assumption in the variational problem.

To express the same condition directly as an orbital gradient, consider an infinitesimal occupied-virtual rotation,

$$\delta\chi_i = \sum_a \chi_a \kappa_{ai} \tag{1.289}$$

where the κ_{ai} are independent complex rotation parameters and the complementary virtual-orbital rotation preserves orthonormality to first order. The first-order energy change is

$$\begin{aligned}\delta E_{\mathrm{HF}} &= 2\,\mathrm{Re}\sum_i \int \delta\chi_i^*(\boldsymbol{x})\hat{f}(\boldsymbol{x})\chi_i(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} \\ &= 2\,\mathrm{Re}\sum_{ia} \kappa_{ai}^* \langle \chi_a | \hat{f} | \chi_i \rangle = 2\,\mathrm{Re}\sum_{ia} \kappa_{ai}^* f_{ai}\end{aligned} \tag{1.290}$$

Both the real and imaginary parts of each allowed rotation may be varied. Stationarity therefore requires

$$\boxed{f_{ai} = 0} \tag{1.291}$$

Equivalently, the projection of the general HF equations onto the virtual space vanishes. Hermiticity gives $f_{ia} = f_{ai}^* = 0$, and the Slater–Condon rule then recovers

$$\left\langle \Psi_0 \middle| \hat{\mathcal{H}}_{\mathrm{elec}} \middle| \Psi_i^a \right\rangle = f_{ia} = 0 \tag{1.292}$$

This is Brillouin's theorem. Occupied-occupied rotations merely change the representation of the determinant, as shown in the next subsection. Finally, a vanishing first variation establishes only a stationary point. Its stability depends on the second variation, so an SCF solution may be a minimum or an unstable stationary solution rather than the global HF minimum.

1.7.2 Unitary invariance and canonical orbitals

Historical perspective: Paul Adrien Maurice Dirac (1902–1984)

Born in Bristol, Dirac trained in engineering and mathematics before becoming one of the founders of quantum mechanics at Cambridge. His 1930 paper on exchange expressed the exchange-corrected self-consistent-field theory in terms of a one-particle density matrix and explicitly examined invariance under changes of occupied orbitals [24]. This formulation makes the occupied subspace, rather than a particular set of orbitals spanning it, the central object. It is distinct from the local exchange approximation for the Thomas atom also developed in that paper.

Biography: University of St Andrews, MacTutor. Source portrait: Nobel Foundation, 1933, via Wikimedia Commons, listed as public domain. Sketch: AI-assisted adaptation of the credited portrait.

Unitary invariance of the occupied orbitals. The HF determinant is not associated with a unique set of occupied orbitals. Consider a unitary transformation among the occupied spin orbitals,

$$\widetilde{\chi}_i(\boldsymbol{x}) = \sum_j \chi_j(\boldsymbol{x}) U_{ji} \qquad \boldsymbol{U}^\dagger \cdot \boldsymbol{U} = \mathbf{1} \tag{1.293}$$

The transformed orbitals remain orthonormal:

$$\langle \widetilde{\chi}_i | \widetilde{\chi}_j \rangle = \sum_{kl} U^*_{ki} U_{lj} \langle \chi_k | \chi_l \rangle = \sum_k U^*_{ki} U_{kj} = \delta_{ij} \tag{1.294}$$

More importantly, the occupied one-particle projector is unchanged:

$$\begin{aligned} \sum_i \widetilde{\chi}_i(\boldsymbol{x}) \widetilde{\chi}^*_i(\boldsymbol{x}') &= \sum_{ijk} \chi_j(\boldsymbol{x}) U_{ji} U^*_{ki} \chi^*_k(\boldsymbol{x}') = \sum_{jk} \chi_j(\boldsymbol{x}) \delta_{jk} \chi^*_k(\boldsymbol{x}') \\ &= \sum_j \chi_j(\boldsymbol{x}) \chi^*_j(\boldsymbol{x}') \end{aligned} \tag{1.295}$$

Since the Fock operator depends only on the occupied one-particle density, it is invariant under this transformation. Exercise 9 establishes this explicitly for its Coulomb and exchange contributions.

The determinant itself transforms as

$$\boxed{|\widetilde{\Psi}_{\mathrm{HF}}\rangle = \det(\boldsymbol{U})\, |\Psi_{\mathrm{HF}}\rangle} \tag{1.296}$$

Because $\boldsymbol{U}$ is unitary,

$$\boldsymbol{U}^\dagger \cdot \boldsymbol{U} = \mathbf{1} \quad \Rightarrow \quad \det\left(\boldsymbol{U}^\dagger \cdot \boldsymbol{U}\right) = 1 \tag{1.297}$$

and therefore

$$|\det(\boldsymbol{U})|^2 = 1 \quad \Rightarrow \quad \det(\boldsymbol{U}) = e^{\mathrm{i}\theta} \tag{1.298}$$

Hence,

$$|\widetilde{\Psi}_{\mathrm{HF}}\rangle = e^{\mathrm{i}\theta} |\Psi_{\mathrm{HF}}\rangle \tag{1.299}$$

The two determinants differ only by an irrelevant global phase and describe the same physical state. Thus, unitary rotations within the occupied space leave the energy E_{HF} and the physical state unchanged, with the wavefunction itself defined only up to that phase.

Exercise 9: Unitary invariance of the HF operators

Show that the total Coulomb, exchange, and Fock operators are unchanged by a unitary transformation of the occupied spin orbitals.

Solution 9: Unitary invariance of the HF operators

Use the occupied-space transformation in Eq. (1.293), allowing complex coefficients. All orbital sums below run over the N occupied spin orbitals. Since $\boldsymbol{U}$ is a square unitary matrix, the contraction needed in both proofs is

$$\sum_i U_{ji} U_{ki}^* = (\boldsymbol{U} \cdot \boldsymbol{U}^\dagger)_{jk} = \delta_{jk} \tag{1.300}$$

Define the total operators by $\hat{\mathcal{J}} = \sum_i \hat{\mathcal{J}}_i$ and $\hat{\mathcal{K}} = \sum_i \hat{\mathcal{K}}_i$; tildes denote operators constructed from the transformed occupied orbitals. To prove an operator identity, let these operators act on an arbitrary one-electron test function $g(\boldsymbol{x})$, not necessarily an occupied orbital. For the Coulomb operator, substitution into Eq. (1.80) gives

$$\begin{aligned}
(\tilde{\hat{\mathcal{J}}} g)(\boldsymbol{x}_1) &= \sum_i \left[\int \tilde{\chi}_i^*(\boldsymbol{x}_2) \frac{1}{r_{12}} \tilde{\chi}_i(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \right] g(\boldsymbol{x}_1) \\
&= \sum_{jk} \underbrace{\left(\sum_i U_{ji} U_{ki}^* \right)}_{\delta_{jk}} \left[\int \chi_k^*(\boldsymbol{x}_2) \frac{1}{r_{12}} \chi_j(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \right] g(\boldsymbol{x}_1) \\
&= \sum_j \left[\int \chi_j^*(\boldsymbol{x}_2) \frac{1}{r_{12}} \chi_j(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \right] g(\boldsymbol{x}_1) \\
&= (\hat{\mathcal{J}} g)(\boldsymbol{x}_1)
\end{aligned} \tag{1.301}$$

Unitarity therefore leaves the total density, and hence its Coulomb potential, unchanged. The exchange operator depends on the nonlocal one-particle density rather than only its diagonal. Using Eq. (1.84), its action becomes

$$\begin{aligned}
(\tilde{\hat{\mathcal{K}}} g)(\boldsymbol{x}_1) &= \sum_i \tilde{\chi}_i(\boldsymbol{x}_1) \int \tilde{\chi}_i^*(\boldsymbol{x}_2) \frac{1}{r_{12}} g(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \\
&= \sum_{jk} \underbrace{\left(\sum_i U_{ji} U_{ki}^* \right)}_{\delta_{jk}} \chi_j(\boldsymbol{x}_1) \int \chi_k^*(\boldsymbol{x}_2) \frac{1}{r_{12}} g(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \\
&= \sum_j \chi_j(\boldsymbol{x}_1) \int \chi_j^*(\boldsymbol{x}_2) \frac{1}{r_{12}} g(\boldsymbol{x}_2)\, \mathrm{d}\boldsymbol{x}_2 \\
&= (\hat{\mathcal{K}} g)(\boldsymbol{x}_1)
\end{aligned} \tag{1.302}$$

Since g is arbitrary, these equalities establish the operator identities

$$\tilde{\hat{\mathcal{J}}} = \hat{\mathcal{J}} \qquad\qquad \tilde{\hat{\mathcal{K}}} = \hat{\mathcal{K}} \tag{1.303}$$

The core Hamiltonian is independent of the occupied orbitals, so

$$\boxed{\tilde{\hat{f}} = \hat{h} + \tilde{\hat{\mathcal{J}}} - \tilde{\hat{\mathcal{K}}} = \hat{h} + \hat{\mathcal{J}} - \hat{\mathcal{K}} = \hat{f}} \tag{1.304}$$

Solution 9 (continued): Unitary invariance of the HF operators

The invariance applies to the occupied sums: individual $\hat{\mathcal{J}}_i$ and $\hat{\mathcal{K}}_i$ generally change when their generating orbitals are rotated. It does not require self-consistency. Rotations among virtual orbitals also leave these operators unchanged because the virtuals do not enter their construction, whereas occupied-virtual mixing generally changes the density and the operators.
Operator invariance must not be confused with invariance of its matrix elements in a changing orbital basis. For occupied indices i, j,

$$\begin{aligned} \widetilde{f}_{ij} &= \langle \widetilde{\chi}_i | \hat{f} | \widetilde{\chi}_j \rangle \\ &= \sum_{kl} U^*_{ki} \langle \chi_k | \hat{f} | \chi_l \rangle U_{lj} \\ &= \sum_{kl} U^*_{ki} f_{kl} U_{lj} \end{aligned} \tag{1.305}$$

Thus the occupied block obeys

$$\widetilde{\boldsymbol{f}}_{\mathrm{oo}} = \boldsymbol{U}^\dagger \cdot \boldsymbol{f}_{\mathrm{oo}} \cdot \boldsymbol{U} \tag{1.306}$$

Its entries need not remain unchanged, even though the underlying Fock operator does. This is precisely the freedom used in canonicalization.

Canonical Hartree–Fock orbitals. This unitary freedom can be exploited to choose a particularly convenient set of occupied orbitals. Since the Lagrange-multiplier matrix is Hermitian, there exists a unitary matrix $\boldsymbol{U}$ such that

$$\widetilde{\boldsymbol{\epsilon}} = \boldsymbol{U}^\dagger \cdot \boldsymbol{\epsilon} \cdot \boldsymbol{U} \tag{1.307}$$

is diagonal, i.e., $\widetilde{\epsilon}_{ij} = \epsilon_i \delta_{ij}$.

Applying the same unitary transformation to the occupied orbitals converts the general HF equations,

$$\hat{f} \, |\chi_i\rangle = \sum_j \epsilon_{ji} \, |\chi_j\rangle \tag{1.308}$$

into

$$\hat{f} \, |\widetilde{\chi}_i\rangle = \epsilon_i \, |\widetilde{\chi}_i\rangle \tag{1.309}$$

Dropping the tildes, one obtains the canonical Hartree–Fock equations,

$$\boxed{\hat{f}(\boldsymbol{x}) \chi_i(\boldsymbol{x}) = \epsilon_i \chi_i(\boldsymbol{x})} \tag{1.310}$$

The eigenfunctions $\{\chi_i\}$ are called canonical spin orbitals, and the corresponding eigenvalues $\{\epsilon_i\}$ are called orbital energies. Canonicalization does not produce a new Hartree–Fock wavefunction. It is only a convenient choice of basis within the same occupied orbital space. At self-consistency, the Fock operator may also be diagonalized in the orthogonal complement of the occupied space. This defines the canonical virtual orbitals. The complete canonical orbital set then satisfies

$$\langle \chi_p | \hat{f} | \chi_q \rangle = \epsilon_p \delta_{pq} \tag{1.311}$$

In particular, the occupied-virtual block vanishes,

$$f_{ia} = \langle i | \hat{f} | a \rangle = 0 \tag{1.312}$$

This is the orbital form of the stationarity condition encountered in Brillouin's theorem:

$$\langle \Psi_0 | \hat{\mathcal{H}}_{\mathrm{elec}} | \Psi_i^a \rangle = f_{ia} = 0 \tag{1.313}$$

Exercise 10: Fock matrix elements in the spin-orbital basis

Express the Fock matrix elements in terms of one- and two-electron integrals,

$$f_{ij} = \langle \chi_i | \hat{f} | \chi_j \rangle$$

Solution 10: Fock matrix elements in the spin-orbital basis

Use orthonormal spin orbitals and keep the occupied-orbital sum explicit; the indices i, j specify the matrix element, while k runs over occupied spin orbitals. Canonical orbitals are not required for this expansion. Expanding the Fock operator gives

$$\begin{aligned} f_{ij} &= \langle \chi_i | \hat{h} + \sum_k \left(\hat{\mathcal{J}}_k - \hat{\mathcal{K}}_k \right) | \chi_j \rangle \\ &= \langle \chi_i | \hat{h} | \chi_j \rangle + \sum_k \left[\langle \chi_i | \hat{\mathcal{J}}_k | \chi_j \rangle - \langle \chi_i | \hat{\mathcal{K}}_k | \chi_j \rangle \right] \end{aligned} \tag{1.314}$$

The Coulomb contribution is

$$\langle \chi_i | \hat{\mathcal{J}}_k | \chi_j \rangle = \langle ik | jk \rangle \tag{1.315}$$

whereas the exchange contribution is

$$\langle \chi_i | \hat{\mathcal{K}}_k | \chi_j \rangle = \langle ik | kj \rangle \tag{1.316}$$

These index arrangements follow directly from the actions of the two operators. Writing $\chi_p(n) = \chi_p(\boldsymbol{x}_n)$, their matrix elements are

$$\langle \chi_i | \hat{\mathcal{J}}_k | \chi_j \rangle = \iint \chi_i^*(1) \chi_k^*(2) \frac{1}{r_{12}} \chi_j(1) \chi_k(2) \, \mathrm{d}\boldsymbol{x}_1 \, \mathrm{d}\boldsymbol{x}_2 \tag{1.317}$$

$$\langle \chi_i | \hat{\mathcal{K}}_k | \chi_j \rangle = \iint \chi_i^*(1) \chi_k^*(2) \frac{1}{r_{12}} \chi_k(1) \chi_j(2) \, \mathrm{d}\boldsymbol{x}_1 \, \mathrm{d}\boldsymbol{x}_2 \tag{1.318}$$

The Coulomb operator leaves χ_j at coordinate 1, whereas exchange places it at coordinate 2 and returns χ_k at coordinate 1. Reading the bra and ket indices in physicists' notation gives the two integrals above; their difference is the antisymmetrized integral. Therefore,

$$\begin{aligned} f_{ij} &= \langle i | \hat{h} | j \rangle + \sum_k \left[\langle ik | jk \rangle - \langle ik | kj \rangle \right] \\ &= \langle i | \hat{h} | j \rangle + \sum_k \langle ik || jk \rangle \end{aligned} \tag{1.319}$$

Thus, each Fock matrix element contains the bare one-electron contribution plus the Coulomb and exchange interactions with all occupied spin orbitals.
For an occupied-virtual matrix element,

$$f_{ia} = h_{ia} + \sum_k \langle ik || ak \rangle \tag{1.320}$$

which is precisely the quantity appearing in Brillouin's theorem. For a self-consistent stationary determinant this occupied-virtual element vanishes, but the formula itself holds for any chosen orthonormal spin-orbital basis with the stated occupied set.

Exercise 11: Orbital energies in the spin-orbital basis

Deduce the energy ϵ_i of a canonical spin orbital from the diagonal Fock matrix element.

Solution 11: Orbital energies in the spin-orbital basis

For a canonical orbital,

$$\hat{f}\,|\chi_i\rangle = \epsilon_i\,|\chi_i\rangle \tag{1.321}$$

Multiplying by $\langle\chi_i|$ gives

$$\begin{aligned}\langle\chi_i|\hat{f}|\chi_i\rangle &= \epsilon_i\,\langle\chi_i|\chi_i\rangle \\ &= \epsilon_i\end{aligned} \tag{1.322}$$

because the orbital is normalized. Therefore,

$$\begin{aligned}\epsilon_i &= f_{ii} \\ &= \langle i|\hat{h}|i\rangle + \sum_k \left[\langle ik|ik\rangle - \langle ik|ki\rangle\right] \\ &= \langle i|\hat{h}|i\rangle + \sum_k \langle ik||ik\rangle\end{aligned} \tag{1.323}$$

The term with $k = i$ vanishes,

$$\langle ii||ii\rangle = \langle ii|ii\rangle - \langle ii|ii\rangle = 0 \tag{1.324}$$

which again reflects the exact cancellation of one-electron self-interaction in Hartree–Fock theory. Here i labels an occupied canonical spin orbital, so its own term in the occupied sum over k cancels as shown. The analogous expression for a virtual canonical orbital a follows by replacing i with a, while k still runs only over occupied orbitals; there is then no $k = a$ term. Canonicalization is essential to interpreting f_{ii} as an orbital eigenvalue: a diagonal expectation value in a noncanonical orbital is not, in general, one of the canonical orbital energies.

1.7.3 Orbital energies and the total Hartree–Fock energy

The orbital energies are Lagrange multipliers associated with the orthonormality constraints. They should therefore not be interpreted simply as independent one-electron contributions to the total energy.

Indeed, summing the occupied orbital energies gives

$$\sum_i \epsilon_i = \sum_i h_{ii} + \sum_{ij} \langle ij||ij\rangle \tag{1.325}$$

By comparison, the Hartree–Fock energy is

$$E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij} \langle ij||ij\rangle \tag{1.326}$$

Thus,

$$\boxed{E_{\mathrm{HF}} = \sum_i \epsilon_i - \frac{1}{2}\sum_{ij} \langle ij||ij\rangle} \tag{1.327}$$

Equivalently,

$$E_{\mathrm{HF}} = \frac{1}{2}\sum_i (h_{ii} + \epsilon_i) \tag{1.328}$$

The reason the total HF energy is not simply the sum of the occupied orbital energies is that each ϵ_i contains the interaction of electron i with every other occupied electron. Summing the orbital energies therefore counts each electron pair twice, whereas the total energy must count each pair only once. Although orbital energies are therefore not independent electron energies, they nevertheless acquire an important physical interpretation within the frozen-orbital approximation. This observation leads directly to Koopmans' theorem, discussed next.

1.7.4 Koopmans' theorem

The canonical Hartree–Fock orbital energies are Lagrange multipliers associated with the orthonormality constraints, but they also acquire an important physical interpretation when electrons are removed from or added to the system while the orbitals are kept fixed. This is the content of Koopmans' theorem [25]. Its central approximation is the frozen-orbital approximation: when the number of electrons is changed, the spin orbitals of the original N-electron HF calculation are not allowed to relax. For fixed nuclear coordinates, Koopmans' theorem therefore provides vertical removal and attachment energies directly from the canonical HF orbital energies.

Koopmans' theorem: frozen-orbital removal and attachment

Tjalling Koopmans

Source portrait: Eric Koch / Anefo, Nationaal Archief, Leuven, 1967 (crop: Wabbuh), CC0 1.0. Sketch: AI-assisted adaptation of the credited portrait.

For fixed nuclei and canonical orbitals of the N-electron HF reference, removing an electron from occupied spin orbital i, or adding one to virtual spin orbital a, gives the frozen-orbital energy differences [25]. These are not relaxed or correlated energy differences.

$$\begin{aligned} \mathrm{IP}_i^{\mathrm{frozen}} &= E_{N-1}^{\mathrm{frozen},i} - E_N^{\mathrm{HF}} = -\epsilon_i \\ \mathrm{EA}_a^{\mathrm{frozen}} &= E_N^{\mathrm{HF}} - E_{N+1}^{\mathrm{frozen},a} = -\epsilon_a \end{aligned} \tag{1.329}$$

Orbital relaxation and correlation modify these estimates, as the complete decompositions below make explicit.

Historical perspective: Tjalling Charles Koopmans (1910–1985)

Koopmans was born in the Netherlands and initially trained in mathematics and physics. His 1934 paper related electron-removal energies to Hartree–Fock orbital energies under the frozen-orbital approximation [25]. He subsequently turned to economics and joined Yale in 1955. His later work on the allocation of resources brought him a share of the 1975 Nobel Memorial Prize in Economic Sciences—an unusual career link between electronic-structure theory and mathematical economics.

Biography: Yale University, Cowles Foundation. Source portrait: Eric Koch / Anefo, Nationaal Archief, Leuven, 1967, crop by Wabbuh, CC0 1.0. Sketch: AI-assisted adaptation of the credited portrait.

Electron removal. Consider the N-electron HF ground state. Its electronic energy is

$$ {}^{N}E_0 = \sum_i h_{ii} + \frac{1}{2}\sum_{ij} \langle ij||ij\rangle \tag{1.330} $$

All sums run over the occupied spin orbitals of the N-electron reference. Suppose now that the electron occupying spin orbital k is removed while all remaining orbitals are kept frozen. The energy of this frozen $(N-1)$-electron determinant is

$$ {}^{N-1}\widetilde{E}_k = \sum_{i\neq k} h_{ii} + \frac{1}{2}\sum_{i\neq k}\sum_{j\neq k} \langle ij||ij\rangle \tag{1.331} $$

where the tilde emphasizes that the orbitals have *not* been reoptimized for the cation. The corresponding frozen-orbital ionization potential is

$$ \mathrm{IP}_k^{\mathrm{K}} = {}^{N-1}\widetilde{E}_k - {}^{N}E_0 \tag{1.332} $$

Subtracting Eqs. (1.330) and (1.331) gives

$$ \begin{aligned} \mathrm{IP}_k^{\mathrm{K}} &= -h_{kk} - \frac{1}{2}\sum_i \langle ik||ik\rangle - \frac{1}{2}\sum_j \langle kj||kj\rangle \\ &= -h_{kk} - \sum_i \langle ki||ki\rangle \end{aligned} \tag{1.333} $$

where the symmetry of the two-electron integrals has been used. From the expression of the canonical orbital energy,

$$ \epsilon_k = h_{kk} + \sum_i \langle ki||ki\rangle \tag{1.334} $$

we immediately obtain

$$ \mathrm{IP}_k^{\mathrm{K}} = -\epsilon_k \tag{1.335} $$

Thus, within the frozen-orbital approximation, the energy required to remove an electron from an occupied canonical HF spin orbital is the negative of its orbital energy.

For the first ionization potential, electron removal usually occurs from the highest occupied molecular orbital (HOMO), giving the familiar estimate

$$ \mathrm{IP} \approx -\epsilon_{\mathrm{HOMO}} \tag{1.336} $$

This result also provides a useful interpretation of the occupied HF orbital energies: $-\epsilon_i$ is the energy required to create a frozen hole in occupied spin orbital i.

Exercise 12: Koopmans' theorem for electron affinity

Derive the frozen-orbital electron affinity for attachment to a vacant canonical spin orbital and relate it to that orbital's energy.

Solution 12: Koopmans' theorem for electron affinity

Consider the attachment of an electron to a vacant canonical spin orbital c. All energies refer to the same nuclear geometry and the occupied orbitals of the N-electron reference remain frozen. The original one-electron and occupied-pair contributions are therefore unchanged. The new orbital contributes h_{cc} and interacts with every occupied spin orbital. In the double-sum energy

Solution 12 (continued): Koopmans' theorem for electron affinity

expression this gives

$$ {}^{N+1}\widetilde{E}^{c} = {}^{N}E_0 + h_{cc} + \frac{1}{2}\sum_i \left[\langle ci||ci\rangle + \langle ic||ic\rangle \right] + \frac{1}{2}\langle cc||cc\rangle \tag{1.337} $$

The two terms for each old–new pair are equal by electron interchange, and the self-pair term is zero by antisymmetry. Thus the factor of one half disappears when each added pair is counted once. The energy of the resulting $(N+1)$-electron determinant is

$$ {}^{N+1}\widetilde{E}^{c} = {}^{N}E_0 + h_{cc} + \sum_i \langle ci||ci\rangle \tag{1.338} $$

The frozen-orbital electron affinity is defined as

$$ \mathrm{EA}_c^{\mathrm{K}} = {}^{N}E_0 - {}^{N+1}\widetilde{E}^{c} \tag{1.339} $$

Consequently,

$$ \begin{aligned} \mathrm{EA}_c^{\mathrm{K}} &= -h_{cc} - \sum_i \langle ci||ci\rangle \\ &= \boxed{-\epsilon_c} \end{aligned} \tag{1.340} $$

For attachment into the lowest unoccupied molecular orbital (LUMO),

$$ \mathrm{EA} \approx -\epsilon_{\mathrm{LUMO}} \tag{1.341} $$

With this sign convention, a positive electron affinity means that the attached determinant lies below the reference in energy. The orbital energy ϵ_c is evaluated using the reference Fock operator, not a Fock operator reoptimized for the anion. Nuclear repulsion cancels from the energy difference because the geometry is fixed.

The algebra is therefore completely analogous to that for electron removal. There is, however, an important practical asymmetry between the two cases. Occupied HF orbitals have been variationally optimized to describe the N-electron ground state, whereas the virtual orbitals have not been optimized to describe an additional electron. Consequently, Koopmans-type electron affinities are usually considerably less reliable than the corresponding ionization potentials.

In particular, a neutral molecule may have a positive HF LUMO energy, leading to a negative Koopmans electron affinity even when the exact anion is bound.

The frozen-orbital approximation. The crucial assumption behind Eqs. (1.335) and (1.340) is not Hartree–Fock theory itself, but the decision to retain the orbitals of the neutral N-electron system after changing the electron number. In reality, removing or adding an electron changes the effective potential felt by all the remaining electrons. The orbitals therefore relax.

For ionization from orbital k, let

$$ {}^{N-1}\widetilde{E}_k \tag{1.342} $$

denote the frozen-orbital cation energy and

$$ {}^{N-1}E_k^{\mathrm{HF}} \tag{1.343} $$

the energy obtained after a separate HF optimization of the cation. All comparisons in this discussion use the same nuclear geometry and one-electron space. Provided that the variational

space for the ion contains the frozen determinant and that its energy is minimized in that space, optimization can only lower the energy:

$$ {}^{N-1}E_k^{\mathrm{HF}} \leq {}^{N-1}\widetilde{E}_k \tag{1.344} $$

It is therefore useful to define the ionization relaxation correction as

$$ \Delta_{\mathrm{relax}}^{\mathrm{IP}} = {}^{N-1}E_k^{\mathrm{HF}} - {}^{N-1}\widetilde{E}_k \leq 0 \tag{1.345} $$

Orbital relaxation thus tends to *lower* the ionization potential relative to the frozen-orbital Koopmans value.

For electron attachment, the relaxed anion satisfies the corresponding bound under the same variational assumption:

$$ {}^{N+1}E_c^{\mathrm{HF}} \leq {}^{N+1}\widetilde{E}^c \tag{1.346} $$

However, because the electron affinity is defined with the opposite energy difference,

$$ \mathrm{EA} = {}^{N}E_0 - {}^{N+1}E_c \tag{1.347} $$

the corresponding relaxation correction is

$$ \Delta_{\mathrm{relax}}^{\mathrm{EA}} = {}^{N+1}\widetilde{E}^c - {}^{N+1}E_c^{\mathrm{HF}} \geq 0 \tag{1.348} $$

Relaxation therefore tends to *increase* the electron affinity relative to the frozen-orbital estimate.

Relaxation, correlation, and error compensation. Koopmans' theorem neglects not only orbital relaxation but also electron correlation beyond Hartree–Fock theory.

In the following decomposition, the superscript M denotes the electron number, and E_{corr}^M is the correlation energy of that system, defined as the difference between its exact and optimized HF electronic energies at the same nuclear geometry [26]. Writing

$$ E_{\mathrm{exact}}^M = E_{\mathrm{HF}}^M + E_{\mathrm{corr}}^M \tag{1.349} $$

the exact ionization potential may be decomposed as

$$ \begin{aligned} \mathrm{IP}_k &= E_{\mathrm{exact}}^{N-1} - E_{\mathrm{exact}}^{N} \\ &= E_{\mathrm{HF}}^{N-1} - E_{\mathrm{HF}}^{N} + E_{\mathrm{corr}}^{N-1} - E_{\mathrm{corr}}^{N} \\ &= \boxed{-\epsilon_k + \underbrace{\Delta_{\mathrm{relax}}^{\mathrm{IP}}}_{\leq 0} + \underbrace{\Delta_{\mathrm{corr}}^{\mathrm{IP}}}_{E_{\mathrm{corr}}^{N-1} - E_{\mathrm{corr}}^{N}}} \end{aligned} \tag{1.350} $$

For many ordinary molecular ionizations, the neutral system has a somewhat larger magnitude of correlation energy than the corresponding cation, so that

$$ \Delta_{\mathrm{corr}}^{\mathrm{IP}} > 0 \tag{1.351} $$

In such cases, relaxation and correlation corrections have opposite signs and partially compensate:

$$ \underbrace{\Delta_{\mathrm{relax}}^{\mathrm{IP}}}_{<0} + \underbrace{\Delta_{\mathrm{corr}}^{\mathrm{IP}}}_{>0} \longrightarrow \text{partial error compensation} \tag{1.352} $$

This error compensation is one reason why Koopmans' theorem can yield surprisingly reasonable ionization potentials despite neglecting both effects. The sign of the correlation correction is,

however, *not* guaranteed by any general theorem. The statement above is a common chemical trend, not a mathematical requirement.

For electron attachment,

$$\begin{aligned} \mathrm{EA}_c &= E_{\mathrm{exact}}^{N} - E_{\mathrm{exact}}^{N+1} \\ &= E_{\mathrm{HF}}^{N} - E_{\mathrm{HF}}^{N+1} + E_{\mathrm{corr}}^{N} - E_{\mathrm{corr}}^{N+1} \\ &= \boxed{-\epsilon_c + \underbrace{\Delta_{\mathrm{relax}}^{\mathrm{EA}}}_{\geq 0} + \underbrace{\Delta_{\mathrm{corr}}^{\mathrm{EA}}}_{E_{\mathrm{corr}}^{N} - E_{\mathrm{corr}}^{N+1}}} \end{aligned} \tag{1.353}$$

For many bound molecular anions, the $(N+1)$-electron system gains more correlation energy than the neutral system, so that

$$\Delta_{\mathrm{corr}}^{\mathrm{EA}} > 0 \tag{1.354}$$

In this common situation, relaxation and correlation act in the same direction rather than compensating one another:

$$\underbrace{\Delta_{\mathrm{relax}}^{\mathrm{EA}}}_{>0} + \underbrace{\Delta_{\mathrm{corr}}^{\mathrm{EA}}}_{>0} \longrightarrow \text{no analogous error compensation} \tag{1.355}$$

Again, the sign of the correlation contribution is system-dependent. Nevertheless, the combination of stronger orbital-relaxation effects, unoptimized virtual orbitals, and the frequent absence of favorable error compensation explains why Koopmans' theorem is usually more reliable for IPs than for EAs.

1.7.5 The auxiliary Hartree–Fock Hamiltonian

The interpretation of orbital energies becomes clearer if we distinguish carefully between the physical electronic Hamiltonian and the auxiliary independent-particle Hamiltonian associated with the converged Hartree–Fock potential.

Define the N-electron Hartree–Fock Hamiltonian as

$$\boxed{\underbrace{\hat{\mathcal{H}}^{\mathrm{HF}}}_{N\text{-electron auxiliary operator}} = \sum_{i=1}^{N} \underbrace{\hat{f}(\boldsymbol{x}_i)}_{\text{one-electron Fock operator}} = \sum_{i=1}^{N} \left[\hat{h}(\boldsymbol{x}_i) + \hat{v}^{\mathrm{HF}}(\boldsymbol{x}_i)\right]} \tag{1.356}$$

Once the Hartree–Fock potential has been converged, $\hat{\mathcal{H}}^{\mathrm{HF}}$ is a sum of one-electron operators. The Slater determinant constructed from the canonical occupied orbitals is therefore an eigenfunction of this *auxiliary* Hamiltonian:

$$\hat{\mathcal{H}}^{\mathrm{HF}} |\Psi_{\mathrm{HF}}\rangle = \left(\sum_{i=1}^{N} \epsilon_i\right) |\Psi_{\mathrm{HF}}\rangle \tag{1.357}$$

Three quantities must not be confused. First,

$$\hat{\mathcal{H}}^{\mathrm{HF}} |\Psi_{\mathrm{HF}}\rangle = \left(\sum_{i=1}^{N} \epsilon_i\right) |\Psi_{\mathrm{HF}}\rangle \qquad \sum_i \epsilon_i \neq E_{\mathrm{HF}} \tag{1.358}$$

The Hartree–Fock determinant is an eigenfunction of the auxiliary Hartree–Fock Hamiltonian, but its eigenvalue is the sum of the occupied orbital energies, not the Hartree–Fock energy. Second,

$$\hat{\mathcal{H}}_{\mathrm{elec}} |\Psi_{\mathrm{HF}}\rangle \neq E_{\mathrm{HF}} |\Psi_{\mathrm{HF}}\rangle \tag{1.359}$$

The Hartree–Fock determinant is, in general, not an eigenfunction of the true interacting electronic Hamiltonian. Third,

$$\langle \Psi_{\mathrm{HF}} | \hat{\mathcal{H}}_{\mathrm{elec}} | \Psi_{\mathrm{HF}} \rangle = E_{\mathrm{HF}} \tag{1.360}$$

The Hartree–Fock energy is the *expectation value* of the physical electronic Hamiltonian over the optimized HF determinant. Thus,

$$\hat{\mathcal{H}}^{\mathrm{HF}} \Psi_{\mathrm{HF}} = \left(\sum_i \epsilon_i \right) \Psi_{\mathrm{HF}} \qquad \langle \Psi_{\mathrm{HF}} | \hat{\mathcal{H}}_{\mathrm{elec}} | \Psi_{\mathrm{HF}} \rangle = E_{\mathrm{HF}} \qquad \hat{\mathcal{H}}_{\mathrm{elec}} \Psi_{\mathrm{HF}} \neq E_{\mathrm{HF}} \Psi_{\mathrm{HF}} \tag{1.361}$$

Why the sum of orbital energies is not the HF energy. For occupied canonical HF orbitals,

$$\epsilon_i = h_{ii} + \sum_j \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \tag{1.362}$$

Summing over all occupied orbitals gives

$$\sum_{i=1}^{N} \epsilon_i = \sum_{i=1}^{N} h_{ii} + \sum_{ij}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \tag{1.363}$$

By contrast,

$$E_{\mathrm{HF}} = \sum_{i=1}^{N} h_{ii} + \frac{1}{2} \sum_{ij}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \tag{1.364}$$

Hence,

$$\sum_{i=1}^{N} \epsilon_i \neq E_{\mathrm{HF}} \tag{1.365}$$

This recovers the double-counting correction in Eq. (1.327): the orbital energy ϵ_i contains the interaction of electron i with every other occupied electron, so their sum counts each pair twice. Here the same correction distinguishes the eigenvalue of the auxiliary Hamiltonian from the expectation value of the physical Hamiltonian. Equivalently,

$$E_{\mathrm{HF}} = \sum_i \epsilon_i - \frac{1}{2} \sum_{ij} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \tag{1.366}$$

1.7.6 Connection with many-body perturbation theory

This distinction between the sum of the orbital energies and the HF energy also provides the natural starting point for Møller–Plesset perturbation theory [19, 20].

Historical perspective: Møller and Plesset

Christian Møller (1904–1980). A Danish theoretical physicist closely associated with Copenhagen and Niels Bohr's institute, Møller worked on relativistic quantum theory as well as many-electron systems. His teaching and research helped shape Danish theoretical physics. The perturbation treatment developed with Plesset in 1934 used a Hartree–Fock reference [19].

Milton S. Plesset (1908–1991). An American physicist, Plesset received his doctorate at Yale and spent 1933–1934 at Bohr's institute after postdoctoral work at Caltech. His later Caltech research centered on applied mechanics, hydrodynamics, and cavitation. His name therefore appears both in electronic-structure perturbation theory and, through the Rayleigh–Plesset equation, in bubble dynamics.

Biographies: Mogens Pihl, Danish Biographical Dictionary; Caltech oral history. Source portraits: Møller, Rochester, 1957, AIP Emilio Segrè Visual Archives, Marshak Collection, existing Commons detail, **draft reproduction: permission pending**; Plesset, Copenhagen, 1963, Friedrich Hund estate / Gerhard Hund (GFHund), via Commons, CC BY 3.0. Sketch: AI-assisted adaptation of the credited portrait.

The electronic Hamiltonian can be partitioned as

$$\boxed{\hat{\mathcal{H}}_{\text{elec}} = \underbrace{\sum_{i=1}^{N} \hat{f}(\boldsymbol{x}_i)}_{\hat{\mathcal{H}}^{(0)}} + \underbrace{\left[\sum_{i<j} \frac{1}{r_{ij}} - \sum_{i=1}^{N} \hat{v}^{\text{HF}}(\boldsymbol{x}_i)\right]}_{\hat{\mathcal{V}}}} \tag{1.367}$$

The zeroth-order Hamiltonian is therefore the sum of the Fock operators,

$$\hat{\mathcal{H}}^{(0)} = \hat{\mathcal{H}}^{\text{HF}} \tag{1.368}$$

and its eigenvalue for the HF reference determinant is

$$E^{(0)} = \sum_{i=1}^{N} \epsilon_i \tag{1.369}$$

For an Aufbau reference occupying the lowest spin orbitals, this is also the ground-state eigenvalue of the auxiliary operator. For a non-Aufbau reference, such as those targeted by the maximum-overlap method below, it is the eigenvalue of the selected determinant instead. The first-order correction is the expectation value of the perturbation over the Hartree–Fock determinant:

$$\begin{aligned} E^{(1)} &= \langle \Psi_{\text{HF}} | \hat{\mathcal{V}} | \Psi_{\text{HF}} \rangle \\ &= \langle \Psi_{\text{HF}} | \sum_{i<j} \frac{1}{r_{ij}} - \sum_{i=1}^{N} \hat{v}^{\text{HF}}(\boldsymbol{x}_i) | \Psi_{\text{HF}} \rangle \\ &= -\frac{1}{2} \sum_{ij}^{N} \left(\mathcal{J}_{ij} - \mathcal{K}_{ij} \right) \end{aligned} \tag{1.370}$$

Consequently,

$$E^{(0)} + E^{(1)} = E_{\text{HF}} \tag{1.371}$$

Thus, in the conventional Møller–Plesset partitioning, the sum of the occupied HF orbital energies is the zeroth-order energy, while the first-order correction removes the double counting of the electron-electron interaction. Correlation beyond Hartree–Fock first appears at second order, leading to the MP2 correlation energy [20].

The MP2 correlation energy. The HF determinant is stationary against single substitutions, but it still couples to doubly substituted determinants through the two-electron interaction. MP2 turns these couplings into the first correction beyond the mean-field energy. For a nondegenerate canonical Hartree–Fock reference, second-order perturbation theory gives

$$E^{(2)} = \sum_{n\neq 0} \frac{|\langle \Psi_n|\hat{\mathcal{V}}|\Psi_0\rangle|^2}{E^{(0)} - E_n^{(0)}} \tag{1.372}$$

Here $|\Psi_n\rangle$ are excited determinants constructed from the canonical orbital basis, eigenfunctions of $\hat{\mathcal{H}}^{(0)}$, and the reference is $|\Psi_0\rangle = |\Psi_{\mathrm{HF}}\rangle$. Brillouin's theorem eliminates singles, and the two-electron character of $\hat{\mathcal{V}}$ eliminates couplings to triples and higher substitutions. Therefore, only doubly-excited determinants remain. For a double substitution, the one-electron Hartree–Fock potential cannot connect the two determinants, so the remaining matrix element is the antisymmetrized ERI. Consequently,

$$E_{\mathrm{c}}^{(2)} = \sum_{i<j}^{\mathrm{occ}} \sum_{a<b}^{\mathrm{virt}} \frac{|\langle ij||ab\rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} = \frac{1}{4} \sum_{ij}^{\mathrm{occ}} \sum_{ab}^{\mathrm{virt}} \frac{|\langle ij||ab\rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} \tag{1.373}$$

These indices label spin orbitals. The unrestricted sums count each pair of occupied labels and each pair of virtual labels twice, explaining the factor $1/4$; coincident labels contribute zero by antisymmetry. The absolute square also makes this expression valid for complex canonical orbitals. The MP2 electronic energy is

$$E_{\mathrm{MP2}} = E_{\mathrm{HF}} + E_{\mathrm{c}}^{(2)} \tag{1.374}$$

For an Aufbau reference with a positive occupied-virtual gap, every denominator in Eq. (1.373) is negative, so the second-order correction is nonpositive. Small excitation gaps, however, make the correction sensitive to the denominators; the nondegenerate perturbative construction should not be interpreted as a remedy for a nearly degenerate HF reference.

For implementation, it is useful to specialize to a closed-shell RHF reference with *real* spatial orbitals. Let $O = N/2$ and $V = K - O$ be the numbers of occupied and virtual spatial orbitals. As usual, i, j label the O occupied spatial orbitals, and a, b the V virtual spatial orbitals. Carrying out the spin sum gives

$$\begin{aligned} E_{\mathrm{c}}^{(2)} &= \sum_{ij}^{O} \sum_{ab}^{V} \frac{\langle ij|ab\rangle \left(2\langle ij|ab\rangle - \langle ij|ba\rangle\right)}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} \\ &= \underbrace{2 \sum_{ij}^{O} \sum_{ab}^{V} \frac{\langle ij|ab\rangle^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}}_{\text{direct contribution}} - \underbrace{\sum_{ij}^{O} \sum_{ab}^{V} \frac{\langle ij|ab\rangle \langle ij|ba\rangle}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}}_{\text{exchange contribution}} \end{aligned} \tag{1.375}$$

The factor of two in the direct term is the result of spin summation, not an additional empirical weight. More explicitly, if the spatial direct and exchanged integrals are denoted temporarily by $A = \langle ij|ab\rangle$ and $B = \langle ij|ba\rangle$, summing the squared spin-orbital integral over its four spin labels gives $4A^2 + 4B^2 - 4AB$. Multiplication by $1/4$ and interchange of the dummy virtual labels in the B^2 sum yield $2A^2 - AB$ after summation.

Why a four-loop energy expression leads to a fifth-order method. If the MO integrals are already available, evaluating Eq. (1.375) requires only a loop over each occupied pair and each virtual pair. One initializes the energy accumulator to zero, forms the denominator for each quartet, and adds its direct-minus-exchange contribution. With

$$\Delta_{ij}^{ab} = \epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b \tag{1.376}$$

the accumulation step is

$$E_{\mathrm{c}}^{(2)} \longleftarrow E_{\mathrm{c}}^{(2)} + \frac{2\langle ij|ab\rangle^2 - \langle ij|ab\rangle\langle ij|ba\rangle}{\Delta_{ij}^{ab}} \tag{1.377}$$

There are O^2V^2 such quartets. Thus the final energy summation costs $\mathcal{O}(O^2V^2)$, or $\mathcal{O}(K^4)$ when both orbital populations grow proportionally to K. The accumulator itself requires only scalar storage. This quartic summation does not explain the familiar fifth-order cost of conventional MP2 because it assumes that the transformed integrals have already been obtained. That preparation is the missing computational step.

The distinction between integral conventions matters here. The spatial integrals required by the MP2 expression are

$$\langle ij|ab\rangle = (ia\,|\,jb) \qquad\qquad \langle ij|ba\rangle = (ib\,|\,ja) \tag{1.378}$$

Thus MP2 needs the occupied-virtual pair block of the chemists' ERI tensor, not the block $(ij\,|\,ab)$. This follows from the physicists'-to-chemists' index conversion established earlier.

The naive AO-to-MO transformation. The integrals generated by a molecular program are initially expressed in the AO basis. Although the finite-basis equations are developed in the next section, the required change of representation can already be understood from $\psi_p = \sum_\mu C_{\mu p}\phi_\mu$. Retaining the real-orbital convention and using chemists' notation, substitution of the four orbital expansions gives

$$\underbrace{(pq\,|\,rs)}_{\text{MO integral}} = \sum_{\mu\nu\lambda\sigma}^{K} C_{\mu p}C_{\nu q}C_{\lambda r}C_{\sigma s}\underbrace{(\mu\nu\,|\,\lambda\sigma)}_{\text{AO integral}} \tag{1.379}$$

Here p, q, r, s are free MO labels, whereas the four Greek AO labels are summed. For a full transformation there are K^4 output integrals, and computing each independently requires K^4 AO terms. A literal algorithm therefore loops over p, q, r, s, initializes that output entry to zero, and accumulates it through four further loops over $\mu, \nu, \lambda, \sigma$. The resulting eight nested loops cost $\mathcal{O}(K^8)$. This is the cost of a poor contraction order, not an intrinsic cost of changing orbital basis.

Four successive one-index transformations. A semi-direct construction avoids repeating the same partial sums by transforming one AO index at a time [27]. The four stages are

$$I_{\mu\nu\lambda s} = \sum_\sigma C_{\sigma s}(\mu\nu\,|\,\lambda\sigma) \tag{1.380}$$

$$J_{\mu\nu rs} = \sum_\lambda C_{\lambda r}I_{\mu\nu\lambda s} \tag{1.381}$$

$$I_{\mu qrs} = \sum_\nu C_{\nu q}J_{\mu\nu rs} \tag{1.382}$$

$$(pq\,|\,rs) = \sum_\mu C_{\mu p}I_{\mu qrs} \tag{1.383}$$

The labels I and J denote temporary four-index arrays, not Coulomb operators or identity matrices. An accumulation implementation allocates and zeros I before the first stage and J before the second. The first stage contracts σ for each fixed (μ, ν, λ, s) and stores the result in I. It has four output loops and one contraction loop, so costs $\mathcal{O}(K^5)$ and requires $\mathcal{O}(K^4)$ storage. The second stage contracts λ and writes J; its cost and storage have the same orders.

Once J is complete, the original contents of I are no longer needed. The third stage therefore reuses that buffer, now with index pattern (μ, q, r, s), to contract ν. Its entries must be reset before accumulation; no new intermediate buffer is required. Finally, contraction of μ writes the MO integral array. Both final stages again have five loops and cost $O(K^5)$. The output is also quartic in size, but it does not require another kind of intermediate. The AO data, output tensor, and temporary buffers together retain a quartic storage requirement.

Substitution of each intermediate into the following stage recovers Eq. (1.379) exactly. The improvement is therefore purely algebraic: four fifth-order contractions replace an eighth-order one without approximating any integral. In this full-tensor construction,

$$\text{transformation work} = O(K^5) \qquad \text{tensor storage} = O(K^4) \tag{1.384}$$

The transformed integrals can then be used in the quartic MP2 energy summation. This explains the conventional fifth-order scaling. In an actual MP2 calculation, only the $(ia\,|\,jb)$ block is required, so the output can be restricted to O^2V^2 entries rather than retaining every MO integral.

2 Finite-basis Hartree–Fock and self-consistent-field methods

The orbital equations derived above define the Hartree–Fock stationary condition, but they are not yet in a form suited to routine molecular calculations. This section first projects the continuous equations into a finite nonorthogonal basis, then develops the iterative procedure required by the density dependence of the Fock operator. The same matrix formulation also provides the natural setting for convergence acceleration, state-specific SCF solutions, and the evaluation and interpretation of molecular properties.

2.1 The Roothaan–Hall equations

Projection onto a finite nonorthogonal basis converts each orbital equation into a generalized matrix eigenvalue problem. The construction proceeds from the basis expansion and overlap metric to orthogonalization, Fock-matrix assembly, and the finite-basis energy expression [28, 29].

Historical perspective: Clemens C. J. Roothaan (1918–2019)

Born in Nijmegen in the Netherlands, Roothaan earned his doctorate at the University of Chicago under Robert Mulliken in 1950. His 1951 formulation of the molecular self-consistent-field problem in a finite basis [28], developed independently of Hall's, helped turn Hartree–Fock theory into a practical computational method. He remained closely involved in scientific computing, directing Chicago's Computation Center from 1962 to 1968.

Biography: University of Chicago obituary. Source portrait: University of Chicago Photographic Archive, apf1-12856, Hanna Holborn Gray Special Collections Research Center, University of Chicago Library. Educational/scholarly reuse; commercial permission required. Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: George Garfield Hall (1925–2018)

Born in Belfast, Hall studied at Cambridge under John Lennard-Jones. His 1951 treatment of molecular orbitals gave an independent derivation of the matrix self-consistent-field equations now known as the Roothaan–Hall equations [29]. He became professor of applied mathematics at Nottingham in 1962. Alongside quantum chemistry, his career included substantial work in mathematical education, illustrating the mathematical breadth behind the molecular-orbital methods.

Biography: IAQMS obituary. Source portrait: IAQMS archive; photographer and date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

2.1.1 The basis-set approximation

The Hartree–Fock equations derived in the previous section are formally equations for unknown functions,

$$\hat{f}(\boldsymbol{r})\psi_i(\boldsymbol{r}) = \epsilon_i\psi_i(\boldsymbol{r}) \tag{2.1}$$

Solving these equations directly in continuous three-dimensional space is possible in special cases, most notably for atoms, but is inconvenient for general molecules. In practical molecular calculations, the spatial orbitals are therefore expanded in a finite set of known one-electron basis functions. This introduces a second approximation, distinct from the Hartree–Fock single-determinant approximation itself. Hartree–Fock restricts the many-electron wavefunction to a single determinant, whereas the basis-set approximation restricts the one-electron functions entering that determinant to a finite-dimensional space.

Expansion of the molecular orbitals. Let

$$\{\phi_\mu(\boldsymbol{r})\}_{\mu=1}^{K} \tag{2.2}$$

be a set of K basis functions. Each molecular orbital is expanded as

$$\psi_i(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}\phi_\mu(\boldsymbol{r}) \qquad \equiv \qquad |i\rangle = \sum_{\mu=1}^{K} C_{\mu i}\,|\mu\rangle \tag{2.3}$$

The unknown functions $\psi_i(\boldsymbol{r})$ have therefore been replaced by the unknown expansion coefficients $C_{\mu i}$. This is the linear combination of atomic orbitals (LCAO) representation when the basis functions are centered on the nuclei, as is usually the case in molecular electronic-structure calculations. More generally, however, the derivation requires only that the functions span a suitable finite one-electron space. For a fixed basis, the Hartree–Fock problem is therefore no longer a variational problem over arbitrary functions, but over the coefficients contained in the matrix $\boldsymbol{C}$.

Nonorthogonality of the basis functions. The molecular orbitals are required to be orthonormal,

$$\langle\psi_i|\psi_j\rangle = \delta_{ij} \tag{2.4}$$

but the underlying basis functions generally are not:

$$S_{\mu\nu} = \langle\mu|\nu\rangle = \int \phi_\mu^*(\boldsymbol{r})\phi_\nu(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{2.5}$$

The matrix $\boldsymbol{S}$ is called the overlap matrix. Substituting the basis expansion into the MO orthonormality condition gives

$$\left\langle \psi_i \middle| \psi_j \right\rangle = \sum_{\mu\nu} C^*_{\mu i} C_{\nu j} \left\langle \mu | \nu \right\rangle = \sum_{\mu\nu} C^*_{\mu i} S_{\mu\nu} C_{\nu j} = \delta_{ij} \tag{2.6}$$

In matrix notation,

$$\boxed{\boldsymbol{C}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C} = \mathbf{1}} \tag{2.7}$$

Thus, the coefficient matrix is generally *not* unitary:

$$\boldsymbol{C}^\dagger \cdot \boldsymbol{C} \neq \mathbf{1} \tag{2.8}$$

Instead, the molecular orbitals are orthonormal with respect to the metric defined by $\boldsymbol{S}$.

If the basis functions are linearly independent, $\boldsymbol{S}$ is Hermitian and positive definite. If two or more basis functions become nearly linearly dependent, one or more eigenvalues of $\boldsymbol{S}$ become very small, which can lead to numerical difficulties. This issue becomes particularly relevant for large and diffuse basis sets.

From the HF equation to the Roothaan–Hall equations. We now insert the finite-basis expansion into the canonical HF equation,

$$\hat{f} \left| i \right\rangle = \epsilon_i \left| i \right\rangle \tag{2.9}$$

Using

$$\left| i \right\rangle = \sum_\nu C_{\nu i} \left| \nu \right\rangle \tag{2.10}$$

gives

$$\hat{f} \sum_\nu C_{\nu i} \left| \nu \right\rangle = \epsilon_i \sum_\nu C_{\nu i} \left| \nu \right\rangle \tag{2.11}$$

Projecting from the left with $\langle \mu |$ yields

$$\sum_\nu C_{\nu i} \left\langle \mu | \hat{f} | \nu \right\rangle = \epsilon_i \sum_\nu C_{\nu i} \left\langle \mu | \nu \right\rangle \tag{2.12}$$

Defining the Fock matrix

$$F_{\mu\nu} = \left\langle \mu | \hat{f} | \nu \right\rangle \tag{2.13}$$

and recalling

$$S_{\mu\nu} = \left\langle \mu | \nu \right\rangle \tag{2.14}$$

we obtain

$$\sum_\nu F_{\mu\nu} C_{\nu i} = \sum_\nu S_{\mu\nu} C_{\nu i} \epsilon_i \tag{2.15}$$

The complete derivation can therefore be summarized as

$$\begin{aligned} \hat{f} \left| i \right\rangle = \epsilon_i \left| i \right\rangle \Rightarrow \quad & \hat{f} \sum_\nu C_{\nu i} \left| \nu \right\rangle = \epsilon_i \sum_\nu C_{\nu i} \left| \nu \right\rangle \\ \xRightarrow{\langle \mu |} \quad & \sum_\nu C_{\nu i} \underbrace{\left\langle \mu | \hat{f} | \nu \right\rangle}_{F_{\mu\nu}} = \epsilon_i \sum_\nu C_{\nu i} \underbrace{\left\langle \mu | \nu \right\rangle}_{S_{\mu\nu}} \\ \Rightarrow \quad & \sum_\nu F_{\mu\nu} C_{\nu i} = \sum_\nu S_{\mu\nu} C_{\nu i} \epsilon_i \end{aligned} \tag{2.16}$$

Collecting all molecular orbitals simultaneously gives

$$\boxed{\boldsymbol{F}\cdot\boldsymbol{C}=\boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{\epsilon}} \tag{2.17}$$

These are the Roothaan–Hall equations. Here,

$$\boldsymbol{C}=\begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1K} \\ C_{21} & C_{22} & \cdots & C_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ C_{K1} & C_{K2} & \cdots & C_{KK} \end{pmatrix} \tag{2.18}$$

contains the MO coefficients, while

$$\boldsymbol{\epsilon}=\begin{pmatrix} \epsilon_1 & 0 & \cdots & 0 \\ 0 & \epsilon_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \epsilon_K \end{pmatrix} \tag{2.19}$$

contains the canonical orbital energies.

Equation (2.17) is the finite-dimensional counterpart of the canonical HF equation. The basis-set expansion has converted the integro-differential orbital problem into a matrix problem.

A generalized eigenvalue problem. If the basis were orthonormal, i.e., $\boldsymbol{S}=\mathbf{1}$, the Roothaan–Hall equations would reduce to the ordinary eigenvalue problem

$$\boldsymbol{F}\cdot\boldsymbol{C}=\boldsymbol{C}\cdot\boldsymbol{\epsilon} \tag{2.20}$$

For an atomic-orbital basis, however, $\boldsymbol{S}\neq\mathbf{1}$, and the equations instead constitute a generalized eigenvalue problem,

$$\boldsymbol{F}\cdot\boldsymbol{C}=\boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{\epsilon} \tag{2.21}$$

For an individual orbital,

$$\boldsymbol{F}\cdot\boldsymbol{C}_i=\epsilon_i\boldsymbol{S}\cdot\boldsymbol{C}_i \tag{2.22}$$

where $\boldsymbol{C}_i$ is the ith column of $\boldsymbol{C}$. Nontrivial solutions satisfy the secular equation

$$\det\left(\boldsymbol{F}-\epsilon_i\boldsymbol{S}\right)=0 \tag{2.23}$$

The presence of $\boldsymbol{S}$ is not a peculiarity of Hartree–Fock theory. It simply reflects the fact that the coordinates used to represent the one-electron functions form a nonorthogonal basis.

2.1.2 Orthogonalization and the generalized eigenvalue problem

A generalized Hermitian eigenvalue problem can be converted into an ordinary Hermitian eigenvalue problem by transforming to an orthonormal basis.

We seek a nonsingular matrix $\boldsymbol{X}$ satisfying

$$\boxed{\boldsymbol{X}^\dagger\cdot\boldsymbol{S}\cdot\boldsymbol{X}=\mathbf{1}} \tag{2.24}$$

The transformed basis functions may be viewed as

$$|\mu'\rangle=\sum_\nu|\nu\rangle\,X_{\nu\mu} \tag{2.25}$$

Their overlaps are

$$\langle \mu' | \nu' \rangle = \left(\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} \right)_{\mu\nu} = \delta_{\mu\nu} \tag{2.26}$$

so the transformed basis is orthonormal by construction.

Writing

$$\boldsymbol{C} = \boldsymbol{X} \cdot \boldsymbol{C}' \tag{2.27}$$

and substituting into the Roothaan–Hall equations gives

$$\begin{aligned} \boldsymbol{F} \cdot \boldsymbol{C} &= \boldsymbol{S} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{F} \cdot \boldsymbol{X} \cdot \boldsymbol{C}' &= \boldsymbol{S} \cdot \boldsymbol{X} \cdot \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \end{aligned} \tag{2.28}$$

Multiplying from the left by $\boldsymbol{X}^\dagger$ gives

$$\boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X} \cdot \boldsymbol{C}' = \boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} \cdot \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \tag{2.29}$$

Using Eq. (2.24),

$$\begin{aligned} \boldsymbol{F}' \cdot \boldsymbol{C}' &= \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \\ \boldsymbol{F}' &= \boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X} \end{aligned} \tag{2.30}$$

Thus,

$$\boldsymbol{F} \cdot \boldsymbol{C} = \underbrace{\boldsymbol{S}}_{\text{metric}} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \qquad \boldsymbol{F}' \cdot \boldsymbol{C}' = \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \tag{2.31}$$

$$\boldsymbol{F}' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X} \qquad \boldsymbol{C} = \boldsymbol{X} \cdot \boldsymbol{C}' \qquad \boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1} \tag{2.32}$$

Every step from the generalized problem to the standard problem is

$$\boldsymbol{F} \cdot \boldsymbol{C} = \underbrace{\boldsymbol{S}}_{\text{metric}} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \qquad \Longleftrightarrow \qquad \text{generalized eigenvalue problem} \tag{2.33}$$

$$\Downarrow \tag{2.34}$$

$$\boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \underbrace{\boldsymbol{C}}_{\boldsymbol{X} \cdot \boldsymbol{C}'} = \boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \underbrace{\boldsymbol{C}}_{\boldsymbol{X} \cdot \boldsymbol{C}'} \cdot \boldsymbol{\epsilon} \tag{2.35}$$

$$\Downarrow \tag{2.36}$$

$$\underbrace{\boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X}}_{\boldsymbol{F}'} \cdot \boldsymbol{C}' = \underbrace{\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X}}_{\mathbf{1}} \cdot \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \tag{2.37}$$

$$\Downarrow \tag{2.38}$$

$$\boldsymbol{F}' \cdot \boldsymbol{C}' = \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \qquad \Longleftrightarrow \qquad \text{standard eigenvalue problem} \tag{2.39}$$

Thus, orthogonalization converts the generalized problem into a standard eigenvalue problem.

Because $\boldsymbol{F}$ is Hermitian and $\boldsymbol{X}$ is nonsingular,

$$\boldsymbol{F}'^\dagger = \left(\boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X} \right)^\dagger = \boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X} = \boldsymbol{F}' \tag{2.40}$$

so $\boldsymbol{F}'$ is also Hermitian. It may therefore be diagonalized by a unitary transformation, and its eigenvalues are real.

Choice of the orthogonalization matrix. The matrix $\boldsymbol{X}$ is not unique. Since the overlap matrix is Hermitian and positive definite, it may be diagonalized as

$$\boldsymbol{S} = \boldsymbol{U} \cdot \boldsymbol{s} \cdot \boldsymbol{U}^{\dagger} \tag{2.41}$$

where

$$\boldsymbol{s} = \mathrm{diag}(s_1, s_2, \ldots, s_K) \qquad\qquad s_p > 0 \tag{2.42}$$

A particularly common choice is the symmetric, or Löwdin, orthogonalizer [30],

$$\boxed{\boldsymbol{X} = \boldsymbol{S}^{-1/2} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2} \cdot \boldsymbol{U}^{\dagger}} \tag{2.43}$$

Indeed,

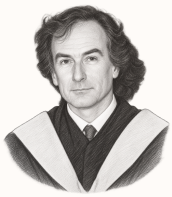
$$\boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \boldsymbol{S}^{-1/2} \cdot \boldsymbol{S} \cdot \boldsymbol{S}^{-1/2} = \boldsymbol{1} \tag{2.44}$$

Another frequently used choice is the canonical orthogonalizer [31],

$$\boldsymbol{X} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2} \tag{2.45}$$

Both choices span the same one-electron space and produce the same generalized eigenvalues. They simply correspond to different orthonormal representations of that space.

Historical perspective: Per-Olov Löwdin (1916–2000)



A Swedish theoretical scientist, Löwdin built influential quantum-chemistry schools at Uppsala and the University of Florida. His 1950 study of nonorthogonal atomic orbitals established the symmetric orthogonalization used here [30]. His research emphasized the mathematical foundations of quantum chemistry, while his international summer and winter schools and the Sanibel symposia helped train successive generations of electronic-structure researchers.

Biography: IAQMS profile. Source portrait: Sorbonne Université, unknown photographer, 25 November 1977, honorary-doctorate ceremony (detail), Licence Ouverte; via Wikimedia Commons. Sketch: AI-assisted adaptation of the credited portrait.

If some eigenvalues s_p of the overlap matrix are extremely small, the factor $s_p^{-1/2}$ becomes very large. In practical implementations, directions associated with sufficiently small overlap eigenvalues may therefore be removed to avoid numerical instabilities caused by near-linear dependencies.

Recovering the molecular-orbital coefficients. After diagonalizing the transformed Fock matrix,

$$\boldsymbol{F}' \cdot \boldsymbol{C}' = \boldsymbol{C}' \cdot \boldsymbol{\epsilon} \tag{2.46}$$

the coefficients in the original nonorthogonal basis are recovered through

$$\boldsymbol{C} = \boldsymbol{X} \cdot \boldsymbol{C}' \tag{2.47}$$

Since the eigenvectors of the Hermitian matrix $\boldsymbol{F}'$ may be chosen orthonormal,

$$\boldsymbol{C}'^{\dagger} \cdot \boldsymbol{C}' = \boldsymbol{1} \tag{2.48}$$

we recover the correct generalized normalization,

$$\begin{aligned} \boldsymbol{C}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{C} &= \boldsymbol{C}'^{\dagger} \cdot \boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X} \cdot \boldsymbol{C}' \\ &= \boldsymbol{C}'^{\dagger} \cdot \boldsymbol{C}' \\ &= \boldsymbol{1} \end{aligned} \tag{2.49}$$

Thus the orthogonalization procedure does not alter the physical molecular orbitals; it merely provides a convenient orthonormal coordinate system in which the generalized eigenvalue problem can be solved using standard Hermitian eigensolvers.

The matrix problem is still self-consistent. At first sight,

$$\boldsymbol{F} \cdot \boldsymbol{C} = \boldsymbol{S} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \tag{2.50}$$

looks like an ordinary generalized eigenvalue problem. There is, however, one essential complication: the Fock matrix depends on the occupied molecular orbitals and therefore on the coefficient matrix itself,

$$\boldsymbol{F} = \boldsymbol{F}(\boldsymbol{C}) \tag{2.51}$$

The actual Roothaan–Hall problem is therefore

$$\boldsymbol{F}(\boldsymbol{C}) \cdot \boldsymbol{C} = \boldsymbol{S} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \tag{2.52}$$

It is a nonlinear generalized eigenvalue problem, not a single matrix diagonalization.

Schematically,

$$\boldsymbol{C} \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{F}[\boldsymbol{P}] \longrightarrow \boldsymbol{C} \tag{2.53}$$

where $\boldsymbol{P}$ is the one-particle density matrix constructed from the occupied molecular orbitals [32].

Historical perspective: Roy McWeeny (1924–2021)

A British theoretical chemist, McWeeny earned his doctorate with Charles Coulson at Oxford and worked in Newcastle, Keele, and Sheffield before moving to Pisa in 1982. His work made reduced density matrices a powerful language for electronic structure. His 1960 account [32] brought together their use in orbital theories, connecting electron densities, energies, and the constraints associated with a determinantal state. His wider contributions included valence-bond theory, multiconfigurational SCF methods, and the response of molecules to external fields.

Biography: Brian Sutcliffe and Michael Robb, IAQMS obituary, 2021. Source portrait: IAQMS archive; photographer and date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

This circular dependence is the finite-basis manifestation of the self-consistency already encountered in the continuous HF equations: How do we solve these HF equations? The answer is the self-consistent-field procedure developed below.

2.1.3 The variational meaning of the finite basis

Restricting the orbitals to a finite basis reduces the variational freedom available to the Hartree–Fock determinant. Consequently, for a given Hamiltonian,

Variational bounds and the Hartree–Fock limit

$$E_{\mathrm{HF}}^{(K)} \geq E_{\mathrm{HF}}^{\mathrm{CBS}} \geq E_{\mathrm{exact}} \tag{2.54}$$

Here, $E_{\mathrm{HF}}^{(K)}$ is the minimum HF energy in a finite K-dimensional one-electron space, whereas $E_{\mathrm{HF}}^{\mathrm{CBS}}$ denotes the complete-basis-set (CBS) HF limit. The bounds concern the global minima within the same chosen determinant class, not arbitrary stationary SCF solutions.

The first inequality reflects basis-set incompleteness: a finite basis cannot describe all possible orbital shapes. The second reflects the single-determinant approximation: even with a complete one-electron basis, Hartree–Fock theory explores only the manifold of Slater determinants, whereas the exact wavefunction belongs to the much larger space of antisymmetric many-electron functions.

For a nested sequence of basis spaces,

$$\mathcal{B}_1 \subset \mathcal{B}_2 \subset \cdots \tag{2.55}$$

the variational principle implies

$$E_{\mathrm{HF}}^{(1)} \geq E_{\mathrm{HF}}^{(2)} \geq \cdots \geq E_{\mathrm{HF}}^{\mathrm{CBS}} \tag{2.56}$$

Enlarging the variational space can lower, or leave unchanged, the optimized HF energy, but cannot raise it.

It is useful to keep these two approximations conceptually separate:

$$\boxed{\text{finite basis} \quad + \quad \text{single determinant} \quad \Longrightarrow \quad \text{practical finite-basis Hartree–Fock}} \tag{2.57}$$

The following sections examine first how the Fock matrix is constructed in this finite basis and then how the resulting nonlinear Roothaan–Hall equations are solved self-consistently.

2.1.4 The Fock matrix

The Roothaan–Hall equations become operational only after the matrix elements of the Fock operator have been expressed in the atomic-orbital basis. This is the point at which the one-electron Hamiltonian, the electron–electron interaction, and the occupied molecular orbitals are brought together in a single matrix quantity.

From the orbital operator to AO matrix elements. The spin-orbital Fock operator is a one-electron operator even though it contains the mean effect of the two-electron interaction. Its matrix element between two AO basis functions therefore separates naturally into a fixed one-electron contribution and a density-dependent contribution,

$$F_{\mu\nu} = H_{\mu\nu} + G_{\mu\nu}[\boldsymbol{P}] \tag{2.58}$$

The following worked example makes the contraction hidden in $G_{\mu\nu}[\boldsymbol{P}]$ explicit. It first uses the general spin-orbital formulation. In this example and the general AO-energy exercise below, $\chi_\mu^{\mathrm{AO}}(\boldsymbol{x})$ denotes a basis function formed from a spatial AO and a spin function. The Greek basis label temporarily collects both labels and runs over $2K$ spin-AO functions; the coefficients $C_{\mu i}$ refer to this spin-AO expansion. The closed-shell specialization subsequently returns to the K spatial AO functions $\phi_\mu(\boldsymbol{r})$ and a spin-summed density. Thus ϕ continues to denote a spatial basis function and χ a spin orbital.

Exercise 13: AO Fock matrix

Express the Fock matrix in the spin-AO basis in terms of the one- and two-electron integrals and the occupied-orbital density matrix.

Solution 13: AO Fock matrix

Use the spin-AO expansion defined above, with one occupied spin orbital per electron and no closed-shell factor of two. Allow complex coefficients, so the coefficient from the bra orbital must be conjugated. Expanding the occupied spin orbitals in the basis exposes how their coefficients

Solution 13 (continued): AO Fock matrix

enter the AO Fock matrix:

$$\begin{aligned} F_{\mu\nu} &= \langle\mu|\hat{h} + \sum_{i=1}^{N}(\hat{\mathcal{J}}_i - \hat{\mathcal{K}}_i)|\nu\rangle \\ &= \underbrace{\langle\mu|\hat{h}|\nu\rangle}_{H_{\mu\nu}} + \sum_{i=1}^{N}\langle\mu|\hat{\mathcal{J}}_i - \hat{\mathcal{K}}_i|\nu\rangle \\ &= H_{\mu\nu} + \sum_{i=1}^{N}\left(\langle\mu\chi_i|r_{12}^{-1}|\nu\chi_i\rangle - \langle\mu\chi_i|r_{12}^{-1}|\chi_i\nu\rangle\right) \\ &= H_{\mu\nu} + \sum_{i=1}^{N}\sum_{\lambda\sigma}C_{\lambda i}C_{\sigma i}^{*}\left(\langle\mu\sigma|r_{12}^{-1}|\nu\lambda\rangle - \langle\mu\sigma|r_{12}^{-1}|\lambda\nu\rangle\right) \\ &= H_{\mu\nu} + \sum_{\lambda\sigma}P_{\lambda\sigma}(\langle\mu\sigma|\nu\lambda\rangle - \langle\mu\sigma|\lambda\nu\rangle) \\ &= H_{\mu\nu} + \underbrace{\sum_{\lambda\sigma}P_{\lambda\sigma}\langle\mu\sigma||\nu\lambda\rangle}_{G_{\mu\nu}} = H_{\mu\nu} + G_{\mu\nu} \end{aligned} \tag{2.59}$$

The fourth line follows by expanding both occurrences of the occupied orbital χ_i,

$$\begin{aligned} \chi_i &= \sum_{\lambda}C_{\lambda i}\chi_{\lambda}^{\mathrm{AO}} \\ \chi_i^{*} &= \sum_{\sigma}C_{\sigma i}^{*}\chi_{\sigma}^{\mathrm{AO}*} \end{aligned} \tag{2.60}$$

The sum over occupied orbitals can then be performed before the integral contraction:

$$\sum_{i=1}^{N}C_{\lambda i}C_{\sigma i}^{*} = P_{\lambda\sigma} \tag{2.61}$$

Thus the occupied orbitals enter the Fock matrix only through their one-particle density matrix. In particular, $P_{\lambda\sigma}$ contains the ket coefficient $C_{\lambda i}$ and the conjugated bra coefficient $C_{\sigma i}^{*}$. This ordering fixes the integral indices even when the coefficients are complex. No overlap matrix is inserted in this contraction: the AO integrals already contain the basis functions, while their overlap metric enters orbital normalization and the generalized eigenvalue problem.
The three quantities introduced in this derivation are the core Hamiltonian, the density matrix, and the two-electron part:

$$\begin{aligned} H_{\mu\nu} &= \langle\mu|\hat{h}|\nu\rangle \\ P_{\lambda\sigma} &= \sum_{i=1}^{N}C_{\lambda i}C_{\sigma i}^{*} \\ G_{\mu\nu} &= \sum_{\lambda\sigma}P_{\lambda\sigma}\langle\mu\sigma||\nu\lambda\rangle \end{aligned} \tag{2.62}$$

The density matrix $\boldsymbol{P}$ thus collects all occupied-orbital coefficients needed for the two-electron contraction. Its explicit form also makes the nonlinearity of the Roothaan–Hall equations apparent.

Solution 13 (continued): AO Fock matrix

The separation

$$\boldsymbol{F} = \boldsymbol{H} + \boldsymbol{G}[\boldsymbol{P}] \tag{2.63}$$

is worth emphasizing. The core Hamiltonian $\boldsymbol{H}$ is fixed once the molecular geometry and basis set have been chosen. By contrast, $\boldsymbol{G}$ must be rebuilt whenever the density changes. Thus, although each diagonalization is an ordinary linear-algebra operation, the complete Hartree–Fock problem remains self-consistent.

Interpretation of the AO contraction. For every pair (μ, ν), the two-electron part is obtained by contracting the four-index integral tensor with the two-index density matrix. The density selects the occupied part of the one-electron space, while the integral tensor supplies the Coulomb interaction and its exchange counterpart. Schematically,

$$\boxed{\text{density} \quad + \quad \text{two-electron integrals} \quad \longrightarrow \quad \text{mean-field potential}} \tag{2.64}$$

This contraction is the central repeated operation in a conventional SCF calculation.

2.1.5 The Hartree–Fock energy

The matrix form of the energy is obtained from the same ingredients. It is particularly useful in an implementation because it allows the energy to be evaluated directly from the density, core Hamiltonian, and the Fock matrix constructed from that density, without transforming all integrals to the molecular-orbital basis.

Exercise 14: AO Hartree–Fock energy

Express the electronic HF energy in the spin-AO basis using the density matrix, and derive its trace form in terms of $\boldsymbol{H}$ and $\boldsymbol{F}$.

Solution 14: AO Hartree–Fock energy

Retain the spin-AO convention of the preceding exercise: $\boldsymbol{P}$ counts each occupied spin orbital once, and the coefficients need not be real. The energy below excludes nuclear repulsion. Starting

Solution 14 (continued): AO Hartree–Fock energy

from the spin-orbital energy and explicitly substituting each basis expansion gives

$$\begin{aligned} E_{\mathrm{HF}} &= \sum_{i=1}^{N} h_{ii} + \frac{1}{2}\sum_{ij}^{N}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right) \\ &= \sum_{i=1}^{N}\left\langle \sum_{\mu} C_{\mu i}\chi_{\mu}^{\mathrm{AO}} \middle| \hat{h} \middle| \sum_{\nu} C_{\nu i}\chi_{\nu}^{\mathrm{AO}} \right\rangle \\ &+ \frac{1}{2}\sum_{ij}^{N}\left\langle \left(\sum_{\mu} C_{\mu i}\chi_{\mu}^{\mathrm{AO}}\right)\left(\sum_{\lambda} C_{\lambda j}\chi_{\lambda}^{\mathrm{AO}}\right) \middle\| \left(\sum_{\nu} C_{\nu i}\chi_{\nu}^{\mathrm{AO}}\right)\left(\sum_{\sigma} C_{\sigma j}\chi_{\sigma}^{\mathrm{AO}}\right)\right\rangle \\ &= \sum_{\mu\nu} P_{\nu\mu}\left(H_{\mu\nu} + \frac{1}{2}\sum_{\lambda\sigma} P_{\sigma\lambda}\left\langle \mu\lambda||\nu\sigma\right\rangle\right) \\ &= \frac{1}{2}\sum_{\mu\nu} P_{\nu\mu}\left(H_{\mu\nu} + F_{\mu\nu}\right) \\ &= \boxed{\frac{1}{2}\,\mathrm{Tr}\left[\boldsymbol{P}\cdot(\boldsymbol{H}+\boldsymbol{F})\right]} \end{aligned} \tag{2.65}$$

The reduction from orbital coefficients to density matrices can be seen separately for the one- and two-electron terms. For the first,

$$\begin{aligned} \sum_{i} h_{ii} &= \sum_{\mu\nu}\left(\sum_{i} C_{\mu i}^{*}C_{\nu i}\right)H_{\mu\nu} \\ &= \sum_{\mu\nu} P_{\nu\mu}H_{\mu\nu} \end{aligned} \tag{2.66}$$

For the second, the two occupied-orbital sums are independent:

$$\begin{aligned} \sum_{ij} C_{\mu i}^{*}C_{\lambda j}^{*}C_{\nu i}C_{\sigma j} &= \left(\sum_{i} C_{\nu i}C_{\mu i}^{*}\right)\left(\sum_{j} C_{\sigma j}C_{\lambda j}^{*}\right) \\ &= P_{\nu\mu}P_{\sigma\lambda} \end{aligned} \tag{2.67}$$

Consequently, the two-electron energy is

$$\begin{aligned} \frac{1}{2}\sum_{ij}\left(\mathcal{J}_{ij} - \mathcal{K}_{ij}\right) &= \frac{1}{2}\sum_{\mu\nu\lambda\sigma} P_{\nu\mu}P_{\sigma\lambda}\left\langle \mu\lambda||\nu\sigma\right\rangle \\ &= \frac{1}{2}\sum_{\mu\nu} P_{\nu\mu}G_{\mu\nu} \end{aligned} \tag{2.68}$$

The prefactor removes the double counting of electron pairs; it does not halve the one-electron energy. These contractions also explain why the index order must be retained rather than silently replacing a Hermitian density by a real symmetric one. To see the origin of the trace expression,

Solution 14 (continued): AO Hartree–Fock energy

define the two-electron matrix $\boldsymbol{G} = \boldsymbol{F} - \boldsymbol{H}$. The preceding result may then be written as

$$\begin{aligned} E_{\mathrm{HF}} &= \sum_{\mu\nu} P_{\nu\mu} H_{\mu\nu} + \frac{1}{2} \sum_{\mu\nu} P_{\nu\mu} G_{\mu\nu} \\ &= \frac{1}{2} \sum_{\mu\nu} P_{\nu\mu} \left(H_{\mu\nu} + H_{\mu\nu} + G_{\mu\nu} \right) \\ &= \frac{1}{2} \sum_{\mu\nu} P_{\nu\mu} \left(H_{\mu\nu} + F_{\mu\nu} \right) \end{aligned} \tag{2.69}$$

which is precisely Eq. (2.65). The ordering of the indices $P_{\nu\mu}$ ensures that the component sum is the matrix trace. This identity applies in a nonorthogonal basis as well: the overlap metric is already accounted for in the occupied-orbital normalization, and the energy is a contraction of the AO density coefficients with the AO Hamiltonian matrix elements. In an SCF implementation, $\boldsymbol{F}$ must be built from the same $\boldsymbol{P}$ used in the energy trace. The spin-summed closed-shell specialization follows below.

Closed-shell specialization. We now carry out the spin sum and return to spatial AO indices. For a closed-shell system, the density matrix is

$$\begin{aligned} P_{\mu\nu} &= 2 \sum_{i=1}^{N/2} C_{\mu i} C_{\nu i}^{*} \\ \boldsymbol{P} &= \boxed{2 \boldsymbol{C}_{\mathrm{o}} \cdot \boldsymbol{C}_{\mathrm{o}}^{\dagger}} \end{aligned} \tag{2.70}$$

where $\boldsymbol{C}_{\mathrm{o}}$ contains the occupied MO coefficients. In chemists' notation, the closed-shell AO Fock matrix becomes

$$\boxed{F_{\mu\nu} = H_{\mu\nu} + \underbrace{\sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\nu \,|\, \lambda\sigma)}_{J_{\mu\nu}=\text{ Coulomb}} - \underbrace{\frac{1}{2} \sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\sigma \,|\, \lambda\nu)}_{K_{\mu\nu}=\text{ exchange}}} \tag{2.71}$$

The factor of two in $\boldsymbol{P}$ accounts for the two electrons occupying each spatial orbital: $\boldsymbol{P}$ here is the spin-summed density, whereas a density for one spin component counts each occupied orbital once. Correspondingly, the exchange term in Eq. (2.71) carries a factor of one half when the spin-summed closed-shell density is used. The corresponding energy is

$$E_{\mathrm{HF}} = \sum_{\mu\nu} P_{\nu\mu} H_{\mu\nu} + \frac{1}{2} \sum_{\mu\nu\lambda\sigma} P_{\nu\mu} \left[(\mu\nu \,|\, \lambda\sigma) - \frac{1}{2} (\mu\sigma \,|\, \lambda\nu) \right] P_{\sigma\lambda} \tag{2.72}$$

The factor of one half multiplying the two-electron contribution removes the double counting that would otherwise arise when the interaction of each electron pair is summed from the one-electron Fock operators. Equivalently, the compact trace formula contains one full copy of the one-electron energy and one half of the mean-field electron-electron energy.

Exercise 15 makes explicit how the overlap metric enters the normalization and projector properties of the density matrix. These identities also provide useful checks on an implementation.

Exercise 15: Density matrices in a nonorthogonal basis

Derive the electron-counting and idempotency relations for RHF and UHF density matrices in a nonorthogonal AO basis.

Solution 15: Density matrices in a nonorthogonal basis

Consider K linearly independent spatial AOs with overlap matrix $\boldsymbol{S}$. For a collinear determinant, let $\boldsymbol{C}_\mathrm{o}^\sigma$ contain the N^σ occupied spatial-orbital coefficients of spin $\sigma = \alpha, \beta$, with

$$\left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o}^\sigma = \mathbf{1} \tag{2.73}$$

Define the single-spin density by

$$\boldsymbol{P}^\sigma = \boldsymbol{C}_\mathrm{o}^\sigma \cdot \left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger \tag{2.74}$$

Taking the Hermitian adjoint reverses the order of the two coefficient matrices and conjugates each factor. Thus

$$\left(\boldsymbol{P}^\sigma\right)^\dagger = \left[\boldsymbol{C}_\mathrm{o}^\sigma \cdot \left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger\right]^\dagger = \boldsymbol{C}_\mathrm{o}^\sigma \cdot \left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger = \boldsymbol{P}^\sigma \tag{2.75}$$

Hermiticity therefore does not require the AOs to be orthogonal or the coefficients to be real.
The spin density in real space is

$$\rho^\sigma(\boldsymbol{r}) = \sum_{i=1}^{N^\sigma} \left|\psi_i^\sigma(\boldsymbol{r})\right|^2 = \sum_{\mu\nu} P_{\mu\nu}^\sigma \phi_\mu(\boldsymbol{r}) \phi_\nu^*(\boldsymbol{r}) \tag{2.76}$$

Integrating the AO products gives $S_{\nu\mu}$, rather than a Kronecker delta. Consequently,

$$\begin{aligned} \int \rho^\sigma(\boldsymbol{r}) \mathrm{d}\boldsymbol{r} &= \sum_{\mu\nu} P_{\mu\nu}^\sigma S_{\nu\mu} \\ &= \mathrm{Tr}(\boldsymbol{P}^\sigma \cdot \boldsymbol{S}) \\ &= \mathrm{Tr}\left[\left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o}^\sigma\right] \\ &= \boxed{N^\sigma} \end{aligned} \tag{2.77}$$

The cyclic trace identity moves the rectangular coefficient matrices into the occupied space, where the identity matrix has dimension N^σ. The unweighted trace $\mathrm{Tr}\,\boldsymbol{P}^\sigma$ is not generally an electron count.
The same occupied-orbital orthonormality gives

$$\begin{aligned} \boldsymbol{P}^\sigma \cdot \boldsymbol{S} \cdot \boldsymbol{P}^\sigma &= \boldsymbol{C}_\mathrm{o}^\sigma \cdot \underbrace{\left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o}^\sigma}_{\mathbf{1}} \cdot \left(\boldsymbol{C}_\mathrm{o}^\sigma\right)^\dagger \\ &= \boxed{\boldsymbol{P}^\sigma} \end{aligned} \tag{2.78}$$

This is metric idempotency: the overlap matrix is required between the two density factors.
For RHF, the spin-summed matrix is Hermitian because it is twice the Hermitian single-spin density. Its electron count is

$$\mathrm{Tr}(\boldsymbol{P} \cdot \boldsymbol{S}) = 2\,\mathrm{Tr}\left(\boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o}\right) = 2\frac{N}{2} = N \tag{2.79}$$

Solution 15 (continued): Density matrices in a nonorthogonal basis

Its metric product instead contains two factors of two:

$$\begin{aligned} \boldsymbol{P}\cdot\boldsymbol{S}\cdot\boldsymbol{P} &= 4\boldsymbol{C}_\mathrm{o}\cdot\underbrace{\boldsymbol{C}_\mathrm{o}^\dagger\cdot\boldsymbol{S}\cdot\boldsymbol{C}_\mathrm{o}}_{\mathbf{1}}\cdot\boldsymbol{C}_\mathrm{o}^\dagger \\ &= 4\boldsymbol{C}_\mathrm{o}\cdot\boldsymbol{C}_\mathrm{o}^\dagger \\ &= \boxed{2\boldsymbol{P}} \end{aligned} \tag{2.80}$$

Even in an orthonormal AO basis, the spin-summed RHF matrix therefore satisfies $\boldsymbol{P}^2 = 2\boldsymbol{P}$, not $\boldsymbol{P}^2 = \boldsymbol{P}$. It is the density divided by two that represents a projector in that orthonormal basis. To see the role of the metric directly, apply the one-spin occupied-space projector to an arbitrary spatial orbital $\psi_v = \sum_\nu v_\nu\phi_\nu$. Its projection is

$$\sum_{i=1}^{N^\sigma}\psi_i^\sigma(\boldsymbol{r})\left\langle\psi_i^\sigma\middle|\psi_v\right\rangle = \sum_{\mu\nu\lambda}\phi_\mu(\boldsymbol{r})P^\sigma_{\mu\lambda}S_{\lambda\nu}v_\nu \tag{2.81}$$

Thus its action on AO coefficient vectors is represented by $\boldsymbol{P}^\sigma\cdot\boldsymbol{S}$, and

$$(\boldsymbol{P}^\sigma\cdot\boldsymbol{S})^2 = \boldsymbol{P}^\sigma\cdot\boldsymbol{S}\cdot\boldsymbol{P}^\sigma\cdot\boldsymbol{S} = \boldsymbol{P}^\sigma\cdot\boldsymbol{S} \tag{2.82}$$

The coefficient-space projector is ordinarily idempotent even though $\boldsymbol{P}^\sigma$ alone generally is not. For RHF, the corresponding projector is $\boldsymbol{P}\cdot\boldsymbol{S}/2$. For a general collinear open-shell determinant the two spin projectors need not coincide, so the RHF relation $\boldsymbol{P}\cdot\boldsymbol{S}\cdot\boldsymbol{P} = 2\boldsymbol{P}$ must not be imposed on their sum.

At this stage the finite-basis Hartree–Fock problem is completely specified: $\boldsymbol{P}$ determines $\boldsymbol{F}$, the generalized eigenvectors of $\boldsymbol{F}$ determine a new $\boldsymbol{P}$, and the energy provides a scalar measure of the resulting state. The remaining task is to solve this circular dependence.

2.2 The SCF algorithm and convergence

The self-consistent-field procedure solves the nonlinear Roothaan–Hall equations by fixed-point iteration. One begins from an approximate density, constructs the Fock matrix generated by that density, solves the associated generalized eigenvalue problem, and uses the occupied eigenvectors to construct a new density. The process is repeated until the input and output densities describe the same mean field. Algorithm 1 collects the working steps in one place, separating the one-time setup from the iterative loop and the final output.

2.2.1 Fixed quantities and the SCF iteration

The geometry, basis functions, overlap matrix, core Hamiltonian, and electron-repulsion integrals remain fixed throughout an SCF calculation. The density, Fock matrix, orbital coefficients, orbital energies, and total energy change from one iteration to the next. Separating these two groups clarifies why, in the conventional stored-integral algorithm described here, the expensive integral generation is performed before the iterative cycle, whereas the Fock build and diagonalization are repeated. Direct SCF, discussed in Section 5.4, instead regenerates the required integrals during each Fock build.

Algorithm 1 | The RHF self-consistent-field procedure

Closed-shell RHF in a fixed spatial AO basis; $\boldsymbol{P}$ is the spin-summed density. The electronic energy excludes nuclear repulsion.

SETUP | Prepare once at a fixed geometry

1. **Specify the calculation.** Choose $\{\boldsymbol{R}_A, Z_A\}$, an even electron number N, the basis $\{\phi_\mu\}$, and convergence thresholds.
2. **Prepare integrals and the AO metric.** Calculate $\boldsymbol{S}$, $\boldsymbol{H}$, and the ERIs; form an orthogonalizer $\boldsymbol{X}$ satisfying $\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1}$.
3. **Choose an initial density.** Supply $\boldsymbol{P}^{(0)}$ and set $n = 0$.

ITERATION | Repeat Steps 4–7 until self-consistency

4. **Build the physical Fock matrix.** Use the current density in Eq. (2.71):
   $\boldsymbol{F}^{(n)} = \boldsymbol{F}[\boldsymbol{P}^{(n)}] = \boldsymbol{H} + \boldsymbol{G}[\boldsymbol{P}^{(n)}]$
5. **Evaluate the energy and test convergence.** Use this same density–Fock pair:
   $E_{\mathrm{HF}}^{(n)} = \frac{1}{2}\,\mathrm{Tr}\left[\boldsymbol{P}^{(n)} \cdot (\boldsymbol{H} + \boldsymbol{F}^{(n)})\right]$
   $\boldsymbol{e}^{(n)} = \boldsymbol{F}^{(n)} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{F}^{(n)}$
   After at least one orbital update, require $\max|\boldsymbol{e}^{(n)}| < \tau_{\mathrm{SCF}}$ and the chosen energy/density change criteria. If satisfied, go to Step 8.
6. **Solve for new orbitals.** Transform, diagonalize, and return to AO coefficients:
   $(\boldsymbol{F}^{(n)})' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F}^{(n)} \cdot \boldsymbol{X}$
   $(\boldsymbol{F}^{(n)})' \cdot (\boldsymbol{C}^{(n+1)})' = (\boldsymbol{C}^{(n+1)})' \cdot \boldsymbol{\epsilon}^{(n+1)}$
   $\boldsymbol{C}^{(n+1)} = \boldsymbol{X} \cdot (\boldsymbol{C}^{(n+1)})'$
7. **Occupy and update the density.** Occupy the $N/2$ lowest spatial orbitals and form
   $\boldsymbol{P}^{(n+1)} = 2\boldsymbol{C}_{\mathrm{o}}^{(n+1)} \cdot (\boldsymbol{C}_{\mathrm{o}}^{(n+1)})^\dagger$
   Set $n \leftarrow n + 1$ and return to Step 4, rebuilding the physical Fock matrix.

OUTPUT | Only after convergence

8. **Evaluate the converged observables.** Report E_{HF}, the density, orbitals, and required properties. Add the fixed nuclear repulsion for the total molecular energy.

The SCF algorithm. In practice, a restricted Hartree–Fock calculation begins by specifying the molecule through $\{\boldsymbol{R}_A\}$ and $\{Z_A\}$ and choosing the basis set $\{\phi_\mu\}$. The integrals $S_{\mu\nu}$, $H_{\mu\nu}$, and $\langle\mu\nu|\lambda\sigma\rangle$ are then calculated, after which $\boldsymbol{S}$ is diagonalized to obtain $\boldsymbol{X}$.

Once an initial density matrix $\boldsymbol{P}$ has been chosen, the iterative cycle begins. The matrix $\boldsymbol{G}$ is calculated and used to form $\boldsymbol{F} = \boldsymbol{H} + \boldsymbol{G}$. The Fock matrix is transformed according to $\boldsymbol{F}' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{X}$ and diagonalized to obtain $\boldsymbol{C}'$ and $\boldsymbol{\epsilon}$. The AO coefficients are recovered as $\boldsymbol{C} = \boldsymbol{X} \cdot \boldsymbol{C}'$, and a new density matrix is formed from $\boldsymbol{P} = 2\boldsymbol{C}_{\mathrm{o}} \cdot \boldsymbol{C}_{\mathrm{o}}^\dagger$. The calculation is then tested for convergence. If it has not converged, the cycle returns to the construction of $\boldsymbol{G}$ and $\boldsymbol{F}$; otherwise, the desired observables, including E_{HF}, are evaluated.

At a given iteration the diagonalization solves exactly the linear eigenvalue problem defined by the current Fock matrix. It does not solve the nonlinear Hartree–Fock problem until the occupied eigenvectors reproduce the density from which that Fock matrix was constructed.

The orthogonalization matrix. Step 2 of Algorithm 1 uses the orthogonalizer derived in Section 2.1.2. For reference, the two constructions and their checks are collected here. The required condition is

$$\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1} \tag{2.83}$$

For the symmetric, or Löwdin, orthogonalization,

$$\boldsymbol{X} = \boldsymbol{S}^{-1/2} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2} \cdot \boldsymbol{U}^{\dagger} \tag{2.84}$$

is one solution. Direct substitution verifies it:

$$\boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \boldsymbol{S}^{-1/2} \cdot \boldsymbol{S} \cdot \boldsymbol{S}^{-1/2} = \mathbf{1} \qquad \checkmark \tag{2.85}$$

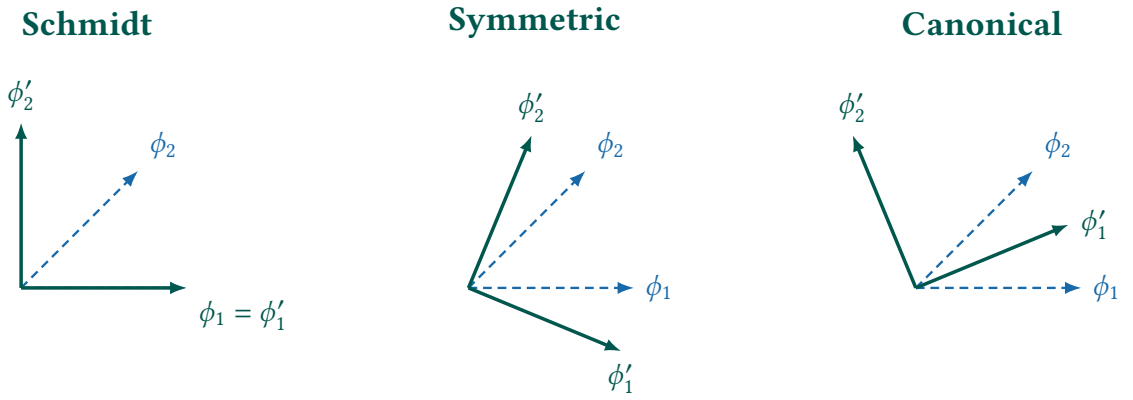
The canonical orthogonalization,

$$\boldsymbol{X} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2} \tag{2.86}$$

is another solution and is useful when linear dependencies are present. In this case,

$$\boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \boldsymbol{s}^{-1/2} \cdot \underbrace{\boldsymbol{U}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{U}}_{\boldsymbol{s}} \cdot \boldsymbol{s}^{-1/2} = \mathbf{1} \qquad \checkmark \tag{2.87}$$

When all overlap eigenvectors are retained, the two constructions differ only by a unitary transformation within the orthonormalized space. Discarding near-dependent directions instead reduces the represented orbital space. The symmetric choice changes the original basis as little as possible in the Löwdin sense, whereas the canonical construction makes it particularly transparent how directions associated with very small overlap eigenvalues can be removed. Figure 7 compares the resulting orthonormal bases with a Gram–Schmidt construction in a two-function example.



Sequential
Retain ϕ_1; subtract its component from ϕ_2 and normalize.

Equal treatment
$\boldsymbol{X} = \boldsymbol{S}^{-1/2}$

Overlap eigenvectors
$\boldsymbol{X} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2}$

Figure 7: Orthogonalization of an illustrative pair of real normalized atomic-orbital functions. Dashed blue vectors show the original functions; solid green vectors show orthonormal functions. The three methods select different orthonormal coordinates in the same space. Adapted from Szabo and Ostlund [21].

2.2.2 Initial guesses

Because the Fock matrix cannot be built before a density is available, every SCF calculation requires an initial approximation (Step 3 of Algorithm 1). The quality of this guess does not change the equations being solved, but it can strongly influence the number of iterations and, when several stationary solutions exist, the solution to which the procedure converges. Several

initial density matrices are possible. Setting $\boldsymbol{P} = \boldsymbol{0}$, so that $\boldsymbol{F} = \boldsymbol{H}$, gives the core guess; it is usually poor, but it is easy to implement. An inexpensive, though now less common, alternative is to use extended Hückel theory or a semi-empirical method. A superposition-of-atomic-densities (SAD) guess is obtained from tabulated atomic densities. Finally, it is very common and useful to read the molecular orbitals from a previous calculation [33].

2.2.3 Convergence criteria

Small changes in the energy alone do not necessarily imply that the orbitals are self-consistent. It is therefore common to monitor both energetic changes and a matrix residual that tests whether the occupied space is stationary with respect to the current Fock operator. This is the convergence test in Step 5 of Algorithm 1; the density update in Step 7 must be followed by a new physical Fock build before that test is repeated. The SCF procedure defines the iterative map

$$\boldsymbol{P}^{(n)} \longrightarrow \boldsymbol{F}[\boldsymbol{P}^{(n)}] \longrightarrow \boldsymbol{P}^{(n+1)} \tag{2.88}$$

Here n labels the SCF iteration and $\boldsymbol{P}^{\star}$ below denotes a self-consistent density. In an orthonormal basis, the corresponding stationarity condition is simply that the Fock and density matrices commute. The factors of $\boldsymbol{S}$ in the AO expression are the metric factors required by the nonorthogonality of the original basis. At self-consistency, the occupied and virtual subspaces therefore do not mix under the Fock operator. At convergence,

$$\boldsymbol{P}^{(n+1)} = \boldsymbol{P}^{(n)} = \boldsymbol{P}^{\star} \tag{2.89}$$

One may check the energy and/or the density matrix. A more direct test of stationarity uses the commutator residual $\boldsymbol{e}^{(n)}$, which vanishes at self-consistency:

$$\boxed{\boldsymbol{e}^{(n)} = \boldsymbol{F}^{(n)} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{(n)} \cdot \boldsymbol{F}^{(n)}} \tag{2.90}$$

Accordingly,

$$\boldsymbol{P}^{(n)} = \boldsymbol{P}^{\star} \qquad \Longrightarrow \qquad \boldsymbol{e}^{(n)} = \boldsymbol{0} \qquad \Longrightarrow \qquad \max|\boldsymbol{e}^{(n)}| < \tau_{\mathrm{SCF}} \tag{2.91}$$

In practice, the largest absolute residual element is compared with the positive convergence threshold τ_{SCF}. Finally, DIIS (direct inversion in the iterative subspace) is usually used to accelerate convergence. It extrapolates the Fock matrix from the m previous iterations according to

$$\boldsymbol{F}^{(n+1)} = \sum_{i=n-m+1}^{n} c_i \boldsymbol{F}^{(i)} \tag{2.92}$$

The ordinary SCF iteration should therefore be understood as the underlying fixed-point map. Damping, level shifting [34], and DIIS modify how successive approximations are generated, but convergence is always judged against the same self-consistency conditions.

Exercise 16: What does SCF convergence mean?

Derive the SCF stationarity condition in a nonorthogonal AO basis. Is a small energy change sufficient to establish convergence?

Solution 16: What does SCF convergence mean?

Assume linearly independent AOs and Hermitian overlap and Fock matrices. For the RHF density $\boldsymbol{P} = 2\boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger$, define the residual

$$\boldsymbol{e} = \boldsymbol{F}[\boldsymbol{P}] \cdot \boldsymbol{P} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P} \cdot \boldsymbol{F}[\boldsymbol{P}] \tag{2.93}$$

At self-consistency the Fock matrix constructed from $\boldsymbol{P}$ has occupied eigenvectors that generate this same density. Writing $\boldsymbol{\epsilon}_\mathrm{o}$ for the real diagonal matrix of occupied orbital energies, the occupied Roothaan–Hall equations and their adjoint are

$$\boldsymbol{F} \cdot \boldsymbol{C}_\mathrm{o} = \boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{\epsilon}_\mathrm{o} \tag{2.94}$$

$$\boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{F} = \boldsymbol{\epsilon}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{S} \tag{2.95}$$

Substitution of the first equation gives

$$\begin{aligned} \boldsymbol{F} \cdot \boldsymbol{P} \cdot \boldsymbol{S} &= 2\boldsymbol{F} \cdot \boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{S} \\ &= 2\boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{\epsilon}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{S} \end{aligned} \tag{2.96}$$

Using the adjoint equation for the other term gives precisely the same matrix:

$$\begin{aligned} \boldsymbol{S} \cdot \boldsymbol{P} \cdot \boldsymbol{F} &= 2\boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{F} \\ &= 2\boldsymbol{S} \cdot \boldsymbol{C}_\mathrm{o} \cdot \boldsymbol{\epsilon}_\mathrm{o} \cdot \boldsymbol{C}_\mathrm{o}^\dagger \cdot \boldsymbol{S} \end{aligned} \tag{2.97}$$

Their difference therefore yields $\boldsymbol{e} = \boldsymbol{0}$. Hermiticity also gives $\boldsymbol{e}^\dagger = -\boldsymbol{e}$, as expected for a commutator represented with the AO metric.

For an arbitrary trial determinant, complete the occupied coefficient matrix with virtual columns to form $\boldsymbol{C} = (\boldsymbol{C}_\mathrm{o}, \boldsymbol{C}_\mathrm{v})$, with $\boldsymbol{C}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{C} = \boldsymbol{1}$. These orbitals need not diagonalize the current $\boldsymbol{F}[\boldsymbol{P}]$. In this orthonormal MO basis, define

$$\boldsymbol{f} = \boldsymbol{C}^\dagger \cdot \boldsymbol{F}[\boldsymbol{P}] \cdot \boldsymbol{C} \tag{2.98}$$

$$\boldsymbol{P}_\mathrm{MO} = \boldsymbol{C}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{P} \cdot \boldsymbol{S} \cdot \boldsymbol{C} = \begin{pmatrix} 2\boldsymbol{1}_\mathrm{o} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{pmatrix} \tag{2.99}$$

The occupied identity has dimension $N/2$ and the virtual block has dimension $K - N/2$. Because the AO basis is linearly independent and $\boldsymbol{C}$ is complete, $\boldsymbol{C} \cdot \boldsymbol{C}^\dagger = \boldsymbol{S}^{-1}$. Inserting this identity between the factors gives

$$\begin{aligned} \boldsymbol{C}^\dagger \cdot \boldsymbol{e} \cdot \boldsymbol{C} &= \left(\boldsymbol{C}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{C}\right) \cdot \left(\boldsymbol{C}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{P} \cdot \boldsymbol{S} \cdot \boldsymbol{C}\right) \\ &\quad - \left(\boldsymbol{C}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{P} \cdot \boldsymbol{S} \cdot \boldsymbol{C}\right) \cdot \left(\boldsymbol{C}^\dagger \cdot \boldsymbol{F} \cdot \boldsymbol{C}\right) \\ &= \boldsymbol{f} \cdot \boldsymbol{P}_\mathrm{MO} - \boldsymbol{P}_\mathrm{MO} \cdot \boldsymbol{f} \\ &= \begin{pmatrix} \boldsymbol{0} & -2\boldsymbol{f}_\mathrm{ov} \\ 2\boldsymbol{f}_\mathrm{vo} & \boldsymbol{0} \end{pmatrix} \end{aligned} \tag{2.100}$$

In particular, for occupied i and virtual a,

$$\left(\boldsymbol{C}^\dagger \cdot \boldsymbol{e} \cdot \boldsymbol{C}\right)_{ai} = +2f_{ai} \qquad \left(\boldsymbol{C}^\dagger \cdot \boldsymbol{e} \cdot \boldsymbol{C}\right)_{ia} = -2f_{ia} \tag{2.101}$$

The residual vanishes exactly when the occupied-virtual Fock blocks vanish. These are the matrix elements that drive first-order changes of the energy under occupied-virtual orbital rotations. The

Solution 16 (continued): What does SCF convergence mean?

condition is therefore one of orbital stationarity, not merely a statement about a small difference between two scalar energies. The occupied-occupied and virtual-virtual blocks can then be diagonalized independently to obtain canonical orbitals without changing the determinant's occupied space.

A small energy change does not impose this condition. For a real orbital rotation parameter q and a small step δq, a Taylor expansion gives

$$E(q+\delta q) - E(q) = \frac{\mathrm{d}E}{\mathrm{d}q}\delta q + \frac{1}{2}\frac{\mathrm{d}^2E}{\mathrm{d}q^2}^2 + \cdots \tag{2.102}$$

The energy change can be arbitrarily small because the step is small, even when the derivative is not. Near a stationary point the leading energy error is generally quadratic in the orbital error, whereas the orbital gradient is generally linear. An apparent energy plateau can therefore hide an appreciable residual. Energy changes, density changes, and a consistently constructed residual provide complementary information.

There is a related distinction within a single iteration. Diagonalizing $\boldsymbol{F}^{(n)} = \boldsymbol{F}[\boldsymbol{P}^{(n)}]$ constructs a new density that satisfies

$$\boldsymbol{F}^{(n)} \cdot \boldsymbol{P}^{(n+1)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{(n+1)} \cdot \boldsymbol{F}^{(n)} = \boldsymbol{0} \tag{2.103}$$

This mixed-iteration expression vanishes for an exact diagonalization, whether or not the SCF problem has converged. The new density must instead be tested against its own rebuilt Fock matrix, or the old density against its own Fock matrix as in the stated residual. The same caution applies when the matrix diagonalized is a DIIS extrapolation rather than the Fock matrix rebuilt from the density being tested.

Finally, vanishing occupied-virtual blocks constrain the first variation of the energy, not the second. A stationary determinant may be a local minimum or an unstable stationary point; stability requires examining the curvature under the allowed orbital rotations. Even a stable local minimum need not be the global minimum, and stability within RHF does not establish stability against the additional variations allowed in UHF or GHF.

2.3 Direct inversion in the iterative subspace (DIIS)

Near a self-consistent solution, several recent Fock matrices and their error vectors contain information about the direction in which the residual is decreasing. DIIS uses this information to form an extrapolated Fock matrix whose estimated residual is as small as possible within the affine space spanned by the stored iterations [35, 36].

Historical perspective: Péter Pulay (b. 1941)

Born in Hungary, Pulay developed methods that made molecular electronic structure more practical, including analytic energy gradients and efficient geometry optimization. His DIIS procedure [35, 36] accelerates self-consistency by combining information from earlier iterations. His career at the University of Arkansas also encompassed vibrational force fields and local correlation, linking algorithmic developments with calculations of molecular structure and properties.

Biography and source portrait: IAQMS profile. Source portrait photographer/date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

2.3.1 Residual minimization and extrapolation

Using the m stored iterations, DIIS constructs an extrapolated Fock matrix and approximates its error vector as follows. Here i labels a stored iteration, not an occupied orbital; $\boldsymbol{e}^{(i)}$ is its commutator residual, and c_i is its extrapolation weight.

$$\boldsymbol{F}_{\mathrm{DIIS}}^{(m+1)} = \sum_{i=1}^{m} c_i \boldsymbol{F}^{(i)} \qquad\qquad \boldsymbol{e}_{\mathrm{DIIS}}^{(m+1)} \approx \sum_{i=1}^{m} c_i \boldsymbol{e}^{(i)} \tag{2.104}$$

The coefficients minimize the squared Frobenius norm of the residual, that is, the sum of the squared magnitudes of its matrix elements

$$\min_{\boldsymbol{c}} \left\| \sum_{i=1}^{m} c_i \boldsymbol{e}^{(i)} \right\|_F^2 \tag{2.105}$$

subject to the affine constraint

$$\sum_{i=1}^{m} c_i = 1 \tag{2.106}$$

Define the DIIS overlap matrix by

$$B_{ij} = \langle \boldsymbol{e}^{(i)} | \boldsymbol{e}^{(j)} \rangle = \mathrm{Tr}\left[\boldsymbol{e}^{(i)\dagger} \cdot \boldsymbol{e}^{(j)} \right] \tag{2.107}$$

The elements of $\boldsymbol{B}$ are therefore pairwise inner products between residual matrices; the extrapolation weights c_i are taken to be real. For the Hermitian Fock, density, and overlap matrices used here, the residual is anti-Hermitian and these inner products are real even when the orbitals are complex. If two iterations have very similar errors, the corresponding rows of $\boldsymbol{B}$ become nearly linearly dependent; this observation explains the possible numerical singularity discussed below. The squared residual norm is then

$$\left\| \boldsymbol{e}_{\mathrm{DIIS}}^{(m+1)} \right\|_F^2 = \sum_{ij} c_i B_{ij} c_j = \boldsymbol{c}^{\mathsf{T}} \cdot \boldsymbol{B} \cdot \boldsymbol{c} \tag{2.108}$$

DIIS weights and extrapolated Fock matrix. The constrained minimization is described by the DIIS Lagrangian, in which λ is the multiplier enforcing the normalization of the weights:

$$\mathcal{L}(\boldsymbol{c}, \lambda) = \boldsymbol{c}^{\mathsf{T}} \cdot \boldsymbol{B} \cdot \boldsymbol{c} - 2\lambda \left(\sum_{i=1}^{m} c_i - 1 \right) \tag{2.109}$$

Its stationarity conditions are

$$\frac{\partial \mathcal{L}(\boldsymbol{c}, \lambda)}{\partial \boldsymbol{c}} = \boldsymbol{0} \qquad\qquad \frac{\partial \mathcal{L}(\boldsymbol{c}, \lambda)}{\partial \lambda} = 0 \tag{2.110}$$

Writing the first condition component by component gives

$$\frac{\partial \mathcal{L}}{\partial c_k} = 2 \sum_{j=1}^{m} B_{kj} c_j - 2\lambda = 0 \qquad\qquad \frac{\partial \mathcal{L}}{\partial \lambda} = -2 \left(\sum_{i=1}^{m} c_i - 1 \right) = 0 \tag{2.111}$$

Hence every row obeys

$$\sum_{j=1}^{m} B_{kj} c_j - \lambda = 0 \tag{2.112}$$

together with the normalization constraint. These relations imply

$$\boldsymbol{B}\cdot\boldsymbol{c}-\lambda\mathbf{1}=\mathbf{0} \qquad\qquad \sum_{i=1}^{m} c_i = 1 \tag{2.113}$$

These equations form the augmented DIIS linear system

$$\boxed{\begin{pmatrix} B_{11} & \cdots & B_{1m} & -1 \\ \vdots & \ddots & \vdots & \vdots \\ B_{m1} & \cdots & B_{mm} & -1 \\ -1 & \cdots & -1 & 0 \end{pmatrix}\cdot\begin{pmatrix} c_1 \\ \vdots \\ c_m \\ \lambda \end{pmatrix}=\begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \end{pmatrix}} \tag{2.114}$$

After solving for the coefficients, the extrapolated Fock matrix is

$$\boldsymbol{F}_{\mathrm{DIIS}}^{(m+1)}=\sum_{i=1}^{m} c_i \boldsymbol{F}^{(i)} \tag{2.115}$$

The linear system has dimension only $(m+1)\times(m+1)$. The normalization constraint $\sum_i c_i = 1$ makes the extrapolation affine rather than arbitrary. In particular, if all stored Fock matrices were equal, their extrapolated combination would reproduce that same matrix. The Lagrange multiplier is the algebraic device that enforces this constraint.

2.3.2 The DIIS algorithm and practical considerations

The complete algorithm starts from the current density $\boldsymbol{P}^{(n)}$, from which the corresponding Fock matrix $\boldsymbol{F}^{(n)}=\boldsymbol{F}[\boldsymbol{P}^{(n)}]$ is constructed. The commutator residual is then evaluated as

$$\boldsymbol{e}^{(n)}=\boldsymbol{F}^{(n)}\cdot\boldsymbol{P}^{(n)}\cdot\boldsymbol{S}-\boldsymbol{S}\cdot\boldsymbol{P}^{(n)}\cdot\boldsymbol{F}^{(n)} \tag{2.116}$$

The pair $\{\boldsymbol{F}^{(n)},\boldsymbol{e}^{(n)}\}$ is stored and the DIIS subspace is updated. The matrix $\boldsymbol{B}$ is constructed, the augmented linear system is solved, and the coefficients $\{c_i\}$ are obtained. These coefficients define the extrapolated Fock matrix

$$\boldsymbol{F}^{\mathrm{DIIS}}=\sum_{i} c_i \boldsymbol{F}^{(i)} \tag{2.117}$$

This is the working matrix used for orbital generation in Step 6 of Algorithm 1; it does not replace $\boldsymbol{F}[\boldsymbol{P}]$ in the convergence test. Finally, the generalized eigenvalue problem

$$\boldsymbol{F}^{\mathrm{DIIS}}\cdot\boldsymbol{C}=\boldsymbol{S}\cdot\boldsymbol{C}\cdot\boldsymbol{\epsilon} \tag{2.118}$$

is solved and the next density $\boldsymbol{P}^{(n+1)}$ is formed. The subspace is normally updated as a rolling history: after its maximum size has been reached, the oldest Fock matrix and residual are discarded when a new pair is added. If the residuals become nearly linearly dependent, restarting with a smaller history restores a well-conditioned extrapolation problem. Figure 8 summarizes the SCF cycle and indicates where convergence acceleration can enter.

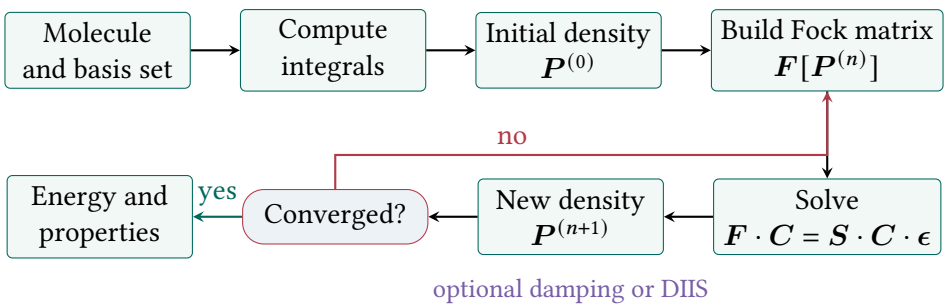


Figure 8: The self-consistent-field cycle: fixed molecular data and integrals, iterative Fock construction, orbital solution, density update, and convergence testing. Damping or DIIS may accelerate the iteration.

Practical considerations. The key idea is that DIIS accelerates SCF convergence by finding a linear combination of Fock matrices. The practical consequences are straightforward. A small rolling subspace, commonly $m \approx 6$–10, is normally sufficient, and DIIS can also be used for many other methods. The acceleration nevertheless requires additional storage of approximately $2mK^2$. Moreover, the extrapolated energy is not guaranteed to decrease monotonically, and $\boldsymbol{B}$ can become singular when the residuals are nearly linearly dependent, in which case a restart may be required.

DIIS changes neither the Hartree–Fock energy functional nor the final self-consistency condition. It is an acceleration strategy for reaching a stationary solution of the same equations; the convergence and stability distinctions in Exercise 16 still apply.

2.4 Excited-state SCF and the maximum-overlap method

The nonlinear character of the Hartree–Fock equations permits multiple stationary determinants. Some correspond to non-Aufbau occupations and can be used as state-specific approximations to electronically excited states. The challenge is not writing such an occupation initially, but preventing the SCF iterations from replacing it by a lower-energy Aufbau occupation. The maximum-overlap method (MOM) provides a practical occupation rule for following such a state-specific solution [37].

2.4.1 Variational collapse

An excited determinant can be generated by promoting one or more electrons, but the usual Aufbau occupation does not preserve that configuration. Figure 9 illustrates the loss of the promoted occupation under an Aufbau SCF iteration.

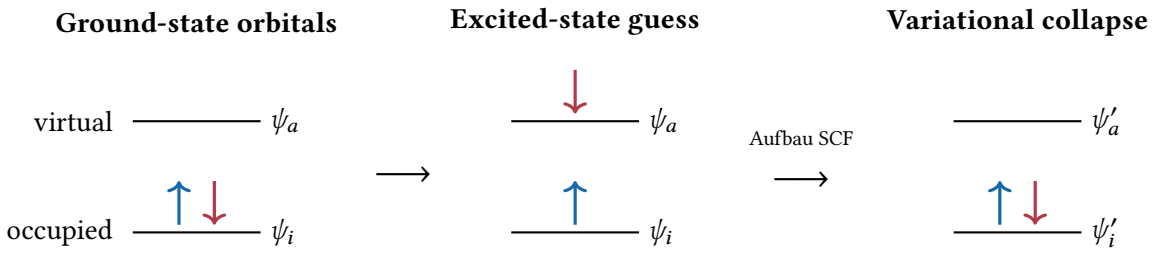


Figure 9: Variational collapse of an excited-state guess. After orbital reoptimization, an Aufbau occupation can refill the lowest orbital and destroy the initially promoted configuration.

After each diagonalization, Aufbau occupies the lowest-energy orbitals. Consequently, the SCF procedure tends to return to a lower-energy solution. This behavior is called variational collapse. The variational principle is not being violated: an unconstrained minimization is doing precisely what it is designed to do. To follow a chosen excited configuration, the occupation rule must retain the character of the targeted orbitals rather than simply selecting the lowest eigenvalues.

2.4.2 The maximum-overlap criterion

Let $\boldsymbol{C}_{\mathrm{o}}^{\mathrm{ref}}$ contain the coefficients of the occupied reference spin orbitals defining the target configuration. For every new orbital χ_p^{new}, compute

$$\begin{aligned} O_{ip} &= \left\langle \chi_i^{\mathrm{ref}} \middle| \chi_p^{\mathrm{new}} \right\rangle = \sum_{\mu\nu} C_{\mu i}^{\mathrm{ref}*} S_{\mu\nu} C_{\nu p}^{\mathrm{new}} \\ w_p &= \sum_{i \in O_{\mathrm{ref}}} \left| O_{ip} \right|^2 \end{aligned} \tag{2.119}$$

In matrix form, the overlap matrix between the reference occupied orbitals and the new orbitals is

$$\boldsymbol{O} = \boldsymbol{C}_{\mathrm{o}}^{\mathrm{ref}\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{C}^{\mathrm{new}} \tag{2.120}$$

The weight w_p is the squared norm of column p of this matrix. It therefore measures how much of the new orbital lies in the occupied reference subspace. The MOM procedure computes w_p for every new orbital and occupies the N spin orbitals with the largest w_p. In MOM, the reference occupied space is taken from the previous iteration, whereas in the initial maximum-overlap method (IMOM) the reference orbitals remain fixed to the initial guess [38]. Figure 10 illustrates selection by overlap weight rather than by orbital energy.

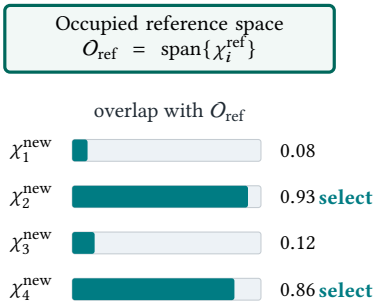


Figure 10: Illustrative maximum-overlap weights w_p for new orbitals projected onto the occupied reference space O_{ref}. The orbitals with the largest weights are selected for occupation.

Thus, MOM selects orbitals by orbital character, not by energy. The weight w_p measures the total squared projection of a new orbital onto the reference occupied subspace. It is insensitive to unitary rotations within that reference space, so the selection is based on the subspace representing the target configuration rather than on an arbitrary labeling of individual reference orbitals.

2.4.3 MOM-based SCF and interpretation

The MOM-SCF cycle is shown in Fig. 11.

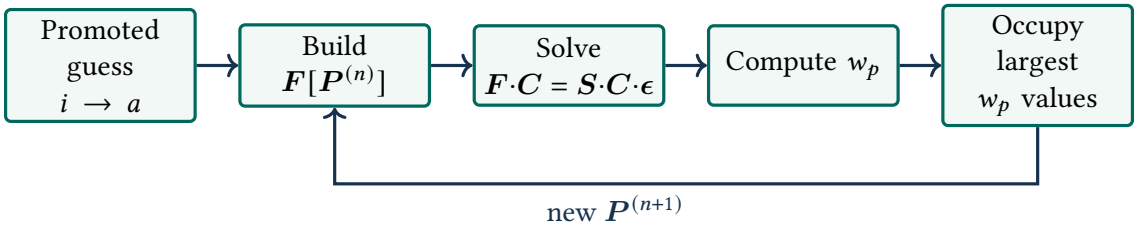


Figure 11: The maximum-overlap SCF cycle. After solving the orbital equations, the new occupied space is selected by its overlap weights with the reference space, rather than by the Aufbau rule.

Conventional SCF occupies the lowest orbital energies,

$$\epsilon_1 \leq \epsilon_2 \leq \cdots \leq \epsilon_N \tag{2.121}$$

and thereby applies the Aufbau occupation rule at each iteration. This rule alone does not guarantee convergence or selection of the global minimum. MOM-SCF instead occupies the largest overlap weights,

$$w_1 \geq w_2 \geq \cdots \geq w_N \tag{2.122}$$

and thereby attempts to follow the chosen occupied space. For collinear calculations with fixed N^α and N^β, the selection is made separately in each spin sector. Thus, MOM changes the occupation to target non-Aufbau stationary solutions of the Hartree–Fock equations.

What does a MOM calculation give? When successful, MOM provides a targeted stationary solution of the HF equations, often higher in energy than the conventional Aufbau solution, with full orbital relaxation and a corresponding state-specific density and set of molecular properties. It can yield useful excitation energies for states dominated by a single electronic configuration. Its formal cost per SCF iteration is the same as that of a conventional calculation, although convergence may be more difficult.

Several cautions remain. The result is still a single determinant, spin symmetry may be broken, and different SCF states are not necessarily orthogonal. Convergence depends on the initially promoted orbitals, and the MOM iteration may drift or converge to an unintended solution. MOM should consequently be viewed as an occupation-following prescription, not as a new electronic-structure approximation. It selects a particular stationary solution of the Hartree–Fock equations and allows the orbitals of that determinant to relax self-consistently.

2.5 Molecular properties and orbital analysis

Once a self-consistent density matrix has been obtained, one-electron properties can be evaluated by contracting that density with the appropriate property integrals. The same density also provides the starting point for population analyses. The HF determinant does not select a unique orbital representation: unitary rotations within the occupied or virtual subspaces preserve the state, although the rotated orbitals need not remain canonical. This freedom motivates localized-orbital representations.

2.5.1 Dipole moments

The dipole moment illustrates the general structure of a one-electron property: a quantum-mechanical electronic expectation value is combined with the corresponding classical nuclear contribution. For point charges q_i at positions $\boldsymbol{r}_i$, the classical dipole vector is

$$\boldsymbol{\mu} = (\mu_x, \mu_y, \mu_z) = \underbrace{\sum_i q_i \boldsymbol{r}_i}_{\text{classical definition}} \tag{2.123}$$

The electronic term follows from the one-particle density. In the spatial AO contraction, $\boldsymbol{P}$ is the spin-summed density; integration over the spin coordinate is implicit in the reduction from the spin-orbital sum:

$$\begin{aligned} -\sum_i \langle \chi_i | \boldsymbol{r} | \chi_i \rangle &= -\sum_i \sum_{\mu\nu} C^*_{\mu i} C_{\nu i} (\mu \,|\, \boldsymbol{r} \,|\, \nu) \\ &= -\sum_{\mu\nu} P_{\nu\mu} (\mu \,|\, \boldsymbol{r} \,|\, \nu) \end{aligned} \tag{2.124}$$

Thus the many-electron expectation value reduces to a contraction of the AO density with one-electron dipole integrals. For a molecule, the corresponding quantum-mechanical expression separates the electronic and nuclear contributions:

$$\begin{aligned}\boldsymbol{\mu} &= (\mu_x, \mu_y, \mu_z) \\ &= \underbrace{\langle \Psi_0 | - \sum_{i=1}^{N} \boldsymbol{r}_i | \Psi_0 \rangle}_{\text{electrons}} + \underbrace{\sum_{A=1}^{M} Z_A \boldsymbol{R}_A}_{\text{nuclei}} \\ &= -\sum_{\mu\nu} P_{\mu\nu} (\nu \,|\, \boldsymbol{r} \,|\, \mu) + \sum_{A=1}^{M} Z_A \boldsymbol{R}_A\end{aligned} \tag{2.125}$$

For example, its x component is

$$\begin{aligned}\mu_x &= -\sum_{\mu\nu} P_{\mu\nu} (\nu \,|\, x \,|\, \mu) + \sum_{A=1}^{M} Z_A X_A \\ \underbrace{(\nu \,|\, x \,|\, \mu)}_{\text{one-electron integrals}} &= \int \phi_\nu^*(\boldsymbol{r})\, x\, \phi_\mu(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r}\end{aligned} \tag{2.126}$$

2.5.2 Charge analysis

An atomic charge is not itself a direct observable of the molecular wavefunction. Population analyses define such charges by adopting a rule for partitioning the density among atom-centered basis functions. Mulliken and Löwdin analyses differ precisely in this partitioning convention. For the population formulas below, the AOs and orbital coefficients are taken to be real.

Historical perspective: Robert Sanderson Mulliken (1896–1986)

An American physicist and chemist based at the University of Chicago, Mulliken helped develop the molecular-orbital description of chemical bonding and spectroscopy. He received the 1966 Nobel Prize in Chemistry for this work. His population analysis [39] connects the orbital coefficients and overlap matrix to a chemical interpretation of the electron distribution, while remaining dependent on the chosen partitioning convention.

Biography: R. Stephen Berry, NAS memoir; Nobel Foundation. Source portrait: Chicago, 1929, Friedrich Hund estate / Gerhard Hund (GFHund), via Wikimedia Commons, CC BY 3.0. Sketch: AI-assisted adaptation of the credited portrait.

The electron density is

$$\rho(\boldsymbol{r}) = \sum_{\mu\nu} \phi_\mu(\boldsymbol{r}) P_{\mu\nu} \phi_\nu^*(\boldsymbol{r}) \qquad \int \rho(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r} = N \tag{2.127}$$

The identity $N = \mathrm{Tr}(\boldsymbol{P} \cdot \boldsymbol{S})$ expresses electron-number conservation in a nonorthogonal basis. A population analysis partitions the diagonal contributions to this trace among the atoms. Consequently,

$$N = \sum_{\mu\nu} P_{\mu\nu} S_{\nu\mu} = \sum_{\mu} (\boldsymbol{P} \cdot \boldsymbol{S})_{\mu\mu} = \mathrm{Tr}(\boldsymbol{P} \cdot \boldsymbol{S}) \tag{2.128}$$

Assuming that the basis functions are atom-centered, the Mulliken population analysis [39] assigns the net charge

$$\underbrace{q_A^{\text{Mulliken}}}_{\text{net charge on } A} = Z_A - \sum_{\mu \in A} (\boldsymbol{P} \cdot \boldsymbol{S})_{\mu\mu} \tag{2.129}$$

For Löwdin population analysis [30], cyclic invariance of the trace, $\mathrm{Tr}(\boldsymbol{A}\cdot\boldsymbol{B}) = \mathrm{Tr}(\boldsymbol{B}\cdot\boldsymbol{A})$, gives, for any real partition parameter α,

$$N = \sum_{\mu} (\boldsymbol{S}^{\alpha}\cdot\boldsymbol{P}\cdot\boldsymbol{S}^{1-\alpha})_{\mu\mu} \tag{2.130}$$

For $\alpha = 1/2$, this becomes

$$N = \sum_{\mu} (\boldsymbol{S}^{1/2}\cdot\boldsymbol{P}\cdot\boldsymbol{S}^{1/2})_{\mu\mu} \qquad \Rightarrow \qquad q_A^{\text{Löwdin}} = Z_A - \sum_{\mu\in A} (\boldsymbol{S}^{1/2}\cdot\boldsymbol{P}\cdot\boldsymbol{S}^{1/2})_{\mu\mu} \tag{2.131}$$

2.5.3 Localized orbitals

We retain the real-orbital convention for the population-based localization criteria below. Canonical orbitals diagonalize the occupied-occupied and virtual-virtual blocks of the Fock matrix, but they are often delocalized over an entire molecule. Because unitary rotations within either subspace leave the Slater determinant invariant, the same Hartree–Fock state can be represented by orbitals chosen to satisfy a localization criterion. At HF convergence and after canonicalization and localization, the Fock matrix passes through the sequence

$$\boldsymbol{f} = \begin{pmatrix} \boldsymbol{f}_{\mathrm{oo}} & \boldsymbol{f}_{\mathrm{ov}} \\ \boldsymbol{f}_{\mathrm{vo}} & \boldsymbol{f}_{\mathrm{vv}} \end{pmatrix} \xrightarrow{\text{HF convergence}} \begin{pmatrix} \boldsymbol{f}_{\mathrm{oo}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{f}_{\mathrm{vv}} \end{pmatrix} \xrightarrow{\text{canonicalization}} \begin{pmatrix} \boldsymbol{\epsilon}_{\mathrm{o}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{\epsilon}_{\mathrm{v}} \end{pmatrix} \xrightarrow{\text{localization}} \begin{pmatrix} \widetilde{\boldsymbol{f}}_{\mathrm{oo}} & \boldsymbol{0} \\ \boldsymbol{0} & \widetilde{\boldsymbol{f}}_{\mathrm{vv}} \end{pmatrix} \tag{2.132}$$

The localized occupied orbitals are obtained through a unitary transformation,

$$\begin{aligned} \widetilde{\boldsymbol{C}}_{\mathrm{o}} &= \boldsymbol{C}_{\mathrm{o}}\cdot\boldsymbol{U} \\ \boldsymbol{U}^{\dagger}\cdot\boldsymbol{U} = \boldsymbol{1} \quad &\Longleftrightarrow \quad \text{unitary transformation} \end{aligned} \tag{2.133}$$

For a closed-shell determinant, the density matrix is unchanged:

$$\widetilde{\boldsymbol{P}} = 2\widetilde{\boldsymbol{C}}_{\mathrm{o}}\cdot\widetilde{\boldsymbol{C}}_{\mathrm{o}}^{\dagger} = 2\boldsymbol{C}_{\mathrm{o}}\cdot\boldsymbol{C}_{\mathrm{o}}^{\dagger} = \boldsymbol{P} \tag{2.134}$$

The determinant changes only by a phase,

$$|\widetilde{\Psi}_0\rangle = \det(\boldsymbol{U})\,|\Psi_0\rangle \tag{2.135}$$

so the density, energy, and every expectation value are unchanged. Figure 12 contrasts schematic canonical and localized representations of the same occupied space.

Figure 12: Schematic localization of occupied orbitals by a unitary transformation. The individual orbital shapes change, but the occupied space, determinant up to a phase, density, and energy do not.

Only occupied-occupied or virtual-virtual rotations are invariant. Mixing occupied and virtual orbitals changes the wavefunction and the energy. Localization functionals are generally nonconvex, so different guesses may converge to different local extrema. This invariance separates two logically distinct operations. Orbital optimization mixes occupied and virtual spaces and changes the physical determinant; localization rotates only within an already optimized subspace and changes its representation without changing the state.

Localization procedures. Different localization procedures choose different scalar measures of orbital compactness. Since all admissible transformations are unitary within the selected subspace, optimizing any of these measures leaves the Hartree–Fock state itself unchanged. The Edmiston–Ruedenberg functional [40] is

$$\Omega_{\mathrm{ER}} = \sum_i (\psi_i\psi_i \,|\, \psi_i\psi_i) \quad \longrightarrow \quad \text{max orbital self-repulsion} \tag{2.136}$$

The Foster–Boys functional [41] may be written in two equivalent forms:

$$\begin{aligned} \Omega_{\mathrm{BF}} &= \frac{1}{2}\sum_i \left[\langle\psi_i\psi_i|r_{12}^2|\psi_i\psi_i\rangle \right] \quad \longrightarrow \quad \text{min spatial extent} \\ &\equiv \sum_i \left[\langle\psi_i|r^2|\psi_i\rangle - |\langle\psi_i|\boldsymbol{r}|\psi_i\rangle|^2 \right] \quad \longrightarrow \quad \text{min sum of orbital variances} \end{aligned} \tag{2.137}$$

Within a fixed orbital subspace, $\sum_i \langle\psi_i|r^2|\psi_i\rangle$ is invariant under unitary rotations. Minimizing Ω_{BF} is therefore equivalent to maximizing $\sum_i |\langle\psi_i|\boldsymbol{r}|\psi_i\rangle|^2$. Since the sum of orbital centroids is also invariant, this maximizes their total pairwise squared separation. Both displayed expressions for Ω_{BF} are minimized; the centroid criterion is maximized. Its strengths are that it requires only one-electron integrals and produces compact bonds and lone pairs. Its limitations are that it does not explicitly preserve σ/π separation and that diffuse virtual orbitals may be difficult to localize.

For an atomic population q_i^A of orbital i on atom A, the original Pipek–Mezey criterion [42] maximizes

$$\Omega_{\mathrm{PM}} = \sum_A \sum_i |q_i^A|^2 \tag{2.138}$$

using the Mulliken populations

$$q_i^A = \sum_{\mu\in A}\sum_\nu C_{\mu i}^* S_{\mu\nu} C_{\nu i} \tag{2.139}$$

Its strengths are that it often preserves the separation between σ and π orbitals and produces intuitive bonds, lone pairs, and core orbitals. Its principal limitation is that the original Mulliken form can be basis-set dependent; Löwdin populations or intrinsic atomic orbitals provide more robust alternatives. There is no universally best localized-orbital set; the criterion should match the intended interpretation or computational use.

Summary of orbital localization. The occupied subspace and physical determinant state remain unchanged, as do the density matrix, electron density, HF energy, and molecular properties. The occupied-virtual Fock block also remains zero:

$$\widetilde{\boldsymbol{f}}_{\mathrm{ov}} = \boldsymbol{f}_{\mathrm{ov}} = \boldsymbol{0} \tag{2.140}$$

What changes is that localized orbitals are generally not eigenfunctions of the Fock operator:

$$\widetilde{\boldsymbol{f}}_{\mathrm{oo}} = \boldsymbol{U}^\dagger \cdot \boldsymbol{\epsilon}_{\mathrm{o}} \cdot \boldsymbol{U} \tag{2.141}$$

Canonical orbital energies therefore cannot be assigned to individual localized orbitals. Localization is useful for identifying bonds, lone pairs, and fragments; for enabling local correlation and embedding methods; for selecting active spaces and domains; and for constructing Wannier functions in periodic systems. In practice, the occupied and virtual spaces should be localized separately, and it is often also advantageous to separate the core and valence orbitals. One must check for multiple extrema and discontinuous solutions and remember that spatial symmetry may be lost.

3 Open-shell and generalized Hartree–Fock methods

The restricted closed-shell ansatz is efficient and preserves spin symmetry, but it is too restrictive whenever the two spin populations require different spatial descriptions. Open-shell species and bond-breaking problems therefore motivate a hierarchy of increasingly flexible single-determinant wavefunctions. Unrestricted Hartree–Fock relaxes the requirement that α and β electrons share spatial orbitals, while generalized Hartree–Fock also removes the requirement that each orbital possess a fixed spin projection.

3.1 Unrestricted Hartree–Fock theory

Unrestricted Hartree–Fock introduces separate variational spaces for the α and β electrons. The resulting energy, density matrices, and coupled Pople–Nesbet–Berthier equations are developed before the unrestricted SCF procedure is applied to bond dissociation.

3.1.1 The unrestricted ansatz

In restricted Hartree–Fock, a doubly occupied spatial orbital is shared by one α and one β electron. Unrestricted Hartree–Fock removes this pairing constraint. For open-shell systems, one uses unrestricted spin orbitals,

$$\chi_p^{\mathrm{UHF}}(\boldsymbol{x}) = \begin{cases} \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) \\ \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \end{cases} \tag{3.1}$$

The α and β spatial orbitals are therefore allowed to differ. As a consequence, UHF may break spin symmetry: the determinant need not be an eigenfunction of $\hat{\mathcal{S}}^2$, and the HF wavefunction is no longer a pure spin function. Singlets may be contaminated by triplets, quintets, and higher-spin states, while doublets may be contaminated by quartets, sextets, and higher-spin states. Generalized Hartree–Fock (GHF) also permits spin mixing and need not preserve $\hat{\mathcal{S}}_z$. Its orbitals have the form

$$\chi_p^{\mathrm{GHF}}(\boldsymbol{x}) = \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) + \beta(\omega)\psi_p^{\beta}(\boldsymbol{r}) \qquad \Rightarrow \qquad \text{two-component spinor} \tag{3.2}$$

The additional variational freedom can lower the energy, but it changes the symmetry content of the determinant. In particular, the distinction between a useful broken-symmetry energy and a spin-pure wavefunction must be kept in mind throughout the following discussion.

3.1.2 The unrestricted Hartree–Fock equations and energy

The UHF determinant contains two independently optimized orthonormal orbital sets. Variation with respect to these two sets gives two coupled orbital equations. To minimize the UHF energy, the unrestricted spatial orbitals must be eigenfunctions of the separate α and β Fock operators:

$$\hat{f}^{\alpha}(\boldsymbol{r}_1)\psi_i^{\alpha}(\boldsymbol{r}_1) = \epsilon_i^{\alpha}\psi_i^{\alpha}(\boldsymbol{r}_1) \qquad\qquad \hat{f}^{\beta}(\boldsymbol{r}_1)\psi_i^{\beta}(\boldsymbol{r}_1) = \epsilon_i^{\beta}\psi_i^{\beta}(\boldsymbol{r}_1) \tag{3.3}$$

The two Fock operators are

$$\begin{aligned} \hat{f}^{\alpha}(\boldsymbol{r}_1) &= \hat{h}(\boldsymbol{r}_1) + \sum_i^{N^{\alpha}} \left[\hat{\mathcal{J}}_i^{\alpha}(\boldsymbol{r}_1) - \hat{\mathcal{K}}_i^{\alpha}(\boldsymbol{r}_1)\right] + \sum_i^{N^{\beta}} \hat{\mathcal{J}}_i^{\beta}(\boldsymbol{r}_1) \\ \hat{f}^{\beta}(\boldsymbol{r}_1) &= \hat{h}(\boldsymbol{r}_1) + \sum_i^{N^{\beta}} \left[\hat{\mathcal{J}}_i^{\beta}(\boldsymbol{r}_1) - \hat{\mathcal{K}}_i^{\beta}(\boldsymbol{r}_1)\right] + \sum_i^{N^{\alpha}} \hat{\mathcal{J}}_i^{\alpha}(\boldsymbol{r}_1) \end{aligned} \tag{3.4}$$

The Coulomb and exchange operators are defined by

$$\begin{aligned} \hat{\mathcal{J}}_i^\sigma(\boldsymbol{r}_1) &= \int \psi_i^{\sigma*}(\boldsymbol{r}_2) r_{12}^{-1} \psi_i^\sigma(\boldsymbol{r}_2)\, \mathrm{d}\boldsymbol{r}_2 \\ \hat{\mathcal{K}}_i^\sigma(\boldsymbol{r}_1)\psi_j^\sigma(\boldsymbol{r}_1) &= \left[\int \psi_i^{\sigma*}(\boldsymbol{r}_2) r_{12}^{-1} \psi_j^\sigma(\boldsymbol{r}_2)\, \mathrm{d}\boldsymbol{r}_2\right] \psi_i^\sigma(\boldsymbol{r}_1) \end{aligned} \tag{3.5}$$

Each spin population contributes a Coulomb field felt by electrons of either spin. Exchange, however, acts only between electrons carrying the same spin function because opposite spin functions are orthogonal. Consequently, the two Fock operators differ whenever the α and β densities differ, yet they remain coupled through their mutual Coulomb contributions.

The UHF energy. The UHF energy comprises three contributions:

$$\boxed{E_{\mathrm{UHF}} = E_{\mathrm{UHF}}^{\alpha\alpha} + E_{\mathrm{UHF}}^{\beta\beta} + E_{\mathrm{UHF}}^{\alpha\beta}} \tag{3.6}$$

Explicitly,

$$E_{\mathrm{UHF}} = \sum_i^{N^\alpha} h_{ii}^\alpha + \frac{1}{2}\sum_{ij}^{N^\alpha}\left(J_{ij}^{\alpha\alpha} - K_{ij}^{\alpha\alpha}\right) + \sum_i^{N^\beta} h_{ii}^\beta + \frac{1}{2}\sum_{ij}^{N^\beta}\left(J_{ij}^{\beta\beta} - K_{ij}^{\beta\beta}\right) + \sum_i^{N^\alpha}\sum_j^{N^\beta} J_{ij}^{\alpha\beta} \tag{3.7}$$

The matrix elements are

$$h_{ii}^\sigma = \langle \psi_i^\sigma | \hat{h} | \psi_i^\sigma \rangle \qquad J_{ij}^{\sigma\sigma'} = \langle \psi_i^\sigma \psi_j^{\sigma'} | \psi_i^\sigma \psi_j^{\sigma'} \rangle \qquad K_{ij}^{\sigma\sigma} = \langle \psi_i^\sigma \psi_j^\sigma | \psi_j^\sigma \psi_i^\sigma \rangle \tag{3.8}$$

There is no exchange between opposite-spin electrons. The factors of one half in the same-spin contributions remove double counting of electron pairs. The opposite-spin Coulomb term is written only once as a double sum over the two populations and therefore requires no such factor.

Exercise 17: UHF energy of the Li atom

Write down the UHF energy of the doublet state of the lithium atom. Figure 13 specifies the occupation for this exercise.

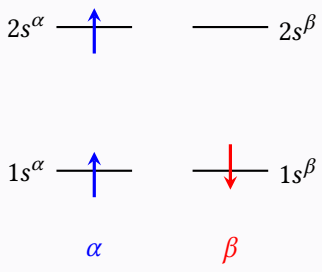


Figure 13: Occupation of the doublet lithium determinant used in the UHF exercise: $1s^\alpha$, $1s^\beta$, and $2s^\alpha$ are occupied, whereas $2s^\beta$ is empty.

Solution 17: UHF energy of the Li atom

The occupied spatial indices are $i = 1, 2$ for α spin and $i = 1$ for β spin, so $N^\alpha = 2$ and $N^\beta = 1$. The functions ψ_1^α and ψ_1^β need not coincide. Each spatial function is normalized, and the two occupied α functions are mutually orthogonal. No spatial orthogonality between opposite-spin orbitals is

Solution 17 (continued): UHF energy of the Li atom

required because their spin functions are orthogonal. The one-electron contribution is

$$E_1 = h_{11}^{\alpha} + h_{11}^{\beta} + h_{22}^{\alpha} \tag{3.9}$$

For the α–α interaction, the diagonal terms cancel through $J_{ii}^{\alpha\alpha} = K_{ii}^{\alpha\alpha}$, while the two ordered off-diagonal pairs give the same contribution:

$$\begin{aligned} E_2^{\alpha\alpha} &= \frac{1}{2}\sum_{i,j=1}^{2}\left(J_{ij}^{\alpha\alpha} - K_{ij}^{\alpha\alpha}\right) \\ &= \frac{1}{2}\left[J_{12}^{\alpha\alpha} - K_{12}^{\alpha\alpha} + J_{21}^{\alpha\alpha} - K_{21}^{\alpha\alpha}\right] \\ &= J_{12}^{\alpha\alpha} - K_{12}^{\alpha\alpha} \end{aligned} \tag{3.10}$$

There is only one β electron, so its same-spin two-electron contribution is zero. The opposite-spin interaction contains two distinct pairs and no exchange:

$$E_2^{\alpha\beta} = \sum_{i=1}^{2} J_{i1}^{\alpha\beta} = J_{11}^{\alpha\beta} + J_{21}^{\alpha\beta} \tag{3.11}$$

Adding these contributions gives, for the occupation shown above,

$$E_{\mathrm{UHF}} = h_{11}^{\alpha} + h_{11}^{\beta} + h_{22}^{\alpha} + J_{12}^{\alpha\alpha} - K_{12}^{\alpha\alpha} + J_{11}^{\alpha\beta} + J_{21}^{\alpha\beta} \tag{3.12}$$

The same-spin interaction between the $1s^{\alpha}$ and $2s^{\alpha}$ electrons contains both Coulomb and exchange contributions. The two interactions with the $1s^{\beta}$ electron are purely Coulombic, exactly as required by the spin selection rule above. As a check, impose the common-spatial-orbital limit $\psi_1^{\alpha} = \psi_1^{\beta} = \psi_1$ and $\psi_2^{\alpha} = \psi_2$. Then $J_{11}^{\alpha\beta} = J_{11}$ and $J_{21}^{\alpha\beta} = J_{12}$, and the energy reduces to

$$E_{\mathrm{UHF}} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12} \tag{3.13}$$

This recovers the earlier lithium exercise. The unrestricted expression differs by allowing the two spin populations to optimize their spatial functions independently, not by changing which electron pairs interact.

3.1.3 The Pople–Nesbet–Berthier matrix equations

Historical perspective: Pople and Nesbet

John Anthony Pople (1925–2004). A British theoretical chemist, Pople developed both semiempirical and ab initio molecular-orbital methods. His work helped establish reproducible computational procedures and widely used Gaussian basis-set families. He shared the 1998 Nobel Prize in Chemistry for the development of computational methods in quantum chemistry.

Robert K. Nesbet (b. 1930). Nesbet earned his Cambridge doctorate under Boys in 1954, using early digital computers for molecular calculations. He later worked with Slater at MIT and joined IBM in 1962. His joint paper with Pople on self-consistent orbitals for radicals [43] underlies the coupled spin-dependent matrix equations developed in this subsection.

Biographies: IAQMS Pople profile; Nesbet's IBM author biography; Nesbet author profile. Source portraits: Pople, IAQMS archive; Nesbet, Science Featured author profile. Photographers/dates unspecified. **Draft reproductions: permissions pending.** Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: Gaston Berthier (1923–2009)

A French theoretical chemist, Berthier was a pioneer of quantum chemistry in France, working with the Pullman group and at the École normale supérieure. In 1954 he developed an open-shell self-consistent-field treatment allowing electrons of opposite spin to occupy different spatial orbitals [44], independently of Pople and Nesbet [43]. His contribution is recognized here in the name *Pople–Nesbet–Berthier equations*. With Bernard Lévy, he later developed a generalized Brillouin theorem for multiconfigurational SCF wavefunctions. Alongside his research, he trained generations of theoretical chemists and helped build the French and wider Latin-language quantum-chemistry community.

Biography: Yves Ellinger, Société Française d'Exobiologie obituary, 2010. Source portrait: Fabriceparis, via Wikimedia Commons, CC BY-SA 3.0; photograph date unspecified. Sketch: AI-assisted adaptation of the credited portrait.

The continuous UHF equations are reduced to matrix form in the same way as the restricted equations, except that a separate coefficient matrix and Fock matrix are required for each spin. The unrestricted spatial orbitals are expanded in the same AO basis with separate spin-dependent coefficients:

$$\psi_i^\alpha(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^\alpha \phi_\mu(\boldsymbol{r}) \qquad \psi_i^\beta(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^\beta \phi_\mu(\boldsymbol{r}) \tag{3.14}$$

The resulting Pople–Nesbet–Berthier equations, independently formulated in 1954 [43, 44], are

$$\boxed{\boldsymbol{F}^\alpha \cdot \boldsymbol{C}^\alpha = \boldsymbol{S} \cdot \boldsymbol{C}^\alpha \cdot \boldsymbol{\epsilon}^\alpha} \tag{3.15}$$

$$\boxed{\boldsymbol{F}^\beta \cdot \boldsymbol{C}^\beta = \boldsymbol{S} \cdot \boldsymbol{C}^\beta \cdot \boldsymbol{\epsilon}^\beta} \tag{3.16}$$

Their spin-dependent Fock matrices are

$$F_{\mu\nu}^\alpha = H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}^\alpha [(\mu\nu\,|\,\sigma\lambda) - (\mu\lambda\,|\,\sigma\nu)] + \sum_{\lambda\sigma} P_{\lambda\sigma}^\beta (\mu\nu\,|\,\sigma\lambda) \tag{3.17}$$

$$F_{\mu\nu}^\beta = H_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}^\beta [(\mu\nu\,|\,\sigma\lambda) - (\mu\lambda\,|\,\sigma\nu)] + \sum_{\lambda\sigma} P_{\lambda\sigma}^\alpha (\mu\nu\,|\,\sigma\lambda) \tag{3.18}$$

Both $\boldsymbol{F}^{\alpha}$ and $\boldsymbol{F}^{\beta}$ depend on $\boldsymbol{C}^{\alpha}$ and $\boldsymbol{C}^{\beta}$. Therefore, the α and β orbitals are coupled. Thus the two generalized eigenvalue problems cannot be solved independently. Updating either spin density changes the Coulomb potential in both Fock matrices and changes the exchange potential in the Fock matrix of the same spin.

Unrestricted density matrices. The spin-up and spin-down density matrices are

$$P^{\alpha}_{\mu\nu} = \sum_{i=1}^{N^{\alpha}} C^{\alpha}_{\mu i} C^{\alpha *}_{\nu i} \quad \Longleftrightarrow \quad \boldsymbol{P}^{\alpha} \qquad P^{\beta}_{\mu\nu} = \sum_{i=1}^{N^{\beta}} C^{\beta}_{\mu i} C^{\beta *}_{\nu i} \quad \Longleftrightarrow \quad \boldsymbol{P}^{\beta} \tag{3.19}$$

For $\sigma = \alpha$ or β, the corresponding real-space density and its normalization are

$$\rho^{\sigma}(\boldsymbol{r}) = \sum_{\mu\nu} \phi_{\mu}(\boldsymbol{r}) P^{\sigma}_{\mu\nu} \phi^{*}_{\nu}(\boldsymbol{r}) \qquad \int \rho^{\sigma}(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r} = N^{\sigma} \tag{3.20}$$

Finally, the total and spin density matrices are

$$\underbrace{\boldsymbol{P}^{\mathrm{T}}}_{\text{charge density}} = \boldsymbol{P}^{\alpha} + \boldsymbol{P}^{\beta} \qquad \underbrace{\boldsymbol{P}^{\mathrm{S}}}_{\text{spin density}} = \boldsymbol{P}^{\alpha} - \boldsymbol{P}^{\beta} \tag{3.21}$$

The sum $\boldsymbol{P}^{\mathrm{T}}$ determines the total charge density. The difference $\boldsymbol{P}^{\mathrm{S}}$ measures the spatial imbalance between the two spin populations and is the matrix representation of the collinear spin density.

3.1.4 The unrestricted SCF procedure

The practical UHF SCF algorithm mirrors the restricted procedure, but both spin densities must be propagated until they are mutually self-consistent. It is summarized in Algorithm 2, using the same step organization as Algorithm 1. The calculation begins by specifying the molecule through $\{\boldsymbol{R}_A\}$ and $\{Z_A\}$ and choosing the basis set $\{\phi_\mu\}$, as in RHF. The integrals $S_{\mu\nu}$, $H_{\mu\nu}$, and $\langle\mu\nu|\lambda\sigma\rangle$ are calculated, and $\boldsymbol{S}$ is diagonalized to obtain $\boldsymbol{X}$.

After initial density matrices $\boldsymbol{P}^{\alpha}$ and $\boldsymbol{P}^{\beta}$ have been chosen (Step 3), the two spin-dependent matrix builds in Step 4 are performed: $\boldsymbol{G}^{\alpha}$ gives $\boldsymbol{F}^{\alpha} = \boldsymbol{H} + \boldsymbol{G}^{\alpha}$, while $\boldsymbol{G}^{\beta}$ gives $\boldsymbol{F}^{\beta} = \boldsymbol{H} + \boldsymbol{G}^{\beta}$. The transformed matrices are $(\boldsymbol{F}^{\alpha})' = \boldsymbol{X}^{\dagger} \cdot \boldsymbol{F}^{\alpha} \cdot \boldsymbol{X}$ and $(\boldsymbol{F}^{\beta})' = \boldsymbol{X}^{\dagger} \cdot \boldsymbol{F}^{\beta} \cdot \boldsymbol{X}$. Diagonalizing them yields $(\boldsymbol{C}^{\alpha})'$ and $\boldsymbol{\epsilon}^{\alpha}$, together with $(\boldsymbol{C}^{\beta})'$ and $\boldsymbol{\epsilon}^{\beta}$. The AO coefficients are then recovered from $\boldsymbol{C}^{\alpha} = \boldsymbol{X} \cdot (\boldsymbol{C}^{\alpha})'$ and $\boldsymbol{C}^{\beta} = \boldsymbol{X} \cdot (\boldsymbol{C}^{\beta})'$. These coefficients define the new density matrices $\boldsymbol{P}^{\alpha}$ and $\boldsymbol{P}^{\beta}$, from which $\boldsymbol{P}^{\mathrm{T}} = \boldsymbol{P}^{\alpha} + \boldsymbol{P}^{\beta}$ is formed. If the convergence test is not satisfied, the cycle returns to the construction of $\boldsymbol{G}^{\alpha}$ and $\boldsymbol{G}^{\beta}$; otherwise, the desired observables, including E_{UHF}, are evaluated.

Convergence therefore requires a fixed point of the coupled map

$$(\boldsymbol{P}^{\alpha}, \boldsymbol{P}^{\beta}) \longrightarrow (\boldsymbol{F}^{\alpha}, \boldsymbol{F}^{\beta}) \longrightarrow (\boldsymbol{P}^{\alpha}, \boldsymbol{P}^{\beta}) \tag{3.22}$$

Different initial spin densities can lead to different stationary solutions, especially where restricted spin symmetry becomes unstable. The joint test in Step 5 is essential: convergence of only one spin channel does not solve the UHF equations. Updating either density in Step 7 requires rebuilding both physical Fock matrices before testing the pair again.

3.2 Symmetry breaking in the dissociation of H_2

The dissociation of the hydrogen molecule is the simplest illustration of both the usefulness and the limitation of unrestricted Hartree–Fock. It shows explicitly how relaxing the equality of the

Algorithm 2 | The UHF self-consistent-field procedure

Collinear UHF with fixed N^α and N^β; $\sigma = \alpha, \beta$. Each $\boldsymbol{P}^\sigma$ counts only its own spin population, with no factor of two.

SETUP | Prepare once at a fixed geometry

1. **Specify the calculation.** Choose the geometry, AO basis, spin populations $N^\alpha + N^\beta = N$, and convergence thresholds.
2. **Prepare integrals and the common AO metric.** Calculate $\boldsymbol{S}$, $\boldsymbol{H}$, and the ERIs; use the same $\boldsymbol{X}$ for both spins, with $\boldsymbol{X}^\dagger \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1}$.
3. **Choose two initial densities.** Supply $\boldsymbol{P}^{\alpha,(0)}$ and $\boldsymbol{P}^{\beta,(0)}$; set $n = 0$.

ITERATION | Repeat Steps 4–7 until self-consistency

4. **Build both physical Fock matrices.** Both densities contribute to Coulomb; exchange is same-spin only:
   $\boldsymbol{F}^{\sigma,(n)} = \boldsymbol{F}^\sigma[\boldsymbol{P}^{\alpha,(n)}, \boldsymbol{P}^{\beta,(n)}]$
5. **Evaluate the energy and test both spin channels.** Use the physical matrices from Step 4:
   $$E_{\mathrm{UHF}}^{(n)} = \frac{1}{2} \sum_{\sigma=\alpha,\beta} \mathrm{Tr}\left[\boldsymbol{P}^{\sigma,(n)} \cdot (\boldsymbol{H} + \boldsymbol{F}^{\sigma,(n)})\right]$$
   $$\boldsymbol{e}^{\sigma,(n)} = \boldsymbol{F}^{\sigma,(n)} \cdot \boldsymbol{P}^{\sigma,(n)} \cdot \boldsymbol{S} - \boldsymbol{S} \cdot \boldsymbol{P}^{\sigma,(n)} \cdot \boldsymbol{F}^{\sigma,(n)}$$
   After at least one orbital update, require $\max_{\sigma=\alpha,\beta} \max |\boldsymbol{e}^{\sigma,(n)}| < \tau_{\mathrm{SCF}}$ and the chosen energy/-density change criteria for both spins. If satisfied, go to Step 8.
6. **Solve for new orbitals in each spin channel.**
   $$(\boldsymbol{F}^{\sigma,(n)})' = \boldsymbol{X}^\dagger \cdot \boldsymbol{F}^{\sigma,(n)} \cdot \boldsymbol{X}$$
   $$(\boldsymbol{F}^{\sigma,(n)})' \cdot (\boldsymbol{C}^{\sigma,(n+1)})' = (\boldsymbol{C}^{\sigma,(n+1)})' \cdot \boldsymbol{\epsilon}^{\sigma,(n+1)}$$
   $$\boldsymbol{C}^{\sigma,(n+1)} = \boldsymbol{X} \cdot (\boldsymbol{C}^{\sigma,(n+1)})'$$
7. **Occupy and update both densities.** Occupy the N^σ lowest orbitals of each spin and form
   $\boldsymbol{P}^{\sigma,(n+1)} = \boldsymbol{C}_{\mathrm{o}}^{\sigma,(n+1)} \cdot (\boldsymbol{C}_{\mathrm{o}}^{\sigma,(n+1)})^\dagger$
   Set $n \leftarrow n + 1$ and return to Step 4, rebuilding both physical Fock matrices.

OUTPUT | Only after convergence

8. **Evaluate the converged observables.** Report E_{UHF}, orbitals, charge/spin densities, and $\langle \hat{\mathcal{S}}^2 \rangle$. Add nuclear repulsion for the total molecular energy.

two spatial orbitals repairs the restricted dissociation energy while producing a wavefunction that is no longer a pure eigenfunction of $\hat{S}^2$. Start from normalized 1s functions ϕ_A and ϕ_B, one on each atom. Their symmetric and antisymmetric combinations define the gerade bonding orbital ψ_g and ungerade antibonding orbital ψ_u:

$$\psi_g = \frac{\phi_A + \phi_B}{\sqrt{2(1+S)}} \qquad \psi_u = \frac{\phi_A - \phi_B}{\sqrt{2(1-S)}} \qquad S = \langle \phi_A | \phi_B \rangle \tag{3.23}$$

Here S is the overlap between the two atomic functions, not the total-spin quantum number; R below is the internuclear distance. For two electrons in two spatial orbitals, the UHF parametrization used below reduces to a single-parameter problem. Figure 14 defines the two-center geometry and the associated atomic basis functions.

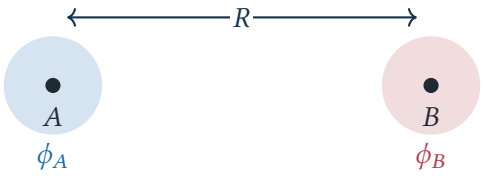


Figure 14: Minimal two-center description of H_2: atomic functions ϕ_A and ϕ_B are centered on nuclei A and B, separated by the internuclear distance R.

3.2.1 The restricted dissociation limit

In the restricted description both electrons occupy the gerade bonding orbital. The RHF wavefunction and energy are

$$|\Psi_{\mathrm{RHF}}\rangle = |\psi_g \bar{\psi}_g\rangle \qquad E_{\mathrm{RHF}} = 2h_{gg} + J_{gg} + \frac{1}{R} \tag{3.24}$$

At large R, where $S \to 0$,

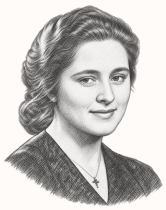
$$\psi_g(1)\psi_g(2) \longrightarrow \frac{1}{2}\Big[\underbrace{\phi_A(1)\phi_B(2) + \phi_B(1)\phi_A(2)}_{\text{covalent}} + \underbrace{\phi_A(1)\phi_A(2) + \phi_B(1)\phi_B(2)}_{\text{ionic}}\Big] \tag{3.25}$$

RHF retains 50% ionic character as $R \to \infty$. Both electrons are forced to share the same spatial orbital, so its energy lies above that of two separated H atoms. Hence, RHF is qualitatively correct near equilibrium but not at dissociation. The defect is not caused by an inadequate optimization of the bonding orbital within RHF. It follows from the restricted ansatz itself: placing both electrons in the same spatial orbital inevitably retains the ionic configurations together with the covalent ones.

3.2.2 Unrestricted symmetry breaking and the Coulson–Fischer point

Historical perspective: Coulson and Fischer-Hjalmars

Charles Alfred Coulson (1910–1974). A British theoretical chemist and mathematician, Coulson held positions in London and Oxford and made molecular-orbital theory a powerful language for discussing chemical bonding. His research and teaching emphasized physical understanding alongside calculation. His collaboration with Fischer examined the limitations of simple orbital descriptions of the hydrogen molecule.



Inga Fischer-Hjalmars (1918–2008). Initially trained in pharmacy, Fischer-Hjalmars became a pioneer of Swedish quantum chemistry and succeeded Oskar Klein as professor of theoretical physics at Stockholm in 1963. She published the 1949 hydrogen-molecule paper as Inga Fischer [45]. The Coulson–Fischer point discussed here preserves her earlier surname in the terminology of Hartree–Fock symmetry breaking.

Biographies: IAQMS Coulson profile; Björn Roos, IAQMS obituary. Source portraits: Coulson, IAQMS archive, photographer/date unspecified, **draft reproduction: permission pending**; Fischer-Hjalmars, unknown photographer, Svenskt Porträttarkiv, existing Commons crop, CC BY 4.0. Sketch: AI-assisted adaptation of the credited portrait.

The two-dimensional orbital space spanned by ψ_g and ψ_u permits the α and β orbitals to rotate in opposite directions. The UHF orbitals can be parametrized as

$$\psi^{\alpha}(\theta) = \cos\theta\,\psi_g + \sin\theta\,\psi_u \qquad \psi^{\beta}(\theta) = \cos\theta\,\psi_g - \sin\theta\,\psi_u \tag{3.26}$$

with $\langle\psi^{\alpha}|\psi^{\beta}\rangle = \cos(2\theta)$. The two choices $\pm\theta$ are degenerate and exchange the spin localization. The parameter θ therefore provides a continuous path from the restricted determinant at $\theta = 0$ to orthogonal spatial orbitals at $\theta = \pi/4$. In the dissociation limit, these endpoint orbitals become localized on separate atoms. The orbital overlap $\cos(2\theta)$ decreases along this path and vanishes at its endpoint; at finite atomic overlap, orthogonality alone does not imply complete localization on individual atoms. Figure 15 illustrates the progressive localization of the two spin channels along this path.

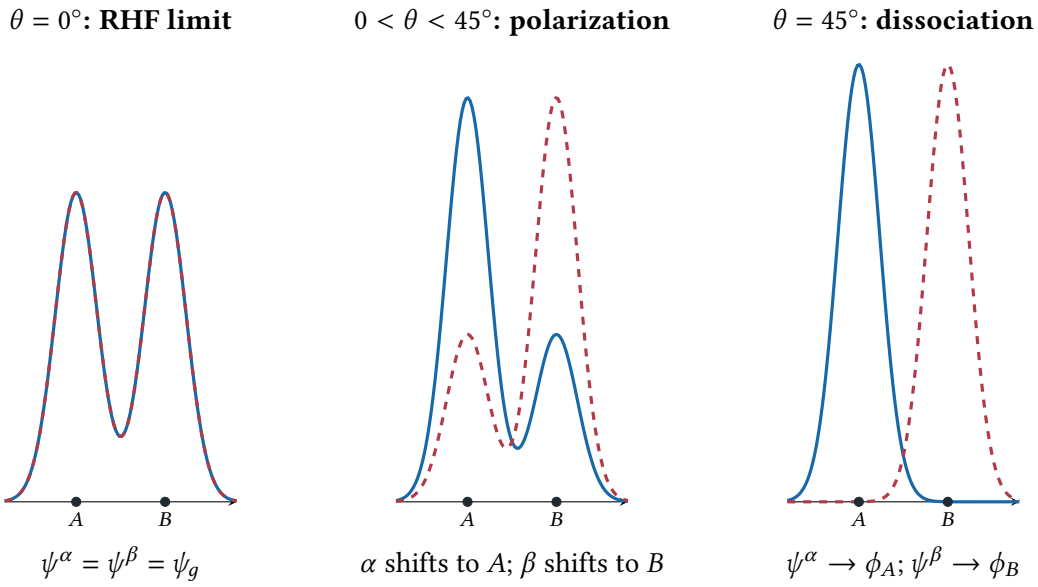


Figure 15: Schematic spatial profiles of the α (solid blue) and β (dashed red) UHF orbitals under opposite rotations by θ. The three panels show the restricted limit, partial spin polarization, and localization on separate atoms at dissociation.

With $J_{pq} = (pp\,|\,qq)$ and $K_{pq} = (pq\,|\,pq)$, the angle-dependent electronic energy is given below. Unlike the preceding RHF total-energy expression, this polynomial omits the nuclear constant $1/R$. That constant does not affect minimization with respect to θ at fixed R, but must be added when comparing total energies as R varies.

$$E_{\mathrm{UHF}}(\theta) = 2\cos^2\theta\, h_{gg} + 2\sin^2\theta\, h_{uu} + \cos^4\theta\, J_{gg} + \sin^4\theta\, J_{uu} + 2\sin^2\theta\cos^2\theta\big(J_{gu} - 2K_{gu}\big) \tag{3.27}$$

Introducing

$$x = \cos^2\theta \qquad\qquad A = J_{gu} - 2K_{gu} \tag{3.28}$$

allows the same energy to be written as

$$E_{\mathrm{UHF}}(x) = 2xh_{gg} + 2(1-x)h_{uu} + x^2J_{gg} + (1-x)^2J_{uu} + 2x(1-x)A \tag{3.29}$$

Its derivative is

$$\frac{\mathrm{d}E_{\mathrm{UHF}}}{\mathrm{d}x} = 2\big[h_{gg} - h_{uu} - J_{uu} + A + x\big(J_{gg} + J_{uu} - 2A\big)\big] \tag{3.30}$$

Because $\mathrm{d}x/\mathrm{d}\theta = -\sin(2\theta)$, stationarity requires either $\sin(2\theta) = 0$ or $\mathrm{d}E_{\mathrm{UHF}}/\mathrm{d}x = 0$. The first condition contains the restricted solution $\theta = 0$; the second gives the broken-symmetry solution. Besides the restricted stationary point $\theta = 0$, a broken-symmetry stationary solution satisfies

$$\cos^2\theta^\star = \frac{h_{uu} - h_{gg} + J_{uu} - J_{gu} + 2K_{gu}}{J_{gg} + J_{uu} - 2J_{gu} + 4K_{gu}} \tag{3.31}$$

At short bond lengths, only the minimum at $\theta = 0$ is physically accessible and UHF reduces to RHF. On the dissociating branch, the coefficient $J_{gg} + J_{uu} - 2J_{gu} + 4K_{gu}$ is positive. When $0 < \cos^2\theta^\star < 1$, two minima appear at $\pm\theta^\star$ and RHF becomes unstable to spin polarization. The endpoint $\cos^2\theta^\star = 1$ is the point at which these solutions meet RHF; positivity of the quadratic coefficient distinguishes a minimum from a maximum of $E_{\mathrm{UHF}}(x)$.

The RHF curvature is

$$\left.\frac{\mathrm{d}^2E}{\mathrm{d}\theta^2}\right|_{\theta=0} = 4\big(h_{uu} - h_{gg} - J_{gg} + J_{gu} - 2K_{gu}\big) \tag{3.32}$$

and the Coulson–Fischer point [45] is defined by

$$R = R_{\mathrm{CF}} \quad\Longleftrightarrow\quad \left.\frac{\mathrm{d}^2E}{\mathrm{d}\theta^2}\right|_{\theta=0} = 0 \tag{3.33}$$

For $R < R_{\mathrm{CF}}$, the RHF curvature is positive and there is one minimum at $\theta = 0$. At $R = R_{\mathrm{CF}}$, it vanishes and the minimum becomes flat. For $R > R_{\mathrm{CF}}$ it is negative and two UHF minima occur at $\pm\theta^\star$. The RHF determinant does not disappear: it changes from a minimum to an unstable stationary point while two degenerate UHF minima emerge. This is a bifurcation of stationary solutions. The vanishing curvature marks the geometry at which an infinitesimal spin-polarizing orbital rotation ceases to raise the energy and begins to lower it. The three regimes are illustrated in Fig. 16.

At dissociation,

$$S \to 0 \qquad\qquad \psi_g \to \frac{\phi_A + \phi_B}{\sqrt{2}} \qquad\qquad U = (AA|AA) > 0 \tag{3.34}$$

Writing E_{H} for the isolated-atom energy represented in the same basis, the restricted and unrestricted total-energy limits are

$$E^{\infty}_{\mathrm{RHF}} = 2E_{\mathrm{H}} + \frac{1}{2}U \qquad\qquad E^{\infty}_{\mathrm{UHF}} = 2E_{\mathrm{H}} \tag{3.35}$$

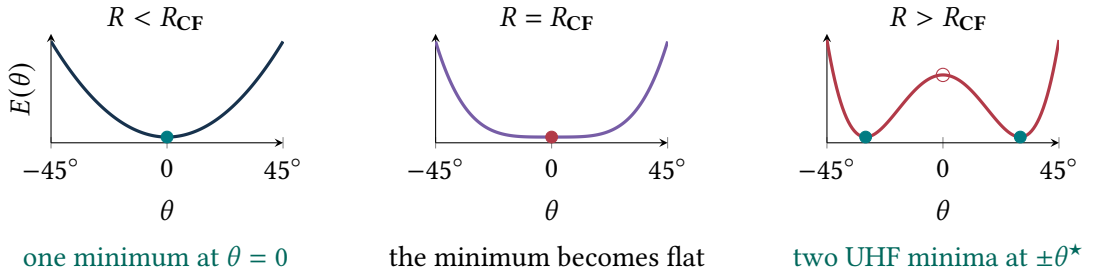


Figure 16: Evolution of the stationary points at the Coulson–Fischer point.

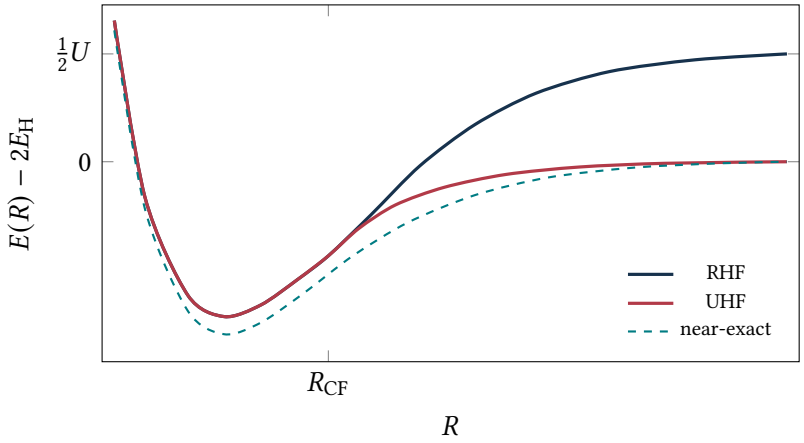


Figure 17: Schematic RHF, UHF, and near-exact dissociation curves for minimal-basis H_2.

The ionic terms therefore leave a finite RHF error, whereas $\psi^\alpha \to \phi_A$ and $\psi^\beta \to \phi_B$ give the correct UHF limit. Figure 17 compares the resulting schematic dissociation curves with a near-exact reference.

For the two-electron $M_S = 0$ determinant,

$$\left\langle \hat{S}^2 \right\rangle_{\text{UHF}} = 1 - \left| \left\langle \psi^\alpha \middle| \psi^\beta \right\rangle \right|^2 = \sin^2(2\theta) \tag{3.36}$$

Near equilibrium, $\theta = 0$ and $\langle \hat{S}^2 \rangle = 0$, so the determinant is a pure singlet. At dissociation, $\theta \to \pi/4$ and $\langle \hat{S}^2 \rangle \to 1$, so it is an equal mixture of a singlet and an $M_S = 0$ triplet:

$$|A\alpha\, B\beta\rangle = \frac{1}{\sqrt{2}}(|\text{covalent singlet}\rangle + |\text{triplet}, M_S = 0\rangle) \qquad \text{as } R \to \infty \tag{3.37}$$

Thus, the UHF energy is physically useful, but the UHF wavefunction is spin-contaminated. Three conclusions follow. A stationary SCF solution need not be stable, the Coulson–Fischer point marks a continuous RHF-to-UHF transition, and symmetry breaking mimics static correlation at the price of a contaminated wavefunction. The example exposes the characteristic compromise of broken-symmetry mean-field theory. The determinant reproduces the correct separated-atom energy through spin localization, but represents the singlet character only through a mixture of spin eigenfunctions.

3.3 Generalized Hartree–Fock theory

Unrestricted Hartree–Fock remains collinear: every occupied orbital is either purely α or purely β with respect to one common quantization axis. Generalized Hartree–Fock removes this remaining

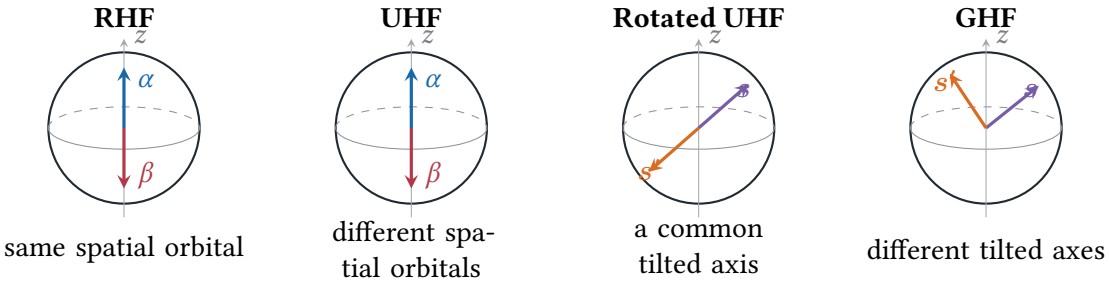
restriction. The key idea is that GHF allows every orbital to choose its spin direction. A GHF orbital is a two-component spinor:

$$\boxed{\chi_p^{\mathrm{GHF}}(\boldsymbol{x}) = \alpha(\omega)\psi_p^{\alpha}(\boldsymbol{r}) + \beta(\omega)\psi_p^{\beta}(\boldsymbol{r})} \tag{3.38}$$

Both components are expanded in the same spatial AO basis,

$$\psi_i^{\sigma}(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu i}^{\sigma}\phi_{\mu}(\boldsymbol{r}) \qquad \boldsymbol{C} = \begin{pmatrix} \boldsymbol{C}^{\alpha} \\ \boldsymbol{C}^{\beta} \end{pmatrix} \qquad \boldsymbol{C} \in \mathbb{C}^{2K\times N} \tag{3.39}$$

RHF pairs electrons in one spatial orbital and preserves both $\hat{S}^2$ and $\hat{S}_z$. UHF uses orbitals that are purely α or β, has a collinear spin density, and preserves $\hat{S}_z$. GHF permits each orbital to mix α and β, so in general neither $\hat{S}^2$ nor $\hat{S}_z$ is preserved. GHF therefore does not require a common spin-quantization axis or fixed values of N^{α} and N^{β}. Figure 18 distinguishes these possibilities from a mere global rotation of the common UHF spin axis. RHF and UHF are consequently contained as special cases of GHF. A GHF coefficient matrix has twice as many rows because every spatial basis function can appear in both spin components of each occupied spinor.



$$\underbrace{\text{UHF} \xrightarrow{\text{global spin rotation}} \text{rotated UHF}}_{\text{physically equivalent collinear solutions}} \neq \underbrace{\text{genuine GHF}}_{\text{non-collinear}} \tag{3.40}$$

Figure 18: RHF, UHF, globally rotated UHF, and genuinely noncollinear GHF spin directions. A tilted spin axis alone does not imply a genuinely noncollinear GHF solution.

The GHF density matrix is

$$\boldsymbol{\mathcal{P}} = \begin{pmatrix} \boldsymbol{P}^{\alpha\alpha} & \boldsymbol{P}^{\alpha\beta} \\ \boldsymbol{P}^{\beta\alpha} & \boldsymbol{P}^{\beta\beta} \end{pmatrix} \qquad P_{\mu\nu}^{\sigma\sigma'} = \sum_{i=1}^{N} C_{\mu i}^{\sigma} C_{\nu i}^{\sigma' *} \tag{3.41}$$

The diagonal blocks describe the two collinear spin populations, whereas the off-diagonal blocks encode coherence between the α and β components. These off-diagonal blocks vanish for an ordinary UHF determinant. Its charge and magnetization blocks are

$$\boldsymbol{P} = \frac{1}{2}\left(\boldsymbol{P}^{\alpha\alpha} + \boldsymbol{P}^{\beta\beta}\right) \qquad \boldsymbol{M}_z = \frac{1}{2}\left(\boldsymbol{P}^{\alpha\alpha} - \boldsymbol{P}^{\beta\beta}\right) \tag{3.42}$$

$$\boldsymbol{M}_y = \frac{1}{2\mathrm{i}}\left(\boldsymbol{P}^{\beta\alpha} - \boldsymbol{P}^{\alpha\beta}\right) \qquad \boldsymbol{M}_x = \frac{1}{2}\left(\boldsymbol{P}^{\beta\alpha} + \boldsymbol{P}^{\alpha\beta}\right) \tag{3.43}$$

Equivalently, using the Pauli spin matrices $\boldsymbol{\sigma}_k$ and the Kronecker product $\otimes$,

$$\boldsymbol{\mathcal{P}} = \begin{pmatrix} \boldsymbol{P} + \boldsymbol{M}_z & \boldsymbol{M}_x - \mathrm{i}\boldsymbol{M}_y \\ \boldsymbol{M}_x + \mathrm{i}\boldsymbol{M}_y & \boldsymbol{P} - \boldsymbol{M}_z \end{pmatrix} = \mathbf{1}_{\mathrm{spin}} \otimes \boldsymbol{P} + \sum_{k=x,y,z} \boldsymbol{\sigma}_k \otimes \boldsymbol{M}_k \tag{3.44}$$

with

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_x \\ \boldsymbol{M}_y \\ \boldsymbol{M}_z \end{pmatrix} \qquad \boldsymbol{\sigma} = \begin{pmatrix} \boldsymbol{\sigma}_x \\ \boldsymbol{\sigma}_y \\ \boldsymbol{\sigma}_z \end{pmatrix} \tag{3.45}$$

Here $\mathbf{1}_{\text{spin}}$ is the 2×2 identity matrix in the spin basis. In this decomposition, $\boldsymbol{P}$ is half of the total charge-density matrix, whereas $\boldsymbol{\mathcal{P}}$ is the full spinor density matrix.

In spinor space, $\boldsymbol{C}$ stacks the two spin components of the orbital coefficient matrix, $\boldsymbol{\mathcal{F}}$ is the spinor Fock matrix, $\boldsymbol{\epsilon}$ contains the orbital energies, and $\boldsymbol{\mathcal{S}}$ is the block-diagonal overlap metric. The Roothaan–Hall equation is

$$\boldsymbol{\mathcal{F}} \cdot \boldsymbol{C} = \boldsymbol{\mathcal{S}} \cdot \boldsymbol{C} \cdot \boldsymbol{\epsilon} \tag{3.46}$$

with

$$\boldsymbol{\mathcal{S}} = \begin{pmatrix} \boldsymbol{S} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{S} \end{pmatrix} \tag{3.47}$$

and normalization

$$\boldsymbol{C}^\dagger \cdot \boldsymbol{\mathcal{S}} \cdot \boldsymbol{C} = \mathbf{1} \tag{3.48}$$

The GHF Fock matrix is given below, using the same AO core-Hamiltonian matrix $\boldsymbol{H}$ as in the finite-basis derivation.

$$\boldsymbol{\mathcal{F}} = \begin{pmatrix} \boldsymbol{H} + \boldsymbol{J}[\boldsymbol{P}^{\alpha\alpha}] + \boldsymbol{J}[\boldsymbol{P}^{\beta\beta}] - \boldsymbol{K}[\boldsymbol{P}^{\alpha\alpha}] & -\boldsymbol{K}[\boldsymbol{P}^{\alpha\beta}] \\ -\boldsymbol{K}[\boldsymbol{P}^{\beta\alpha}] & \boldsymbol{H} + \boldsymbol{J}[\boldsymbol{P}^{\alpha\alpha}] + \boldsymbol{J}[\boldsymbol{P}^{\beta\beta}] - \boldsymbol{K}[\boldsymbol{P}^{\beta\beta}] \end{pmatrix} \tag{3.49}$$

where

$$J_{\mu\nu}[\boldsymbol{P}] = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\nu \,|\, \lambda\sigma) \qquad K_{\mu\nu}[\boldsymbol{P}] = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\lambda \,|\, \sigma\nu) \tag{3.50}$$

The AO functions in these contractions are real, but the spinor coefficients and density blocks may be complex. The exchange indices follow the definition $P_{\lambda\sigma} = \sum_i C_{\lambda i} C^*_{\sigma i}$; exchanging λ and σ in the integral without also transposing the density would be incorrect for complex orbitals. The off-diagonal exchange blocks couple the α and β components self-consistently. The SCF procedure builds all four density blocks, builds all four Fock blocks, solves a $2K \times 2K$ generalized eigenvalue problem, occupies the N lowest spinors, and iterates.

The mathematical structure is the same nonlinear generalized eigenvalue problem as before, now expressed in a doubled spinor space. The off-diagonal exchange blocks are precisely the new terms that permit the spin direction to rotate away from the common UHF axis.

3.3.1 Variational hierarchy and stability

For the corresponding global minima in the same one-electron basis,

$$\{\Phi_{\text{RHF}}\} \subset \{\Phi_{\text{UHF}}\} \subset \{\Phi_{\text{GHF}}\} \qquad \Longrightarrow \qquad \boxed{E_{\text{GHF}} \leq E_{\text{UHF}} \leq E_{\text{RHF}}} \tag{3.51}$$

A converged UHF determinant may remain unstable to occupied-virtual rotations that mix α and β orbitals; following a negative-curvature direction gives a lower-energy GHF solution. Not every GHF solution is noncollinear: RHF and UHF are special cases, and a global spin rotation of a UHF determinant changes its apparent quantization axis while remaining physically collinear.

The energy inequalities compare minima over nested variational spaces. They do not imply that every numerical GHF stationary point lies below every UHF or RHF stationary point. Establishing whether a converged solution is a local minimum requires a stability analysis with respect to the orbital rotations allowed in the larger space [46, 47].

Lowering the variational energy may require breaking exact symmetries. UHF may break $\hat{\mathcal{S}}^2$; GHF may break both $\hat{\mathcal{S}}^2$ and $\hat{\mathcal{S}}_z$; and complex GHF may also break time-reversal symmetry. GHF remains a single determinant, so correlation is not included, and S and M_S are generally not good quantum numbers. The many stationary solutions make the initial guess and stability analysis important. A putative GHF solution may be a globally rotated UHF solution, requiring a collinearity test. Equilateral H_3 and triangular magnetic clusters are typical genuine noncollinear examples. GHF is the most general number-conserving single-determinant HF ansatz; particle-number symmetry can instead be broken within Hartree–Fock–Bogoliubov (HFB) theory [48, 49].

Practical interpretation. The charge and magnetization matrices provide a direct diagnostic of the solution. Collinear determinants admit one global spin axis along which only one magnetization component remains. A genuinely noncollinear GHF solution cannot be reduced to that form by a single global spin rotation. This is why a tilted axis, by itself, is insufficient evidence of noncollinearity.

3.4 Classification of Hartree–Fock solutions

The labels RHF, UHF, and GHF describe broad variational classes, but each class contains further possibilities depending on whether the orbitals are real or complex and which spin and time-reversal symmetries are retained. The Stuber–Paldus notation [50] organizes these stationary solutions by symmetry and by the block structure of their coefficient matrices. For the Fukutome designations used in the table, see Fukutome [51]. The Stuber–Paldus classification is summarized in Table 2. In this table, $\hat{\mathcal{K}}$ denotes complex conjugation, rather than the exchange operator used earlier, and $\hat{\Theta}$ denotes time reversal.

Historical perspective: Josef Paldus (1935–2023)

A Czech-born Canadian theoretical chemist, Paldus trained in Prague and joined the University of Waterloo in 1968. With Jiří Čížek, he developed the analysis of Hartree–Fock stability, including the distinction between singlet and triplet instabilities in closed-shell systems [52]. This work clarified why a self-consistent solution need not be an energy minimum. Their collaboration also helped establish coupled-cluster theory in molecular quantum chemistry. Paldus developed unitary-group methods for treating spin-adapted many-electron states [53], bringing algebraic structure to configuration-interaction calculations. His later account with Jason Stuber [50] places the symmetry-broken determinants discussed here in a systematic framework. His work thus connects the classification of independent-particle states with the treatment of electron correlation beyond a single determinant.

Biography: University of Waterloo, 2023 obituary; IAQMS profile. Source portrait: IAQMS archive; photographer and date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

Here TICS denotes time-reversal-invariant closed-shell; CCW, charge current wave; ASCW, axial spin current wave; ASDW, axial spin density wave; ASW, axial spin wave; TSCW, torsional spin current wave; TSDW, torsional spin density wave; and TSW, torsional spin wave. Figure 19 displays the connections between these classes as successive constraints are released.

The table makes the variational hierarchy concrete. Moving from restricted to unrestricted and then generalized orbitals removes constraints from the coefficient matrix; allowing complex coefficients removes additional antiunitary symmetries. These larger spaces can contain lower stationary energies, but identifying their physical symmetry requires examining the solution rather than relying only on its numerical energy.

Table 2: Stuber–Paldus classification of Hartree–Fock solutions. Reproduced in part from Table I of Henderson, Jiménez-Hoyos, and Scuseria [54], with the density-matrix columns omitted and notation adjusted to the present chapter.

Fukutome designation	Stuber–Paldus designation	Symmetries preserved	Structure of the orbital coefficient matrix C
Restricted Hartree–Fock			
TICS[a]	real RHF	$\hat{\mathcal{S}}^2, \hat{\mathcal{S}}_z, \hat{\mathcal{K}}, \hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma\sigma} \end{pmatrix}, C \in \mathbb{R}$
CCW[b]	complex RHF	$\hat{\mathcal{S}}^2, \hat{\mathcal{S}}_z$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma\sigma} \end{pmatrix}$
Unrestricted Hartree–Fock			
ASCW[c]	paired UHF	$\hat{\mathcal{S}}_z, \hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C^*_{\sigma\sigma} \end{pmatrix}$
ASDW[d]	real UHF	$\hat{\mathcal{S}}_z, \hat{\mathcal{K}}$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma'\sigma'} \end{pmatrix}, C \in \mathbb{R}$
ASW[e]	complex UHF	$\hat{\mathcal{S}}_z$	$\begin{pmatrix} C_{\sigma\sigma} & \mathbf{0} \\ \mathbf{0} & C_{\sigma'\sigma'} \end{pmatrix}$
Generalized Hartree–Fock			
TSCW[f]	paired GHF	$\hat{\Theta}$	$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ -C^*_{\sigma\sigma'} & C^*_{\sigma\sigma} \end{pmatrix}$
TSDW[g]	real GHF	$\hat{\mathcal{K}}$	$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ C_{\sigma'\sigma} & C_{\sigma'\sigma'} \end{pmatrix}, C \in \mathbb{R}$
TSW[h]	complex GHF	—	$\begin{pmatrix} C_{\sigma\sigma} & C_{\sigma\sigma'} \\ C_{\sigma'\sigma} & C_{\sigma'\sigma'} \end{pmatrix}$

Space	Real coefficients	Time-reversal paired	Complex coefficients
RHF	RHF: $\hat{\mathcal{S}}^2, \hat{\mathcal{S}}_z, \hat{\mathcal{K}}, \hat{\Theta}$	Not a separate class here	c-RHF: $\hat{\mathcal{S}}^2, \hat{\mathcal{S}}_z$
UHF	UHF: $\hat{\mathcal{S}}_z, \hat{\mathcal{K}}$	p-UHF: $\hat{\mathcal{S}}_z, \hat{\Theta}$	c-UHF: $\hat{\mathcal{S}}_z$
GHF	GHF: $\hat{\mathcal{K}}$	p-GHF: $\hat{\Theta}$	c-GHF: none of these

Cells name the retained symmetries. Real and paired classes are alternative constraints.

Constraint-release connections

RHF → c-RHF: release $\hat{\Theta}, \hat{\mathcal{K}}$
RHF → UHF: release $\hat{\mathcal{S}}^2, \hat{\Theta}$
RHF → p-UHF: release $\hat{\mathcal{S}}^2, \hat{\mathcal{K}}$
c-RHF → c-UHF: release $\hat{\mathcal{S}}^2$
UHF → GHF: release $\hat{\mathcal{S}}_z$
UHF → c-UHF: release $\hat{\mathcal{K}}$
p-UHF → p-GHF: release $\hat{\mathcal{S}}_z$
p-UHF → c-UHF: release $\hat{\Theta}$
c-UHF → c-GHF: release $\hat{\mathcal{S}}_z$
GHF → c-GHF: release $\hat{\mathcal{K}}$
p-GHF → c-GHF: release $\hat{\Theta}$

Figure 19: Stuber–Paldus classification of Hartree–Fock solutions, retaining all eight classes and eleven constraint-release connections. New comparison layout based on Fig. 1 of Burton, Thom, and Loos, *Parity-Time Symmetry in Hartree–Fock Theory* [55].

4 One-electron basis sets

The Hartree–Fock formalism determines the optimal orbitals only within the one-electron space made available to it. A practical calculation therefore requires a second set of choices: the functions used to represent the orbitals and the algorithms used to evaluate their matrix elements. The remainder of the chapter develops these ingredients, beginning with basis-set construction and continuing with Gaussian molecular integrals and their low-rank representations.

4.1 Basis-set expansions and design principles

The matrix equations in Section 2 assumed that the orbital basis had already been chosen. We now examine that choice: which functions can represent the radial and angular variations of molecular orbitals accurately with a manageable number of coefficients? In practical molecular calculations, each molecular orbital is expanded in a finite set of known one-electron functions,

$$\psi_p(\boldsymbol{r}) = \sum_{\mu=1}^{K} C_{\mu p}\phi_\mu(\boldsymbol{r}) \tag{4.1}$$

The basis functions ϕ_μ are fixed, whereas the coefficients $C_{\mu p}$ are determined variationally. Enlarging a nested sequence of basis spaces increases the variational freedom and lowers, or leaves unchanged, the minimum Hartree–Fock energy as the complete-basis-set (CBS) limit is approached. Basis exponents and contraction coefficients are fixed numerical parameters, often obtained from atomic electronic-structure calculations. Their presence does not by itself make the method empirical; they specify the finite representation of the otherwise ab initio problem. Orbital optimization therefore changes the molecular combination of the building blocks, not their shapes or nonlinear parameters. At a fixed geometry these are selected before the SCF calculation begins; the centers move with the nuclei when atom-centered calculations are repeated at new geometries.

A useful basis must be accurate, compact, numerically stable, and practical. These requirements compete: a larger and more flexible basis improves the description of the orbitals, but increases both the computational cost and the risk of linear dependence.

4.1.1 The LEGO principle

Basis-set construction follows a LEGO-like philosophy: a small collection of well-designed building blocks is combined to describe chemically diverse systems, as illustrated in Fig. 20.

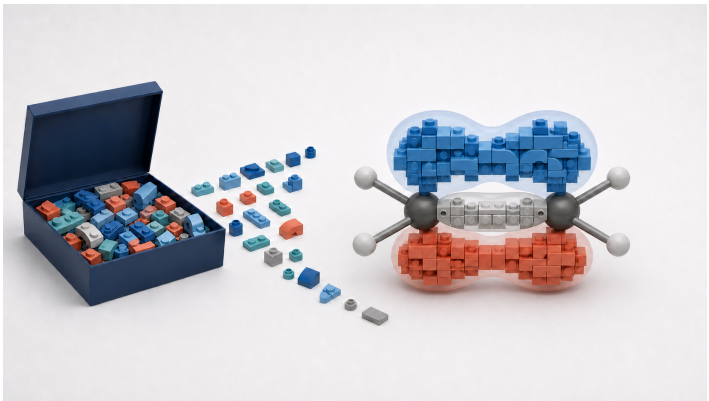


Figure 20: The basis functions are the elementary building blocks from which molecular orbitals are assembled.

The analogy also captures transferability. Atom-centered functions designed for individual elements can be assembled in different combinations without redesigning the basis for every molecule. The molecular environment enters through the variational coefficients that combine these fixed building blocks.

4.1.2 Principal families of basis functions

Several families of one-electron functions are routinely used. Gaussian-type orbitals (GTOs), which are proportional to $\exp(-\alpha r^2)$, are atom centered and are the standard choice for molecules because their multicenter integrals can be evaluated efficiently. Here r is the distance from the function's center, and α is a positive Gaussian exponent. They are used, for example, in ORCA [56], Q-CHEM [57], MOLCAS [58], MOLPRO [59], PySCF [60], Psi4 [61], CFOUR [62], MRCC [63], and GAMESS [64].

Slater-type orbitals (STOs) are also atom centered, but are proportional to $\exp(-\zeta r)$. Their radial form can reproduce a nuclear cusp and an exponential long-range tail, and they are used, for example, in ADF [65]. The positive exponent ζ controls their radial extent. By contrast, plane waves (PWs), proportional to $\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r})$, are delocalized, mutually orthonormal, and naturally adapted to periodic boundary conditions. Here $\boldsymbol{k}$ is the wavevector and i the imaginary unit. Their basis size is controlled by an energy cutoff, and they are used in QUANTUM ESPRESSO [66], VASP [67], ABINIT [68], and CASTEP [69]. Finally, numerical atomic orbitals (NAOs) have tabulated radial parts. Their compactness and flexibility make them useful in programs such as FHI-AIMS [70] and SIESTA [71].

4.2 Plane-wave basis sets

The four families balance radial accuracy, integral efficiency, periodicity, and numerical flexibility differently. We first examine the delocalized plane-wave representation, then return to atom-centered functions.

Plane waves are the natural one-electron functions of a translationally invariant system. Under periodic boundary conditions, only reciprocal-lattice wavevectors are compatible with the periodicity of the simulation cell. In a unit cell of volume Ω, a normalized plane wave is

$$\phi_{\boldsymbol{k}}(\boldsymbol{r}) = \boxed{\frac{1}{\sqrt{\Omega}}\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r})} \qquad |\boldsymbol{k}| \le k_{\mathrm{cut}} \qquad E_{\mathrm{cut}} = \frac{k_{\mathrm{cut}}^2}{2} \tag{4.2}$$

The reciprocal-lattice vector is indexed by three integers:

$$\boldsymbol{k} = k_1\boldsymbol{b}_1 + k_2\boldsymbol{b}_2 + k_3\boldsymbol{b}_3 \tag{4.3}$$

where, for direct-lattice vectors $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$,

$$\Omega = \boldsymbol{a}_1\cdot(\boldsymbol{a}_2\times\boldsymbol{a}_3) \qquad \boldsymbol{A} = (\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3) \tag{4.4}$$

$$\boldsymbol{b}_1 = 2\pi\frac{\boldsymbol{a}_2\times\boldsymbol{a}_3}{\Omega} \qquad \boldsymbol{b}_2 = 2\pi\frac{\boldsymbol{a}_3\times\boldsymbol{a}_1}{\Omega} \qquad \boldsymbol{b}_3 = 2\pi\frac{\boldsymbol{a}_1\times\boldsymbol{a}_2}{\Omega} \tag{4.5}$$

The reciprocal vectors satisfy $\boldsymbol{a}_i\cdot\boldsymbol{b}_j = 2\pi\delta_{ij}$. The energy cutoff selects reciprocal-lattice vectors inside a sphere of radius k_{cut} and therefore supplies a single systematic parameter for enlarging the basis. The overlap integrals are diagonal:

$$\begin{aligned}\langle\boldsymbol{k}_1|\boldsymbol{k}_2\rangle &= \frac{1}{\Omega}\int_\Omega \exp[-\mathrm{i}\boldsymbol{k}_1\cdot\boldsymbol{r}]\exp[\mathrm{i}\boldsymbol{k}_2\cdot\boldsymbol{r}]\,\mathrm{d}\boldsymbol{r}\\ &= \frac{1}{\Omega}\int_\Omega \exp[\mathrm{i}(\boldsymbol{k}_2-\boldsymbol{k}_1)\cdot\boldsymbol{r}]\,\mathrm{d}\boldsymbol{r} = \delta_{\boldsymbol{k}_1\boldsymbol{k}_2}\end{aligned} \tag{4.6}$$

The kinetic-energy integrals are diagonal as well:

$$\begin{aligned}\langle \boldsymbol{k}_1|-\frac{1}{2}\nabla^2|\boldsymbol{k}_2\rangle &= \frac{1}{\Omega}\int_\Omega \exp[-\mathrm{i}\boldsymbol{k}_1\cdot\boldsymbol{r}]\left(-\frac{1}{2}\nabla^2\right)\exp[\mathrm{i}\boldsymbol{k}_2\cdot\boldsymbol{r}]\mathrm{d}\boldsymbol{r}\\ &= \frac{k_2^2}{2\Omega}\int_\Omega \exp[\mathrm{i}(\boldsymbol{k}_2-\boldsymbol{k}_1)\cdot\boldsymbol{r}]\mathrm{d}\boldsymbol{r} = \frac{k_2^2}{2}\delta_{\boldsymbol{k}_1\boldsymbol{k}_2}\end{aligned} \tag{4.7}$$

For the plane-wave ERI, we use the physicists' index grouping: the first and third indices belong to electron 1, and the second and fourth to electron 2. The integral begins as

$$\langle \boldsymbol{k}_1\boldsymbol{k}_2|\boldsymbol{k}_3\boldsymbol{k}_4\rangle = \frac{1}{\Omega^2}\iint_\Omega \frac{\exp[\mathrm{i}(\boldsymbol{k}_3-\boldsymbol{k}_1)\cdot\boldsymbol{r}_1]\exp[\mathrm{i}(\boldsymbol{k}_4-\boldsymbol{k}_2)\cdot\boldsymbol{r}_2]}{r_{12}}\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2 \tag{4.8}$$

Under periodic boundary conditions, the Coulomb kernel is understood as its periodic counterpart. With its zero Fourier component omitted, the reciprocal-space resolution is

$$v_\Omega(\boldsymbol{r}_1-\boldsymbol{r}_2) = \frac{1}{\Omega}\sum_{\boldsymbol{k}\neq\boldsymbol{0}}\frac{4\pi}{k^2}\exp[\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{r}_1-\boldsymbol{r}_2)] \tag{4.9}$$

The $\boldsymbol{k}=\boldsymbol{0}$ term requires a specified electrostatic convention, for example a neutralizing background. The symbol r_{12}^{-1} in the preceding integral represents this periodic kernel in the following calculation. Thus, we have

$$\begin{aligned}\langle \boldsymbol{k}_1\boldsymbol{k}_2|\boldsymbol{k}_3\boldsymbol{k}_4\rangle &= \frac{1}{\Omega^3}\sum_{\boldsymbol{k}\neq\boldsymbol{0}}\frac{4\pi}{k^2}\iint_\Omega \exp[\mathrm{i}(\boldsymbol{k}_3-\boldsymbol{k}_1+\boldsymbol{k})\cdot\boldsymbol{r}_1]\exp[\mathrm{i}(\boldsymbol{k}_4-\boldsymbol{k}_2-\boldsymbol{k})\cdot\boldsymbol{r}_2]\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2\\ &= \frac{1}{\Omega}\sum_{\boldsymbol{k}\neq\boldsymbol{0}}\frac{4\pi}{k^2}\delta_{\boldsymbol{k}_3-\boldsymbol{k}_1+\boldsymbol{k},\boldsymbol{0}}\delta_{\boldsymbol{k}_4-\boldsymbol{k}_2-\boldsymbol{k},\boldsymbol{0}}\\ &= \frac{4\pi}{\Omega|\boldsymbol{k}_1-\boldsymbol{k}_3|^2}\delta_{\boldsymbol{k}_1+\boldsymbol{k}_2,\boldsymbol{k}_3+\boldsymbol{k}_4}\end{aligned} \tag{4.10}$$

The Kronecker delta expresses conservation of crystal momentum. The last expression applies when $\boldsymbol{k}_1\neq\boldsymbol{k}_3$. For zero momentum transfer, the omitted zero Fourier component gives zero under the convention used here; the fraction must not be evaluated with a zero denominator. The simplicity of these expressions follows because differentiation and convolution become multiplication in reciprocal space. Overlap and kinetic energy are diagonal, while a two-electron integral survives only when the total incoming and outgoing wavevectors are equal.

The free-electron solutions provide the building blocks; their use as a basis does not require the molecular or crystalline potential to be a weak perturbation. Plane waves are orthonormal and independent of the atomic positions; naturally compatible with periodic boundary conditions; systematically convergent through the single parameter E_{cut}; and diagonalize the kinetic and Coulomb operators in reciprocal space. They also permit efficient fast-Fourier-transform evaluation and have neither basis-set superposition error nor atom-centered Pulay forces. Their limitations are large basis sizes,

$$N_{\mathrm{PW}} \propto \Omega E_{\mathrm{cut}}^{3/2} \tag{4.11}$$

Core orbitals require prohibitively large cutoffs, motivating pseudopotentials; the cost increases with supercell size; delocalized functions provide little real-space sparsity; and the large virtual spaces make correlated methods expensive. Plane waves are therefore especially effective for periodic valence-electron calculations, usually in conjunction with pseudopotentials [72].

4.3 Slater-type and Gaussian-type orbitals

For isolated atoms and molecules, atom-centered functions provide a compact description by placing radial flexibility where the electronic density is concentrated. The central choice is between radial forms that reproduce the exact local behavior and forms that permit efficient multicenter integration. In the following Slater- and Gaussian-type orbitals, n labels the radial form, ℓ and m are the angular-momentum and magnetic quantum numbers, and $Y_{\ell m}(\hat{\boldsymbol{r}})$ is a spherical harmonic evaluated in the direction of $\boldsymbol{r}$. The Slater radial form is associated historically with screened atomic orbitals [14]. The general forms used in the comparison are

$$\begin{aligned} \phi^{\mathrm{STO}}_{\zeta n\ell m}(\boldsymbol{r}) &\propto r^{n-1}e^{-\zeta r}Y_{\ell m}(\hat{\boldsymbol{r}}) \\ \phi^{\mathrm{GTO}}_{\alpha n\ell m}(\boldsymbol{r}) &\propto r^{2(n-1)-\ell}e^{-\alpha r^2}Y_{\ell m}(\hat{\boldsymbol{r}}) \end{aligned} \tag{4.12}$$

For their 1s representatives, the STO has a nonzero radial slope at the nucleus,

$$\phi^{\mathrm{STO}}_{1\mathrm{s}}(r) \propto e^{-\zeta r} \qquad \left.\frac{\mathrm{d}\phi^{\mathrm{STO}}_{1\mathrm{s}}(r)}{\mathrm{d}r}\right|_{r=0} = -\zeta\phi^{\mathrm{STO}}_{1\mathrm{s}}(0) \tag{4.13}$$

The logarithmic slope is $-\zeta$; the electron–nucleus cusp condition requires $-Z$ for a point nucleus of charge Z [73], and is satisfied by this representative when $\zeta = Z$. The factor $\phi(0)$ accounts for the normalization dependence of the radial derivative. By contrast, the GTO is flat at the nucleus,

$$\phi^{\mathrm{GTO}}_{1\mathrm{s}}(r) \propto e^{-\alpha r^2} \qquad \left.\frac{\mathrm{d}\phi^{\mathrm{GTO}}_{1\mathrm{s}}(r)}{\mathrm{d}r}\right|_{r=0} = 0 \tag{4.14}$$

The STO also has the correct exponential long-range behavior of a molecular orbital. For the HOMO, its leading radial exponential factor in the frozen-orbital interpretation is

$$\psi^{\mathrm{STO}}_{\mathrm{HOMO}}(r) \propto e^{-\sqrt{-2\epsilon_{\mathrm{HOMO}}}r} \qquad \mathrm{IP} = -\epsilon_{\mathrm{HOMO}} \tag{4.15}$$

A Gaussian instead decays too rapidly. Multicenter STO integrals are substantially more difficult to evaluate than their Gaussian counterparts. This computational advantage usually favors GTOs, although more Gaussian functions are required to reproduce comparable radial flexibility. The practical success of Gaussian basis sets is therefore computational. A linear combination of several Gaussians is used to recover the radial flexibility that a single Slater function supplies more naturally. The Gaussian strategy was established in Boys's early molecular calculations and developed further through systematic atomic expansions [74, 75].

Historical perspective: Samuel Francis Boys (1911–1972)

A British theoretical chemist at Cambridge, Boys made Gaussian functions central to practical molecular calculations through his 1950 work on electronic wavefunctions [74]. Efficient multicenter integrals were a major part of their appeal. His later contributions included orbital localization and, with Fernando Bernardi, the counterpoise correction for basis-set superposition error [41, 76]. He was elected a Fellow of the Royal Society in 1972.

Biography: IAQMS profile; Royal Society record. Source portrait: IAQMS archive; photographer and date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

Historical perspective: Sigeru Huzinaga

A Japanese quantum chemist and professor at the University of Alberta, Huzinaga helped turn Gaussian basis functions into systematic tools for molecular calculations. His 1965 atomic Gaussian expansions [75] provided an influential foundation for practical basis construction. His work connected the choice and contraction of atomic functions with the accuracy and computational demands of molecular electronic-structure calculations.

Biography: University of Alberta emeriti record; Huzinaga's account of computational chemistry, 2007. Source portrait: Yamada Science Foundation report, printed p. 16 (detail); photographer/date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

4.3.1 Primitive and contracted Gaussian functions

A Gaussian basis function has two layers of structure. A primitive specifies one exponent and one Cartesian polynomial. A contracted function combines several primitives of the same center and angular momentum with fixed coefficients, so that many primitive shapes share a single variational MO coefficient [77]. A primitive Cartesian Gaussian centered at $\boldsymbol{A}$ is labeled by $\boldsymbol{a} = (a_x, a_y, a_z)$, a vector of nonnegative integer powers. Its total angular momentum is $a = a_x + a_y + a_z$. In the integral discussion, these angular-momentum labels replace the earlier use of a, b, c, d for virtual orbital indices. The square ket denotes a primitive function:

$$\boxed{|\boldsymbol{a}] = (x - A_x)^{a_x}(y - A_y)^{a_y}(z - A_z)^{a_z} \exp\left[-\alpha|\boldsymbol{r} - \boldsymbol{A}|^2\right]} \tag{4.16}$$

An ordinary ket denotes a contracted Gaussian, a fixed linear combination of primitives with coefficients D_k:

$$\boxed{|\boldsymbol{a}\rangle \equiv \phi_{\boldsymbol{a}}^{\boldsymbol{A}}(\boldsymbol{r}) = \sum_{k=1}^{K} D_k |\boldsymbol{a}_k]} \tag{4.17}$$

The square-ket definition is unnormalized. Its normalization factor can be absorbed into D_k; coefficients tabulated for normalized primitives must be combined with those factors when this unnormalized convention is used. Here α is the primitive exponent. All primitives in a contraction have the same center and polynomial powers but may have different exponents. The upper limit K in this local contraction sum counts primitives; elsewhere K denotes the number of orbital basis functions.

An s shell contains $(0,0,0)$; a p shell contains $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$; a Cartesian d shell contains $(2,0,0)$, $(0,2,0)$, $(0,0,2)$, $(1,1,0)$, $(1,0,1)$, and $(0,1,1)$; and a Cartesian f shell contains $(3,0,0)$, $(0,3,0)$, $(0,0,3)$, $(2,1,0)$, $(2,0,1)$, $(1,2,0)$, $(0,2,1)$, $(0,1,2)$, $(1,0,2)$, and $(1,1,1)$. In general, a Cartesian shell of total angular momentum ℓ contains

$$N_\ell^{\text{cart}} = \frac{(\ell+1)(\ell+2)}{2} \tag{4.18}$$

which gives $1, 3, 6, 10$ functions for s, p, d, f. Spherical harmonics instead give $N_\ell^{\text{sph}} = 2\ell + 1$, namely $1, 3, 5, 7$. Thus

$$6d = 5d + 1s \qquad\qquad 10f = 7f + 3p \tag{4.19}$$

because, for example, $x^2 + y^2 + z^2 = r^2$. Most molecular calculations use pure spherical functions. The distinction appears from d functions onward. Cartesian shells are convenient for integral generation, whereas transformation to pure spherical functions removes the lower-angular-momentum combinations and restores the expected $2\ell + 1$ components.

Contraction reduces the number of variational coefficients, but it does not remove the primitive-integral work. For four contracted functions,

$$(\boldsymbol{ab}\,|\,\boldsymbol{cd}) = \sum_{i=1}^{K_a}\sum_{j=1}^{K_b}\sum_{k=1}^{K_c}\sum_{l=1}^{K_d} D_i^a D_j^b D_k^c D_l^d\,[\boldsymbol{a}_i\boldsymbol{b}_j\,|\,\boldsymbol{c}_k\boldsymbol{d}_l] \tag{4.20}$$

For example, a cc-pVTZ carbon 1*s* function contracted from ten primitives requires 10^4 primitive *ssss* integrals for a single four-center integral. The complete contraction in this example is

$$(1s1s\,|\,1s1s) = \sum_{i=1}^{10}\sum_{j=1}^{10}\sum_{k=1}^{10}\sum_{l=1}^{10} D_i D_j D_k D_l\,[s_i^{\alpha_i} s_j^{\alpha_j}\,|\,s_k^{\alpha_k} s_l^{\alpha_l}] \tag{4.21}$$

The corresponding exponents and contraction coefficients are listed in Table 3. Basis-set definitions can be inspected and downloaded from the Basis Set Exchange. The contraction thus reduces the dimensions of the Fock and density matrices, while integral algorithms must still reorganize the underlying primitive sums efficiently.

Table 3: Primitive exponents and coefficients of one ten-primitive *s* contraction for carbon in cc-pVTZ. The shell scale is 1.00. Values are extracted from the Basis Set Exchange; the basis family is due to Dunning [78, 79].

Primitive i	Exponent α_i	Coefficient D_i
1	8.236000×10^{3}	5.310000×10^{-4}
2	1.235000×10^{3}	4.108000×10^{-3}
3	2.808000×10^{2}	2.108700×10^{-2}
4	7.927000×10^{1}	8.185300×10^{-2}
5	2.559000×10^{1}	2.348170×10^{-1}
6	8.997000×10^{0}	4.344010×10^{-1}
7	3.319000×10^{0}	3.461290×10^{-1}
8	9.059000×10^{-1}	3.937800×10^{-2}
9	3.643000×10^{-1}	-8.983000×10^{-3}
10	1.285000×10^{-1}	2.385000×10^{-3}

4.4 Gaussian basis-set design and selection

No single enlargement of a Gaussian basis improves every physical feature in the same way. Different physical effects demand different forms of flexibility. Radial flexibility is provided by multiple-ζ functions, as in double-ζ, triple-ζ, and quadruple-ζ bases. Angular flexibility is supplied by polarization functions, such as *d* functions on second-row atoms and *p* functions on hydrogen. Long-range flexibility is supplied by diffuse functions with small exponents. Such functions are important for anions, Rydberg states, noncovalent interactions, multipoles, and (hyper)polarizabilities. Core flexibility, in turn, is supplied by tight functions and is needed for core correlation, nuclear magnetic shielding, and spin-spin coupling constants.

The phrase *larger basis* therefore needs a physical qualification: the added functions must address the radial, angular, diffuse, or core flexibility that controls the target energy or property. Increasing the function count alone does not identify which of these deficiencies is being reduced. Figure 21 shows the Basis Set Exchange interface through which the definitions of these basis families can be accessed.

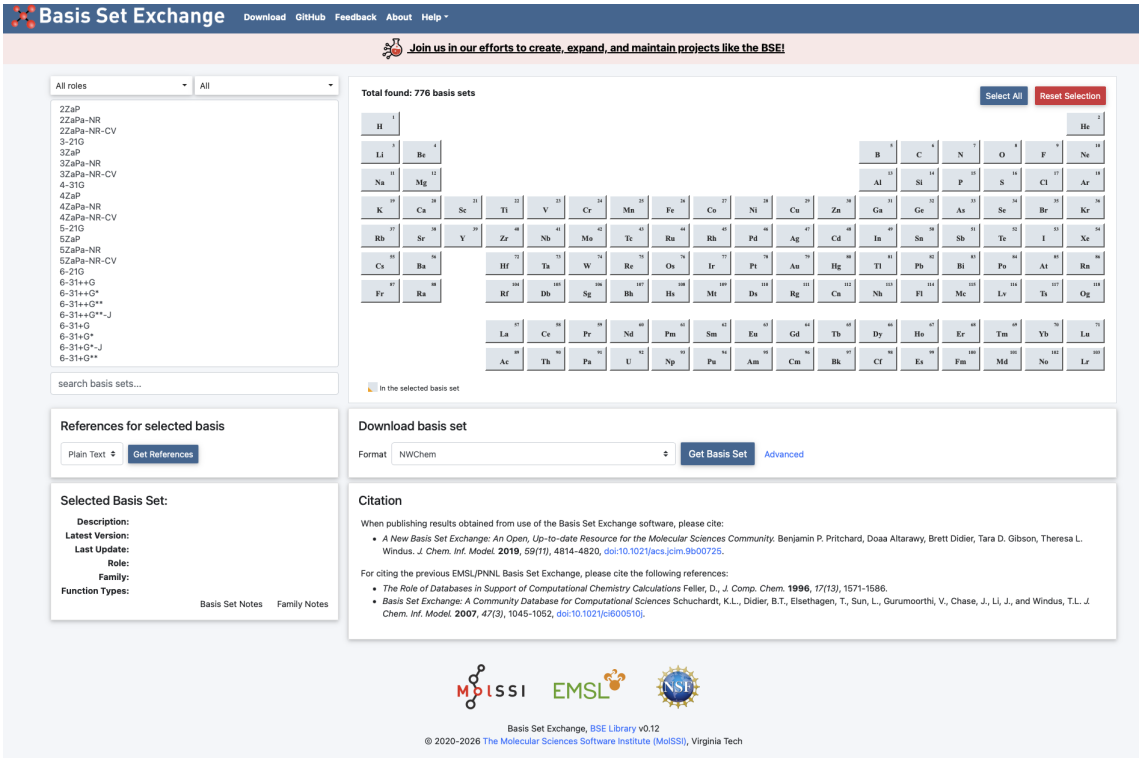


Figure 21: The Basis Set Exchange provides standardized definitions for a large number of orbital and auxiliary basis sets [79].

4.4.1 STO-nG minimal basis sets

In an STO-nG basis, there is one contracted basis function for every occupied orbital in the minimal atomic description. Each contracted GTO contains n primitive GTOs, whose contraction coefficients and exponents are selected so that the contraction approximates an STO. STO-nG bases are therefore minimal, single-ζ basis sets; STO-3G is the most familiar member [80]. For hydrogen, Table 4 lists the three primitives in the contraction, and Fig. 22 shows how their sum approximates the Slater-type 1s orbital.

Table 4: Primitive exponents and contraction coefficients for the hydrogen 1s function in STO-3G, with shell scale 1.00. The dimensionless coefficients D_i multiply normalized primitive Gaussians; a_0 is the Bohr radius. Values are extracted from the Basis Set Exchange [79, 80].

Primitive i	Exponent α_i (a_0^{-2})	Coefficient D_i
1	3.425250914	0.1543289673
2	0.6239137298	0.5353281423
3	0.1688554040	0.4446345422

The quality of the contraction is controlled by n, but the molecular orbital space still contains only the minimal number of functions. Such a basis is useful for illustrating the construction, while its lack of extra radial and angular flexibility limits quantitative accuracy.

4.4.2 Pople split-valence basis sets

Pople basis sets [81] extend the minimal description by splitting the valence space and by adding optional diffuse and polarization shells. Their notation explicitly records the contractions and

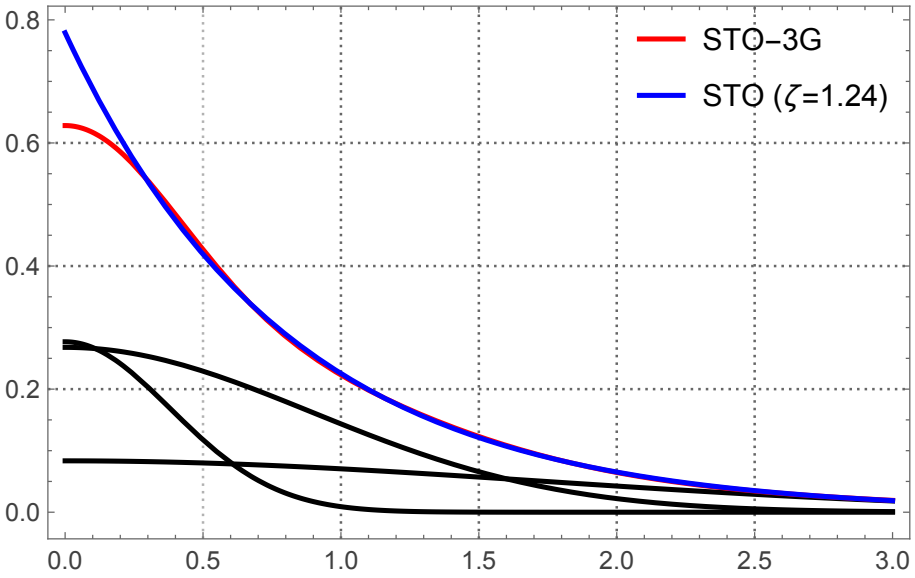


Figure 22: Radial dependence of the hydrogen 1*s* STO with $\zeta = 1.24\, a_0^{-1}$ (blue) and its STO-3G approximation (red). The black curves are the three coefficient-weighted normalized Gaussian primitives whose sum forms the contraction, using the parameters in Table 4. The horizontal coordinate is the distance r from the nucleus in Bohr radii; the curves show spatial-orbital amplitudes, not radial probability densities. The plot is retained from the original lecture material.

augmentations. In 6-31+G*, the core function is contracted from six primitives; the valence space is double-ζ, with contractions of three and one primitives; the "+" adds diffuse functions to non-hydrogen atoms [82]; and "*" adds polarization functions to non-hydrogen atoms, conventionally a d shell. In 6-311++G**, the valence space is triple-ζ with contractions 3, 1, 1 [83]; the second "+" adds diffuse functions to hydrogen; and the second "*" adds polarization functions to hydrogen. Finally, 6-311G(2df,2pd) adds two d and one f polarization shells to non-hydrogen atoms, and two p and one d shell to hydrogen [84].

The notation describes how the basis is assembled: the contraction pattern is followed by symbols or parentheses identifying diffuse and polarization functions. It is therefore more detailed than a simple label of basis size.

4.4.3 Dunning correlation-consistent basis sets

Historical perspective: Thom H. Dunning Jr. (b. 1943)

An American theoretical chemist trained at Caltech, Dunning developed Gaussian basis sets first for Hartree–Fock calculations and subsequently for correlated wavefunctions. His correlation-consistent construction [78] organizes basis functions into balanced hierarchical sequences. This made systematic convergence toward the complete-basis limit a practical strategy, linking basis design to the reliable assessment of electronic-structure methods.

Biography and source portrait: IAQMS profile. Source portrait photographer/date unspecified. **Draft reproduction: permission pending.** Sketch: AI-assisted adaptation of the credited portrait.

The correlation-consistent construction [78] organizes functions into successive balanced layers designed to recover angular correlation systematically. The main sequence is

$$\text{cc-pVXZ}, \qquad X = \text{D, T, Q, 5, 6} \tag{4.22}$$

In the basis names, D, T, and Q denote double-, triple-, and quadruple-ζ quality. The numerical cardinal number X takes the corresponding values $2, 3, 4, 5, 6$. A standard correlation-energy extrapolation [85] is

$$\boxed{E_{\mathrm{c}}(X) \approx E_{\mathrm{c}}^{\mathrm{CBS}} + AX^{-3}} \tag{4.23}$$

Here $E_{\mathrm{c}}^{\mathrm{CBS}}$ is the complete-basis-set correlation energy and A is an extrapolation coefficient. Diffuse augmentation gives aug-cc-pVXZ, d-aug-cc-pVXZ, and t-aug-cc-pVXZ [86]. Core correlation is described by cc-pCVXZ, optimized for core-core and core-valence correlation [87], or by cc-pwCVXZ, which weights the core-valence contribution [88].

Increasing the cardinal number adds both radial flexibility and a new layer of higher angular momentum, as the explicit shell patterns below demonstrate. The inverse-cubic expression concerns correlation energy, not the Hartree–Fock contribution; the polarization-consistent discussion below uses a different Hartree–Fock extrapolation form. For a second-row atom (the second period, Li–Ne), Table 5 lists the valence correlation-consistent sequence and its diffuse augmentations.

Table 5: Shell contents of correlation-consistent basis sets for second-row atoms, with successive diffuse augmentations. The entries count contracted radial functions in each angular-momentum shell [78, 86].

Basis	Shell content
Unaugmented valence sets	
cc-pVDZ	$3s2p1d$
cc-pVTZ	$4s3p2d1f$
cc-pVQZ	$5s4p3d2f1g$
Single diffuse augmentation	
aug-cc-pVDZ	$4s3p2d$
aug-cc-pVTZ	$5s4p3d2f$
aug-cc-pVQZ	$6s5p4d3f2g$
Double diffuse augmentation	
d-aug-cc-pVDZ	$5s4p3d$
d-aug-cc-pVTZ	$6s5p4d3f$
d-aug-cc-pVQZ	$7s6p5d4f3g$

The successive entries make the layered construction explicit. For the core-valence sequence, the corresponding decomposition is

$$\begin{aligned}
\text{cc-pVDZ: } 3s2p1d &= 1s + 2s2p1d \\
\text{cc-pVTZ: } 4s3p2d1f &= 1s + 3s3p2d1f \\
\text{cc-pVQZ: } 5s4p3d2f1g &= 1s + 4s4p3d2f1g \\
\text{cc-pCVDZ: } 4s3p1d &= 2s1p + 2s2p1d \\
\text{cc-pCVTZ: } 6s5p3d1f &= 3s2p1d + 3s3p2d1f \\
\text{cc-pCVQZ: } 8s7p5d3f1g &= 4s3p2d1f + 4s4p3d2f1g
\end{aligned} \tag{4.24}$$

4.4.4 Jensen polarization-consistent basis sets

Jensen's polarization-consistent sets [89] apply systematic improvement to Hartree–Fock and Kohn–Sham calculations, with functions selected according to their energetic importance. The

sequence has shell contents

$$\text{pc-0} : 3s2p \qquad \text{pc-1} : 3s2p1d \tag{4.25}$$

$$\text{pc-2} : 4s3p2d1f \qquad \text{pc-3} : 6s5p4d2f1g \tag{4.26}$$

Thus pc-0 is unpolarized, pc-1 adds one polarization layer, and the later members add systematically more radial and angular flexibility. Diffuse variants are denoted aug-pc-n. Their Hartree–Fock convergence is described by the extrapolation form [90]

$$E_{\mathrm{HF}}(L) = E_{\mathrm{HF}}^{\mathrm{CBS}} + A(L+1)\exp\left(-B\sqrt{L}\right) \tag{4.27}$$

Here L is the highest angular momentum present in the basis, while A and B are extrapolation coefficients. This Hartree–Fock extrapolation is distinct from the inverse-cubic correlation-energy extrapolation above.

These hierarchies illustrate why Gaussian bases are widely used: they are compact, atom centered, chemically transferable, supported by analytic integral technology, and available in systematic hierarchies. They make all-electron calculations practical for light and medium elements, and their locality supports integral screening and sparse algorithms. The physical picture is that a molecule is assembled from atoms whose electron distributions are distorted by their environment. Because the basis functions move with the nuclei, nuclear derivatives contain Pulay terms in addition to the derivatives of the electronic Hamiltonian. Their principal limitations are the absent nuclear cusp, the incorrect long-range Gaussian decay, basis-set superposition error, and possible linear dependence when many diffuse functions are used. There is consequently no single best Gaussian basis: the appropriate choice depends on the property, the electronic-structure method, and the desired accuracy.

4.4.5 Basis-set superposition error and counterpoise correction

Historical perspective: Fernando Bernardi

An Italian theoretical organic chemist at the University of Bologna, Bernardi studied with Boys in Cambridge in 1969–1971. Their joint 1970 paper introduced the counterpoise procedure discussed here [76], making the comparison of fragment basis spaces an explicit part of intermolecular-energy calculations. His later research connected electronic-structure methods with organic reactivity and photochemistry, including work with Massimo Olivucci and Michael Robb on potential-energy-surface crossings. The Boys–Bernardi name acknowledges both contributors to a correction that remains central to the practical use of atom-centered basis sets.

Biography: author profile in Bernardi, Olivucci, and Robb, Chemical Society Reviews, 1996, p. 321.

The locality that makes atom-centered bases compact also creates a possible imbalance when fragment energies are compared. Atom-centered basis functions belong formally to the complete calculation rather than to immutable fragments. When two fragments approach one another, each gains access to functions centered on the other, and the dimer and isolated-fragment energies no longer possess equivalent variational flexibility. Let E_X^Y denote the energy of system X evaluated in basis Y. In a dimer AB, each fragment can borrow the basis functions of the other fragment as ghost functions. The variational principle therefore gives

$$E_A^{AB} \le E_A^A \qquad E_B^{AB} \le E_B^B \tag{4.28}$$

The corresponding uncorrected interaction energy is

$$\Delta E_{\mathrm{int}} = E_{AB}^{AB} - E_A^A - E_B^B \tag{4.29}$$

which is too attractive relative to a comparison in the common dimer basis. This basis imbalance is distinct from other basis-incompleteness or electronic-structure errors; counterpoise does not establish an exact interaction energy. The Boys–Bernardi counterpoise correction [76] uses the full dimer basis for every term,

$$\boxed{\Delta E_{\text{int}}^{\text{CP}} = E_{AB}^{AB} - E_{A}^{AB} - E_{B}^{AB}} \tag{4.30}$$

and the basis-set superposition error (BSSE) is

$$\text{BSSE} = \Delta E_{\text{int}}^{\text{CP}} - \Delta E_{\text{int}} = (E_{A}^{A} - E_{A}^{AB}) + (E_{B}^{B} - E_{B}^{AB}) \geq 0 \tag{4.31}$$

The partner's basis functions are retained as ghost functions: they provide variational flexibility but carry neither electrons nor nuclear charge. The counterpoise expression thereby compares the dimer and both fragments in the same one-electron space. Figure 23 illustrates the difference between an isolated fragment basis and the full dimer basis retained in a counterpoise calculation.

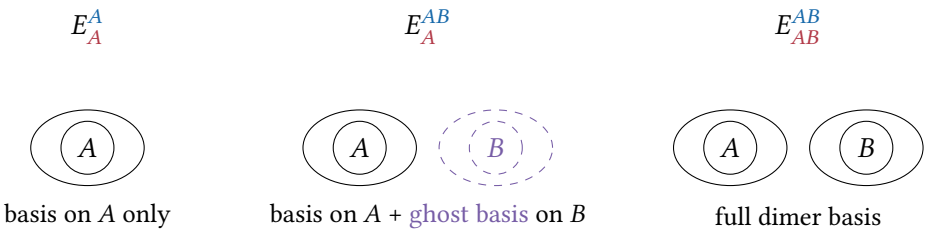


Figure 23: A fragment in its own basis, the same fragment supplemented by ghost functions, and the full dimer. Dashed violet shapes retain basis functions without adding nuclear charge or electrons.

4.5 Effective core potentials and pseudopotentials

Core electrons are numerous and spatially compact, yet they often participate only indirectly in chemical bonding. Effective core potentials (ECPs) exploit this separation by replacing selected core electrons and their nuclear interaction with an operator acting on the remaining explicit electrons. The key idea of an effective core potential [91] is to treat only the chemically relevant electrons explicitly. It replaces the nuclear attraction and the chemically inactive core electrons

$$\hat{h}(\boldsymbol{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|} \tag{4.32}$$

by an effective one-electron operator

$$\hat{h}^{\text{ECP}}(\boldsymbol{r}) = -\frac{\nabla^2}{2} - \sum_A \frac{Q_A}{|\boldsymbol{r} - \boldsymbol{R}_A|} + \sum_A \hat{U}_A^{\text{ECP}}(\boldsymbol{r}) \qquad Q_A = Z_A - N_A^{\text{core}} \tag{4.33}$$

Here N_A^{core} is the number of electrons removed by the ECP on atom A, and Q_A is the charge of the remaining ionic core. It is not the population-analysis charge q_A introduced earlier. This lowers the cost by reducing the number of explicit electrons and removing the basis functions needed for core orbitals. It can incorporate scalar-relativistic effects and, with a suitable operator, spin-orbit coupling, and it can be used in both molecular and periodic calculations with Gaussian or plane-wave bases. The price is a new class of one-electron integrals that can be difficult to evaluate; core properties are not described explicitly, and the accuracy depends on the ECP quality.

For one center, with r measured from that center and L labeling the local channel, a common semilocal form is

$$\hat{U}^{\mathrm{ECP}} = U_L(r) + \sum_{\ell=0}^{L-1} \sum_{m=-\ell}^{\ell} |Y_{\ell m}\rangle\, U_\ell(r)\, \langle Y_{\ell m}| \tag{4.34}$$

The first term is local, or unprojected; the remaining angular-momentum channels are nonlocal, or projected. In this convention, each projected $U_\ell(r)$ is a correction relative to the local potential $U_L(r)$, not the full potential in channel ℓ. Their radial parts are generally expanded as

$$U_\ell(r) = \sum_k D_{\ell k} r^{n_{\ell k}} \exp\left(-\eta_{\ell k} r^2\right) \qquad n_{\ell k} \in \{-2, -1, 0\} \tag{4.35}$$

The $D_{\ell k}$ and $\eta_{\ell k}$ are fitted parameters. Although these are one-electron operators, their angular projectors make the integrals nontrivial.

For silver, a large-core ECP removes 36 electrons and treats explicitly the $4d^{10}5s^1$ shells, leaving 11 electrons. A small-core ECP removes 28 electrons and retains $4s^2 4p^6 4d^{10} 5s^1$, leaving 19 explicit electrons; the semicore $4s$ and $4p$ shells are therefore retained, providing more flexibility at higher cost. Several Gaussian-basis ECP families are commonly used. LANL [91] and SBK/SBKJC [92] are established families; in particular, LANL2DZ combines an ECP with a double-ζ valence basis. The Stuttgart/Cologne energy-consistent pseudopotentials are offered with different core sizes and dedicated bases [93], while the def2-ECPs are paired with the def2 bases of Weigend and Ahlrichs [94]. The Burkatzki–Filippi–Dolg (BFD) potentials were developed for quantum Monte Carlo calculations [95], whereas the correlation-consistent ECPs (ccECPs) were developed for high-accuracy correlated many-body methods [96]. Most ECPs have dedicated basis sets.

The core size is consequently a controlled compromise. A large-core potential is cheaper, whereas a small-core potential retains more semicore response and flexibility. Consistency requires the ECP and its accompanying valence basis to be treated as a matched pair.

5 Molecular integrals over Gaussian functions

Once the orbital basis and any effective core potential have been selected, the abstract matrix elements of the finite-basis Hartree–Fock equations must be evaluated. Gaussian functions are especially valuable because their products and derivatives can be reduced to a small set of fundamental integrals. The discussion below develops that reduction from overlap and one-electron terms to four-center electron-repulsion integrals, and then shows how recurrence relations and screening turn the resulting formulas into a practical algorithm [97].

The Cartesian Gaussian functions in this section are real. Primitive formulas use the unnormalized square-ket convention introduced above; normalization and contraction factors are applied separately. This scope is important when using the permutation symmetries and screening bounds below: the more general complex-orbital symmetries were stated earlier.

5.1 The Gaussian product theorem and overlap integrals

The Gaussian product rule reduces multicenter products to single-center quantities: the product of two s-type Gaussian factors is another Gaussian times a constant. Polynomial factors in higher-angular-momentum functions are handled by the recurrence relations developed later. This closure property underlies the classical Gaussian integral formulation of Boys [74]. Writing the s-type Gaussian factors as

$$G_{\alpha,\boldsymbol{A}}(\boldsymbol{r}) = \exp\left[-\alpha|\boldsymbol{r} - \boldsymbol{A}|^2\right] \qquad G_{\beta,\boldsymbol{B}}(\boldsymbol{r}) = \exp\left[-\beta|\boldsymbol{r} - \boldsymbol{B}|^2\right] \tag{5.1}$$

then

Gaussian product theorem

For positive exponents α and β, the product of the two s-type Gaussian factors is a Gaussian centered at the exponent-weighted point $\boldsymbol{P}$, multiplied by a separation-dependent prefactor:

$$G_{\alpha,\boldsymbol{A}}(\boldsymbol{r})G_{\beta,\boldsymbol{B}}(\boldsymbol{r}) = K_{\zeta,\boldsymbol{P}}G_{\zeta,\boldsymbol{P}}(\boldsymbol{r}) \tag{5.2}$$

with

$$\zeta = \alpha + \beta \qquad \boldsymbol{P} = \frac{\alpha\boldsymbol{A} + \beta\boldsymbol{B}}{\alpha + \beta} \tag{5.3}$$

and

$$K_{\zeta,\boldsymbol{P}} = \exp\left[-\frac{\alpha\beta}{\alpha + \beta}|\boldsymbol{A} - \boldsymbol{B}|^2\right] \tag{5.4}$$

The product center $\boldsymbol{P}$ lies on the line joining $\boldsymbol{A}$ and $\boldsymbol{B}$ and is weighted toward the Gaussian with the larger exponent. The prefactor contains the dependence on the separation of the original centers: its Gaussian decay makes distant pairs rapidly negligible. The bold $\boldsymbol{P}$ here denotes a coordinate, not the density matrix used in the Hartree–Fock equations. For two s Gaussians, the overlap illustrates Cartesian factorization:

$$\begin{aligned}\int G_{\alpha,\boldsymbol{A}}(\boldsymbol{r})G_{\beta,\boldsymbol{B}}(\boldsymbol{r})\mathrm{d}\boldsymbol{r} &= K_{\zeta,\boldsymbol{P}}\int G_{\zeta,\boldsymbol{P}}(\boldsymbol{r})\mathrm{d}\boldsymbol{r} \\ &= K_{\zeta,\boldsymbol{P}}\left(\int_{-\infty}^{+\infty} e^{-\zeta x^2}\mathrm{d}x\right)\left(\int_{-\infty}^{+\infty} e^{-\zeta y^2}\mathrm{d}y\right)\left(\int_{-\infty}^{+\infty} e^{-\zeta z^2}\mathrm{d}z\right) \\ &= K_{\zeta,\boldsymbol{P}}\left(\frac{\pi}{\zeta}\right)^{3/2}\end{aligned} \tag{5.5}$$

The product formula above uses unnormalized Gaussian factors. Exercise 18 shows how normalization changes the overlap and provides useful limiting cases for checking an integral routine.

Exercise 18: Normalized Gaussian overlap

Normalize two s-type Gaussian primitives and derive their overlap for arbitrary exponents and centers.

Solution 18: Normalized Gaussian overlap

For $\alpha > 0$, define a normalized s-type primitive centered at $\boldsymbol{A}$ by

$$\phi_{\alpha,\boldsymbol{A}}(\boldsymbol{r}) = \mathcal{N}_\alpha \exp\left[-\alpha|\boldsymbol{r} - \boldsymbol{A}|^2\right] \tag{5.6}$$

Normalization requires the integral of the squared function to be one. Translating the integration variable to the center removes any dependence on $\boldsymbol{A}$. The three Cartesian integrals then factorize:

$$\begin{aligned}1 &= \mathcal{N}_\alpha^2\int \exp\left[-2\alpha|\boldsymbol{r} - \boldsymbol{A}|^2\right]\mathrm{d}\boldsymbol{r} \\ &= \mathcal{N}_\alpha^2\left(\int_{-\infty}^{+\infty} e^{-2\alpha x^2}\mathrm{d}x\right)^3 \\ &= \mathcal{N}_\alpha^2\left(\frac{\pi}{2\alpha}\right)^{3/2}\end{aligned} \tag{5.7}$$

Solution 18 (continued): Normalized Gaussian overlap

Choosing the positive root gives

$$\boxed{\mathcal{N}_\alpha = \left(\frac{2\alpha}{\pi}\right)^{3/4}} \tag{5.8}$$

The exponent has dimensions of inverse length squared, so the normalized three-dimensional function has dimensions of inverse length to the power 3/2, as required by its normalization integral.

For two such real functions, write

$$\begin{aligned} S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}} &\equiv \langle \phi_{\alpha,\boldsymbol{A}} | \phi_{\beta,\boldsymbol{B}} \rangle \\ &= \mathcal{N}_\alpha \mathcal{N}_\beta \int \exp\left[-\alpha|\boldsymbol{r}-\boldsymbol{A}|^2 - \beta|\boldsymbol{r}-\boldsymbol{B}|^2\right] \mathrm{d}\boldsymbol{r} \end{aligned} \tag{5.9}$$

Using $\zeta = \alpha + \beta$ and $\boldsymbol{P} = (\alpha\boldsymbol{A} + \beta\boldsymbol{B})/\zeta$, the exponent can be completed to a square:

$$\alpha|\boldsymbol{r}-\boldsymbol{A}|^2 + \beta|\boldsymbol{r}-\boldsymbol{B}|^2 = \zeta|\boldsymbol{r}-\boldsymbol{P}|^2 + \frac{\alpha\beta}{\zeta}|\boldsymbol{A}-\boldsymbol{B}|^2 \tag{5.10}$$

Thus the separation-dependent factor leaves the integral:

$$\begin{aligned} S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}} &= \mathcal{N}_\alpha \mathcal{N}_\beta \exp\left[-\frac{\alpha\beta}{\zeta}|\boldsymbol{A}-\boldsymbol{B}|^2\right] \int \exp\left[-\zeta|\boldsymbol{r}-\boldsymbol{P}|^2\right] \mathrm{d}\boldsymbol{r} \\ &= \mathcal{N}_\alpha \mathcal{N}_\beta \left(\frac{\pi}{\zeta}\right)^{3/2} \exp\left[-\frac{\alpha\beta}{\zeta}|\boldsymbol{A}-\boldsymbol{B}|^2\right] \end{aligned} \tag{5.11}$$

Substituting the normalization factors yields

$$\boxed{S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}} = \left(\frac{2\sqrt{\alpha\beta}}{\alpha+\beta}\right)^{3/2} \exp\left[-\frac{\alpha\beta}{\alpha+\beta}|\boldsymbol{A}-\boldsymbol{B}|^2\right]} \tag{5.12}$$

The result is dimensionless and symmetric under exchange of the two functions. Both factors are at most one: the first follows from $2\sqrt{\alpha\beta} \leq \alpha + \beta$, and the second from the nonpositive exponent. The positive s primitives therefore have an overlap between zero and one.

For coincident centers the exponential is one, but the overlap is

$$S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}}\Big|_{\boldsymbol{B}=\boldsymbol{A}} = \left(\frac{2\sqrt{\alpha\beta}}{\alpha+\beta}\right)^{3/2} \tag{5.13}$$

It is strictly less than one when $\alpha \neq \beta$: sharing a center does not make two different radial functions identical. If the exponents also coincide, the functions are identical, and the overlap is exactly one, recovering the normalization condition. For equal exponents at arbitrary separation, the prefactor is one and

$$S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}}\Big|_{\beta=\alpha} = \exp\left[-\frac{\alpha}{2}|\boldsymbol{A}-\boldsymbol{B}|^2\right] \tag{5.14}$$

For any fixed positive exponents, the overlap tends to zero as the centers are separated without bound. This decay is Gaussian in the separation; more diffuse primitives retain appreciable overlap over longer distances.

Solution 18 (continued): Normalized Gaussian overlap

As a numerical illustration, take $\alpha = 1\,a_0^{-2}$, $\beta = 2\,a_0^{-2}$, $\boldsymbol{A} = (0,0,0)\,a_0$, and $\boldsymbol{B} = (1,0,0)\,a_0$, where a_0 is the Bohr radius. The combined exponent and center are

$$\zeta = 3\,a_0^{-2} \tag{5.15}$$

$$\boldsymbol{P} = \left(\frac{2}{3}, 0, 0\right) a_0 \tag{5.16}$$

The center lies closer to $\boldsymbol{B}$, which carries the tighter primitive. The overlap is

$$S_{\alpha\boldsymbol{A},\beta\boldsymbol{B}} = \left(\frac{2\sqrt{2}}{3}\right)^{3/2} e^{-2/3} \simeq 0.4700 \tag{5.17}$$

The product theorem determines the shape and center of the product, not its normalization. Even when both original functions are normalized, their product is not generally a normalized primitive; its prefactor must be retained when evaluating the overlap.

5.2 One-electron Gaussian integrals

Once the product theorem has combined two basis functions, overlap, kinetic, and nuclear-attraction integrals reduce to Gaussian moments about the product center. The s-type formulas expose the essential structure before angular momentum is introduced.

5.2.1 Kinetic-energy integrals

The Laplacian of an s Gaussian is

$$-\frac{1}{2}\nabla^2 G_{\beta,\boldsymbol{B}}(\boldsymbol{r}) = \left[3\beta - 2\beta^2|\boldsymbol{r}-\boldsymbol{B}|^2\right] G_{\beta,\boldsymbol{B}}(\boldsymbol{r}) \tag{5.18}$$

Consequently, using the Gaussian product theorem (5.2), its kinetic-energy integral becomes

$$\begin{aligned} -\frac{1}{2}\int G_{\alpha,\boldsymbol{A}}(\boldsymbol{r})\nabla^2 G_{\beta,\boldsymbol{B}}(\boldsymbol{r})\mathrm{d}\boldsymbol{r} &= \int G_{\alpha,\boldsymbol{A}}(\boldsymbol{r})\left[3\beta - 2\beta^2|\boldsymbol{r}-\boldsymbol{B}|^2\right] G_{\beta,\boldsymbol{B}}(\boldsymbol{r})\mathrm{d}\boldsymbol{r} \\ &= \frac{\alpha\beta}{\zeta}\left[3 - 2\frac{\alpha\beta}{\zeta}|\boldsymbol{A}-\boldsymbol{B}|^2\right] K_{\zeta,\boldsymbol{P}}\left(\frac{\pi}{\zeta}\right)^{3/2} \end{aligned} \tag{5.19}$$

The result has the same overlap prefactor as the preceding integral. The additional polynomial factor is the consequence of differentiating the Gaussian centered at $\boldsymbol{B}$ twice. It contains both a local curvature term, 3β, and a term that depends on the separation of the two centers.

5.2.2 Nuclear-attraction integrals

For a nucleus of charge Z_C located at $\boldsymbol{C}$, the nuclear-attraction integral introduces the function F_0, the zeroth-order Boys function defined in Eq. (5.29):

$$\begin{aligned} -Z_C\int G_{\alpha,\boldsymbol{A}}(\boldsymbol{r})\frac{1}{|\boldsymbol{r}-\boldsymbol{C}|}G_{\beta,\boldsymbol{B}}(\boldsymbol{r})\mathrm{d}\boldsymbol{r} &= -Z_C K_{\zeta,\boldsymbol{P}}\int \frac{e^{-\zeta|\boldsymbol{r}-\boldsymbol{P}|^2}}{|\boldsymbol{r}-\boldsymbol{C}|}\mathrm{d}\boldsymbol{r} \\ &= -\frac{2Z_C}{\sqrt{\pi}}K_{\zeta,\boldsymbol{P}}\int_0^\infty \mathrm{d}t\int e^{-\zeta|\boldsymbol{r}-\boldsymbol{P}|^2}e^{-t^2|\boldsymbol{r}-\boldsymbol{C}|^2}\mathrm{d}\boldsymbol{r} \\ &= -\frac{2\pi Z_C}{\zeta}K_{\zeta,\boldsymbol{P}}F_0\left(\zeta|\boldsymbol{P}-\boldsymbol{C}|^2\right) \end{aligned} \tag{5.20}$$

5.3 Four-center electron-repulsion integrals

Electron-repulsion integrals are the most numerous quantities in conventional molecular Hartree–Fock theory. Four basis functions and the nonseparable Coulomb kernel appear, but the product theorem reduces the four original centers to two Gaussian product centers.

5.3.1 Gaussian-pair reduction and shell structure

For four unnormalized primitive s Gaussians, applying the Gaussian product rule on both electron coordinates gives the following reduction, with $\mathbf{0} = (0,0,0)$ denoting zero angular momentum. Higher-angular-momentum primitives also contain polynomial factors, which are generated through the recurrences in Section 5.3.4.

$$\begin{aligned}[\mathbf{00}\,|\,\mathbf{00}] &= \iint G_{\alpha,\boldsymbol{A}}(\boldsymbol{r}_1)G_{\beta,\boldsymbol{B}}(\boldsymbol{r}_1)\frac{1}{r_{12}}G_{\gamma,\boldsymbol{C}}(\boldsymbol{r}_2)G_{\delta,\boldsymbol{D}}(\boldsymbol{r}_2)\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2 \\ &= K_{\zeta,\boldsymbol{P}}K_{\eta,\boldsymbol{Q}}\iint G_{\zeta,\boldsymbol{P}}(\boldsymbol{r}_1)\frac{1}{r_{12}}G_{\eta,\boldsymbol{Q}}(\boldsymbol{r}_2)\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2\end{aligned} \tag{5.21}$$

The second pair has combined exponent $\eta = \gamma + \delta$ and product center $\boldsymbol{Q} = (\gamma\boldsymbol{C} + \delta\boldsymbol{D})/\eta$, with a prefactor $K_{\eta,\boldsymbol{Q}}$ defined in the same way as for the first pair. There are formally $O(K^4)$ ERIs. At a fixed numerical threshold and with a fixed local basis per atom, the significant subset can approach $O(K^2)$ in sufficiently large, spatially extended systems. This is a locality-based screening estimate, not an exact reduction in the number of nonzero Coulomb integrals.

Functions, shell pairs, and shell quartets. The notation distinguishes individual functions from shells and primitives from contractions. Thus $[\boldsymbol{ab}|$ is one primitive bra, whereas $[ab|$ is a complete primitive shell pair; $[\boldsymbol{ab}\,|\,\boldsymbol{cd}]$ is one primitive integral, whereas $[ab\,|\,cd]$ is a primitive shell quartet. Likewise, $(\boldsymbol{ab}|$ and $(ab|$ denote a contracted bra and contracted shell pair, and $(\boldsymbol{ab}\,|\,\boldsymbol{cd})$ and $(ab\,|\,cd)$ denote a contracted integral and a contracted shell quartet.

5.3.2 The fundamental *ssss* integral

Higher-angular-momentum ERIs can be generated from a hierarchy of fundamental s-type quantities. The derivation begins by replacing the Coulomb operator with its Laplace representation, after which every Cartesian coordinate enters through an ordinary Gaussian integral. For four primitive s functions,

$$\begin{aligned}[\boldsymbol{ss}\,|\,\boldsymbol{ss}] &= K_{\zeta,\boldsymbol{P}}K_{\eta,\boldsymbol{Q}}\iint G_{\zeta,\boldsymbol{P}}(\boldsymbol{r}_1)\frac{1}{r_{12}}G_{\eta,\boldsymbol{Q}}(\boldsymbol{r}_2)\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2 \\ &= K_{\zeta,\boldsymbol{P}}K_{\eta,\boldsymbol{Q}}\iint e^{-\zeta|\boldsymbol{r}_1-\boldsymbol{P}|^2}\frac{1}{r_{12}}e^{-\eta|\boldsymbol{r}_2-\boldsymbol{Q}|^2}\mathrm{d}\boldsymbol{r}_1\mathrm{d}\boldsymbol{r}_2\end{aligned} \tag{5.22}$$

The Laplace transform

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}}\int_0^\infty e^{-t^2r_{12}^2}\mathrm{d}t = \frac{2}{\sqrt{\pi}}\int_0^\infty e^{-t^2(x_1-x_2)^2}e^{-t^2(y_1-y_2)^2}e^{-t^2(z_1-z_2)^2}\mathrm{d}t \tag{5.23}$$

turns the six-dimensional integral into three Cartesian factors:

$$\begin{aligned}[\boldsymbol{ss}\,|\,\boldsymbol{ss}] = \frac{2}{\sqrt{\pi}}K_{\zeta,\boldsymbol{P}}K_{\eta,\boldsymbol{Q}}\int_0^\infty \mathrm{d}t &\left[\iint e^{-\zeta(x_1-P_x)^2}e^{-t^2(x_1-x_2)^2}e^{-\eta(x_2-Q_x)^2}\mathrm{d}x_1\mathrm{d}x_2\right] \\ \cdot &\left[\iint e^{-\zeta(y_1-P_y)^2}e^{-t^2(y_1-y_2)^2}e^{-\eta(y_2-Q_y)^2}\mathrm{d}y_1\mathrm{d}y_2\right] \\ \cdot &\left[\iint e^{-\zeta(z_1-P_z)^2}e^{-t^2(z_1-z_2)^2}e^{-\eta(z_2-Q_z)^2}\mathrm{d}z_1\mathrm{d}z_2\right]\end{aligned} \tag{5.24}$$

Defining

$$\vartheta = \frac{\zeta\eta}{\zeta+\eta} \tag{5.25}$$

the Cartesian integrations yield, step by step,

$$\begin{aligned}[ss\,|\,ss] &= \frac{2}{\sqrt{\pi}} K_{\zeta,\boldsymbol{P}} K_{\eta,\boldsymbol{Q}} \int_0^\infty \prod_{i=x,y,z} \left[\frac{\pi}{\sqrt{\zeta\eta}} \frac{\exp\!\left[-\vartheta (P_i - Q_i)^2 t^2/(t^2+\vartheta)\right]}{\sqrt{1+t^2/\vartheta}} \right] \mathrm{d}t \\ &= \frac{2\pi^{5/2}}{(\zeta\eta)^{3/2}} K_{\zeta,\boldsymbol{P}} K_{\eta,\boldsymbol{Q}} \int_0^\infty \frac{\exp\!\left[-\vartheta |\boldsymbol{P}-\boldsymbol{Q}|^2 t^2/(t^2+\vartheta)\right]}{(1+t^2/\vartheta)^{3/2}} \mathrm{d}t \end{aligned} \tag{5.26}$$

With the substitution $t^2 = \vartheta u^2/(1-u^2)$,

$$[ss\,|\,ss] = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta,\boldsymbol{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta,\boldsymbol{Q}} \underbrace{\int_0^1 e^{-\vartheta|\boldsymbol{P}-\boldsymbol{Q}|^2 u^2} \mathrm{d}u}_{F_0(\vartheta|\boldsymbol{P}-\boldsymbol{Q}|^2)} \tag{5.27}$$

The zero angular-momentum vector $\mathbf{0} = (0,0,0)$ denotes an s function in this integral notation. The fundamental auxiliary notation is therefore

$$\boxed{[\mathbf{00}\,|\,\mathbf{00}]^{(0)} = \sqrt{\frac{4\vartheta}{\pi}} \left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta,\boldsymbol{P}} \left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta,\boldsymbol{Q}} F_0\!\left(\vartheta|\boldsymbol{P}-\boldsymbol{Q}|^2\right)} \tag{5.28}$$

The original six-dimensional Coulomb integral has thus been reduced to elementary Gaussian prefactors and a one-dimensional function. This reduction is the foundation of the recurrence construction below.

5.3.3 Boys functions and auxiliary integrals

The remaining one-dimensional integral is collected into the generalized Boys function. Here m is a nonnegative integer order and x a nonnegative, dimensionless argument, not a Cartesian coordinate:

$$F_m(x) = \boxed{\int_0^1 t^{2m} \exp\!\left(-x t^2\right) \mathrm{d}t} \qquad\qquad F_0(x) = \sqrt{\frac{\pi}{4x}}\, \mathrm{erf}\!\left(\sqrt{x}\right) \tag{5.29}$$

At the origin, the exponential becomes unity and the limiting value follows directly:

$$F_m(0) = \int_0^1 t^{2m} \mathrm{d}t = \frac{1}{2m+1} \tag{5.30}$$

This limit is particularly useful because the expression for $F_0(x)$ in terms of the error function has an apparent singularity at $x = 0$, although the Boys function itself is perfectly finite there.

In implementations, $F_m(x)$ is evaluated from pretabulated grid values and a Taylor expansion, often with a Chebyshev interpolation table [97]. Lower orders are then obtained by the downward recurrence

$$F_m(x) = \frac{2x F_{m+1}(x) + \exp(-x)}{2m+1} \tag{5.31}$$

This downward recurrence is numerically stable, whereas upward recursion can become unstable for small x. At large x,

$$F_m(x) \sim \frac{(2m-1)!!}{2^{m+1}} \sqrt{\frac{\pi}{x^{2m+1}}} \tag{5.32}$$

The physical *ssss* integral contains only F_0, but differentiation to generate angular momentum produces higher Boys orders. It is therefore useful to define the complete hierarchy

$$[\mathbf{00}\,|\,\mathbf{00}]^{(m)} = \sqrt{\frac{4\vartheta}{\pi}}\left(\frac{\pi}{\zeta}\right)^{3/2} K_{\zeta,\boldsymbol{P}}\left(\frac{\pi}{\eta}\right)^{3/2} K_{\eta,\boldsymbol{Q}} F_m\left(\vartheta|\boldsymbol{P}-\boldsymbol{Q}|^2\right) \tag{5.33}$$

The $m = 0$ member is the physical *ssss* ERI; the $m > 0$ members are auxiliary intermediates. Angular-momentum recurrences couple orders m and $m+1$. For total angular momentum $L = a+b+c+d$, fundamental auxiliary integrals are needed through $m = L$. The recurrence construction must therefore retain the necessary Boys orders alongside the angular-momentum labels.

5.3.4 Obara–Saika recurrence relations

Deriving a separate closed expression for every angular-momentum combination would be inefficient. Obara–Saika instead uses differentiation identities to generate the required quantities recursively from lower-angular-momentum integrals [98].

Generating angular momentum vertically. Here i labels a Cartesian direction, x, y, or z, rather than an occupied orbital. The vector $\mathbf{1}_i$ has unit component in direction i and zero components in the other two directions. For an unnormalized primitive Cartesian Gaussian, the key identity is

$$\frac{\partial|\boldsymbol{a}]}{\partial A_i} = 2\alpha|\boldsymbol{a}+\mathbf{1}_i] - a_i|\boldsymbol{a}-\mathbf{1}_i] \qquad\qquad |\boldsymbol{a}+\mathbf{1}_i] = \frac{1}{2\alpha}\left[\frac{\partial|\boldsymbol{a}]}{\partial A_i} + a_i|\boldsymbol{a}-\mathbf{1}_i]\right] \tag{5.34}$$

Applying this identity to auxiliary ERIs creates the Obara–Saika vertical recurrence relations (VRRs), so that integrals are built recursively rather than independently:

$$[\mathbf{00}\,|\,\mathbf{00}]^{(m)} \longrightarrow [\boldsymbol{a}\mathbf{0}\,|\,\mathbf{00}]^{(m)} \longrightarrow [\boldsymbol{a}\mathbf{0}\,|\,\boldsymbol{c}\mathbf{0}]^{(m)} \tag{5.35}$$

On the bra pair, angular momentum is built on $\boldsymbol{A}$ according to

$$\begin{aligned}[(\boldsymbol{a}+\mathbf{1}_i)\mathbf{0}\,|\,\mathbf{00}]^{(m)} &= (P_i - A_i)[\boldsymbol{a}\mathbf{0}\,|\,\mathbf{00}]^{(m)} - \frac{\eta}{\zeta+\eta}(P_i - Q_i)[\boldsymbol{a}\mathbf{0}\,|\,\mathbf{00}]^{(m+1)} \\ &\quad + \frac{a_i}{2\zeta}\left[[(\boldsymbol{a}-\mathbf{1}_i)\mathbf{0}\,|\,\mathbf{00}]^{(m)} - \frac{\eta}{\zeta+\eta}[(\boldsymbol{a}-\mathbf{1}_i)\mathbf{0}\,|\,\mathbf{00}]^{(m+1)}\right]\end{aligned} \tag{5.36}$$

On the ket pair, angular momentum is built on $\boldsymbol{C}$ according to

$$\begin{aligned}[\boldsymbol{a}\mathbf{0}\,|\,(\boldsymbol{c}+\mathbf{1}_i)\mathbf{0}]^{(m)} &= (Q_i - C_i)[\boldsymbol{a}\mathbf{0}\,|\,\boldsymbol{c}\mathbf{0}]^{(m)} - \frac{\zeta}{\zeta+\eta}(Q_i - P_i)[\boldsymbol{a}\mathbf{0}\,|\,\boldsymbol{c}\mathbf{0}]^{(m+1)} \\ &\quad + \frac{c_i}{2\eta}\left[[\boldsymbol{a}\mathbf{0}\,|\,(\boldsymbol{c}-\mathbf{1}_i)\mathbf{0}]^{(m)} - \frac{\zeta}{\zeta+\eta}[\boldsymbol{a}\mathbf{0}\,|\,(\boldsymbol{c}-\mathbf{1}_i)\mathbf{0}]^{(m+1)}\right] \\ &\quad + \frac{a_i}{2(\zeta+\eta)}[(\boldsymbol{a}-\mathbf{1}_i)\mathbf{0}\,|\,\boldsymbol{c}\mathbf{0}]^{(m+1)}\end{aligned} \tag{5.37}$$

The signs of the product-center terms can be checked directly from the fundamental integral. With $T = \vartheta|\boldsymbol{P}-\boldsymbol{Q}|^2$, the definition of the Boys function gives $\mathrm{d}F_m(T)/\mathrm{d}T = -F_{m+1}(T)$. Since $\partial P_i/\partial A_i = \alpha/\zeta$, differentiating the fundamental integral and dividing by 2α yields

$$[\mathbf{1}_i\mathbf{0}\,|\,\mathbf{00}]^{(m)} = (P_i - A_i)[\mathbf{00}\,|\,\mathbf{00}]^{(m)} - \frac{\vartheta}{\zeta}(P_i - Q_i)[\mathbf{00}\,|\,\mathbf{00}]^{(m+1)} \tag{5.38}$$

Here $\vartheta/\zeta = \eta/(\zeta+\eta)$. The ket-pair result follows by exchanging the two pairs. The derivative argument fixes both product-center terms to have negative signs.

The required angular-momentum build and contraction are organized as

$$[00\,|\,00]^{(m)} \xrightarrow{\text{VRR}} [a0\,|\,c0]^{(m)} \xrightarrow{\text{contraction}} (a0\,|\,c0) \xrightarrow{\text{HRR}} (ab\,|\,cd) \tag{5.39}$$

The VRRs create angular momentum on one center of each Gaussian pair; contraction combines primitive integrals; and the horizontal recurrence relations (HRRs) redistribute angular momentum within each pair.

Redistributing angular momentum horizontally. Once angular momentum has been generated on one center, it is shifted to the other through

$$[\boldsymbol{a}(\boldsymbol{b}+\mathbf{1}_i)\,|\,\boldsymbol{cd}] = [(\boldsymbol{a}+\mathbf{1}_i)\boldsymbol{b}\,|\,\boldsymbol{cd}] + (A_i - B_i)[\boldsymbol{ab}\,|\,\boldsymbol{cd}] \tag{5.40}$$

An analogous relation transfers angular momentum from $\boldsymbol{C}$ to $\boldsymbol{D}$. Because the HRR depends only on $A_i - B_i$, it is independent of the Gaussian exponents and applies directly to contracted integrals:

$$\boxed{(\boldsymbol{a}(\boldsymbol{b}+\mathbf{1}_i)\,|\,\boldsymbol{cd}) = ((\boldsymbol{a}+\mathbf{1}_i)\boldsymbol{b}\,|\,\boldsymbol{cd}) + (A_i - B_i)(\boldsymbol{ab}\,|\,\boldsymbol{cd})} \tag{5.41}$$

The late-contraction algorithm begins by computing the required Boys functions $F_m(T)$ from Section 5.3.3 and forming the fundamental auxiliary integrals $[00\,|\,00]^{(m)}$. The VRRs are then used to build angular momentum on $\boldsymbol{A}$ and $\boldsymbol{C}$, after which the primitive integrals are contracted. Finally, the exponent-independent HRRs transfer angular momentum according to $\boldsymbol{A} \to \boldsymbol{B}$ and $\boldsymbol{C} \to \boldsymbol{D}$. This combination of vertical and horizontal recurrences is the Head–Gordon–Pople late-contraction strategy [99]; the vertical recurrences are those of Obara–Saika. Thus the expensive exponent-dependent recurrences are performed before contraction, whereas the cheap exponent-independent HRRs are performed after contraction. The resulting reduction in the number of quantities handled by the HRRs complements the screening savings considered next. Figure 24 summarizes this path, distinguishing the primitive and contracted stages.

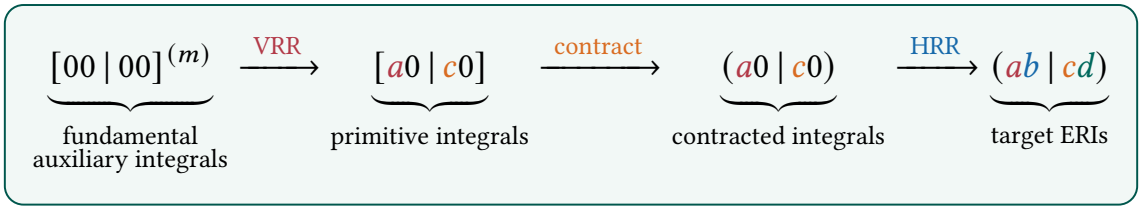


Figure 24: Late-contraction path for ERI evaluation, adapted from slide 38 of the basis-set lecture. Square brackets denote primitive integrals and round brackets contracted integrals; 0 denotes an s-type shell and m the auxiliary Boys-function order. The labels a, b, c, d specify shell angular momenta. The intermediate sets include all angular momenta required to reach the target shell quartet $(ab\,|\,cd)$. The exponent-dependent VRRs precede contraction of the required $m = 0$ primitive integrals, while the exponent-independent HRRs act afterwards [99].

5.4 Integral bounds, screening, and direct SCF

A formal count of K^4 ERIs greatly overestimates the useful work for large local systems because most products of distant basis functions are negligible. Screening converts this locality into a computational saving by testing a cheap upper bound before evaluating an integral or shell quartet.

5.4.1 Cauchy–Schwarz bounds

A useful ERI upper bound must be simple and inexpensive, strong enough to be a good estimate, and scaling consistent, meaning $K_{\mathrm{sig}} = O(N_{\mathrm{upper}})$. Here K_{sig} counts significant integrals and N_{upper} counts those retained by the bound at the chosen threshold. The Cauchy–Schwarz inequality gives

$$\boxed{|(\boldsymbol{ab}\,|\,\boldsymbol{cd})| \le \sqrt{(\boldsymbol{ab}\,|\,\boldsymbol{ab})}\sqrt{(\boldsymbol{cd}\,|\,\boldsymbol{cd})}} \tag{5.42}$$

or, in terms of the Gaussian product centers,

$$|(\boldsymbol{P}\,|\,\boldsymbol{Q})| \le \sqrt{(\boldsymbol{P}\,|\,\boldsymbol{P})}\sqrt{(\boldsymbol{Q}\,|\,\boldsymbol{Q})} \tag{5.43}$$

The integral $(\boldsymbol{P}\,|\,\boldsymbol{Q})$ should decay as $1/|\boldsymbol{P}-\boldsymbol{Q}|$ or faster. The product on the right has no information about the separation between the two charge distributions, so the Cauchy–Schwarz bound is rigorous and cheap but not tight for well-separated shell pairs. Even so, a cheap upper bound avoids evaluating negligible ERIs. The practical use of integral bounds in direct-SCF screening is discussed by Häser and Ahlrichs [100].

5.4.2 Shell-quartet screening

At the individual-integral level, only $O(K^2)$ diagonal factors $(\boldsymbol{ab}\,|\,\boldsymbol{ab})$ are required to bound $O(K^4)$ four-center integrals. More efficiently, define a shell-pair bound

$$Q_{ab} = \max_{\substack{\boldsymbol{a}\in a\\ \boldsymbol{b}\in b}} \sqrt{(\boldsymbol{ab}\,|\,\boldsymbol{ab})} \tag{5.44}$$

Then an entire shell class obeys

$$|(ab\,|\,cd)| \le Q_{ab}Q_{cd} \tag{5.45}$$

For a positive screening threshold τ, if

$$\boxed{Q_{ab}Q_{cd} < \tau} \tag{5.46}$$

the entire shell quartet can be discarded. This is consequential: even the simple $(dd\,|\,dd)$ class contains 1296 Cartesian integrals.

The number of significant integrals may be represented by

$$(\boldsymbol{ab}\,|\,\boldsymbol{cd}) \equiv (\boldsymbol{ab}\,|\,\hat{O}_2\,|\,\boldsymbol{cd}) \tag{5.47}$$

and $K_{\mathrm{sig}} = cK^{\alpha}$. Here K is the number of basis functions, c is a proportionality factor, and α is the effective scaling exponent. The numerical data in Table 6 contrast the long-range Coulomb operator with a short-range Gaussian operator:

The Coulomb interaction remains long ranged, so its effective exponent falls more slowly than that of the short-range Gaussian operator. Nevertheless, the data show that the number of numerically significant Coulomb integrals grows much more slowly than the formal quartic count in sufficiently large systems.

The complete integral-evaluation strategy has three components, summarized in Fig. 25. The target quantities are the two-electron integrals $(\boldsymbol{ab}\,|\,\boldsymbol{cd})$. Their construction requires the fundamental integrals $(\boldsymbol{00}\,|\,\boldsymbol{00})^{(m)}$, the recurrence relations, represented schematically by $((\boldsymbol{a}+\boldsymbol{1}_i)\boldsymbol{0}\,|\,\boldsymbol{00}) = (\boldsymbol{a0}\,|\,\boldsymbol{00}) + ((\boldsymbol{a}-\boldsymbol{1}_i)\boldsymbol{0}\,|\,\boldsymbol{00})$, and an upper bound $|(\boldsymbol{ab}\,|\,\boldsymbol{cd})| \le B$ that permits insignificant integrals to be discarded before evaluation.

Table 6: Numbers of significant integrals and effective scaling exponents for Coulomb and short-range Gaussian operators. Numerical data reproduced from the lecture slides.

		$\hat{O} = r_{12}^{-1}$		$\hat{O} = e^{-r_{12}^2}$	
Molecule	K	K_{sig}	α	K_{sig}	α
propene	12	1,625	—	1,650	—
butadiene	16	5,020	3.9	5,020	3.9
hexatriene	24	24,034	3.9	23,670	3.8
octatetraene	32	63,818	3.4	52,808	2.8
decapentaene	40	119,948	2.8	81,404	1.9
dodecahexaene	48	192,059	2.6	109,965	1.6

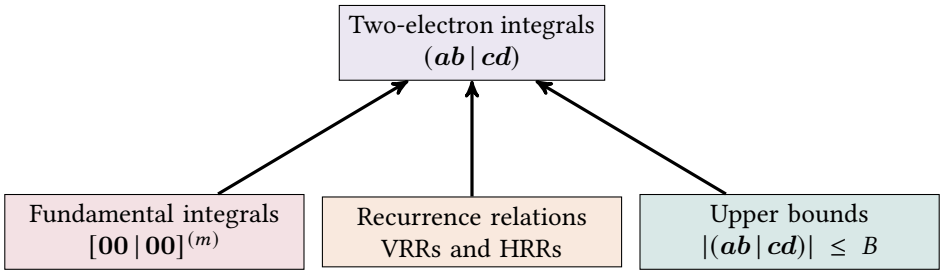


Figure 25: The three ingredients of integral evaluation: fundamental integrals, recurrence relations, and screening bounds.

5.4.3 Direct SCF evaluation

Once screening has identified the significant shell quartets, the resulting integrals need not be stored. Direct SCF contracts each batch with the current density immediately after it is generated. For the real AO functions and real symmetric density used in the following direct-SCF contractions, the Coulomb and exchange matrices are

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\nu \,|\, \lambda\sigma) \qquad\qquad K_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\lambda \,|\, \nu\sigma) \tag{5.48}$$

Direct SCF [101] trades disk storage for repeated integral evaluation. Instead of storing all $O(K^4)$ ERIs, it loops over the shell quartets $(\mu\nu \,|\, \lambda\sigma)$ and uses an upper bound to discard negligible quartets before their integrals are evaluated. The significant ERIs are computed, their contributions are accumulated immediately in $\boldsymbol{J}$ and $\boldsymbol{K}$, and the integrals are then discarded. The density-matrix elements can themselves be included in the screening criterion. In compact form, the algorithm is

$$\text{screen} \longrightarrow \text{compute} \longrightarrow \text{accumulate} \longrightarrow \text{discard} \tag{5.49}$$

Because the Fock build contains products of density-matrix elements and ERIs, screening can be tightened by bounding the contribution to the matrix rather than the integral alone. A shell quartet whose bound is below the integral threshold is a candidate for omission, while a density-weighted criterion can also discard a formally non-negligible quartet when the density elements multiplying it are small at the current SCF iteration. An integral threshold alone does not bound the accumulated error in the Fock matrix or energy: the density weights and the number of discarded contributions also matter. The threshold must therefore be chosen consistently with the target accuracy of the SCF energy and residual.

6 Resolution of the identity and low-rank ERI representations

Screening avoids work associated with small individual integrals, but it does not alter the four-index structure of the ERI tensor. Resolution of the identity (RI), also called density fitting (DF), addresses this complementary problem by replacing four-center quantities with contractions over a smaller auxiliary index. Its starting point is that products of orbital basis functions span a large, highly redundant space. Representing that space economically connects the integral technology developed above to less costly Fock-matrix construction. In this section, orbital and auxiliary functions are taken to be real, as in the Gaussian integral formulas above. An ERI is the Coulomb interaction between two pair densities,

$$(\mu\nu \,|\, \lambda\sigma) = \iint \underbrace{\phi_\mu(\boldsymbol{r}_1)\phi_\nu(\boldsymbol{r}_1)}_{\rho_{\mu\nu}(\boldsymbol{r}_1)} \frac{1}{r_{12}} \underbrace{\phi_\lambda(\boldsymbol{r}_2)\phi_\sigma(\boldsymbol{r}_2)}_{\rho_{\lambda\sigma}(\boldsymbol{r}_2)} \mathrm{d}\boldsymbol{r}_1 \mathrm{d}\boldsymbol{r}_2 \tag{6.1}$$

With K orbital basis functions, there are formally $O(K^2)$ different pair densities. RI/DF replaces this large product space by a compact auxiliary basis $\{\varphi_A^{\mathrm{aux}}\}$ [102–104]:

$$\rho_{\mu\nu}(\boldsymbol{r}) \approx \widetilde{\rho}_{\mu\nu}(\boldsymbol{r}) = \sum_A C_{\mu\nu}^A \varphi_A^{\mathrm{aux}}(\boldsymbol{r}) \tag{6.2}$$

with $K_{\mathrm{aux}} = O(K)$. Here $\widetilde{\rho}_{\mu\nu}$ is the fitted pair density and K_{aux} the number of auxiliary functions. The coefficients $C_{\mu\nu}^A$ are fitting coefficients, not the MO coefficients $C_{\mu p}$. The auxiliary basis is therefore not a second orbital basis. It is designed to represent products of orbital functions, and its index A labels functions in this approximate product space. The reduction from $O(K^2)$ pair densities to $O(K)$ auxiliary functions is the source of the lower-rank representation.

6.1 Fitting pair densities in the Coulomb metric

The fitted coefficients depend on the norm used to measure the residual pair density. For electron-repulsion integrals, the natural choice weights fitting errors by their Coulomb self-interaction. Define the Coulomb inner product

$$(f \,|\, g) = \iint f(\boldsymbol{r}_1) \frac{1}{r_{12}} g(\boldsymbol{r}_2) \mathrm{d}\boldsymbol{r}_1 \mathrm{d}\boldsymbol{r}_2 \tag{6.3}$$

The optimal fitting coefficients minimize the Coulomb norm of the error,

$$\Delta_{\mu\nu} = (\rho_{\mu\nu} - \widetilde{\rho}_{\mu\nu} \,|\, \rho_{\mu\nu} - \widetilde{\rho}_{\mu\nu}) \tag{6.4}$$

We abbreviate an auxiliary function by its index inside an integral: $(A \,|\, B)$ is the Coulomb interaction between two auxiliary functions, and $(A \,|\, \mu\nu)$ couples an auxiliary function to an AO pair density. Substituting the auxiliary expansion and using the symmetry of the Coulomb metric gives

$$\Delta_{\mu\nu} = (\rho_{\mu\nu} \,|\, \rho_{\mu\nu}) - 2\sum_A C_{\mu\nu}^A (A \,|\, \mu\nu) + \sum_{AB} C_{\mu\nu}^A \underbrace{(A \,|\, B)}_{V_{AB}} C_{\mu\nu}^B \tag{6.5}$$

The first term is independent of the fitting coefficients. Differentiating the remaining two terms with respect to one coefficient gives the stationarity condition,

$$\frac{\partial \Delta_{\mu\nu}}{\partial C_{\mu\nu}^A} = -2(A \,|\, \mu\nu) + 2\sum_B V_{AB} C_{\mu\nu}^B = 0 \tag{6.6}$$

Hence the least-squares equations are

$$\sum_B \underbrace{(A\,|\,B)}_{V_{AB}} C^B_{\mu\nu} = \underbrace{(A\,|\,\mu\nu)}_{\text{three-center integral}} \qquad C^A_{\mu\nu} = \sum_B (\boldsymbol{V}^{-1})_{AB}(B\,|\,\mu\nu) \tag{6.7}$$

Thus the Coulomb metric determines the optimal auxiliary expansion. The stationarity condition also states that the residual is Coulomb-orthogonal to every auxiliary function:

$$\boxed{(A\,|\,\rho_{\mu\nu} - \widetilde{\rho}_{\mu\nu}) = 0} \tag{6.8}$$

The inverse metric accounts for the nonorthogonality of the auxiliary functions. These expressions assume that the retained auxiliary functions are linearly independent in the Coulomb metric. If the metric has near-zero eigenvalues, the inverse is formed in the retained eigenspace, using the same principle as canonical orthogonalization of the orbital overlap matrix. The resulting coefficients are optimal within that auxiliary space under the Coulomb norm.

Because a fitted density belongs to the auxiliary space, Eq. (6.8) gives

$$(\widetilde{\rho}_{\mu\nu}\,|\,\widetilde{\rho}_{\lambda\sigma}) = (\rho_{\mu\nu}\,|\,\widetilde{\rho}_{\lambda\sigma}) = (\widetilde{\rho}_{\mu\nu}\,|\,\rho_{\lambda\sigma}) \tag{6.9}$$

Thus fitting both densities is equivalent to fitting either one in this Coulomb projection. Inserting the fitted pair densities converts the four-center ERI to three-center quantities:

$$\begin{aligned}(\mu\nu\,|\,\lambda\sigma) &\approx \sum_A C^A_{\lambda\sigma}(\mu\nu\,|\,A) \\ &= \sum_{AB} (\mu\nu\,|\,A)(\boldsymbol{V}^{-1})_{AB}(B\,|\,\lambda\sigma)\end{aligned} \tag{6.10}$$

Equivalently,

$$\boxed{(\mu\nu\,|\,\lambda\sigma) \approx \sum_{AB} (\mu\nu\,|\,A)(\boldsymbol{V}^{-1})_{AB}(B\,|\,\lambda\sigma)} \tag{6.11}$$

With the symmetrically orthonormalized quantities, where P labels an orthonormalized auxiliary direction,

$$B^P_{\mu\nu} = \sum_A (\mu\nu\,|\,A)(\boldsymbol{V}^{-1/2})_{AP} \tag{6.12}$$

the factorization becomes

$$\boxed{(\mu\nu\,|\,\lambda\sigma) \approx \sum_P B^P_{\mu\nu} B^P_{\lambda\sigma}} \tag{6.13}$$

The four-index tensor has thereby been replaced by products of three-index tensors. This last form follows by splitting the inverse metric symmetrically, $\boldsymbol{V}^{-1} = \boldsymbol{V}^{-1/2} \cdot \boldsymbol{V}^{-1/2}$. The explicit matrix multiplication symbol emphasizes that the auxiliary indices are contracted through the metric rather than multiplied element by element.

6.2 RI, Cholesky decomposition, and practical contractions

Both density fitting and Cholesky decomposition expose the low numerical rank of the ERI matrix in compound pair indices. They differ in how the three-index representation is constructed. In RI/DF, the AO products are expanded in a predefined auxiliary basis,

$$\phi_\mu(\boldsymbol{r})\phi_\nu(\boldsymbol{r}) \approx \sum_A C^A_{\mu\nu}\varphi^{\text{aux}}_A(\boldsymbol{r}) \quad \Longrightarrow \quad (\mu\nu\,|\,\lambda\sigma) \approx \sum_P B^P_{\mu\nu} B^P_{\lambda\sigma} \tag{6.14}$$

The quality of the approximation therefore depends on the auxiliary basis. Alternatively, the ERI tensor $\boldsymbol{V}_{(\mu\nu),(\lambda\sigma)} = (\mu\nu \,|\, \lambda\sigma)$ is a positive-semidefinite matrix in the compound indices. Here $\boldsymbol{V}$ denotes the full AO-pair ERI matrix, rather than the auxiliary Coulomb metric V_{AB} of the preceding derivation. A pivoted Cholesky decomposition constructs an auxiliary space adaptively [105],

$$(\mu\nu \,|\, \lambda\sigma) \approx \sum_{P=1}^{N_{\mathrm{CD}}} L^{P}_{\mu\nu} L^{P}_{\lambda\sigma} \tag{6.15}$$

until the residual is below a prescribed threshold τ_{CD}. RI/DF therefore uses a predefined auxiliary space, whereas Cholesky decomposition constructs one adaptively.

For the Coulomb matrix,

$$\begin{aligned} J_{\mu\nu} &= \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu \,|\, \lambda\sigma) \\ &\approx \sum_{AB} (\mu\nu \,|\, A) (\boldsymbol{V}^{-1})_{AB} \underbrace{\sum_{\lambda\sigma} P_{\lambda\sigma} (B \,|\, \lambda\sigma)}_{d_B} \end{aligned} \tag{6.16}$$

The practical sequence is

$$\boldsymbol{P} \longrightarrow \{d_A\} \longrightarrow \{c_A\} = \boldsymbol{V}^{-1} \cdot \{d_A\} \longrightarrow \boldsymbol{J} \tag{6.17}$$

The full $O(K^4)$ ERI tensor need not be stored; only two-center $(A \,|\, B)$ and three-center $(\mu\nu \,|\, A)$ integrals are needed; and optimized auxiliary bases make the fitting error very small [106]. RI/DF thus greatly reduces the cost and memory requirements of Coulomb contractions. The computational gain is that contractions can be ordered through the auxiliary index instead of four orbital indices [107]. The approximation is controlled by the auxiliary basis in RI/DF or by the decomposition threshold in Cholesky decomposition.

Taken together, the basis-set and integral constructions complete the practical formulation of Hartree–Fock theory. The basis determines the variational one-electron space, Gaussian integral algorithms provide the matrix elements required within that space, and screening or low-rank factorization controls the cost of assembling the mean-field operators. These components lead back to the Roothaan–Hall and SCF equations developed in Section 2.

7 Concluding perspective

The developments in this chapter connect a variational approximation to a practical molecular calculation. Hartree–Fock theory restricts the electronic wavefunction to a single determinant while retaining antisymmetry and the Coulomb and exchange contributions to its energy. Orbital optimization gives a Fock operator that depends on the occupied orbitals themselves, and hence a self-consistent problem. Restricted, unrestricted, and generalized formulations impose different constraints on that optimization (Section 3). A converged solution is not, by convergence alone, guaranteed to be the lowest-energy determinant or to possess the desired spin and spatial symmetries.

The basis-set approximation is a separate restriction. Expanding the orbitals in a finite one-electron space converts the orbital equations into the Roothaan–Hall matrix problem of Section 2. The basis families discussed in Section 4 offer different ways of approaching the required radial and angular flexibility. Improving that space reduces basis-set incompleteness, but does not remove the single-determinant restriction or recover the correlation missing from HF.

Integral evaluation makes this finite-basis formulation operational. The Gaussian product theorem and recurrence relations provide the one- and two-electron matrix elements needed to build the Fock matrix (Section 5). Screening avoids negligible contributions, while density fitting and Cholesky decomposition replace the four-index ERI representation by more economical contractions (Section 6). Their thresholds or auxiliary spaces control numerical approximations distinct from the choice of orbital basis.

Interpreting a calculation therefore requires keeping these levels separate: the chosen HF variational space, the orbital basis, the integral representation, and the convergence of the SCF solution. The resulting total energy, orbital energies, and molecular properties must then be read with their respective assumptions in mind. In particular, a stationary HF reference supplies a useful starting point for correlated methods, not a substitute for them. The link between these theoretical assumptions and their numerical realization is what makes Hartree–Fock theory both a foundational model and a practical framework for electronic structure.

Further reading

Useful general references include Jensen's *Introduction to Computational Chemistry* [108], Cramer's *Essentials of Computational Chemistry* [109], Szabo and Ostlund's *Modern Quantum Chemistry* [21], and Helgaker, Jørgensen, and Olsen's *Molecular Electronic Structure Theory* [110].

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